TEJAS200

BITSAT

Mathematics

Chapter-wise Question Book

50 MCQs per chapter / One answer-key page after every chapter

32 chapter units

1,600 questions

Revised practice edition V2 / October 2026

Using this book

Each chapter contains 50 single-correct, four-option questions. Select one option, A–D, for each question. The next page after the chapter contains its complete answer key. Solutions are not included. Questions 1–10 retain the strongest ten questions selected from the previous middle block, 11–40 are newly written middle-to-hard replacements, and 41–50 preserve the previous hard block. All previous easy questions and ten weaker middle questions have been removed. These are intended practice levels, not official difficulty grades or empirical calibration. Work through the chapter before opening its answer key. Use the questions to check understanding, selection of the first step, calculation, and the stated conditions. Constants and idealisations given in a question take priority for that question. Chapter units follow the updated Tejas200 chapter engine. They are study groupings, rather than a claim that the official syllabus uses the same chapter headings. This is original practice material, not an official BITS question paper. **Reference conventions.** Angles are in radians unless degrees are specified. ln, and log without a stated base, denote the natural logarithm unless a question specifies otherwise. An inverse trigonometric function uses its principal-value branch unless another branch is explicitly stated.

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Chapter 1

Sets Relations & Functions

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. In a universe of 100 elements, three sets have sizes $50, 45, 40$. Their pairwise intersections have sizes $20, 15, 10$, and the triple intersection has size 5. How many elements lie outside all three sets?

(A) 5 (B) 0

(C) 15 (D) 10

02. If $|A| = 3$, $|B| = 4$ and $b \in B$ is fixed, how many functions $f : A \to B$ have b in their range?

(A) 27 (B) 64

(C) 37 (D) 36

03. How many functions $f : \{1, 2, 3\} \to \{1, 2, 3\}$ have exactly two fixed points?

(A) 9 (B) 3

(C) 6 (D) 2

04. A real function satisfies $f(x) + 2f(-x) = 3x + 1$ for every real x. Find $f(2)$.

(A) $\frac{19}{3}$ (B) $-\frac{19}{3}$

(C) $\frac{17}{3}$ (D) $-\frac{17}{3}$

05. How many integers $x \in [0, 20]$ satisfy $\lfloor x/2 \rfloor + \lfloor x/3 \rfloor = 5$?

(A) 2 (B) 3

(C) 4 (D) 1

06. Let $f : [1, \infty) \to [0, \infty)$ be $f(x) = \sqrt{x - 1}$ and let $g(x) = x^2 + 1$ on $\mathbb{R}$. Which statement is correct?

(A) $f(g(x)) = x$ for every real x (B) $g(f(x)) = |x|$ for $x \geq 1$

(C) f has no inverse on its stated domain (D) $g(f(x)) = x$ for $x \geq 1$, but $f(g(x)) = |x|$

07. For $a * b = a + b - ab$ on $\mathbb{R}$, the identity is 0. The inverse of the element 2 is

(A) 2 (B) $\frac{1}{2}$

(C) 0 (D) -2

08. How many distinct values does $\lfloor x \rfloor + \lfloor 3 - x \rfloor$ take for $0 \le x \le 3$?

(A) 4 (B) 2

(C) 1 (D) 3

09. Let $A = \{z \in \mathbb{C} : z^4 = 1\}$ and $B = \{z \in \mathbb{C} : |z - 1| = \sqrt{2}\}$. Find $|A \cap B|$.

(A) 0 (B) 1

(C) 2 (D) 4

10. For $x > 2$, define $f(x) = \log_2 x + \log_2(x - 2)$. If $f(x) = 3$, find x.

(A) 2 (B) 4

(C) 8 (D) -2

11–40 / New middle-to-hard practice

11. In a universe of 100 elements, $|A| = 55$, $|B| = 48$, and the number in neither set is twice the number in $A \cap B$ minus 18. Find $|A \triangle B|$.

(A) 74 (B) 75

(C) 72 (D) 73

12. Sets A, B have $|A \cup B| = 40$. Exactly 24 elements belong to just one of them, and $|A \setminus B| = 3|B \setminus A|$. Find $|A|$.

(A) 34 (B) 36

(C) 33 (D) 35

13. How many bijections of $\{1, 2, 3, 4, 5\}$ to itself fix 1 and have exactly two fixed points in total?

(A) 20 (B) 6

(C) 24 (D) 8

14. How many onto functions $f : \{1, \ldots, 6\} \to \{a, b, c\}$ satisfy $f(1) = f(2)$ and $f(3) \neq f(1)$?

(A) 108 (B) 114

(C) 162 (D) 72

15. A real function satisfies $f(x) + 2f(1 - x) = 3x^2 - 2x + 4$ for all real x. How many integer inputs satisfy $f(x) \le 10$?

(A) 7 (B) 9

(C) 8 (D) 6

16. A function on the integers satisfies $f(m+n) = f(m) + f(n) + 2mn$ and $f(1) = 3$. Find $\sum_{k=1}^{10} f(k)$.

(A) 385 (B) 440

(C) 605 (D) 495

17. How many integers belong to the real domain of $\sqrt{(x-1)/(x-3)} + \ln(5 - x^2)$?

(A) 4 (B) 6

(C) 3 (D) 5

18. How many integers in $[-5, 5]$ belong to the real domain of $\sqrt{(x-2)/(x-3)} + \ln(4-x)/\ln(x^2-1)$?

(A) 6 (B) 4

(C) 7 (D) 5

19. Let $f(x) = x/\sqrt{1+2x^2}$. If $f(f(f(t))) = 1/3$, find $t + 1/t$.

(A) 4 (B) $\sqrt{3}$

(C) $\frac{4\sqrt{3}}{3}$ (D) $\frac{2\sqrt{3}}{3}$

20. For $f(x) = (x-1)/(x+1)$, let $f^{\circ j}$ mean j compositions. Find $\sum_{j=1}^{2027} f^{\circ j}(2)$.

(A) $-\frac{1187}{2}$ (B) $-\frac{1193}{2}$

(C) $-\frac{1181}{2}$ (D) $-\frac{1177}{2}$

21. Let $x = k/10$ with integer $0 \le k \le 100$. How many such x satisfy both $\lfloor x/2 \rfloor + \lfloor x/3 \rfloor = 5$ and $\lfloor x \rfloor + \lfloor 3 - x \rfloor = 2$?

(A) 18 (B) 20

(C) 19 (D) 17

22. Find the sum of all real solutions in $[0, 4]$ of $\lfloor x \rfloor = x^2 - 2x$.

(A) $\sqrt{3} + 5$ (B) $\sqrt{2} + 4$

(C) $\sqrt{3} + 4$ (D) $\sqrt{3} + 3$

23. On $\{1, 2, \ldots, 60\}$ define aRb when $a - b$ is divisible by both 3 and 7. How many ordered pairs belong to R?

(A) 162 (B) 168

(C) 180 (D) 174

24. For $a, b \in \{0, 1, \ldots, 15\}$, define aRb if $a^2 \equiv b^2 \pmod 8$. How many pairs in R have $a \neq b$?

(A) 96 (B) 88

(C) 64 (D) 80

25. Let $f(x) = (x^2 - 4x + 1)^2$. Find the sum of all distinct real elements of $f^{-1}(9)$.

(A) 8 (B) 2

(C) 4 (D) 6

26. Let $f(x) = |x^2 - 4|$ and $g(x) = |x - 3|$. Find the sum of the squares of all real solutions of $(g \circ f)(x) = 2$.

(A) 26 (B) 42

(C) 34 (D) 36

27. For how many integers k does $(x^2 + 2x + 3)/(x^2 + 2x + 2) = k/5$ have exactly two distinct real solutions?

(A) 6 (B) 4

(C) 5 (D) 3

28. For how many integers k does $(2x^2 + 3x + 2)/(x^2 + x + 1) = k/3$ have exactly one real solution?

(A) 3 (B) 5

(C) 4 (D) 2

29. On $\mathbb{R} \setminus \{-1\}$ define $a * b = a + b + ab$. For every real solution u of $u * u = 8$, take its inverse under $*$. Find the sum of those inverses.

(A) 2 (B) $-\frac{2}{3}$

(C) -2 (D) $-\frac{4}{3}$

30. Define $a * b = a + b - 2ab$. A real number u satisfies $(u * u) * u = 7/16$. What is its inverse under $*$?

(A) $-\frac{1}{2}$ (B) $-\frac{1}{4}$

(C) -2 (D) $\frac{1}{2}$

31. How many integers in $[-5, 5]$ satisfy $|(x^2 - 4) + (x - 1)| < |x^2 - 4| + |x - 1|$?

(A) 4 (B) 3

(C) 5 (D) 6

32. Find the product of the distinct real solutions of $|x - 1| + |x - 4| = 2|2x - 5|$.

(A) $\frac{9}{16}$ (B) $\frac{91}{16}$

(C) $\frac{39}{4}$ (D) $\frac{25}{4}$

33. If $|A| = 3$, $|B| = 4$, and $|\mathcal{P}(A) \cap \mathcal{P}(B)| = 4$, how many subsets of $A \cup B$ meet both A and B?

(A) 27 (B) 24

(C) 28 (D) 30

34. A set S has five elements. How many ordered pairs (X, Y) of subsets satisfy $X \subseteq Y \subseteq S$, $|Y \setminus X| = 2$, and $X \neq \varnothing$?

(A) 90 (B) 70

(C) 60 (D) 80

35. For which starting points $b \in [-2\pi, 2\pi]$ is $f(x) = \sin x + \sqrt{3}\cos x$, restricted to $[b, b+\pi]$, bijective onto $[-2, 2]$? Find the sum of all such b.

(A) $-\frac{4\pi}{3}$ (B) $-\frac{2\pi}{3}$

(C) 0 (D) $\frac{4\pi}{3}$

36. For every $a \in [0, 2\pi]$ such that $x \mapsto \sin 2x$ is bijective from $[a, a+\pi/2]$ onto $[-1, 1]$, retain a. Find their sum.

(A) 2π (B) 4π

(C) 3π (D) 5π

37. Let $f(y) = y^2 - 4y + 3$. A strictly increasing function $g : \mathbb{R} \to \mathbb{R}$ satisfies $f(g(x)) = x^2 + 2x - 3$. Find $g^{-1}(f(3))$.

(A) 0 (B) -1

(C) 3 (D) -3

38. Let $f(y) = y^2 + y - 2$. A function h satisfies $f(h(x)) = 4x^2 - 6x$ and $h(x) \geq -1/2$ for every real x. Find $h(0) + h(1) + h(2)$.

(A) 4 (B) 2

(C) 3 (D) 5

39. For integer $a \in \{-2, -1, 0, 1, 2, 3, 4\}$, the rule $f(x) = x^2 - 2ax$ must be a bijection from $[0, 3]$ onto $[0, b]$, where $b > 0$. Find the sum of all possible b.

(A) 54 (B) 45

(C) 36 (D) 60

40. An integer $a \in [-3, 7]$ is retained if $x \mapsto |x - a|$ is injective on $[0, 4]$ and its range contains 2. Find the sum of all retained a.

(A) 18 (B) 15

(C) 12 (D) 9

41–50 / Preserved hard practice

41. How many equivalence relations on a five-element set have exactly three equivalence classes?

(A) 20 (B) 25

(C) 30 (D) 15

42. Let $f(x) = (x-1)/(x+1)$ and let $f^{\circ n}$ denote n-fold composition. Find $f^{\circ 2026}(2)$. All iterates involved are defined.

(A) 2 (B) $-\frac{1}{2}$

(C) $\frac{1}{3}$ (D) -3

43. The real domain of $\sqrt{\log_2(x^2-3x+2)}$ is

(A) $\left(-\infty, \frac{3-\sqrt{5}}{2}\right] \cup \left[\frac{3+\sqrt{5}}{2}, \infty\right)$ (B) $\left[\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}\right]$

(C) $(-\infty, 1) \cup (2, \infty)$ (D) $(-\infty, 0] \cup [3, \infty)$

44. On $\{0, 1, 2\}$, how many commutative and associative binary operations have 0 as identity?

(A) 27 (B) 9

(C) 6 (D) 12

45. Let $f(x) = ax^2 + bx + c$ satisfy $f(x+1) - f(x) = 6x + 5$ for every real x, and $f(-1) = 4$. How many real solutions does $f(f(x)) = 33$ have?

(A) 0 (B) 2

(C) 1 (D) 4

46. A function $f : \{1, 2, 3, 4\} \to \{1, 2, 3\}$ is chosen uniformly from the onto functions. The probability that $f(1) = f(2)$ is

(A) $\frac{1}{9}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) $\frac{1}{6}$

47. Let $f(x) = \sin(2\sin^{-1} x)$ on $[-1, 1]$, with principal inverse sine. The sum of all real solutions of $f(x) = \sqrt{3}/2$ is

(A) $-\frac{1}{2} + \frac{\sqrt{3}}{2}$ (B) 1

(C) $\frac{1}{2} + \frac{\sqrt{3}}{2}$ (D) $\sqrt{3}$

48. The real domain of $\sqrt{\ln\left(\log_2\left(\frac{x^2-5x+6}{x-1}\right)\right)}$ is

(A) $(-\infty, 1) \cup [2, 3]$ (B) $(1, 2) \cup (3, \infty)$

(C) $\left(1, \frac{7-\sqrt{17}}{2}\right] \cup \left[\frac{7+\sqrt{17}}{2}, \infty\right)$ (D) $\left[\frac{7-\sqrt{17}}{2}, \frac{7+\sqrt{17}}{2}\right]$

49. An involution is a function $f : A \to A$ satisfying $f(f(x)) = x$. An involution on a six-element labelled set is chosen uniformly. The probability that it has exactly two fixed points is

(A) $\frac{1}{2}$ (B) $\frac{45}{64}$

(C) $\frac{15}{76}$ (D) $\frac{45}{76}$

50. The function $f : [0, 2] \to [1, 11]$, $f(x) = x^3 + x + 1$, is bijective. Evaluate $\displaystyle\int_1^{11} f^{-1}(y)\, dy$.

(A) 12 (B) 16

(C) 14 (D) 18

Answer key

Sets Relations & Functions

M-SRF / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**D**	21	**A**	31	**B**	41	**B**
02	**C**	12	**A**	22	**C**	32	**B**	42	**B**
03	**C**	13	**D**	23	**D**	33	**A**	43	**A**
04	**D**	14	**B**	24	**D**	34	**B**	44	**B**
05	**A**	15	**A**	25	**D**	35	**A**	45	**B**
06	**D**	16	**D**	26	**C**	36	**B**	46	**D**
07	**A**	17	**A**	27	**B**	37	**D**	47	**C**
08	**B**	18	**D**	28	**A**	38	**C**	48	**C**
09	**C**	19	**C**	29	**C**	39	**B**	49	**D**
10	**B**	20	**A**	30	**A**	40	**C**	50	**C**

Chapter 2

Complex Numbers

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For $z \neq -1$, the number $(z-1)/(z+1)$ is purely imaginary (including zero) precisely when

(A) $\operatorname{Im} z = 0$ (B) $|z| = 1$

(C) $|z-1| = 1$ (D) $\operatorname{Re} z = 0$

02. How many complex numbers satisfy $z\overline{z} = 4$ and $z + \overline{z} = 2$?

(A) 2 (B) 4

(C) 0 (D) 1

03. If $z^2 = 3 - 4i$ and $\operatorname{Re} z > 0$, find $\operatorname{Im} z$.

(A) -1 (B) 2

(C) 1 (D) -2

04. If $\omega \neq 1$, $\omega^3 = 1$, then $\dfrac{1}{1-\omega} + \dfrac{1}{1-\omega^2}$ equals

(A) $\frac{1}{3}$ (B) 2

(C) 0 (D) 1

05. If $z \neq 0$ and $z + z^{-1} = 1$, then $z^3 + z^{-3}$ equals

(A) -1 (B) 1

(C) 2 (D) -2

06. If $|z_1| = |z_2| = 1$ and $|z_1 - z_2| = \sqrt{3}$, then $|z_1 + z_2|$ equals

(A) 1 (B) 2

(C) $\sqrt{3}$ (D) $\sqrt{2}$

07. The two intersection points of $|z| = 2$ and $|z - 1| = |z + 1|$ are z_1, z_2. Their product is

(A) -2 (B) -4

(C) 4 (D) 2

08. How many complex numbers satisfy $|z| = 1$, $z \neq -1$, and $(z - 1)/(z + 1) \in \mathbb{R}$?

(A) 1 (B) 2

(C) 4 (D) 0

09. Among the four fourth roots of unity, how many unordered pairs of distinct roots are at distance 2?

(A) 3 (B) 4

(C) 2 (D) 1

10. Let z_1, z_2 be the distinct points satisfying $|z| = 1$ and $\operatorname{Re} z + \operatorname{Im} z = 1$. The area of the triangle with vertices $0, z_1, z_2$ is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{\sqrt{2}}{2}$ (D) 1

11–40 / New middle-to-hard practice

11. Let $w = (z - 1)/(z + 1)$ be purely imaginary, $z \neq -1$. Also $|z - 2| = \sqrt{13/5}$ and $\operatorname{Im} z > 0$. Find $\operatorname{Im} w$.

(A) $-\frac{1}{2}$ (B) $\frac{4}{5}$

(C) $\frac{1}{2}$ (D) 2

12. Let $w = (z - 2)/(z + 2)$ be real and $|w| < 1$, with $z \neq -2$. If $|z - i| = \sqrt{10}$, find w.

(A) $\frac{1}{5}$ (B) $\frac{1}{3}$

(C) 5 (D) $-\frac{1}{5}$

13. Let z be the root of $z^2 - (3 + i)z + 2 + 2i = 0$ with larger modulus. Find $|z - (1 + i)|^2$.

(A) 0 (B) 4

(C) 2 (D) 1

14. Let z be the root of $z^2 - (1 + 4i)z - 5 + 5i = 0$ with larger modulus. Find $\operatorname{Im}(z/(2 + i))$.

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$

(C) $\frac{7}{5}$ (D) $\frac{7}{10}$

15. Let $\zeta_1, \zeta_2, \zeta_3$ be the fourth roots of unity other than 1. Find $\sum_{j=1}^{3} 1/(2 - \zeta_j)$.

(A) $\frac{2}{3}$ (B) $\frac{17}{15}$

(C) $\frac{32}{15}$ (D) $\frac{7}{15}$

16. If $\omega^3 = 1$, $\omega \neq 1$, find $1/(2-\omega)^2 + 1/(2-\omega^2)^2$.

(A) $\frac{5}{7}$ (B) $\frac{11}{7}$

(C) $\frac{25}{49}$ (D) $\frac{11}{49}$

17. For $w = (1 + \sin(\pi/12) + i\cos(\pi/12))/(1 + \sin(\pi/12) - i\cos(\pi/12))$, find the smallest positive integer n for which w^n is negative and purely imaginary.

(A) 18 (B) 6

(C) 12 (D) 24

18. Let $w = (1+i)/(\sqrt{3}+i)$. At the smallest positive integer n for which w^n is positive and purely imaginary, find $|w^n|$.

(A) $\frac{1}{8}$ (B) $\frac{1}{64}$

(C) $\frac{1}{4}$ (D) $\frac{1}{16}$

19. A real z satisfies $z + z^{-1} = 3$ and $z > 1$. Find $z^3 - z^{-3}$.

(A) $4\sqrt{5}$ (B) $8\sqrt{5}$

(C) 8 (D) $-8\sqrt{5}$

20. If $z + z^{-1} = 1$ and $\operatorname{Im} z > 0$, find $\operatorname{Im}(z^{2026} + 2z^{2027})$.

(A) $-\frac{3\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$

(C) $\frac{3\sqrt{3}}{2}$ (D) $-\sqrt{3}$

21. Let $|w| = 2$ and $z = w - 1/w$. If $\operatorname{Re} z = 3/4$ and $\operatorname{Im} z > 0$, find $\operatorname{Im} z$.

(A) $-\frac{5\sqrt{3}}{4}$ (B) $\frac{5\sqrt{3}}{4}$

(C) $\frac{5\sqrt{3}}{2}$ (D) $\frac{3\sqrt{3}}{4}$

22. Let $|w| = 2$ and $z = w + 1/w$. Find the least value of $|z-3|^2$.

(A) $\frac{9}{4}$ (B) $\frac{1}{2}$

(C) $\frac{5}{4}$ (D) $\frac{1}{4}$

23. If $z + 2\bar{z} = 6 - 3i$, find $|(z-i)/(z+1)|^2$.

(A) $\frac{4}{9}$ (B) $\frac{9}{4}$

(C) $\frac{2}{3}$ (D) $\frac{8}{9}$

24. If $z^2 + |z|^2 = 6 + 4i$ and $\operatorname{Re} z > 0$, find $\operatorname{Re}(1/z)$.

(A) $-\frac{3\sqrt{3}}{13}$ (B) $\frac{3\sqrt{13}}{13}$

(C) $\frac{\sqrt{3}}{13}$ (D) $\frac{3\sqrt{3}}{13}$

25. If $\operatorname{Re}(z+2) = |z-2|$, $|z|^2 = 20$ and $\operatorname{Im} z > 0$, find $\operatorname{Im} z$.

(A) 2 (B) 4

(C) -4 (D) $2\sqrt{5}$

26. A nonzero z satisfies $|z-1| = \operatorname{Re} z + 1$ and $\operatorname{Im} z = \operatorname{Re} z$. Find $|z|$.

(A) 8 (B) $4\sqrt{2}$

(C) 4 (D) $2\sqrt{2}$

27. A complex z satisfies $|z| = |z-4|$ and $\operatorname{Arg}((z-4)/z) = \pi/2$, with principal argument in $(-\pi, \pi]$. Find the area of the triangle with vertices $0, 4, z$.

(A) 8 (B) 4

(C) 16 (D) 2

28. Let $a = 1+i$, $b = 4+2i$ and $(z-a)/(b-a) = it$ with $t > 0$. If $|z-b|^2 = 20$, find the triangle area with vertices a, b, z.

(A) $\frac{5}{2}$ (B) 10

(C) 5 (D) 20

29. A complex z has $|z| = 5$, $\operatorname{Re} z = 3$ and $\operatorname{Im} z > 0$. Find $\operatorname{Arg}(e^{(1+i)z})$, in $(-\pi, \pi]$.

(A) 7 (B) $7 - 2\pi$

(C) $7 - \pi$ (D) $-7 + 2\pi$

30. A complex z has $|z| = \sqrt{5}$, $\operatorname{Re} z = 1$ and $\operatorname{Im} z < 0$. Find $\operatorname{Arg}(e^{(2-i)z})$ in $(-\pi, \pi]$.

(A) $-5 + \pi$ (B) $-5 + 2\pi$

(C) -5 (D) $5 - 2\pi$

31. Choose an unordered pair of distinct twelfth roots of unity uniformly, conditional on the pair not being antipodal. Find the probability that their distance is $\sqrt{2}$.

(A) $\frac{2}{11}$ (B) $\frac{1}{5}$

(C) $\frac{1}{10}$ (D) $\frac{1}{6}$

32. Three distinct eighth roots of unity are chosen uniformly as an unordered set. Find the probability that the origin is strictly inside their triangle.

(A) $\frac{3}{7}$ (B) $\frac{1}{4}$

(C) $\frac{1}{7}$ (D) $\frac{2}{7}$

33. If $|a| = 1$ and $az^2 + z + 1 = 0$ has a root $z = it$ with $t > 0$, find $\operatorname{Im} a$.

(A) $-\frac{1}{2} + \frac{\sqrt{5}}{2}$ (B) $-\frac{\sqrt{-2+2\sqrt{5}}}{2}$

(C) $\frac{\sqrt{-2+2\sqrt{5}}}{2}$ (D) $\frac{\sqrt{2+2\sqrt{5}}}{2}$

34. If $|a| = 1$ and $az^2 + 2z + 1 = 0$ has a purely imaginary root, find the modulus of its other root.

(A) $\sqrt{2+\sqrt{5}}$ (B) $\sqrt{-2+\sqrt{5}}$

(C) $-2+\sqrt{5}$ (D) $2\sqrt{-2+\sqrt{5}}$

35. If $|z-1| = 2|z+2|$, find the greatest possible value of $|z-i|$.

(A) $\sqrt{14}$ (B) $-2+\sqrt{10}$

(C) $1+\sqrt{10}$ (D) $2+\sqrt{10}$

36. If $|z-2| = |z|/2$, find the least possible value of $|z+1|$.

(A) $\frac{4}{3}$ (B) $\frac{7}{3}$

(C) 5 (D) $\frac{11}{3}$

37. For $|z| = 1$, find the least possible value of $|z^2 + 2z + 2|$.

(A) $\frac{1}{2}$ (B) 1

(C) $\sqrt{2}$ (D) $\frac{\sqrt{2}}{2}$

38. For $|z| = 1$ and $\operatorname{Re} z \geq 0$, find the greatest possible value of $|z^2 - 3z + 2|$.

(A) 3 (B) $2\sqrt{2}$

(C) $\sqrt{10}$ (D) 6

39. If $z^2 = -7 + 24i$ and $\operatorname{Re} z > 0$, find $((z-i)/(z+1))^4$.

(A) $\frac{256}{81}$ (B) $\frac{27}{64}$

(C) $\frac{81}{256}$ (D) $\frac{9}{16}$

40. If $z^3 = 8i$ and $\operatorname{Re} z < 0$, find $\operatorname{Re}(z^2/(1+i\sqrt{3}))$.

(A) -1 (B) -2

(C) 1 (D) 2

41–50 / Preserved hard practice

41. How many complex numbers satisfy $z^2 = \overline{z}$?

(A) 5 (B) 3

(C) 2 (D) 4

42. How many complex solutions of $z^3 = \overline{z}^2$ also satisfy $\operatorname{Re} z \geq 0$?

(A) 6 (B) 4

(C) 3 (D) 5

43. A complex number satisfies $z + \overline{z} = 2$ and $|z^2 - 2z| = 5$. Find $|z|$.

(A) $\sqrt{5}$ (B) $\sqrt{6}$

(C) 2 (D) 5

44. Let $|z| = 1$, $\operatorname{Re} z = 1/2$, $\operatorname{Im} z > 0$, and $w = (z - 1/2)/(1 - z/2)$. Find $|w - 1|$.

(A) $\sqrt{2}$ (B) 1

(C) $\sqrt{3}$ (D) 2

45. For $z \neq 1$, suppose $(z + 1)/(z - 1) = -i\sqrt{3}$. Evaluate $z^{2026} + \overline{z}^{2026}$.

(A) 1 (B) -2

(C) -1 (D) 2

46. How many integers $a \in [-5, 5]$ make both roots of $z^2 - 2az + 9 = 0$ have modulus 3?

(A) 5 (B) 9

(C) 6 (D) 7

47. How many monic polynomials of degree four with real coefficients have four distinct roots, all chosen from the non-real twelfth roots of unity?

(A) 10 (B) 15

(C) 5 (D) 20

48. Two distinct eighth roots of unity z_1, z_2 are chosen uniformly as an unordered pair. Given $z_1 + z_2 \neq 0$, the probability that $z_1 z_2$ is real is

(A) $\frac{1}{3}$ (B) $\frac{3}{14}$

(C) $\frac{1}{4}$ (D) $\frac{1}{6}$

49. The region of complex numbers satisfying $|z - 1|^2 + |z + 1|^2 \leq 10$, $|z - 1| \leq |z + 1|$ and $\operatorname{Im} z \geq \operatorname{Re} z$ has area

(A) 2π (B) $\frac{\pi}{4}$

(C) π (D) $\frac{\pi}{2}$

50. Let $z_0 = 0$ and $z_{n+1} = (1 + i)z_n + 2 - i$ for $n \geq 0$. Find $|z_8|$.

(A) $30\sqrt{5}$ (B) $16\sqrt{5}$

(C) 15 (D) $15\sqrt{5}$

Answer key

Complex Numbers

M-CONU / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**C**	21	**B**	31	**B**	41	**D**
02	**A**	12	**A**	22	**D**	32	**C**	42	**B**
03	**A**	13	**C**	23	**A**	33	**C**	43	**A**
04	**D**	14	**C**	24	**D**	34	**B**	44	**C**
05	**D**	15	**B**	25	**B**	35	**D**	45	**C**
06	**A**	16	**D**	26	**B**	36	**B**	46	**D**
07	**C**	17	**A**	27	**B**	37	**D**	47	**A**
08	**A**	18	**A**	28	**C**	38	**C**	48	**D**
09	**C**	19	**B**	29	**B**	39	**C**	49	**D**
10	**A**	20	**A**	30	**B**	40	**A**	50	**D**

Chapter 3

Quadratic Equations

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. If α, β solve $x^2 - 5x + 6 = 0$, find $1/(\alpha+1) + 1/(\beta+1)$.

(A) $\frac{7}{12}$ (B) $\frac{5}{6}$

(C) $\frac{1}{2}$ (D) $\frac{7}{6}$

02. The smallest integer p for which $x^2 - px + p + 2 = 0$ has two positive real roots is

(A) 7 (B) 6

(C) 5 (D) 4

03. The roots of $x^2 + px + q = 0$ are both negative. If $q = 8$ and one root is twice the other, find p.

(A) 4 (B) 6

(C) -6 (D) 8

04. Let $m = k/10$ with integer k. How many values of k make both roots of $x^2 - 2mx + m^2 - 1/16 = 0$ lie strictly between 0 and 1?

(A) 4 (B) 5

(C) 6 (D) 7

05. For how many real values of a does $(a-1)x^2 + 2x + 1 = 0$ have exactly one distinct real solution?

(A) 2 (B) 1

(C) 0 (D) 3

06. The polynomials $x^2 - 3x + a$ and $x^2 - 5x + b$ have the common root 2. The product of their other roots is

(A) 6 (B) 4

(C) 2 (D) 3

07. The equation $|x^2 - 4x + 3| = 1$ has how many distinct real solutions?

(A) 2 (B) 1

(C) 3 (D) 4

08. The consecutive terms u, v, w of an arithmetic progression satisfy $u + w = 10$ and $uw = 16$. Find $(w - u)^2$.

(A) 25 (B) 16

(C) 36 (D) 9

09. How many integers k make $x^2 - 7x + k = 0$ have two distinct positive integer roots?

(A) 4 (B) 6

(C) 2 (D) 3

10. An integer k is chosen uniformly from 1 through 10. The probability that $x^2 - 2kx + 9 = 0$ has real roots is

(A) $\frac{4}{5}$ (B) $\frac{7}{10}$

(C) $\frac{9}{10}$ (D) $\frac{3}{5}$

11–40 / New middle-to-hard practice

11. Let $a = k/10$, where $k \in \{0, 1, \ldots, 40\}$. For how many values of k does $x^2 - 2ax + a + 2 = 0$ have two distinct roots strictly between 0 and 4?

(A) 7 (B) 4

(C) 5 (D) 6

12. For $t = k/2$ with integer $-4 \le k \le 8$, how many values of k make $x^2 - (t + 3)x + 2t + 2 = 0$ have two distinct roots in $(0, 4)$?

(A) 5 (B) 6

(C) 8 (D) 7

13. If α, β are the roots of $x^2 - 5x + 3 = 0$, set $u = (\alpha + 1)/(\alpha - 1)$ and $v = (\beta + 1)/(\beta - 1)$. Find $u^2 + v^2$.

(A) 34 (B) 18

(C) 16 (D) 25

14. Real numbers α, β satisfy $\alpha\beta = 2$ and $\alpha/(\alpha + 1) + \beta/(\beta + 1) = 7/6$, with both denominators nonzero. Find $\alpha^2 + \beta^2$.

(A) $\frac{25}{4}$ (B) 9

(C) 7 (D) 5

15. The quadratics $x^2 - 5x + a$ and $x^2 - 3x + b$ have a common root, $a + b = 6$, and all their roots are strictly positive. Find the product of their two noncommon roots.

(A) 12 (B) 4

(C) 6 (D) 8

16. The quadratics $x^2 - px + 12$ and $x^2 - (p+1)x + 18$ share a root. Find the sum of the squares of their noncommon roots.

(A) 25 (B) 10

(C) 13 (D) 36

17. For distinct roots α, β of $x^2 - 3x + 1 = 0$, find $(\alpha^5 - \beta^5)/(\alpha - \beta)$.

(A) 76 (B) 123

(C) 47 (D) 55

18. If α, β solve $x^2 - 4x + 1 = 0$, find $\alpha^3/(\alpha^2 + 1) + \beta^3/(\beta^2 + 1)$.

(A) 7 (B) $\frac{7}{2}$

(C) 4 (D) $\frac{15}{4}$

19. For real t, the equation $x^2 - 2tx + 2t^2 - 4t + 3 = 0$ has real roots α, β. The largest possible value of $\alpha^2 + \beta^2$ is

(A) 14 (B) 18

(C) 20 (D) 22

20. Let $-1 \le t \le 3$ and require both roots of $x^2 - 2tx + t^2 + t - 2 = 0$ to be real. The difference between the greatest and least possible values of their squared sum is

(A) 4 (B) $\frac{7}{2}$

(C) 5 (D) $\frac{9}{2}$

21. Among integers $a \in [-4, 4]$, take those for which $(a-1)x^2 - 2ax + a + 1 = 0$ has two distinct positive roots. The product of all such a is

(A) 576 (B) -144

(C) -576 (D) -288

22. Let $a = k/4$, where integer k satisfies $-20 \le k \le 20$. For how many k does $(a-2)x^2 - 2(a-2)x + a = 0$ have one positive and one negative root?

(A) 7 (B) 9

(C) 8 (D) 6

23. How many integers $k \in \{1, 2, \ldots, 10\}$ make $|x^2 - 6x + 5| = k$ have four distinct real solutions?

(A) 3 (B) 5

(C) 2 (D) 4

24. The product of all distinct real solutions of $|x^2 - 4x| = 4$ is

(A) 8 (B) 0

(C) -16 (D) -8

25. The sum of the distinct real roots of $x^4 - 5x^3 + 8x^2 - 5x + 1 = 0$ is

(A) 4 (B) 5

(C) 3 (D) 6

26. For $x^4 - 7x^3 + 14x^2 - 7x + 1 = 0$, find the product of those real roots that exceed 1.

(A) $\frac{(3-\sqrt{5})(\sqrt{3}+2)}{2}$ (B) $\frac{(2-\sqrt{3})(\sqrt{5}+3)}{2}$

(C) $\frac{(\sqrt{3}+2)(\sqrt{5}+3)}{2}$ (D) 6

27. Find the sum of all real solutions of $\left((x^2-4)/(x-2)\right)^2 - 7(x^2-4)/(x-2) + 12 = 0$, using the domain of the displayed expression.

(A) 1 (B) 3

(C) 4 (D) 2

28. How many integers x satisfy $4^x - 6 \cdot 2^x + 8 \le 0$ while $\log_2(x-1)$ is real?

(A) 2 (B) 0

(C) 1 (D) 3

29. Positive coefficients p, q, r are in AP, with q/p an integer from 1 through 10. How many possible ratios q/p make $px^2 + qx + r = 0$ have real roots?

(A) 4 (B) 2

(C) 5 (D) 3

30. Positive coefficients p, q, r are in GP and $r/p = 9$. If α, β solve $px^2 + qx + r = 0$, find $\alpha^3 + \beta^3$.

(A) 27 (B) 81

(C) 54 (D) -54

31. Two positive real roots α, β satisfy $\alpha^2 + \beta^2 = 25$ and $\alpha^3 + \beta^3 = 91$. Find $\alpha\beta$.

(A) 12 (B) 14

(C) 6 (D) 10

32. Two positive real roots α, β satisfy $\alpha^2 + \beta^2 = 20$ and $\alpha^3 + \beta^3 = 72$. Find $(\alpha - \beta)^2$.

(A) 16 (B) 4

(C) 12 (D) 8

33. Find the sum of all integers $k \in \{1, \ldots, 10\}$ for which $x^2 - 2kx + k^2 - k = 0$ has two distinct positive integer roots.

(A) 13 (B) 14

(C) 10 (D) 9

34. For $x^2 - kx + 12 = 0$, both roots must be distinct positive integers and at least one root must be prime. Find the product of all possible integer values of k.

(A) 104 (B) 91

(C) 728 (D) 56

35. For each real a such that the distinct roots of $x^2 - 2ax + a + 2 = 0$ have ratio $2 : 1$ in some order, retain that value of a. The product of all retained values is

(A) $\frac{9}{8}$ (B) $-\frac{9}{4}$

(C) $-\frac{9}{8}$ (D) $\frac{9}{4}$

36. The real roots α, β of $x^2 - mx + 4 = 0$ satisfy $\alpha = 3 + 1/\beta$. Find $\alpha^4 + \beta^4$.

(A) 257 (B) 81

(C) 625 (D) 256

37. Choose a uniformly from $\{2, 3, 4\}$ and independently choose b uniformly from $\{3, 4, \ldots, 10\}$. The probability that $x^2 - 2ax + b = 0$ has two distinct roots both greater than 1 is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{3}{8}$

38. Choose a, b independently and uniformly from $\{1, 2, 3, 4, 5\}$. What is the probability that $x^2 - ax + b = 0$ has two distinct roots strictly inside $(0, 3)$?

(A) $\frac{2}{25}$ (B) $\frac{1}{5}$

(C) $\frac{4}{25}$ (D) $\frac{3}{25}$

39. If α, β are all real solutions of $\sqrt{x+1} + \sqrt{5-x} = 2\sqrt{2}$, find $1/(\alpha+2) + 1/(\beta+2)$.

(A) $\frac{1}{2}$ (B) 1

(C) $\frac{3}{2}$ (D) 2

40. The product of all positive real solutions of $\sqrt{5 - \log_2 x} = \log_2 x - 1$ is

(A) $2^{\frac{1}{2} + \frac{\sqrt{21}}{2}}$ (B) $2^{\frac{1}{2} + \frac{\sqrt{17}}{2}}$

(C) $2^{\frac{1}{2} - \frac{\sqrt{17}}{2}}$ (D) 4

41–50 / Preserved hard practice

41. How many integers $a \in [-5, 5]$ make $x^2 - (a+2)x + 2a - 1 = 0$ have exactly one positive real root?

(A) 8 (B) 7

(C) 6 (D) 5

42. Let $m = k/8$, where k is an integer. How many values of k make $x^2 - 2mx + m - 1/4 = 0$ have two distinct roots in $(-1, 1)$?

(A) 6 (B) 8

(C) 5 (D) 7

43. For how many integers a does $x^2 - 2(a+1)x + a^2 - 1 = 0$ have two positive real roots whose difference is at most 6?

(A) 3 (B) 4

(C) 2 (D) 1

44. How many distinct real solutions does $|x^2 - 4x + 1| = 2x - 1$ have?

(A) 1 (B) 2

(C) 3 (D) 4

45. How many integers a make $\dfrac{x^2 - 2ax + a + 2}{x^2 + 1} > 0$ for every real x?

(A) 4 (B) 2

(C) 1 (D) 3

46. Let $P(x) = ax^2 + bx + c$ satisfy $\lim_{x \to 1} P(x)/(x-1) = 6$ and $P(2) = 8$. Find abc.

(A) 16 (B) 8

(C) -8 (D) -16

47. How many real numbers x satisfy $\log_{x-1}(x^2 - 5x + 7) = 2$?

(A) 3 (B) 2

(C) 0 (D) 1

48. Two fair dice produce numbers a, b. The probability that $x^2 - ax + b = 0$ has two distinct positive integer roots is

(A) $\frac{5}{18}$ (B) $\frac{1}{9}$

(C) $\frac{1}{6}$ (D) $\frac{5}{36}$

49. How many solutions in $[0, 2\pi)$ satisfy $\sin^2 x + \sin x \cos x = \cos^2 x$?

(A) 6 (B) 4

(C) 2 (D) 3

50. Let f be a quadratic polynomial with $f(0) = 1$, $f(1) = 2$ and $\int_0^1 f(x)\,dx = 0$. The total area between its graph and the x-axis on $[0, 1]$ is

(A) $\frac{14\sqrt{7}}{81}$ (B) $\frac{7\sqrt{7}}{27}$

(C) $\frac{28\sqrt{7}}{243}$ (D) $\frac{56\sqrt{7}}{243}$

Answer key

Quadratic Equations

M-QUEQ / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**C**	21	**C**	31	**A**	41	**C**
02	**B**	12	**B**	22	**A**	32	**B**	42	**A**
03	**B**	13	**A**	23	**A**	33	**A**	43	**C**
04	**B**	14	**D**	24	**D**	34	**D**	44	**B**
05	**A**	15	**D**	25	**A**	35	**B**	45	**B**
06	**D**	16	**C**	26	**C**	36	**A**	46	**D**
07	**C**	17	**D**	27	**A**	37	**C**	47	**C**
08	**C**	18	**B**	28	**C**	38	**A**	48	**D**
09	**D**	19	**B**	29	**D**	39	**B**	49	**B**
10	**A**	20	**D**	30	**C**	40	**B**	50	**D**

Chapter 4

Logarithms

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Solve in real numbers: $\log_2(x-1)+\log_2(x+1)=3$.

(A) -3 (B) 4

(C) 3 (D) 2

02. Solve $\log_3 x=\log_9(x+6)$ in real numbers.

(A) -2 (B) 9

(C) 3 (D) 6

03. The sum of all real solutions of $\log_2(x^2-5x+6)=1$ is

(A) 3 (B) 5

(C) 4 (D) 6

04. The real solution set of $\log_{1/2}(x-1)>-1$ is

(A) $(0,3)$ (B) $(1,3)$

(C) $[1,3]$ (D) $(3,\infty)$

05. How many integers $x\in\{1,2,\dots,10\}$ make $\log_x(x^2)=2$ a legal true equation?

(A) 1 (B) 10

(C) 8 (D) 9

06. The function $f(x)=\log_{a-1}x$ is defined for every $x>0$ and is strictly decreasing precisely when

(A) $1<a<2$ (B) $a\neq 2$

(C) $0<a<1$ (D) $a>2$

07. The maximum value of $\log_2 x + \log_2(4-x)$ for $0 < x < 4$ is

(A) 1 (B) 2

(C) 3 (D) 4

08. Positive numbers a, b satisfy $a + b = 6$ and $\log_2 a + \log_2 b = 3$. Find $|\log_2 a - \log_2 b|$.

(A) 1 (B) 0

(C) 3 (D) 2

09. An integer x is chosen uniformly from 1 through 8. The probability that $\log_2(x-2)$ is real is

(A) $\frac{3}{4}$ (B) $\frac{7}{8}$

(C) $\frac{1}{2}$ (D) $\frac{5}{8}$

10. The sum of all solutions in $[0, \pi]$ of $\log_2(2\sin x) = 0$ is

(A) $\frac{3\pi}{2}$ (B) 2π

(C) $\frac{\pi}{2}$ (D) π

11–40 / New middle-to-hard practice

11. Positive a, b satisfy $\log_5((a+b)/4) = (\log_5 a + \log_5 b)/2$. Find $(a^3+b^3)^2/(a^3b^3)$.

(A) 194 (B) 256

(C) 2704 (D) 2197

12. Positive a, b satisfy $a + b = 12$ and $\log_3 a + \log_3 b = 3$. Find the greatest possible value of $\log_3 a + \log_9 b$.

(A) $\frac{3}{2}$ (B) 3

(C) $\frac{5}{2}$ (D) 2

13. Find the sum of real solutions $x > 1$ of $2^{2\log_3 x} - 10 \cdot 2^{\log_3 x} + 16 = 0$.

(A) 9 (B) 30

(C) 84 (D) 28

14. For $x > 0$, $3^{\log_2 x} + 3^{\log_2(4/x)} = 10$. Find the product of all real solutions.

(A) 8 (B) 4

(C) 16 (D) 2

15. Let $x = k/4$, where integer $4 < k < 20$. For how many k is $\log_{x-1}(x+5) \geq 2$ a legal true inequality?

(A) 9 (B) 8

(C) 7 (D) 10

16. Let $x = k/4$, where k is a positive integer. How many legal values satisfy $\log_x(4 - x) > 2$?

(A) 2 (B) 1

(C) 4 (D) 3

17. Find the product of all nonzero real solutions of $\sqrt{8 - \log_3 |x|} = \log_3 |x| - 2$.

(A) -6561 (B) 6561

(C) -81 (D) -729

18. Find the sum of all positive real solutions of $\sqrt{10 - \log_2 x} = 4 - \log_2 x$.

(A) 66 (B) 2

(C) 8 (D) 64

19. For $x > 0$, $x \neq 1$, solve $\log_2 x + \log_x 16 = 5$. Find the ratio of its larger solution to its smaller solution.

(A) 16 (B) 4

(C) 8 (D) 2

20. Numbers $x, y > 1$ satisfy $\log_2 x + \log_2 y = 5$ and $\log_x y + \log_y x = 17/4$. Find $x + y$.

(A) 18 (B) 20

(C) 32 (D) 10

21. Let $x = k/2$ with integer k. How many x satisfy $\log_{1/2}(x^2 - 5x + 6) \geq -1$?

(A) 4 (B) 6

(C) 3 (D) 5

22. How many half-integers $x = k/2$ satisfy both $\log_2(x - 1) + \log_2(7 - x) \geq 3$ and $\log_{x-2} 4 > 1$?

(A) 5 (B) 4

(C) 6 (D) 3

23. Among $a = k/2$ with integer $0 \leq k \leq 10$, how many make $\log_2(x(6 - x)) \leq a$ true for every $0 < x < 6$?

(A) 3 (B) 6

(C) 4 (D) 5

24. For $1/2 \leq x \leq 8$, let $F(x) = \log_2 x \cdot \log_2(16/x)$. Find $\max F - \min F$.

(A) 10 (B) 5

(C) 8 (D) 9

25. How many integers $1 \le n \le 1000$ satisfy $\lfloor \log_2 n \rfloor + \lfloor \log_2(1000/n) \rfloor = 8$?

(A) 20 (B) 18

(C) 21 (D) 19

26. How many integers $1 \le n \le 100$ satisfy $\lfloor \log_2 n \rfloor = 1 + \lfloor \log_3 n \rfloor$?

(A) 17 (B) 19

(C) 15 (D) 21

27. If $a = \log_2 3$ and $b = \log_3 5$, express $\log_{45} 120$ in terms of a, b.

(A) $\frac{ab+a+3}{ab+a+2}$ (B) $\frac{ab+a+3}{a(2b+1)}$

(C) $\frac{a+b+3}{a+2b}$ (D) $\frac{ab+a+2}{a(2b+1)}$

28. If $a = \log_2 3$, $b = \log_3 5$, and $c = \log_5 7$, express $\log_{105} 56$.

(A) $\frac{abc+3}{ab(c+1)}$ (B) $\frac{abc+3}{ab+a+1}$

(C) $\frac{abc+3}{a(bc+b+1)}$ (D) $\frac{a+b+c+3}{a+b+c}$

29. Let $a = k/2$, with integer $-4 \le k \le 8$. How many k make $\log_2((x-a)^2+1) = \log_2(2x)$ have two distinct positive real solutions?

(A) 8 (B) 7

(C) 6 (D) 9

30. For how many integers $a \in \{-2, -1, 0, 1, 2, 3, 4\}$ does $\log_2(x-a) + \log_2(4-x) = 1$ have two distinct real solutions?

(A) 5 (B) 3

(C) 4 (D) 6

31. Positive a, b, c are in GP, $abc = 512$, and $a + c = 20$. Find $(\log_2 a)^2 + (\log_2 b)^2 + (\log_2 c)^2$.

(A) 27 (B) 29

(C) 36 (D) 25

32. Positive a, b, c are in GP, $ac = 64$, and $\log_2 a + 2\log_2 b + 3\log_2 c = 20$. Find $\log_2(a^2 + c^2)$.

(A) $\frac{\log(48)}{\log(2)}$ (B) 8

(C) $\frac{\log(80)}{\log(2)}$ (D) $\frac{\log(272)}{\log(2)}$

33. Legal bases $a, b > 1$ satisfy $\log_b a = 2$ and $\log_b(a+b) = 3$. Find $b + 1/b$.

(A) $1 + \sqrt{5}$ (B) $\sqrt{3}$

(C) 2 (D) $\sqrt{5}$

34. Numbers $a, b, c > 1$ satisfy $\log_a b = 2$, $\log_b c = 3/2$, and $a + b + c = 14$. Find abc.

(A) 64 (B) 128

(C) 32 (D) 56

35. Find the sum of the squares of all real solutions of $\log_2 |x - 1| + \log_2 |x + 1| = 3$.

(A) 16 (B) 20

(C) 9 (D) 18

36. Find the product of all real solutions of $\log_2 |x| + \log_2 |x - 3| = 2$.

(A) -2 (B) 4

(C) -8 (D) -4

37. Let $a = k/2$, where k is an integer. For how many a does $\log_2 x + \log_2(8 - x) = a$ have two distinct solutions both in $(1, 7)$?

(A) 3 (B) 2

(C) 1 (D) 4

38. Find the product of all legal real solutions of $\log_x 2 + \log_{4x} 2 = 3/4$.

(A) $2\sqrt[3]{2}$ (B) $\frac{\sqrt[3]{2}}{2}$

(C) 4 (D) $2^{\frac{2}{3}}$

39. Numbers $x, y > 1$ satisfy $x + y = 12$ and $\log_2 x + \log_2 y = 5$. Find the largest possible value of $\log_x y$.

(A) $\frac{3}{2}$ (B) 2

(C) 3 (D) $\frac{2}{3}$

40. Numbers $x, y > 1$ satisfy $\log_2 x + \log_2 y = 6$ and $\log_x y + \log_y x = 5/2$. Find $(x - y)/(\log_2 x - \log_2 y)$.

(A) 8 (B) 6

(C) 4 (D) 12

41–50 / Preserved hard practice

41. How many real solutions does $\log_{x-1}(x^2 - 3x + 3) = 1$ have?

(A) 3 (B) 1

(C) 0 (D) 2

42. The product of all real solutions of $\log_x 16 + \log_{4x} 16 = 3$ is

(A) 4 (B) $\sqrt[3]{2}$

(C) $2^{\frac{2}{3}}$ (D) $\frac{\sqrt[3]{2}}{2}$

43. How many integers $n \in \{1, 2, \ldots, 31\}$ satisfy $\lfloor \log_2 n \rfloor + \lfloor \log_2 (n+1) \rfloor = 7$?

(A) 1 (B) 2

(C) 8 (D) 0

44. How many real numbers x satisfy both $(\log_2 x)^2 - 5 \log_2 x + 6 = 0$ and $\log_x (8 - x) > 0$?

(A) 2 (B) 1

(C) 3 (D) 0

45. The unique real solution of $\log_2 x + \log_2 (\log_2 x) = 3$ is

(A) 16 (B) 8

(C) 2 (D) 4

46. How many unordered pairs of distinct numbers a, b from $\{2, 2^2, \ldots, 2^{10}\}$ satisfy $\log_2 (ab) = 11$ and $(\log_2 a)(\log_2 b) \geq 24$?

(A) 2 (B) 4

(C) 3 (D) 5

47. Evaluate $\displaystyle\lim_{x \to 0} \frac{\ln(1+x) - \ln(1-x)}{\sin 3x}$.

(A) 3 (B) $\frac{1}{3}$

(C) $\frac{3}{2}$ (D) $\frac{2}{3}$

48. A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n^2$. Taking $\log_{10} 2 = 0.3010$, the least n for which a_n has at least 101 decimal digits is

(A) 9 (B) 8

(C) 10 (D) 11

49. For how many integers $a \in \{0, 1, \ldots, 10\}$ does $|\log_2 x| + |\log_2 (x/4)| = a$ have exactly two solutions in $[1/4, 16]$?

(A) 4 (B) 8

(C) 5 (D) 3

50. Two fair dice give a, b. The probability that $\log_x a + \log_x b = 2$ has a real solution with $1 < x < 3$ is

(A) $\frac{4}{9}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{5}{12}$

Answer key

Logarithms

M-LOGA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**C**	21	**A**	31	**B**	41	**C**
02	**C**	12	**C**	22	**B**	32	**D**	42	**C**
03	**B**	13	**B**	23	**C**	33	**D**	43	**A**
04	**B**	14	**B**	24	**D**	34	**A**	44	**B**
05	**D**	15	**B**	25	**D**	35	**D**	45	**D**
06	**A**	16	**A**	26	**A**	36	**D**	46	**C**
07	**B**	17	**A**	27	**B**	37	**B**	47	**D**
08	**A**	18	**B**	28	**C**	38	**D**	48	**C**
09	**A**	19	**C**	29	**A**	39	**A**	49	**A**
10	**D**	20	**A**	30	**C**	40	**B**	50	**D**

Chapter 5

Matrices & Determinants

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. If A is 3×3 and $\det A = 2$, find $\det(3A^{-1})$.

(A) $\frac{3}{2}$ (B) $\frac{27}{2}$

(C) 6 (D) 9

02. If A, B are 2×2 matrices with determinants $3, 5$, find $\det(ABA^T)$.

(A) 15 (B) 45

(C) 225 (D) 75

03. If $A^2 = A$ and A is invertible, find $\det A$.

(A) 0 (B) -1

(C) 2 (D) 1

04. If A is 2×2, $\det A = 3$ and $\operatorname{tr} A = 5$, find $\det(A + I)$.

(A) 8 (B) 15

(C) 9 (D) 6

05. An invertible 2×2 matrix satisfies $A^2 - 3A + 2I = 0$ and $\operatorname{tr} A = 3$. Find $\operatorname{tr}(A^{-1})$.

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) 2 (D) 3

06. For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, find $\det(A + A^{-1})$.

(A) 2 (B) 1

(C) 4 (D) 0

07. The determinant $\begin{vmatrix} x & 1 \\ 1 & x-1 \end{vmatrix}$ vanishes at α, β. Find $\alpha^2+\beta^2$.

(A) 5 (B) 3

(C) 2 (D) 1

08. Each entry of a 2×2 matrix is independently chosen as 0 or 1, equally likely. The probability that the matrix is invertible is

(A) $\frac{1}{4}$ (B) $\frac{5}{8}$

(C) $\frac{1}{2}$ (D) $\frac{3}{8}$

09. A 3×3 permutation matrix has exactly one 1 in each row and column. How many such matrices have determinant 1?

(A) 3 (B) 6

(C) 4 (D) 2

10. The linear map $\binom{u}{v} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\binom{x}{y}$ sends $(0,0),(1,0),(0,1)$ to a triangle. Find its area.

(A) 3 (B) 5

(C) $\frac{5}{2}$ (D) $\frac{3}{2}$

11–40 / New middle-to-hard practice

11. A real 2×2 matrix satisfies $\det(A+I)=12$ and $\det(A-I)=2$. Find $\det(A^2+I)$.

(A) 49 (B) 50

(C) 61 (D) 36

12. A real 2×2 matrix satisfies $\det(A+I)=15$ and $\det(A-I)=3$. Find $\operatorname{tr}(A^{-2})$.

(A) $\frac{5}{16}$ (B) $\frac{1}{8}$

(C) $\frac{5}{8}$ (D) $\frac{3}{4}$

13. For which real a does the system $(a+1)x+y+z=1$, $x+(a+1)y+z=1$, $x+y+(a+1)z=1$ have infinitely many solutions?

(A) 3 (B) 0

(C) -1 (D) -3

14. For how many integers $a\in[-3,3]$ does $(a+1)x+y+z=1$, $x+(a+1)y+z=2$, $x+y+(a+1)z=3$ have a unique solution with $x+y+z>0$?

(A) 5 (B) 4

(C) 3 (D) 6

15. An invertible real 3×3 matrix has $\det(\operatorname{adj}(\operatorname{adj} A)) = 256$ and $\det(-A) < 0$. Find $\det(2A^{-1} \operatorname{adj} A)$.

(A) -32 (B) 64

(C) 128 (D) 32

16. A real 3×3 matrix satisfies $\det(\operatorname{adj} A) = 25$ and $\det A < 0$. Find $\det(A - \operatorname{adj}(\operatorname{adj} A))$.

(A) -1080 (B) -320

(C) 320 (D) 1080

17. Let $A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$. If $(A^2)_{13} = 10$ and $(A^3)_{13} = 24$, find $(A^5)_{13}$.

(A) 60 (B) 70

(C) 50 (D) 80

18. Let $A = \begin{pmatrix} 2 & a & 0 \\ 0 & 2 & b \\ 0 & 0 & 2 \end{pmatrix}$. If $(A^2)_{12} = 12$ and $(A^3)_{13} = 24$, find $(A^4)_{12} + (A^4)_{23}$.

(A) 160 (B) 128

(C) $\frac{208}{3}$ (D) $\frac{416}{3}$

19. A real 3×3 matrix satisfies $A^2 = A$ and $\operatorname{tr} A = 2$. Find $\det(3A + 2(I - A))$.

(A) 18 (B) 12

(C) 27 (D) 8

20. A real 3×3 matrix satisfies $A^2 - 3A + 2I = 0$ and $\det A = 4$. Find $\det(A + I)$.

(A) 16 (B) 27

(C) 12 (D) 18

21. The matrix $A = \frac{1}{5}\begin{pmatrix} 3 & 4 & 0 \\ 4 & -3 & 0 \\ 0 & 0 & a \end{pmatrix}$ is orthogonal and $\det A > 0$. Find $\operatorname{tr} A$.

(A) 0 (B) 5

(C) 1 (D) -1

22. Let $v = (1, a, 2)^T$ and $H = I - 2vv^T/(v^Tv)$. If $H_{11} = 2/3$ and $H_{12} < 0$, find H_{23}.

(A) $-\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $-\frac{2}{3}$ (D) $\frac{1}{3}$

23. A real 2×2 matrix satisfies $A + A^T = \begin{pmatrix} 4 & 6 \\ 6 & 8 \end{pmatrix}$ and $\det A = 3$. Find $\det(A - A^T)$.

(A) 16 (B) 20

(C) -16 (D) 36

24. A real 2×2 matrix has $\det A = 5$ and $A - A^T = \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}$. Find $\det(A + A^T)$.

(A) 20 (B) 4

(C) -4 (D) 16

25. The system $x + 2y + z = 1$, $2x + 5y + 3z = 2$, $3x + 7y + kz = a$ has infinitely many solutions. Find $k + a$.

(A) 9 (B) 7

(C) 8 (D) 6

26. How many integers $t \in [-3, 5]$ make $x + y + z = 1$, $x + ty + 2z = 2$, $x + 2y + tz = 2$ have a unique solution with $x, y, z \geq 0$?

(A) 5 (B) 3

(C) 4 (D) 2

27. Let $A(x) = \begin{pmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{pmatrix}$. Find the sum of all distinct real x for which $\det A(x) = 4$.

(A) 1 (B) -1

(C) 2 (D) 0

28. Let $A(t) = \begin{pmatrix} t & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{pmatrix}$. If $t > 0$ and $\det(A(t) - 2I) = 0$, find $\det A(t)$.

(A) 16 (B) 0

(C) 20 (D) 25

29. Let $A = \begin{pmatrix} 1 & 2 & a \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$. If $(A^{-1})_{13} = A_{13}$, find $(A^4)_{13}$.

(A) 48 (B) 60

(C) 36 (D) 24

30. Let $A = \begin{pmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & a \end{pmatrix}$ with $a > 0$. If $(A^{-1})_{13} = 1/8$, find $(A^{-2})_{13}$.

(A) $\frac{1}{8}$ (B) $\frac{3}{16}$
(C) $\frac{3}{8}$ (D) $\frac{1}{16}$

31. A real 2×2 matrix satisfies $A^2 - 5A + 6I = 0$ and $\operatorname{tr} A = 5$. If $(A - I)^{-1} = uA + vI$, find $u + v$.

(A) $\frac{1}{2}$ (B) 2
(C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

32. A nonscalar 2×2 matrix satisfies $A^2 - 4A + 2I = 0$. Write $A^{-2} = uA + vI$. Find $u + v$.

(A) $\frac{7}{2}$ (B) $\frac{5}{2}$
(C) -1 (D) $\frac{3}{2}$

33. Let $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$. If $\operatorname{tr}(AB) = 5$, find $\det(AB + BA)$.

(A) 25 (B) 13
(C) 9 (D) 1

34. For $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$ with $a, b > 0$, suppose $\det(AB - BA) = -16$. Find $\operatorname{tr}((AB)^2)$.

(A) 32 (B) 18
(C) 34 (D) 36

35. A 4×4 permutation matrix is chosen uniformly among those with no diagonal entry equal to 1. What is the probability that its determinant is positive?

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

36. Each entry of a 2×2 matrix is chosen independently from $\{-1, 1\}$, equally likely. Given that its trace is zero, what is the probability that its determinant is negative?

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$
(C) 1 (D) $\frac{1}{2}$

37. A linear map sends $(1, 1)$ to $(4, 2)$ and $(1, -1)$ to $(0, 4)$. Find the area of the image of the triangle with vertices $(0, 0), (2, 1), (1, 3)$.

(A) 10 (B) 8
(C) 40 (D) 20

38. A linear map sends $(1,1,0),(0,1,1),(1,0,1)$ to $(3,0,0),(0,2,0),(0,0,4)$, respectively. Find the volume of the image of the tetrahedron with vertices $0, e_1, e_2, e_3$.

(A) 4 (B) 24

(C) 2 (D) 12

39. Let $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, $0 < \theta < \pi/2$, with $\operatorname{tr} A = \sqrt{3}$. Find $\det(I + A^5)$.

(A) 3 (B) 1

(C) $\sqrt{3}+2$ (D) $2-\sqrt{3}$

40. Let $A = \begin{pmatrix} 1 & -t \\ t & 1 \end{pmatrix}$, where $t > 0$ and $AA^T = 4I$. Find $\operatorname{tr}(A^6)$.

(A) 64 (B) 0

(C) 128 (D) 256

41–50 / Preserved hard practice

41. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, the $(1,2)$ entry of A^6 is

(A) 365 (B) 363

(C) 364 (D) 728

42. A 2×2 matrix satisfies $A^2 - 5A + 6I = 0$. If $A - 2I$ is invertible, find $\det(A + I)$.

(A) 16 (B) 12

(C) 9 (D) 25

43. Let $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. The $(1,3)$ entry of A^{10} is

(A) 10 (B) 55

(C) 100 (D) 45

44. For $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, find the $(1,2)$ entry of A^8.

(A) 55 (B) 13

(C) 21 (D) 34

45. For $M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & k \end{pmatrix}$, the system $M\mathbf{x} = \mathbf{0}$ has a nonzero solution when k equals

(A) 5 (B) 6

(C) 4 (D) 7

46. For how many $\theta \in [0, 2\pi)$ is $\begin{pmatrix} \sin\theta & 1 \\ 1 & 2\sin\theta \end{pmatrix}$ singular?

(A) 2 (B) 8

(C) 4 (D) 3

47. Let $A(z) = \begin{pmatrix} z & 1 \\ -1 & z \end{pmatrix}$. The sum of the squared moduli of all complex z for which $A(z)$ is singular is

(A) 2 (B) 0

(C) 1 (D) 4

48. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The $(1,2)$ entry of $I + A + A^2 + \cdots + A^{20}$ is

(A) 420 (B) 190

(C) 210 (D) 200

49. Let $N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. The $(1,3)$ entry of $(2I + N)^8$ is

(A) 1792 (B) 3584

(C) 2048 (D) 896

50. For $0 \le t \le 1$, let $A(t) = \begin{pmatrix} t & 1-t \\ -(1-t) & t \end{pmatrix}$. The minimum of $\det A(t)$ is

(A) $\frac{1}{4}$ (B) 0

(C) 1 (D) $\frac{1}{2}$

Answer key

Matrices & Determinants

M-MADE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**D**	31	**D**	41	**C**
02	**B**	12	**A**	22	**C**	32	**B**	42	**A**
03	**D**	13	**B**	23	**A**	33	**B**	43	**D**
04	**C**	14	**A**	24	**B**	34	**C**	44	**C**
05	**B**	15	**D**	25	**B**	35	**A**	45	**A**
06	**C**	16	**A**	26	**B**	36	**D**	46	**C**
07	**B**	17	**B**	27	**A**	37	**D**	47	**A**
08	**D**	18	**D**	28	**C**	38	**C**	48	**C**
09	**A**	19	**A**	29	**A**	39	**D**	49	**A**
10	**C**	20	**D**	30	**B**	40	**C**	50	**D**

Chapter 6

Permutations & Combinations

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Six distinct people stand in a row. Two specified people must not stand together. How many orders are possible?

(A) 480 (B) 360

(C) 240 (D) 600

02. Six distinct people sit around a circular table, with rotations identical and reflections distinct. Two specified people must be adjacent. How many seatings exist?

(A) 48 (B) 120

(C) 24 (D) 96

03. How many distinct arrangements of MISSISSIPPI are possible?

(A) 34650 (B) 34649

(C) 34651 (D) 34652

04. How many binary strings of length five have no adjacent 1s?

(A) 13 (B) 12

(C) 8 (D) 16

05. Three distinct men and three distinct women stand in a row with the sexes alternating. How many orders exist?

(A) 48 (B) 144

(C) 36 (D) 72

06. How many distinct arrangements of AABBCC have no two identical letters adjacent?

(A) 29 (B) 31

(C) 30 (D) 32

07. Three people are selected from four married couples. No selected pair may be spouses. How many selections are possible?

(A) 48 (B) 56

(C) 32 (D) 24

08. Four letters are placed uniformly at random into four addressed envelopes, one per envelope. The probability that no letter goes to its correct envelope is

(A) $\frac{1}{3}$ (B) $\frac{3}{8}$

(C) $\frac{1}{2}$ (D) $\frac{1}{4}$

09. How many onto functions exist from a four-element set to a three-element set?

(A) 36 (B) 24

(C) 30 (D) 42

10. Nine points are the lattice points (i, j) with $i, j \in \{0, 1, 2\}$. How many non-degenerate triangles have vertices among them?

(A) 72 (B) 76

(C) 78 (D) 84

11–40 / New middle-to-hard practice

11. How many distinct arrangements of BALLOON have no adjacent vowels and do not have the two L letters adjacent?

(A) 864 (B) 576

(C) 288 (D) 144

12. How many distinct arrangements of MISSISSIPPI start with M and have no two I letters adjacent?

(A) 1050 (B) 525

(C) 262 (D) 1575

13. How many four-digit numbers can be formed from distinct digits in $\{0, 2, 4, 6, 8\}$ and are divisible by 12?

(A) 72 (B) 48

(C) 24 (D) 12

14. How many four-digit numbers greater than 3000 use four distinct digits from $\{0, 1, 2, 3, 4, 5, 6\}$ and are divisible by 6?

(A) 106 (B) 212

(C) 53 (D) 318

15. Six distinct people A–F sit at a round table, with rotations identical and reflections distinct. A is opposite B and C is not adjacent to A. How many seatings are possible?

(A) 6 (B) 18

(C) 24 (D) 12

16. Eight distinct people A–H sit at a round table; rotations are identical and reflections distinct. A is opposite B, C is adjacent to D, and C is not adjacent to A. How many seatings are possible?

(A) 192 (B) 144

(C) 288 (D) 120

17. How many permutations of six labels have exactly two fixed points and contain the 2-cycle (1 2)?

(A) 6 (B) 12

(C) 8 (D) 24

18. How many permutations of $1, 2, \ldots, 6$ have no fixed points and place label 1 to the left of label 2?

(A) 132 (B) 134

(C) 133 (D) 128

19. How many binary strings of length 10 begin and end with 0, contain exactly four 1 s, and have exactly three runs of consecutive 1 s?

(A) 40 (B) 60

(C) 15 (D) 30

20. How many binary strings of length 12 begin with 1, end with 0, contain five 1 s in exactly two runs, and have no run of four 1 s?

(A) 10 (B) 12

(C) 18 (D) 24

21. In how many ways can 12 identical sweets be distributed among four distinct children, each receiving from 1 through 5, with the first child receiving an even number?

(A) 18 (B) 36

(C) 72 (D) 108

22. Distribute 10 identical balls among four distinct boxes, with at most four in each box and exactly one empty box. How many distributions are possible?

(A) 12 (B) 24

(C) 36 (D) 48

23. From five married couples, choose a committee of six containing three women, three men and exactly one complete married couple. How many committees are possible?

(A) 30 (B) 20

(C) 90 (D) 60

24. From five married couples, choose a committee of six containing two women, four men and exactly one complete married couple. How many committees are possible?

(A) 30 (B) 40

(C) 10 (D) 20

25. How many onto functions $f : \{1, \ldots, 7\} \to \{a, b, c\}$ have fibre sizes 2,2,3 in some order and satisfy $f(1) = f(2)$, $f(3) \neq f(1)$?

(A) 264 (B) 132

(C) 66 (D) 396

26. How many functions from six distinct labels to three distinct labels have every fibre of size 2 and satisfy $f(1) \neq f(2)$?

(A) 18 (B) 60

(C) 72 (D) 90

27. Among the nine points (i, j) with $i, j \in \{0, 1, 2\}$, how many triangles have positive integer area?

(A) 20 (B) 80

(C) 40 (D) 120

28. How many rectangles with vertices among (i, j), $i, j \in \{0, 1, 2, 3\}$, have sides not parallel to the coordinate axes?

(A) 8 (B) 4

(C) 24 (D) 16

29. Ten distinct friends sit around two tables seating 4 and 6. Rotations at each table are identical; reflections are distinct. A and B must be at different tables, and C and D at the same table. How many seatings are possible?

(A) 18720 (B) 112320

(C) 74880 (D) 37440

30. Ten distinct friends sit around tables seating 4 and 6, with rotations identical and reflections distinct. A and B sit at different tables; C and D sit at the same table and next to each other. How many seatings are possible?

(A) 17280 (B) 51840

(C) 8640 (D) 34560

31. Seven distinct people A–G stand in a row. A is before B, B before C, and B is adjacent to neither A nor C. How many orders are possible?

(A) 360 (B) 480

(C) 240 (D) 120

32. Eight distinct people A–H stand in a row with A before B before C. Exactly one of the pairs A–B and B–C is adjacent. How many orders are possible?

(A) 3000 (B) 4800

(C) 2400 (D) 3600

33. Rearrange the multiset $1, 1, 2, 2, 3, 3, 4, 4$ into eight-digit numbers. All odd digits must occupy odd positions, and the number must be divisible by 4. How many numbers result?

(A) 18 (B) 24

(C) 36 (D) 12

34. How many eight-digit numbers can be formed from $0, 0, 1, 1, 2, 2, 4, 4$, using all digits, with the two zeros separated and the number divisible by 4?

(A) 1800 (B) 600

(C) 300 (D) 1200

35. How many ordered triples of integers (x, y, z) satisfy $x + y + z = 15$, $x \geq 0$, x even, $y \geq 2$, $z \geq 3$, and $x < y$?

(A) 69 (B) 46

(C) 11 (D) 23

36. How many triples of positive integers $a < b < c$ satisfy $abc = 72$ and have at least one member that is a perfect square?

(A) 14 (B) 7

(C) 21 (D) 3

37. Four married couples, all eight people distinct, stand in a row. In how many orders are exactly two couples adjacent?

(A) 4032 (B) 8064

(C) 17280 (D) 10080

38. Four married couples A–B, C–D, E–F, G–H stand in a row of eight distinct people. A is first and no couple is adjacent. How many orders are possible?

(A) 1536 (B) 1728

(C) 1488 (D) 1440

39. How many shortest right/up lattice paths from $(0, 0)$ to $(4, 4)$ stay on or below $y = x$ and pass through $(2, 1)$?

(A) 8 (B) 14

(C) 10 (D) 12

40. How many shortest right/up lattice paths from $(0,0)$ to $(5,4)$ stay on or below $y=x$, pass through $(3,2)$ and avoid $(2,2)$?

(A) 20 (B) 25

(C) 15 (D) 10

41–50 / Preserved hard practice

41. Seven distinct people, including A, B and C, stand in a row. No two of A, B, C may be adjacent. How many orders exist?

(A) 2160 (B) 2880

(C) 1440 (D) 720

42. Four married couples sit around a circular table. Rotations are identical, reflections distinct, and nobody may sit beside their spouse. How many seatings exist?

(A) 1488 (B) 1440

(C) 1680 (D) 1536

43. Ten identical sweets are distributed among three distinct children, each receiving at least two and at most five. How many distributions exist?

(A) 15 (B) 12

(C) 18 (D) 9

44. How many six-digit numbers greater than 300000, divisible by 5, can be formed from $0,1,2,3,4,5,6$ without repetition?

(A) 1080 (B) 720

(C) 960 (D) 840

45. How many distinct arrangements of AAABBBCC have no identical adjacent letters?

(A) 73 (B) 76

(C) 75 (D) 74

46. A four-person committee is chosen uniformly from five men and four women. Given that at least two women are chosen, the probability of exactly three women is

(A) $\frac{4}{9}$ (B) $\frac{1}{4}$

(C) $\frac{10}{63}$ (D) $\frac{20}{81}$

47. How many onto functions $f:\{1,2,3,4,5\}\to\{1,2,3\}$ satisfy $f(1)\neq f(2)$?

(A) 120 (B) 108

(C) 114 (D) 150

48. From the eight vertices of a regular octagon, how many triangles contain the centre strictly inside, rather than on an edge?

(A) 8 (B) 32

(C) 16 (D) 24

49. How many 3×3 matrices with entries 0 or 1 have every row sum and column sum equal to 2, and determinant 2?

(A) 4 (B) 6

(C) 2 (D) 3

50. How many monic polynomials $x^2 - sx + p$ with integers $1 \le s, p \le 20$ have two distinct positive integer roots?

(A) 31 (B) 29

(C) 30 (D) 32

Answer key

Permutations & Combinations

M-PECO / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**C**	21	**B**	31	**C**	41	**C**
02	**A**	12	**B**	22	**B**	32	**D**	42	**A**
03	**A**	13	**C**	23	**A**	33	**A**	43	**B**
04	**A**	14	**A**	24	**D**	34	**B**	44	**D**
05	**D**	15	**D**	25	**B**	35	**D**	45	**D**
06	**C**	16	**B**	26	**C**	36	**B**	46	**D**
07	**C**	17	**A**	27	**C**	37	**B**	47	**C**
08	**B**	18	**D**	28	**A**	38	**B**	48	**A**
09	**A**	19	**D**	29	**D**	39	**C**	49	**D**
10	**B**	20	**B**	30	**A**	40	**C**	50	**C**

Chapter 7

Binomial Theorem

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The coefficient of x^3 in $(1+x)^4(1-x)^3$ is

(A) -3 (B) -2

(C) -4 (D) -1

02. The coefficient of x^5 in $(1+x)^3(1+x^2)^4$ is

(A) 21 (B) 23

(C) 22 (D) 24

03. The constant term in $(x+2x^{-2})^6$ is

(A) 59 (B) 62

(C) 60 (D) 61

04. The coefficient of x^4 in $(1+x+x^2)^4$ is

(A) 20 (B) 18

(C) 19 (D) 21

05. How many coefficients of $(1+2x)^5$ attain the greatest coefficient value?

(A) 1 (B) 3

(C) 2 (D) 5

06. The coefficient of x^5 in $(2x^3-x^{-2})^5$ is

(A) 80 (B) -80

(C) 40 (D) 160

07. Evaluate $\sum_{r=0}^{6} r^2 \binom{6}{r}$.

(A) 672 (B) 768

(C) 192 (D) 480

08. Evaluate $\sum_{r=0}^{4} \binom{4}{r}\binom{5}{4-r}$.

(A) 144 (B) 126

(C) 84 (D) 120

09. A geometric progression has $a_1 = 1$, $a_3 = 4$ and positive common ratio. Evaluate $\sum_{r=0}^{10} \binom{10}{r} a_{r+1}$.

(A) 59049 (B) 1024

(C) 19683 (D) 65536

10. Evaluate $\sum_{r=0}^{4} (-1)^r \binom{8}{2r}$.

(A) 8 (B) 0

(C) 16 (D) -16

11–40 / New middle-to-hard practice

11. For a positive integer n, the coefficient of x^2 in $(1+x)^n$ is six times the coefficient of x. Find the coefficient of x^3 in $(1-x)^n$.

(A) -364 (B) -286

(C) 286 (D) -220

12. In $(1+x)^n$, the coefficient of x^3 divided by that of x^2 is $5/3$. Find the greatest coefficient in $(1+2x)^n$.

(A) 672 (B) 560

(C) 448 (D) 128

13. Find the coefficient of x^{19} in $(1+x)^{13}(1-x+x^2)^{12}$.

(A) 792 (B) 924

(C) 462 (D) 1716

14. Find the coefficient of x^{13} in $(1-x)^9(1+x+x^2)^8$.

(A) 56 (B) 70

(C) -70 (D) -56

15. In $(x^{1/2} + 2x^{-1/3})^{10}$, find the sum of coefficients of all terms whose exponent of x is an integer.

(A) 6721 (B) 13442

(C) 13441 (D) 13440

16. Find the sum of coefficients of the terms with nonnegative integer powers of x in $(x^{2/3} - x^{-1/2})^{12}$.

(A) 924 (B) 923

(C) 925 (D) 926

17. For an integer $1 \le n \le 9$, the constant term of $(x^2 + x^{-1})^n$ is 84. Find the coefficient of x^3 in that expansion.

(A) 84 (B) 126

(C) 36 (D) 252

18. In $(2x^3 - x^{-2})^n$, where n is a positive integer, the coefficient of x^{3n-5} is -80. Find the constant term.

(A) 80 (B) -40

(C) -80 (D) 40

19. Evaluate $\sum_{\substack{0\le r\le 8 \\ r \text{ even}}} r(r-1)\binom{8}{r}$.

(A) 2048 (B) 3584

(C) 896 (D) 1792

20. Evaluate $\sum_{\substack{0\le r\le 7 \\ r \text{ odd}}} (r-1)^2\binom{7}{r}$.

(A) 448 (B) 576

(C) 1024 (D) 512

21. Evaluate $\sum_{r=1}^{7} (-1)^r (r^2 + 3r)\binom{8}{r}$.

(A) 88 (B) -88

(C) -64 (D) 0

22. Evaluate $\sum_{r=0}^{8} (-1)^r \binom{8}{r} (1/(r+1) - 1/(r+2))$.

(A) $\frac{1}{10}$ (B) $\frac{1}{9}$

(C) $\frac{1}{90}$ (D) $\frac{1}{11}$

23. For a positive integer n, $\sum_{r=1}^{n} r\binom{n}{r}/\binom{n}{r-1} = 210$. Find $\sum_{r=1}^{n} r^2\binom{n}{r}/\binom{n}{r-1}$.

(A) 1330 (B) 210

(C) 2870 (D) 1540

24. In $(1+x)^n$, let $C_r = \binom{n}{r}$. If $C_1/C_0 + 2C_2/C_1 = 17$, find $\sum_{r=0}^{n} (-1)^r r^2 C_r + \sum_{r=0}^{n} rC_r$.

(A) 4608 (B) 2304

(C) 2048 (D) 1152

25. Evaluate $\sum_{r=1}^{4} r\binom{5}{r}\binom{6}{4-r}$.

(A) 720 (B) 120

(C) 660 (D) 600

26. Evaluate $\sum_{r=0}^{5} r(r-1)\binom{6}{r}\binom{7}{5-r}$.

(A) 5940 (B) 990

(C) 4950 (D) 3960

27. The sum of coefficients of $(1+x)^n$ is 64. Evaluate $\sum_{\substack{0\le i,j\le n \\ i\neq j}} \binom{n}{i}\binom{n}{j}$.

(A) 924 (B) 1586

(C) 4096 (D) 3172

28. Evaluate $\sum_{\substack{0\le i,j\le 5 \\ i\neq j}} (i-j)^2\binom{5}{i}\binom{5}{j}$.

(A) 1280 (B) 3200

(C) 5120 (D) 2560

29. Find the sum of the coefficients of powers divisible by 3 in $(1+x)^{10}$.

(A) 342 (B) 512

(C) 341 (D) 340

30. An integer $n \geq 6$ satisfies $\binom{n}{4} = \binom{n}{6}$. Find $\sum_{0\le 4r\le n}(-1)^r\binom{n}{4r}$.

(A) -164 (B) 164

(C) -254 (D) 256

31. In $(1+tx)^6$, $t>0$, the coefficients of x^4 and x^5 are equal. Find the coefficient of x^3.

(A) 625 (B) $\frac{625}{2}$

(C) $\frac{625}{4}$ (D) $\frac{125}{2}$

32. In $(1+tx)^8$, $t>0$, the coefficients of x^2 and x^3 coincide. Find the sum of all coefficients except those two greatest coefficients.

(A) $\frac{6561}{256}$ (B) $\frac{4769}{256}$

(C) $\frac{2977}{256}$ (D) $\frac{2977}{128}$

33. The coefficient of x in $(1+ax+x^2)^4$ is 12. Find its coefficient of x^4.

(A) 81 (B) 201

(C) 189 (D) 195

34. In $(1+x+x^2)^n$, the coefficient of x^2 exceeds the coefficient of x by 10. Find its coefficient of x^6.

(A) 45 (B) 35

(C) 50 (D) 40

35. For $a>0$, the coefficient of x^2 in $(x+a/x)^6$ is four times the coefficient of x^{-2}. Find its constant term.

(A) $\frac{5}{2}$ (B) 5

(C) 20 (D) $\frac{5}{4}$

36. For $a>0$, the constant term of $(x+a/x)^8$ is 1120. Find the coefficient of x^4.

(A) 224 (B) 56

(C) 448 (D) 112

37. The coefficient of x^{n-1} in $\prod_{k=1}^{n}(x-k)$ is -45. Find the coefficient of x^{n-2}.

(A) 870 (B) 2025

(C) 1740 (D) 285

38. For $P(x)=\prod_{k=1}^{n}(x-(2k-1))$, the coefficient of x^{n-1} is -25. Find the coefficient of x^{n-2}.

(A) 460 (B) 165

(C) 250 (D) 230

39. An integer $n\geq 3$ satisfies $\binom{n}{2}+3\binom{n}{1}+3=66$. Find $\binom{n}{3}+3\binom{n}{2}+3\binom{n}{1}+1$.

(A) 165 (B) 220

(C) 120 (D) 286

40. For an integer $n\geq 5$, $\sum_{r=0}^{4}\binom{4}{r}\binom{n}{5-r}=1287$. Find $\sum_{r=0}^{2}\binom{2}{r}\binom{n}{3-r}$.

(A) 220 (B) 120

(C) 286 (D) 165

41–50 / Preserved hard practice

41. The coefficient of x^9 in $(1+x)^6(1+x^2)^5$ is

(A) 290 (B) 300

(C) 280 (D) 320

42. The constant term in $(2x^2-x^{-1})^9$ is

(A) 336 (B) -672

(C) 1344 (D) 672

43. For an integer $n \geq 4$, the coefficient of x^4 in $(1+x)^n$ is $5/2$ times the coefficient of x^2. Find n.

(A) 6 (B) 7

(C) 8 (D) 9

44. The coefficient of x^7 in $(1+x)^8(1-x)^6$ is

(A) 20 (B) -40

(C) 40 (D) -20

45. The greatest coefficient in $(1+3x)^{19}$ occurs at two powers x^r. The sum of these two exponents is

(A) 30 (B) 31

(C) 28 (D) 29

46. Evaluate $\sum_{r=0}^{10} \binom{10}{r}/(r+1)$.

(A) 93 (B) $\frac{2047}{11}$

(C) $\frac{2047}{10}$ (D) $\frac{2048}{11}$

47. If $a \in \mathbb{R}$ and $\lim_{x\to 0}((1+ax)^8 - 1)/\sin 4x = 6$, find the coefficient of x^2 in $(1+ax)^8$.

(A) 126 (B) 252

(C) 504 (D) 168

48. Six independent trials each succeed with probability $1/3$. Given at least one success, the probability of exactly two successes is

(A) $\frac{48}{133}$ (B) $\frac{15}{64}$

(C) $\frac{16}{45}$ (D) $\frac{80}{243}$

49. Evaluate $\sum_{k=0}^{4} \binom{12}{3k}$.

(A) 1366 (B) 1365

(C) 1376 (D) 1367

50. Evaluate $\sum_{r=0}^{6} (-1)^r \binom{12}{2r} 3^r$.

(A) -4096 (B) 4096

(C) 2048 (D) 1024

Answer key

Binomial Theorem

M-BITH / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**B**	21	**B**	31	**B**	41	**A**
02	**C**	12	**A**	22	**A**	32	**C**	42	**D**
03	**C**	13	**B**	23	**D**	33	**D**	43	**C**
04	**C**	14	**C**	24	**B**	34	**A**	44	**B**
05	**C**	15	**C**	25	**D**	35	**A**	45	**D**
06	**A**	16	**C**	26	**C**	36	**D**	46	**B**
07	**A**	17	**B**	27	**D**	37	**A**	47	**B**
08	**B**	18	**B**	28	**D**	38	**D**	48	**A**
09	**A**	19	**D**	29	**C**	39	**B**	49	**A**
10	**C**	20	**D**	30	**A**	40	**D**	50	**B**

Chapter 8

Sequences & Series

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. In an AP, the third term is 11 and the seventh term is 27. Find the sum of its first ten terms.

(A) 210 (B) 220

(C) 230 (D) 200

02. Three positive increasing numbers are in GP. Their sum is 21 and product is 216. Their common ratio is

(A) $\frac{1}{2}$ (B) 4

(C) 3 (D) 2

03. Evaluate $\sum_{k=1}^{5} k2^k$.

(A) 256 (B) 258

(C) 320 (D) 260

04. Evaluate $\sum_{k=1}^{\infty} k/2^k$.

(A) 3 (B) 4

(C) 2 (D) 1

05. How many numbers below 100 belong to both $2, 4, 8, \ldots$ and $4, 16, 64, \ldots$?

(A) 3 (B) 4

(C) 2 (D) 5

06. Evaluate $\sum_{n=1}^{\infty} n2^n/n!$.

(A) $2e^2$ (B) $2e$

(C) $4e^2$ (D) e^2

07. In an AP, $a_5/a_9 = 2/3$ and $a_1 = 4$. Find the common difference.

(A) $\frac{1}{2}$ (B) 2

(C) 1 (D) 3

08. A positive GP with $0 < r < 1$ has sum 12 and sum of squared terms 24. Find r.

(A) $\frac{5}{7}$ (B) $\frac{1}{2}$

(C) $\frac{3}{4}$ (D) $\frac{2}{3}$

09. Three positive increasing terms of a GP have sum 7 and product 8. Find the greatest term.

(A) 3 (B) 4

(C) 2 (D) 6

10. How many three-element subsets of $\{1, 2, \ldots, 7\}$ form an arithmetic progression when written in increasing order?

(A) 9 (B) 10

(C) 8 (D) 7

11–40 / New middle-to-hard practice

11. An AP has $S_5 = 35$ and $S_{10} = 120$. Find the least positive integer n for which $S_n > 300$.

(A) 16 (B) 18

(C) 15 (D) 17

12. An AP satisfies $S_8 = 3S_4$ and $a_2 + a_6 = 22$. Find S_8.

(A) 88 (B) 72

(C) 104 (D) 96

13. A positive infinite GP has sum 15 and sum of squared terms 45. Find the index of its first term less than $1/2$.

(A) 5 (B) 7

(C) 8 (D) 6

14. A positive infinite GP has sum 12, and the sum of its odd-indexed terms is 8. Find $\sum_{n=1}^{\infty} a_n a_{n+1}$.

(A) 24 (B) 18

(C) 48 (D) 36

15. Four nonzero real terms A, B, C, A have their first three in GP and last three in AP of common difference 6. Find $A^2 + B^2 + C^2 + A^2$.

(A) 144 (B) 84

(C) 148 (D) 168

16. Four nonzero real terms A, B, C, A have their first three in AP of common difference 3 and last three in GP. Find $A^3 + B^3 + C^3 + A^3$.

(A) -129 (B) -57

(C) -121 (D) 121

17. A harmonic progression has $h_1 = 3$ and $h_{10} = 6$. Find the least positive integer n for which h_n is finite and negative.

(A) 18 (B) 21

(C) 20 (D) 19

18. A harmonic progression has $h_1 = 2$ and $h_9 = 6$. Among $h_1, \ldots, h_{24}$, sum the reciprocals of only the finite negative terms.

(A) $-\frac{13}{4}$ (B) $\frac{11}{4}$

(C) $-\frac{11}{2}$ (D) $-\frac{11}{4}$

19. A positive harmonic progression has $h_1 = 2$, $h_6 = 1/2$. Find $\sum_{n=1}^{5} h_n h_{n+1}$.

(A) $\frac{15}{2}$ (B) 10

(C) 3 (D) 5

20. A positive harmonic progression has $h_1 = 6$ and $h_4 = 3$. Find $\sum_{n=1}^{7} h_n h_{n+1}$.

(A) $\frac{189}{5}$ (B) 72

(C) $\frac{378}{5}$ (D) 84

21. For $0 < r < 1$, $\sum_{k=0}^{\infty} r^k = 3$. Evaluate $\sum_{k=1}^{\infty} k(k+1)r^k$.

(A) 36 (B) 24

(C) 18 (D) 54

22. A positive real r satisfies $1 + r + r^2 + r^3 + r^4 = 31$. Evaluate $\sum_{k=1}^{5} kr^k$.

(A) 258 (B) 260

(C) 240 (D) 256

23. Evaluate $\sum_{n=1}^{\infty} \sum_{k=1}^{n} k/2^{n+k}$.

(A) $\frac{4}{9}$ (B) $\frac{8}{9}$

(C) $\frac{2}{3}$ (D) $\frac{16}{9}$

24. Evaluate $\sum_{\substack{m,n \geq 1 \\ m+n \text{ even}}} (m+n)/2^{m+n}$.

(A) $\frac{40}{27}$ (B) $\frac{28}{27}$

(C) $\frac{64}{27}$ (D) $\frac{56}{27}$

25. For real a, b, the coefficient of x in $e^{ax} + 2e^{bx}$ is zero and that of x^2 is 3. Find its coefficient of x^4.

(A) $\frac{1}{4}$ (B) $\frac{9}{4}$

(C) $\frac{3}{4}$ (D) $\frac{3}{2}$

26. The coefficients of x^3 and x^4 in $(1 + ax + bx^2)e^x$ are both zero. Find its coefficient of x^6.

(A) $-\frac{1}{1440}$ (B) $\frac{1}{1440}$

(C) $\frac{1}{2880}$ (D) $\frac{1}{720}$

27. Let $u_1 = 1$ and $u_{n+1} = u_n/(1 + 2u_n)$. Find $\sum_{n=1}^{10} u_n u_{n+1}$.

(A) $\frac{20}{21}$ (B) $\frac{10}{19}$

(C) $\frac{10}{21}$ (D) $\frac{5}{21}$

28. Let $u_1 = 2$ and $u_{n+1} = u_n/(1 + u_n)$. Find the least n such that $u_n < 1/10$.

(A) 10 (B) 11

(C) 21 (D) 12

29. Positive a, b, c satisfy $a + b + c = 6$ and $a + 2b + 3c = 12$. Find the greatest value of a^2bc.

(A) $\frac{2187}{64}$ (B) 16

(C) $\frac{729}{64}$ (D) $\frac{2187}{128}$

30. Positive, pairwise distinct a, b, c are in GP and $a + b + c = 12$. Find the supremum of ab^2c.

(A) 256 (B) 144

(C) 128 (D) 192

31. An increasing AP has $a_2a_4 = 45$ and $a_1a_5 = 33$, and all its first five terms are positive. Find S_{10}.

(A) 110 (B) 100

(C) 120 (D) 130

32. How many strictly increasing triples of positive integers (a, b, c) are in GP, have $abc = 216$, and satisfy $a + b + c \leq 40$?

(A) 2 (B) 5

(C) 3 (D) 4

33. Find $\left|\sum_{k=0}^{2025} ki^k\right|^2$, where $i^2 = -1$.

(A) 2050313 (B) 2050314

(C) 2052338 (D) 2048288

34. Find $\left|\sum_{k=0}^{100}((1 + i)/\sqrt{2})^k\right|^2$.

(A) $3 - 2\sqrt{2}$ (B) 4

(C) $\sqrt{2} + 2$ (D) $2\sqrt{2} + 3$

35. An increasing positive GP satisfies $a_1 + a_3 = 10$, $a_2 = 3$. Find $\sum_{k=0}^{5} \binom{5}{k} a_{k+1}$.

(A) 1024 (B) 729

(C) 243 (D) 4096

36. A decreasing positive GP satisfies $a_1 + a_3 = 10$, $a_2 = 3$. Find $\sum_{k=0}^{4} (-1)^k \binom{4}{k} a_{k+1}$.

(A) 16 (B) $\frac{16}{3}$

(C) $\frac{256}{9}$ (D) $\frac{16}{9}$

37. How many three-element subsets of $\{1, \ldots, 20\}$ form an AP and have sum divisible by 9?

(A) 30 (B) 28

(C) 36 (D) 32

38. Four-element AP subsets of $\{1, \ldots, 25\}$ have sum 50. Find the product of their distinct positive common differences.

(A) 35 (B) 210

(C) 105 (D) 945

39. A positive GP has common ratio 2 and $a_3 + a_5 = 80$. Find the least n such that $S_n > 1000$.

(A) 7 (B) 10

(C) 9 (D) 8

40. A GP has common ratio -2 and $S_3 = 15$. For how many integers $1 \le n \le 10$ is $S_n > 100$?

(A) 5 (B) 2

(C) 3 (D) 4

41–50 / Preserved hard practice

41. For every nonnegative integer n, $\sum_{k=0}^{n} (2k+1)3^k$ equals

(A) $1 + (n+1)3^n$ (B) $(2n+1)3^n$

(C) $1 + n3^n$ (D) $1 + n3^{n+1}$

42. For a nonnegative integer n, evaluate $\sum_{k=0}^{n} (k^2 + k + 1)k!$.

(A) $(n+1)(n+1)! - 1$ (B) $(n+1)(n+1)!$

(C) $(n+1)!$ (D) $(n+2)! - 1$

43. How many integer pairs (r, n) with $r \ge 2$, $n \ge 2$ satisfy $1 + r + \cdots + r^{n-1} = 63$?

(A) 3 (B) 2

(C) 4 (D) 1

44. A sequence satisfies $a_1 = 1$ and $a_{n+1} = 3a_n + 2n + 1$. Find a_6.

(A) 720 (B) 722

(C) 723 (D) 728

45. A positive GP with ratio $0 < r < 1$ has infinite sum 4 and sum of cubed terms $64/7$. Find its first term.

(A) 3 (B) 2

(C) 4 (D) 1

46. Evaluate $\lim_{n\to\infty} \dfrac{\sum_{k=1}^{n} k2^k}{n2^n}$.

(A) 1 (B) 4

(C) 0 (D) 2

47. Let $a_k = 2^k$. The least positive integer n satisfying $\sum_{k=1}^{n} \log_2 a_k \geq 100$ is

(A) 13 (B) 14

(C) 16 (D) 15

48. Three distinct terms are chosen uniformly from the first eight terms of a nonconstant AP. The probability that the selected terms, in increasing index order, themselves form an AP is

(A) $\frac{1}{7}$ (B) $\frac{2}{7}$

(C) $\frac{3}{14}$ (D) $\frac{1}{4}$

49. An AP satisfies $a_0 = 1$ and $a_3 = 10$. Evaluate $\sum_{r=0}^{8} \binom{8}{r} a_r$.

(A) 4096 (B) 3328

(C) 3072 (D) 3584

50. Let $\binom{u_{n+1}}{v_{n+1}} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \binom{u_n}{v_n}$ with $u_0 = 1, v_0 = 0$. Evaluate $\sum_{k=0}^{5} u_k$.

(A) 185 (B) 182

(C) 184 (D) 186

Answer key

Sequences & Series

M-SESE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**D**	21	**A**	31	**C**	41	**D**
02	**D**	12	**D**	22	**A**	32	**C**	42	**B**
03	**B**	13	**B**	23	**B**	33	**A**	43	**B**
04	**C**	14	**A**	24	**D**	34	**D**	44	**B**
05	**A**	15	**C**	25	**C**	35	**A**	45	**B**
06	**A**	16	**C**	26	**B**	36	**D**	46	**D**
07	**C**	17	**C**	27	**C**	37	**A**	47	**B**
08	**A**	18	**D**	28	**B**	38	**C**	48	**C**
09	**B**	19	**D**	29	**D**	39	**D**	49	**B**
10	**A**	20	**C**	30	**A**	40	**B**	50	**A**

Chapter 9

Mathematical Reasoning

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. How many truth assignments to P, Q, R make $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ true?

(A) 3 (B) 5

(C) 6 (D) 4

02. How many truth assignments to P, Q, R make both "exactly one of P, Q" and "exactly one of Q, R" true?

(A) 3 (B) 2

(C) 4 (D) 1

03. How many truth assignments to P, Q, R make $\neg(P \wedge Q) \Rightarrow (\neg P \vee R)$ true?

(A) 8 (B) 9

(C) 7 (D) 6

04. Which statement about real numbers is true?

(A) $\forall y \in \mathbb{R},\ \exists x \in \mathbb{R} : x + y = 1$ and $x + y = 0$ (B) $\forall x, y \in \mathbb{R} : x + y = 0$

(C) $\forall x \in \mathbb{R},\ \exists y \in \mathbb{R} : x + y = 0$ (D) $\exists y \in \mathbb{R},\ \forall x \in \mathbb{R} : x + y = 0$

05. Let $A = \{1, 2, 3\}$. Which statement is true?

(A) $\exists x \in A,\ \forall y \in A : x > y$ (B) $\forall x \in A,\ \exists y \in A : x < y$

(C) $\forall x, y \in A : x = y$ (D) $\exists y \in A,\ \forall x \in A : x \leq y$

06. The negation of "exactly one of P, Q is true" is

(A) $P \Rightarrow Q$ (B) $P \Leftrightarrow Q$

(C) $P \wedge Q$ (D) $\neg P \wedge \neg Q$

07. For two sets A, B, the statement $\forall x:\ x \in A \Rightarrow x \in B$ is equivalent to

(A) $B \setminus A = \varnothing$ (B) $A \cap B = \varnothing$

(C) $A \cup B = \varnothing$ (D) $A \setminus B = \varnothing$

08. Independent propositions P, Q, R have equally likely truth values. The probability that $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ is true is

(A) $\frac{1}{4}$ (B) $\frac{3}{8}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

09. For which real a is the statement "$x^2 - 2ax + 1 \geq 0$ for every real x" true?

(A) $a \leq -1$ or $a \geq 1$ (B) $-1 < a < 1$

(C) $-1 \leq a \leq 1$ (D) $a = 0$ only

10. The statement "$x > a$ for every real $x > 2$" is true precisely when

(A) $a \leq 2$ (B) $a > 2$

(C) $a < 2$ (D) $a \geq 2$

11–40 / New middle-to-hard practice

11. How many assignments to P, Q, R, S make $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \wedge (P \text{ XOR } S)$ true? Here XOR means exactly one is true.

(A) 5 (B) 8

(C) 4 (D) 3

12. How many assignments to P, Q, R, S make $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \wedge (R \Rightarrow P) \wedge (R \vee S)$ true?

(A) 4 (B) 3

(C) 2 (D) 6

13. Exactly two of P, Q, R, S are true. Also $P \Rightarrow Q$ and $Q \Rightarrow R$ are true. How many assignments satisfy all these conditions?

(A) 4 (B) 1

(C) 3 (D) 2

14. For P, Q, R, S, T, no two consecutive entries in this displayed order may both be true, and at least one of P, T is true. How many assignments are possible?

(A) 10 (B) 6

(C) 8 (D) 7

15. Assume $(P \Rightarrow Q)$, $(R \Rightarrow \neg Q)$ and $(P \vee R)$ are all true. Which conclusion must be true?

(A) $P \Leftrightarrow Q$ (B) $P \wedge R$

(C) $Q \Leftrightarrow R$ (D) $\neg(P \vee R)$

16. Assume $((P \vee Q) \Rightarrow R)$, $(R \Rightarrow S)$ and $\neg S$ are true. Which conclusion must follow?

(A) $P \vee Q$ (B) $\neg R \wedge S$

(C) $\neg P \wedge \neg Q$ (D) $P \wedge Q$

17. Which is the direct formal negation, using quantifier and De Morgan rules, of $\forall x \in \mathbb{R}\ \exists y > 0 : (x + y > 1) \wedge (y^2 > x)$?

(A) $\exists x \in \mathbb{R}\ \forall y > 0 :\ (x + y \leq 1) \wedge (y^2 \leq x)$ (B) $\forall x \in \mathbb{R}\ \exists y > 0 :\ (x + y \leq 1) \vee (y^2 \leq x)$

(C) $\exists x \in \mathbb{R}\ \exists y > 0 :\ (x + y \leq 1) \vee (y^2 \leq x)$ (D) $\exists x \in \mathbb{R}\ \forall y > 0 :\ (x + y \leq 1) \vee (y^2 \leq x)$

18. Which is the direct formal negation, with negations pushed to atomic statements, of $\exists a \in \mathbb{R} : [(\forall x \in \mathbb{R},\ x^2 + ax + 1 > 0) \wedge (\exists y \in \mathbb{R},\ y^2 = a)]$.

(A) $\exists a : [(\exists x,\ x^2 + ax + 1 \leq 0) \wedge (\forall y,\ y^2 \neq a)]$

(B) $\forall a : [(\forall x,\ x^2 + ax + 1 \leq 0) \vee (\forall y,\ y^2 \neq a)]$

(C) $\forall a : [(\exists x,\ x^2 + ax + 1 \leq 0) \wedge (\exists y,\ y^2 \neq a)]$

(D) $\forall a \in \mathbb{R} : [(\exists x \in \mathbb{R},\ x^2 + ax + 1 \leq 0) \vee (\forall y \in \mathbb{R},\ y^2 \neq a)]$

19. Let A be the least $a \geq 0$ for which $\exists y \in \mathbb{R}\ \forall x \in [-2, 4] : |x - y| \leq a$. Let B be the least $a \geq 0$ after reversing these two quantifiers. Find $A - B$.

(A) 1 (B) 3

(C) 6 (D) 0

20. Let A be the least $a \geq 0$ making $\exists y \in \{-1, 1\}\ \forall x \in [-1, 1] : |x - y| \leq a$. Let B be the least $a \geq 0$ for $\forall x \in [-1, 1]\ \exists y \in \{-1, 1\} : |x - y| \leq a$. Find $A - B$.

(A) 0 (B) 1

(C) 2 (D) 3

21. For $U = \{1, 2, 3\}$, how many ordered pairs (A, B) of subsets satisfy $A \subseteq B$, $A \neq B$, and $|B| = 2$?

(A) 9 (B) 12

(C) 6 (D) 8

22. For $U = \{1, 2, 3\}$, how many ordered subset pairs (A, B) satisfy $A \cup B = U$, $A \cap B \neq \varnothing$, $A \not\subseteq B$ and $B \not\subseteq A$?

(A) 12 (B) 3

(C) 9 (D) 6

23. For integer n define P: $12 \mid n^2$, Q: $6 \mid n$, and R: $12 \mid n$. Which describes their exact implication structure?

(A) $P \Leftrightarrow Q$, and $R \Rightarrow P$, while $P \not\Rightarrow R$ (B) $P \Leftrightarrow Q \Leftrightarrow R$

(C) $Q \Rightarrow R$, while $R \not\Rightarrow P$ (D) $P \Rightarrow R$, while $R \not\Rightarrow Q$

24. For integer n define P: $15 \mid n(n+1)$, Q: $3 \mid n(n+1)$ and $5 \mid n(n+1)$, R: $15 \mid n$ or $15 \mid (n+1)$. Which is the exact structure?

(A) $P \Leftrightarrow Q$, and $R \Rightarrow P$, while $P \not\Rightarrow R$

(B) $P \Rightarrow R$, while $R \not\Rightarrow Q$

(C) $Q \Rightarrow R$, while $R \not\Rightarrow P$

(D) $P \Leftrightarrow R$, while $Q \not\Rightarrow P$

25. For how many integers $a \in [-3, 3]$ is $\forall x \in \mathbb{R}: \ (x^2 < a \Rightarrow x^2 < 1)$ true?

(A) 4

(B) 5

(C) 6

(D) 3

26. For how many integers $a \in [-2, 3]$ is $\forall x \in \mathbb{R}: \ (x^2 - 2x + a < 0 \Rightarrow x > 0)$ true?

(A) 4

(B) 3

(C) 5

(D) 2

27. The truth values of statements A, B, C obey: A says B is false; B says C is false; C says exactly one of A, B is true. How many statements are true in the consistent assignment?

(A) 1

(B) 0

(C) 2

(D) 3

28. Boolean variables obey $A \Leftrightarrow (B \wedge C)$, $B \Leftrightarrow \neg D$, $C \Leftrightarrow (A \vee D)$, and D is true exactly when two of A, B, C are true. How many consistent assignments exist?

(A) 0

(B) 2

(C) 1

(D) 4

29. Assume $A \Leftrightarrow B$, $C \Leftrightarrow D$, $A \Rightarrow C$ and $E \Rightarrow F$. What is the least number of individual variables that must be asserted true to force all six true?

(A) 3

(B) 2

(C) 1

(D) 4

30. Assume $A \Leftrightarrow B$, $C \Leftrightarrow D$, $D \Leftrightarrow E$, $A \Rightarrow C$ and $F \Leftrightarrow G$. How many smallest sets of individual variables can be asserted true to force all seven true?

(A) 3

(B) 6

(C) 4

(D) 2

31. Independent P, Q, R each have equally likely truth values. Given $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$, find the probability that P is true.

(A) $\frac{1}{4}$

(B) $\frac{3}{4}$

(C) $\frac{1}{8}$

(D) $\frac{1}{2}$

32. Independent P, Q, R are true with probabilities $1/3, 1/2, 2/3$. Given $P \Rightarrow Q$, find the probability that $R \Leftrightarrow (P \vee Q)$ is true.

(A) $\frac{5}{9}$

(B) $\frac{2}{3}$

(C) $\frac{8}{15}$

(D) $\frac{4}{9}$

33. Which is equivalent to $[(P \vee Q) \wedge (\neg P \vee Q)] \vee [P \wedge \neg Q \wedge R]$?

(A) $Q \vee (P \wedge \neg R)$ (B) $Q \vee (P \wedge R)$

(C) $Q \wedge (P \vee R)$ (D) $(P \vee Q) \wedge R$

34. Let XOR mean odd parity. How many variables are essential in $(P \text{ XOR } Q) \text{ XOR } (Q \text{ XOR } R) \text{ XOR } (R \text{ XOR } S)$? A variable is essential if changing it alone can change the output.

(A) 1 (B) 2

(C) 3 (D) 4

35. On $U = \{1, 2, 3\}$, how many nonempty binary relations are symmetric and transitive but not reflexive on all of U?

(A) 10 (B) 6

(C) 14 (D) 9

36. On $U = \{1, 2, 3\}$, how many binary relations satisfy reflexivity, symmetry and transitivity, but fail $\forall x, y \in U : xRy$?

(A) 4 (B) 5

(C) 6 (D) 3

37. The real values of a making $\forall x \in \mathbb{R} : [x^2 - 4x + 3 \leq 0 \Rightarrow (x - a)^2 \leq 4]$ true form an interval. Find its length.

(A) 2 (B) 4

(C) 1 (D) 3

38. For how many integers $a \in [-3, 3]$ is $\forall x \in \mathbb{R} : [x^2 \leq 1 \Rightarrow x^2 - 2ax + 2a \geq 0]$ true?

(A) 2 (B) 5

(C) 3 (D) 4

39. Among the four implications $P \Rightarrow Q$, $Q \Rightarrow R$, $R \Rightarrow S$, $S \Rightarrow P$, exactly one is false. How many truth assignments to P, Q, R, S are possible?

(A) 12 (B) 8

(C) 4 (D) 14

40. For P, Q, R, S, T, exactly two of $P \Rightarrow Q$, $Q \Rightarrow R$, $R \Rightarrow S$, $S \Rightarrow T$ are false. How many assignments exist?

(A) 6 (B) 3

(C) 4 (D) 8

41–50 / Preserved hard practice

41. How many truth assignments to $P_1, \ldots, P_5$ make all five cyclic implications $P_i \Rightarrow P_{i+1}$ true, where $P_6 = P_1$?

(A) 5 (B) 2

(C) 10 (D) 1

42. How many truth assignments to P, Q, R, S make exactly one of $P \Rightarrow Q$, $Q \Rightarrow R$, $R \Rightarrow S$, $S \Rightarrow P$ false?

(A) 10 (B) 8

(C) 12 (D) 14

43. If $P \Leftrightarrow \neg Q$, $Q \Leftrightarrow R$ and $R \Rightarrow P$ are all true, then (P, Q, R) must be

(A) (T, T, F) (B) (F, F, T)

(C) (T, F, F) (D) (F, T, T)

44. How many truth assignments to P, Q, R, S satisfy $(P \vee Q) \wedge (Q \vee R) \wedge (R \vee S) \wedge (S \vee P) \wedge \neg(P \wedge R)$?

(A) 2 (B) 3

(C) 5 (D) 4

45. How many assignments to P, Q, R, S satisfy $P \Leftrightarrow (Q \vee R)$, $S \Leftrightarrow (Q \wedge R)$ and $P \neq S$ in truth value?

(A) 2 (B) 3

(C) 1 (D) 4

46. All eight assignments to P, Q, R are equally likely. Given that $P \Rightarrow Q$ and $Q \Rightarrow R$ are true, the probability that $P \Leftrightarrow R$ is true is

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{3}{4}$ (D) $\frac{1}{4}$

47. A truth table in three propositions P, Q, R has the property that complementing all three inputs reverses the output. If its output at (F, F, F) is fixed as F, how many such truth tables exist?

(A) 8 (B) 16

(C) 32 (D) 4

48. For which real a is the following statement true: "For every $\mathbf{b} \in \mathbb{R}^3$, there exists $\mathbf{x} \in \mathbb{R}^3$ satisfying $A\mathbf{x} = \mathbf{b}$", where $A = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}$?

(A) $a \in \{1, -2\}$ (B) $a \neq 1$ only

(C) $a \neq -2$ only (D) $a \notin \{1, -2\}$

49. For $f_a(x) = x^3 - 3ax$, the statement "for all real $x_1 < x_2$, $f_a(x_1) < f_a(x_2)$" is true precisely when

(A) $a \geq 0$ (B) $a < 0$

(C) $a > 0$ (D) $a \leq 0$

50. For a real parameter a, both statements hold: (i) for every real x there exists real y with $(y - x)^2 + a = 0$; (ii) $(y - x)^2 + a \geq 0$ for every real pair (x, y). Find a.

(A) 1 (B) 2

(C) -1 (D) 0

Answer key

Mathematical Reasoning

M-MREA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**A**	31	**A**	41	**B**
02	**B**	12	**B**	22	**D**	32	**C**	42	**C**
03	**C**	13	**D**	23	**A**	33	**B**	43	**C**
04	**C**	14	**C**	24	**A**	34	**B**	44	**B**
05	**D**	15	**A**	25	**B**	35	**D**	45	**A**
06	**B**	16	**C**	26	**A**	36	**A**	46	**B**
07	**D**	17	**D**	27	**C**	37	**A**	47	**A**
08	**C**	18	**D**	28	**B**	38	**D**	48	**D**
09	**C**	19	**B**	29	**B**	39	**A**	49	**D**
10	**A**	20	**B**	30	**C**	40	**A**	50	**D**

Chapter 10

Mathematical Induction

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. After assuming $\sum_{j=1}^{k} j^2 = k(k+1)(2k+1)/6$, adding $(k+1)^2$ simplifies to

(A) $k(k+1)(2k+3)/6$

(B) $(k+1)(k+2)(2k+1)/6$

(C) $(k+1)^2(k+2)/6$

(D) $(k+1)(k+2)(2k+3)/6$

02. The least positive integer N for which $2^n \geq n^2$ holds for every integer $n \geq N$ is

(A) 5

(B) 2

(C) 4

(D) 3

03. If $P(0)$ is known and $P(k) \Rightarrow P(k+2)$ is proved for every nonnegative integer k, the guaranteed indices are

(A) all nonnegative even integers

(B) only 0 and 2

(C) all nonnegative integers

(D) all positive odd integers

04. Knowing $P(1)$ and $P(k) \Rightarrow P(2k)$ for every positive integer k alone proves $P(n)$ for

(A) all odd positive integers

(B) every positive integer

(C) all multiples of 3

(D) powers of 2

05. Suppose $P(1), Q(1)$ are true, and for every $k \geq 1$ one proves $P(k) \Rightarrow Q(k+1)$ and $Q(k) \Rightarrow P(k+1)$. What follows?

(A) both $P(n)$ and $Q(n)$ for every positive integer n

(B) only $P(n)$ for odd n

(C) only $Q(n)$ for even n

(D) neither statement beyond $n = 1$

06. An invalid proof claims every group of n horses has one colour. For $k \geq 1$, its step compares two k-horse subgroups of a $(k+1)$-horse group, relying on overlap. At which smallest k is the required overlap absent?

(A) 2 (B) 0

(C) 1 (D) 3

07. The least positive integer N such that $n! > 3^n$ for every $n \geq N$ is

(A) 6 (B) 5

(C) 7 (D) 8

08. The largest positive integer dividing $25^n - 1$ for every positive integer n is

(A) 12 (B) 48

(C) 25 (D) 24

09. If $S_n = n(n+1)(n+2)/3$, then $S_{k+1} - S_k$ simplifies to

(A) $k(k+1)$ (B) $(k+1)^2$

(C) $(k+1)(k+2)$ (D) $3(k+1)(k+2)$

10. For $f(x) = 2x + 1$, induction on composition gives $f^{\circ n}(0) =$

(A) $2n - 1$ (B) n^2

(C) $2^n - 1$ (D) 2^n

11–40 / New middle-to-hard practice

11. $u_1 = 2$ and $u_{n+1} = 3u_n + 2$. Least positive n with $u_n > 1000$ is

(A) 6 (B) 9

(C) 8 (D) 7

12. $u_0 = 0$ and $u_{n+1} = 2u_n + 3^n$ for $n \geq 0$. Least n with $u_n > 500$ is

(A) 6 (B) 7

(C) 8 (D) 5

13. $a_1 = 1, a_2 = 4$ and $a_{n+2} - 2a_{n+1} + a_n = 2n + 2$ for $n \geq 1$. Find a_{10}.

(A) 385 (B) 330

(C) 1000 (D) 340

14. $a_0 = 0, a_1 = 1$ and $a_{n+2} - 2a_{n+1} + a_n = 6n + 6$ for $n \geq 0$. Least integer n with $a_n > 10^4$ is

(A) 20 (B) 22

(C) 23 (D) 21

15. Largest positive integer dividing $25^n - 1 - 24n$ for every integer $n \geq 1$ is

(A) 576 (B) 24

(C) 288 (D) 1152

16. Largest positive integer dividing $n^5 - n$ for every positive integer n is

(A) 20 (B) 30

(C) 60 (D) 10

17. If 2^r is the greatest power of two dividing $\prod_{k=1}^{32}(2k)$, find r.

(A) 31 (B) 63

(C) 64 (D) 32

18. $a_1 = 3$ and $a_{n+1} = a_n^2 - 2$. Greatest power of two dividing $a_{20} + 1$ is 2^r. Find r.

(A) 40 (B) 21

(C) 22 (D) 20

19. $a_0 = 0$ and $a_{n+1} = 2a_n + 1$. Remainder of a_{2026} upon division by 7 is

(A) 1 (B) 3

(C) 6 (D) 0

20. $a_1 = a_2 = 1$ and $a_{n+2} = a_{n+1} + a_n$. How many of $a_1, \ldots, a_{2026}$ are odd?

(A) 1013 (B) 1352

(C) 1351 (D) 1350

21. Least positive N for which $3^n > n^4$ holds for every integer $n \geq N$ is

(A) 1 (B) 9

(C) 7 (D) 8

22. Least positive N for which $n! > 5^n$ holds for every integer $n \geq N$ is

(A) 13 (B) 11

(C) 10 (D) 12

23. Only $P(0)$ is known. The available implications are $P(k) \Rightarrow P(k+4)$ and $P(k) \Rightarrow P(k+7)$ for nonnegative k. Number of indices $0 \leq n \leq 30$ guaranteed by repeated use of these implications is

(A) 22 (B) 21

(C) 31 (D) 23

24. Only $P(1)$ is known, with implications $P(k) \Rightarrow P(3k)$ and $P(k) \Rightarrow P(3k+2)$. Number of guaranteed positive indices at most 100 is

(A) 15 (B) 19

(C) 20 (D) 21

25. $P_0 = 1, Q_0 = 0$, $P_{n+1} = 2P_n + Q_n$ and $Q_{n+1} = P_n + 2Q_n$. Find P_6.

(A) 729 (B) 365

(C) 366 (D) 364

26. $u_0 = 2, v_0 = 1$, $u_{n+1} = u_n + 2v_n$ and $v_{n+1} = 2u_n + v_n$. Find u_4/v_4.

(A) $\frac{123}{121}$ (B) $\frac{41}{40}$

(C) $\frac{122}{121}$ (D) $\frac{121}{122}$

27. $x_1 = 1$ and $x_{n+1} = x_n/(1 + 2x_n)$. Find $\sum_{k=1}^{10} x_k x_{k+1}$.

(A) $\frac{5}{11}$ (B) $\frac{1}{2}$

(C) $\frac{20}{21}$ (D) $\frac{10}{21}$

28. $x_0 = 1$, $x_{n+1} > 0$, and $x_{n+1}^2 = x_n^2 + 2n + 3$. Find $\prod_{k=1}^{10}(x_k - 1)/(x_k + 1)$.

(A) $\frac{1}{11}$ (B) $\frac{1}{12}$

(C) $\frac{1}{132}$ (D) $\frac{1}{66}$

29. For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, the $(1, 2)$ entry of $A^6 - 4A^4$ is

(A) 192 (B) 128

(C) 64 (D) 32

30. Let $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = I + A$. The $(1, 2)$ entry of B^{2026} divided by 2^{1013} is

(A) 2 (B) 1

(C) -1 (D) 0

31. Find $\sum_{k=0}^{7}(-1)^k \binom{8}{k}(k + 1)(k + 2)k$.

(A) -720 (B) 0

(C) -504 (D) 720

32. Find $\sum_{k=0}^{9} \binom{9}{k} k/(k + 1)$.

(A) $\frac{1023}{10}$ (B) $\frac{4097}{10}$

(C) $\frac{4607}{10}$ (D) $\frac{511}{10}$

33. Let $P = \prod_{j=0}^{4}(1 + 2^{2^j})$. If $P + 1 = 2^r$, find r.

(A) 32 (B) 64

(C) 31 (D) 16

34. Let $P = \prod_{k=1}^{4} (k+1)^k / k^{k+1}$. Find $(4!)^2 P$.

(A) 625 (B) 3125

(C) 125 (D) 576

35. An arrangement of n lines has exactly one parallel pair, no other parallel lines and no three concurrent. It divides the plane into 91 regions. Number of intersection points is

(A) 90 (B) 77

(C) 78 (D) 91

36. All chords joining n points on a circle are drawn, with no three chords meeting inside. They divide the disc into 57 regions. Number of interior intersection points is

(A) 56 (B) 35

(C) 21 (D) 36

37. For positive integer n, $\sum_{k=1}^{n} k(k+1) = 440$. Find $\sum_{k=1}^{n} (2k-1)$.

(A) 81 (B) 100

(C) 110 (D) 121

38. For positive integer n, $\sum_{k=1}^{n} k^3 = 36 \sum_{k=1}^{n} k$. Find $\sum_{k=1}^{n} k^2$.

(A) 204 (B) 1296

(C) 140 (D) 285

39. Least real C for which $\sum_{k=1}^{n} 1/[k(k+1)] \le Cn/(n+2)$ for every positive integer n is

(A) 1 (B) 2

(C) $\frac{4}{3}$ (D) $\frac{3}{2}$

40. Least real C for which $\sum_{k=1}^{n} k^2 \le Cn(n+1)(n+2)$ for every positive integer n is

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{1}{6}$

41–50 / Preserved hard practice

41. If $\sum_{k=1}^{n} k^4 = an^5 + bn^4 + cn^3 + dn^2 + en$ for all positive integers, coefficient comparison in the induction step gives $d =$

(A) $-\frac{1}{12}$ (B) $\frac{1}{12}$

(C) $\frac{1}{6}$ (D) 0

42. The largest positive integer dividing $n(n+1)(n+2)(n+3)$ for every positive integer n is

(A) 48 (B) 24

(C) 120 (D) 12

43. The largest positive integer dividing $3^{2n+1} + 2^{n+2}$ for every positive integer n is

(A) 7 (B) 35

(C) 14 (D) 5

44. The least positive integer N such that $3^n > n^3$ for every $n \geq N$ is

(A) 5 (B) 4

(C) 2 (D) 3

45. Let $u_0 = 0, u_1 = 1$ and $u_{n+2} = 6u_{n+1} - u_n$. The remainder of u_{2026} on division by 8 is

(A) 6 (B) 2

(C) 0 (D) 4

46. For eight independent events each having probability $p \in [0, 1]$, the probability that none occurs is $(1-p)^8$. The least constant c for which $(1-p)^8 \geq 1 - cp$ holds for every such p is

(A) 8 (B) 16

(C) 9 (D) 7

47. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Induction on n gives the $(1, 2)$ entry of $\sum_{k=0}^{n} A^k$ as

(A) $(n+1)2^n - 1$ (B) $(n-1)2^n + 1$

(C) $2^n - 1$ (D) $n2^{n-1}$

48. Using Pascal identity and induction, $\sum_{k=3}^{12} \binom{k}{3}$ equals

(A) 924 (B) 792

(C) 715 (D) 495

49. A function satisfies $f(0) = 0$ and $f(x+1) - f(x) = 2x + 1$ for all real x. Evaluate $\sum_{n=1}^{10} 1/[f(n) + \sqrt{f(n)}]$.

(A) $\frac{1}{11}$ (B) $\frac{11}{12}$

(C) $\frac{9}{10}$ (D) $\frac{10}{11}$

50. Let α, β be the roots of $x^2 - 3x + 1 = 0$, and set $v_n = \alpha^{2^n} + \beta^{2^n}$. Find v_3.

(A) 2209 (B) 2207

(C) 2303 (D) 2305

Answer key

Mathematical Induction

M-MIND / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**D**	21	**D**	31	**A**	41	**D**
02	**C**	12	**A**	22	**D**	32	**B**	42	**B**
03	**A**	13	**D**	23	**A**	33	**A**	43	**A**
04	**D**	14	**B**	24	**C**	34	**A**	44	**B**
05	**A**	15	**A**	25	**B**	35	**B**	45	**A**
06	**C**	16	**B**	26	**C**	36	**B**	46	**A**
07	**C**	17	**B**	27	**D**	37	**B**	47	**B**
08	**D**	18	**B**	28	**D**	38	**A**	48	**C**
09	**C**	19	**A**	29	**C**	39	**D**	49	**D**
10	**C**	20	**C**	30	**C**	40	**C**	50	**B**

Chapter 11

Linear Inequalities

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For which real a does $ax + 1 > 0$ hold for every $x \geq 0$?

(A) $a \geq 0$ (B) $a < 0$

(C) $a > 0$ (D) $a \leq 0$

02. Solve $|x - 1| + |x + 1| \leq 4$.

(A) $(-2, 2)$ (B) $[-2, 2]$

(C) $(-\infty, -2] \cup [2, \infty)$ (D) $[-1, 1]$

03. The feasible region $x, y \geq 0$, $x + y \leq 4$, $x + 2y \leq 6$ has how many vertices?

(A) 3 (B) 6

(C) 4 (D) 5

04. The bounded region $x, y \geq 0$, $x + 2y \leq 4$ has area

(A) 8 (B) 6

(C) 2 (D) 4

05. The real numbers for which exactly one of $2x > 3$ and $x < 1$ holds form

(A) $(-\infty, 1) \cup (3/2, \infty)$ (B) $[1, 3/2]$

(C) $(-\infty, 1] \cup [3/2, \infty)$ (D) $(1, 3/2)$

06. Every solution of $|x - a| \leq 2$ must lie in $[0, 5]$. The necessary and sufficient condition on a is

(A) $2 \leq a \leq 3$ (B) $2 < a < 3$

(C) $-2 \leq a \leq 7$ (D) $0 \leq a \leq 5$

07. The intervals $(a, a+3)$ and $[1, 2]$ have nonempty intersection precisely when

(A) $-2 \le a \le 2$ (B) $-2 < a < 2$

(C) $a \le -2$ or $a \ge 2$ (D) $-3 < a < 1$

08. The inequality $1 < \log_2(x+1) \le 3$ is equivalent to

(A) $0 < x \le 7$ (B) $1 < x \le 7$

(C) $2 < x \le 8$ (D) $1 \le x < 7$

09. The real domain of $f(x) = \sqrt{2x-1} + \sqrt{5-x}$ is

(A) $(1/2, 5)$ (B) $[1/2, \infty)$

(C) $(-\infty, 5]$ (D) $[1/2, 5]$

10. Let $A = \{x \in \mathbb{Z} : -5 \le x \le 5,\ 3x < 8\}$ and $B = \{x \in \mathbb{Z} : 2x \ge -3\}$. Find $|A \cap B|$.

(A) 5 (B) 8

(C) 4 (D) 3

11–40 / New middle-to-hard practice

11. Let $a = k/5$, with integer $-10 \le k \le 10$. For how many k is $(a-1)x + 2a + 1 \ge 0$ true for every $x \in [-2, 3]$?

(A) 8 (B) 9

(C) 7 (D) 10

12. Let $a = k/3$, where integer $-6 \le k \le 9$. For how many k is $(a-2)x + 3a + 1 > 0$ true for every $0 \le x \le 4$?

(A) 6 (B) 5

(C) 7 (D) 8

13. An integer $a \in [-3, 6]$ is retained if every real solution of $|2x - a| \le 3$ also satisfies $|x - 1| \le 2$. Find the sum of retained a.

(A) 9 (B) 12

(C) 6 (D) 3

14. Let $a = k/2$ with integer $-6 \le k \le 6$. How many a make every solution of $|x - a| < 2$ satisfy $|2x + 1| \le 7$?

(A) 7 (B) 5

(C) 8 (D) 6

15. Let $a = k/2$, where k is an integer. The intersection of $[a, a+3]$ with $[1, 5]$ has length at least 2, but $[a, a+3]$ does not contain 2. How many a qualify?

(A) 3 (B) 2

(C) 5 (D) 4

16. Let $a = k/2$, with integer $-10 \le k \le 10$. For how many a does $(a, a+4) \cap [-1, 2]$ contain exactly one integer?

(A) 4 (B) 5

(C) 3 (D) 6

17. How many integers in $[-5, 5]$ satisfy exactly one of $|x-1| < 3$ and $|2x+1| \le 5$?

(A) 4 (B) 3

(C) 5 (D) 6

18. How many integers in $[-5, 5]$ satisfy exactly two of $2x \ge -3$, $3x < 8$, and $|x-1| > 2$?

(A) 8 (B) 9

(C) 10 (D) 11

19. How many half-integers $x = k/2$ satisfy $(|x-2|-1)/(|x-2|-3) \le 0$?

(A) 6 (B) 8

(C) 10 (D) 9

20. How many integers $-5 \le x \le 5$ satisfy $(|2x+1|-2)/(|2x+1|-5) > 0$?

(A) 9 (B) 6

(C) 8 (D) 7

21. For a real parameter a, the solution interval of $|x-1| + 2|x-3| \le a$ has length 4. Find a.

(A) 5 (B) 6

(C) 8 (D) 4

22. An integer k is retained if $|x+2| + |x-2| + |x-5| \le k$ holds for every $x \in [-1, 3]$ but fails at $x = 6$. Find the sum of retained k.

(A) 33 (B) 30

(C) 39 (D) 36

23. Let $a = k/3$ with integer $-6 \le k \le 6$. How many a allow some real $x \in [-1, 2]$ satisfying both $2x + a > 0$ and $x - a < 1$?

(A) 8 (B) 9

(C) 7 (D) 10

24. For each real a such that $|x-a| \le 1$, $|2x+1| \le 3$, and $x \ge a/2$ have exactly one common solution, retain a. Find the product of retained a.

(A) 0 (B) 4

(C) -4 (D) -2

25. For how many integers $a \in \{1, \ldots, 12\}$ does $|3x - a| < 4$ have exactly three integer solutions x?

(A) 4 (B) 3

(C) 6 (D) 8

26. Let $a = k/2$, with integer $0 \le k \le 20$. For how many a does $|2x - a| \le 5$ have exactly six integer solutions, all nonnegative?

(A) 5 (B) 3

(C) 2 (D) 4

27. How many ordered pairs of nonnegative integers (x, y) satisfy $2x+y \le 8$, $x+2y \le 7$, and $x+y \ge 3$?

(A) 11 (B) 10

(C) 9 (D) 8

28. Over all ordered pairs of positive integers (x, y) satisfying $|x - y| \le 2$ and $x + 2y \le 12$, find the sum of their x coordinates.

(A) 39 (B) 42

(C) 41 (D) 40

29. Let $a = k/4$, where integer $0 \le k \le 12$. For how many a does $1 < (x + a)/(x + 1) \le 2$ hold for every $x > 0$?

(A) 5 (B) 3

(C) 6 (D) 4

30. For how many integers a does $1 \le (x + a)/(2x - 1) \le 2$ hold for every $x \in [4.9, 5.1]$?

(A) 8 (B) 10

(C) 9 (D) 7

31. For an integer a, every solution of $|x-a| \le 1$ must satisfy both $\log_2(x-2) > 0$ and $\log_2(10-x) \ge 1$. Find the sum of all such a.

(A) 12 (B) 15

(C) 18 (D) 21

32. For how many integers a is every real solution of $|2x-a| < 3$ in the real domain of $\sqrt{4x-1}+\sqrt{9-2x}$?

(A) 4 (B) 5

(C) 3 (D) 2

33. Let $a = k/5$, where integer $0 \le k \le 10$. For how many a is $(a+\varepsilon)x + (a-2\varepsilon) \ge 1$ true for every $x \in [0, 3]$ and every $|\varepsilon| \le 1/5$?

(A) 6 (B) 4

(C) 5 (D) 3

34. Positive x, y have $x + y = 10$ and $|x - y| \le 2$. Find the least real a for which $ax + y \ge 18$ holds for every such pair.

(A) 2 (B) 3

(C) $\frac{7}{2}$ (D) $\frac{5}{2}$

35. For $a_n = an + b$ with positive integer a, suppose $a_3 = 7$ and exactly six terms with $1 \le n \le 20$ satisfy $10 < a_n \le 28$. Find a_{20}.

(A) 58 (B) 55

(C) 60 (D) 61

36. For a positive integer $1 \le a \le 6$, let $u_n = an - 2$ for $n = 1, 2, \ldots$. Exactly four terms satisfy $5 < u_n \le 30$. Find u_{10}.

(A) 48 (B) 60

(C) 62 (D) 58

37. Choose a, b independently and uniformly from $\{1, \ldots, 6\}$. Given that the interval $[a - b, a + b]$ contains 0, what is the probability it does not contain 6?

(A) $\frac{1}{3}$ (B) $\frac{2}{7}$

(C) $\frac{3}{7}$ (D) $\frac{1}{6}$

38. Choose a, b independently and uniformly from $\{1, \ldots, 5\}$. Find the probability that $ax + b < 0$ holds at $x = -2$ but fails at $x = -1$.

(A) $\frac{2}{5}$ (B) $\frac{8}{25}$

(C) $\frac{11}{25}$ (D) $\frac{9}{25}$

39. Find the least real M for which $|x - 1| \le M$ and $2|x - 4| \le M$ have a common real solution.

(A) $\frac{5}{2}$ (B) $\frac{3}{2}$

(C) 3 (D) 2

40. Find the least real M for which $|x + 1| \le M$, $2|x - 2| \le M$, and $3|x - 5| \le M$ have a common real solution.

(A) $\frac{9}{2}$ (B) 4

(C) 3 (D) 5

41–50 / Preserved hard practice

41. How many integer points (x, y) satisfy $x, y \geq 0$, $x + 2y \leq 8$ and $2x + y \leq 8$?

(A) 16 (B) 17

(C) 18 (D) 19

42. How many integers $a \in [-10, 10]$ allow an integer x satisfying $a < x \leq 2a + 1$, $x \leq 4$ and $x > -2$?

(A) 5 (B) 6

(C) 3 (D) 4

43. The region $x \geq 1$, $y \geq 2$, $x + y \leq a$ has positive area 2. Find the real parameter a.

(A) 1 (B) 3

(C) 5 (D) 7

44. How many nonnegative integer pairs satisfy $x + 2y \leq 10$ and $x + y \geq 6$?

(A) 14 (B) 16

(C) 17 (D) 15

45. The least positive integer k for which $y > 2x + 1$, $y \leq kx - 1$, $0 \leq x \leq 2$ have a common real solution is

(A) 4 (B) 3

(C) 2 (D) 5

46. The system $\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \binom{x}{y} = \binom{a}{3}$ must have $x \geq 0, y > 0, x + y \leq 6$. How many integer a satisfy these conditions?

(A) 4 (B) 7

(C) 5 (D) 6

47. Let $\alpha < \beta$ be the roots of $x^2 - 5x + 3 = 0$. How many integers k satisfy $\alpha < 2k + 1 \leq \beta$?

(A) 4 (B) 3

(C) 1 (D) 2

48. The total length of the solution set in $[0, \pi]$ of $1 \leq 2\sin x \leq \sqrt{3}$ is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

49. A point (X, Y) is chosen uniformly from $[0, 2] \times [0, 2]$. The probability that $X + Y \leq 2$ and $X + 2Y \leq 3$ is

(A) $\frac{5}{16}$ (B) $\frac{3}{8}$

(C) $\frac{1}{2}$ (D) $\frac{7}{16}$

50. For $x, y \geq 0$ with $3x + 2y \leq 18$ and $x + 3y \leq 15$, the maximum of $2x + 5y$ is

(A) 27 (B) 25

(C) $\frac{183}{7}$ (D) $\frac{180}{7}$

Answer key

Linear Inequalities

M-LINQ / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**B**	21	**B**	31	**C**	41	**B**
02	**B**	12	**A**	22	**A**	32	**C**	42	**D**
03	**C**	13	**C**	23	**A**	33	**B**	43	**C**
04	**D**	14	**A**	24	**C**	34	**B**	44	**D**
05	**A**	15	**B**	25	**A**	35	**A**	45	**A**
06	**A**	16	**A**	26	**B**	36	**D**	46	**C**
07	**B**	17	**B**	27	**C**	37	**B**	47	**D**
08	**B**	18	**C**	28	**D**	38	**D**	48	**C**
09	**D**	19	**B**	29	**D**	39	**D**	49	**D**
10	**C**	20	**D**	30	**A**	40	**A**	50	**C**

Chapter 12

Limits

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Evaluate $\lim_{x\to\infty}(\sqrt{x^2+3x}-x)$.

(A) $\frac{7}{2}$ (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{5}{2}$

02. Evaluate $\lim_{x\to 0^+} x\lfloor 1/x\rfloor$.

(A) 2 (B) 0

(C) -1 (D) 1

03. The two-sided limit $\lim_{x\to 0}(|x|+x)/x$ is

(A) 0 (B) nonexistent because the left limit is 0 and the right limit is 2

(C) 1 (D) 2

04. Evaluate $\lim_{x\to 0^+} \sin x/\sqrt{1-\cos x}$.

(A) $-1+\sqrt{2}$ (B) $1+\sqrt{2}$

(C) $\sqrt{2}$ (D) $\sqrt{2}+2$

05. Evaluate $\lim_{x\to 0}(\sin x - x\cos x)/x^3$.

(A) $-\frac{2}{3}$ (B) $\frac{4}{3}$

(C) $\frac{7}{3}$ (D) $\frac{1}{3}$

06. Evaluate $\lim_{x\to 0}(\tan x - \sin x)/x^3$.

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$

(C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

07. Evaluate $\lim_{x\to 0}(\ln(1+x)-x)/x^2$.

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

08. Evaluate $\lim_{x\to\infty} x(\sqrt{x^2+1}-x)$.

(A) $-\frac{1}{2}$ (B) $\frac{3}{2}$

(C) $\frac{5}{2}$ (D) $\frac{1}{2}$

09. The two-sided limit $\lim_{x\to 2}\lfloor x\rfloor$ is

(A) 1 (B) 2

(C) nonexistent because the left and right limits are 1 and 2 (D) 3/2

10. If $\lim_{x\to 2}(x^2+kx-6)/(x-2)$ is finite, its value is

(A) 6 (B) 4

(C) 3 (D) 5

11–40 / New middle-to-hard practice

11. For $a>0$, $\lim_{x\to 0}(e^{ax}-1-bx)/x^2 = 9/2$. Find $a+b$.

(A) -6 (B) 9

(C) 6 (D) 3

12. For $a<0$, $\lim_{x\to 0}(\ln(1+ax)-bx)/x^2 = -8$. Find $a+b$.

(A) -8 (B) 8

(C) 4 (D) -4

13. Evaluate $\lim_{x\to 0}[1-\cos(\sin 2x)-2x^2]/x^4$.

(A) $-\frac{10}{3}$ (B) $-\frac{8}{3}$

(C) $\frac{10}{3}$ (D) $-\frac{2}{3}$

14. Evaluate $\lim_{x\to 0}[\sin(\tan x)-\tan(\sin x)]/x^5$.

(A) 0 (B) $-\frac{1}{30}$

(C) $\frac{1}{6}$ (D) $\frac{1}{30}$

15. Evaluate $\lim_{x\to\infty} x[\sqrt{x^2+3x+1}-x-3/2]$.

(A) $-\frac{5}{8}$ (B) $-\frac{9}{8}$

(C) $\frac{5}{8}$ (D) $\frac{1}{2}$

16. Evaluate $\lim_{x\to\infty} x^2[\sqrt{x^2+2x+2}-\sqrt{x^2+2x+1}-1/(2x)]$.

(A) $\frac{1}{2}$ (B) -1

(C) $-\frac{1}{2}$ (D) $\frac{1}{4}$

17. Evaluate $\lim_{x\to0}[(1+x)^{1/x}e^{x/2-1}-1]/x^2$.

(A) $\frac{11}{24}$ (B) $\frac{1}{6}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

18. Evaluate $\lim_{n\to\infty} n[(1+1/n)^{n+2}-e]/e$.

(A) 2 (B) $\frac{3}{2}$

(C) $\frac{1}{2}$ (D) $-\frac{3}{2}$

19. For positive integers n, let $a_n=\sqrt{n^2+4n+3}$. Evaluate $\lim_{n\to\infty} n[1-(a_n-\lfloor a_n\rfloor)]$.

(A) $\frac{1}{4}$ (B) 0

(C) $\frac{1}{2}$ (D) 1

20. For positive integers n, let $b_n=\sqrt{n^2+4n+5}$. Evaluate $\lim_{n\to\infty} n(b_n-\lfloor b_n\rfloor)$.

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$

(C) 0 (D) 1

21. Evaluate $\lim_{x\to0}[\sqrt{x^2+x^4}-|x|]/|x|^3$.

(A) 0 (B) $\frac{1}{2}$

(C) $-\frac{1}{2}$ (D) 1

22. For $f(x)=[\sqrt{x^2+x^3}-x]/x^2$ near 0, which gives $(\lim_{x\to0^+} f(x), \lim_{x\to0^-} f(x))$?

(A) $(1/2,-\infty)$ (B) $(1/2,+\infty)$

(C) $(1/2,1/2)$ (D) $(-1/2,+\infty)$

23. For real a, b, $\lim_{x\to\infty}[(2x^3+3x^2+5)/(x^2+x+1)-ax-b]=4$. Find ab.

(A) -6 (B) 6

(C) 2 (D) -3

24. For real a, b, $\lim_{x\to-\infty}[\sqrt{4x^2+3x+1}-ax-b]=2$. Find $a+b$.

(A) $\frac{19}{4}$ (B) $-\frac{19}{4}$

(C) $-\frac{11}{4}$ (D) $-\frac{3}{4}$

25. Evaluate $\lim_{x\to0}[\ln(1+x)-\sin x+x^2/2]/x^3$.

(A) $-\frac{1}{2}$ (B) $\frac{1}{6}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

26. Evaluate $\lim_{x\to 0}[\ln(\cos x)+x^2/2]/x^4$.

(A) $\frac{1}{12}$ (B) $-\frac{1}{24}$

(C) $-\frac{1}{12}$ (D) $-\frac{1}{8}$

27. Evaluate $\lim_{n\to\infty} n\sum_{k=1}^{n} 1/[(n+k)(n+k+1)]$.

(A) 1 (B) $\frac{1}{4}$

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

28. Evaluate $\lim_{n\to\infty} n^2\sum_{k=1}^{n} 1/[(n+k)(n+k+1)(n+k+2)]$.

(A) $\frac{3}{8}$ (B) $\frac{1}{2}$

(C) $\frac{1}{8}$ (D) $\frac{3}{4}$

29. Evaluate $\lim_{n\to\infty}\sum_{k=1}^{2n}(2k-1)/[n^2+(2k-1)^2]$.

(A) $\frac{\log(17)}{4}$ (B) $\frac{\log(17)}{2}$

(C) $\log(17)$ (D) $\frac{\log(5)}{4}$

30. Evaluate $\lim_{n\to\infty}[\prod_{k=1}^{n}(1+k/n)]^{1/n}$.

(A) 2 (B) $\frac{2}{e}$

(C) $\frac{4}{e}$ (D) $4e$

31. Let g be the inverse of $f(x)=x+x^3$. Evaluate $\lim_{y\to 0}[g(y)-y]/y^3$.

(A) -1 (B) 0

(C) -3 (D) 1

32. Let g be the inverse on $[0,\infty)$ of $f(x)=x^2+x^3$. Evaluate $\lim_{y\to 0^+}[g(y)-\sqrt{y}]/y$.

(A) -1 (B) $-\frac{1}{2}$

(C) 0 (D) $\frac{1}{2}$

33. Let $h=x-\pi/4$. Evaluate $\lim_{x\to\pi/4}[(\tan x)^{\sin 2x}-1-2h-2h^2]/h^3$.

(A) $-\frac{8}{3}$ (B) $-\frac{4}{3}$

(C) $\frac{8}{3}$ (D) $\frac{4}{3}$

34. Evaluate $\lim_{x\to 1}[\arctan\sqrt{x}-\arctan x+(x-1)/4]/(x-1)^2$.

(A) $\frac{1}{16}$ (B) $-\frac{1}{8}$

(C) $\frac{1}{4}$ (D) $\frac{1}{8}$

35. For $a,b>0$, the limits of $(a^x-b^x)/x$ and $(a^x+b^x-2)/x$ as $x\to 0$ are $\ln 4$ and $\ln 9$. Find $a+b$.

(A) $\frac{15}{2}$ (B) $\frac{21}{2}$

(C) $\frac{9}{2}$ (D) 9

36. For $a, b > 0$, the limits of $[\ln(1+ax) - \ln(1+bx)]/x$ and $[\ln(1+ax) + \ln(1+bx)]/x$ are 2 and 4. Find $\lim_{x\to 0}[\ln(1+ax) - \ln(1+bx) - 2x]/x^2$.

(A) -2 (B) -4

(C) 4 (D) -8

37. What is the set of all finite subsequential limits of $x^2 \sin(1/x)/(1-\cos x)$ as $x \to 0$?

(A) $[-1, 1]$ (B) $\{-2, 2\}$

(C) $\mathbb{R}$ (D) $[-2, 2]$

38. What is the set of all finite subsequential limits of $[x \sin(1/x)]^2/(1-\cos x)$ as $x \to 0$?

(A) $[0, 1]$ (B) $\{0, 2\}$

(C) $[0, 2]$ (D) $[-2, 2]$

39. Let $a_1 = 2$ and $a_{n+1} = a_n/(1+a_n)$. Evaluate $\lim_{n\to\infty} n^2(a_n - 1/n)$.

(A) $\frac{1}{2}$ (B) 0

(C) $-\frac{1}{2}$ (D) 1

40. Let $a_1 = 2$, $a_n > 0$ and $a_{n+1}^2 = a_n^2 + 2n + 1$. Evaluate $\lim_{n\to\infty} n(a_n - n)$.

(A) $\frac{9}{2}$ (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) 3

41–50 / Preserved hard practice

41. Let $a > 0$ and b be real. If $\lim_{x\to 0}(e^{ax} - 1 - bx)/x^2 = 2$, find $a + b$.

(A) 4 (B) 6

(C) 3 (D) 2

42. A positive integer a satisfies $\lim_{x\to 0}(\sin ax - a \sin x)/x^3 = -4$. Find a.

(A) 2 (B) 5

(C) 4 (D) 3

43. Evaluate $\lim_{x\to 0}(\sqrt{1+x} - \sqrt{1-x} - x)/x^3$.

(A) $\frac{1}{8}$ (B) $\frac{9}{8}$

(C) $\frac{17}{8}$ (D) $-\frac{7}{8}$

44. Evaluate $\lim_{n\to\infty} n\big((1+1/n)^n/e - 1\big)$.

(A) $-\frac{3}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{1}{2}$

45. Evaluate $\lim_{x\to 0}(\ln(1+x)-\ln(1-x)-2x)/x^3$.

(A) $-\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{5}{3}$ (D) $\frac{8}{3}$

46. Define $f(0)=a$, $f(x)=(e^{2x}-1)/x$ for $x<0$, and $f(x)=(\sqrt{1+bx}-1)/x$ for $x>0$. If f is continuous at 0, find $a+b$.

(A) 4 (B) 2

(C) 8 (D) 6

47. Evaluate $\displaystyle\lim_{x\to 0}\frac{\int_0^x \ln(1+t)\,dt - x^2/2}{x^3}$.

(A) $-\frac{1}{6}$ (B) $\frac{5}{6}$

(C) $\frac{11}{6}$ (D) $-\frac{7}{6}$

48. Evaluate $\lim_{x\to 0}(\cos x)^{1/x^2}$.

(A) $-1+e^{-\frac{1}{2}}$ (B) $e^{-\frac{1}{2}}$

(C) $e^{-\frac{1}{2}}+1$ (D) $e^{-\frac{1}{2}}+2$

49. Evaluate $\displaystyle\lim_{x\to 0}\frac{\det\begin{pmatrix}\sin x & x\\ x & \sin x\end{pmatrix}}{x^4}$.

(A) $\frac{5}{3}$ (B) $\frac{2}{3}$

(C) $-\frac{4}{3}$ (D) $-\frac{1}{3}$

50. A positive integer n satisfies $\lim_{x\to 0}((1+x)^n-(1-x)^n-2nx)/x^3=40$. Find n.

(A) 5 (B) 4

(C) 6 (D) 7

Answer key

Limits

M-LIMI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**C**	21	**B**	31	**A**	41	**A**
02	**D**	12	**A**	22	**B**	32	**B**	42	**D**
03	**B**	13	**A**	23	**A**	33	**B**	43	**A**
04	**C**	14	**A**	24	**B**	34	**D**	44	**B**
05	**D**	15	**A**	25	**C**	35	**A**	45	**B**
06	**D**	16	**C**	26	**C**	36	**B**	46	**D**
07	**B**	17	**D**	27	**D**	37	**D**	47	**A**
08	**D**	18	**B**	28	**A**	38	**C**	48	**B**
09	**C**	19	**C**	29	**A**	39	**A**	49	**D**
10	**D**	20	**B**	30	**C**	40	**C**	50	**C**

Chapter 13

Continuity & Differentiability

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. If $f(x) = (x^2+1)/(x+1)$, find $f''(1)$.

(A) $\frac{5}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{1}{2}$

02. If $y = x^2 y + 1$, find dy/dx at $x = 1/2$.

(A) $\frac{25}{9}$ (B) $\frac{34}{9}$

(C) $\frac{7}{9}$ (D) $\frac{16}{9}$

03. For $x = t^2 + 1$ and $y = t^3 - t$, find dy/dx at $t = 2$.

(A) $\frac{19}{4}$ (B) $\frac{11}{4}$

(C) $\frac{7}{4}$ (D) $\frac{15}{4}$

04. For $f(0) = 0$ and $f(x) = x\sin(1/x)$ for $x \neq 0$, $f'(0)$

(A) equals 0 (B) does not exist

(C) equals 1 (D) equals -1

05. If $f(x) = e^x \sin x$, find $f''(0)$.

(A) 3 (B) 2

(C) 4 (D) 1

06. If $f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases}$ find $f'(0)$.

(A) -1 (B) 0

(C) 2 (D) 1

07. For $f(0)=0$ and $f(x)=x^2\sin(1/x)$ for $x\neq 0$, which is true?

(A) f is discontinuous at 0

(B) $f\prime$ is continuous at 0

(C) $f\prime(0)$ exists, but $f\prime$ is discontinuous at 0

(D) $f\prime(0)$ does not exist

08. For $f(x)=|x-a|+|x-b|$ with distinct real a,b, how many real points of nondifferentiability are there?

(A) 2

(B) 4

(C) 3

(D) 1

09. Let $f(x)=x^2+3x+1$ and let g be its local inverse near $x=1$. Find $g\prime(5)$.

(A) $\frac{11}{5}$

(B) $\frac{6}{5}$

(C) $-\frac{4}{5}$

(D) $\frac{1}{5}$

10. Let $f(x)=(1+x)^8$. The coefficient of x^2 in $f\prime(x)$ is

(A) 170

(B) 169

(C) 167

(D) 168

11–40 / New middle-to-hard practice

11. Let $f(x)=ax^2+bx+1$ for $x\leq 1$ and $f(x)=x^3+x+2$ for $x>1$. If f is differentiable at 1, find the right second derivative minus the left second derivative there.

(A) 4

(B) 0

(C) 6

(D) 2

12. For $x>0$, let $f(x)=a\ln x+b$ when $x\leq 1$ and $f(x)=x^2+cx$ when $x>1$. It is differentiable at 1 and $f(1)=3$. Find $f\prime(1/e)$.

(A) $2e$

(B) $\frac{4}{e}$

(C) $4e$

(D) $3e$

13. For a positive integer n define $f(0)=0$, $f(x)=x^n\sin(1/x^2)$ for $x\neq 0$. Find the least n for which f has a continuous first derivative at 0.

(A) 2

(B) 3

(C) 4

(D) 5

14. For positive integer n, let $f(0)=0$ and $f(x)=x^n\sin(1/x)$ otherwise. For which n is f twice differentiable at 0 but its second derivative not continuous there?

(A) 5

(B) 4

(C) 2

(D) 3

15. Let g be the inverse of $f(x) = x^3 + x$. Find $g''(2)$.

(A) $-\frac{1}{4}$ (B) $\frac{3}{32}$

(C) $-\frac{3}{8}$ (D) $-\frac{3}{32}$

16. Let g be the inverse of $f(x) = e^x + x$. Find $g''(1)$.

(A) $-\frac{1}{8}$ (B) $\frac{1}{8}$

(C) $-\frac{1}{4}$ (D) $-\frac{1}{2}$

17. The curve $x^2 + xy + y^2 = 7$ defines y near $(1, 2)$. Find d^2y/dx^2 there.

(A) $-\frac{21}{125}$ (B) $\frac{42}{125}$

(C) $-\frac{42}{125}$ (D) $-\frac{6}{25}$

18. Near $(4, -2)$, y satisfies $x\sqrt{4+y} + y\sqrt{4+x} = 0$. Find d^2y/dx^2 at that point.

(A) $\frac{1}{32}$ (B) $-\frac{1}{16}$

(C) $\frac{1}{16}$ (D) $\frac{1}{8}$

19. For $x = t^2 + t$ and $y = t^3 - t$, find d^2y/dx^2 at $t = 1$.

(A) $\frac{14}{27}$ (B) $\frac{2}{3}$

(C) $\frac{7}{27}$ (D) $\frac{14}{9}$

20. For $x = a\cos^3 t$, $y = a\sin^3 t$, $a > 0$, consider its first-quadrant point $(1, 1)$. Find d^2y/dx^2 there.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $-\frac{2}{3}$ (D) $\frac{4}{3}$

21. Define $f(x) = (1 + x)^{1/x}$ near 0 for $x \neq 0$, and $f(0) = e$. Find $f''(0)/e$.

(A) $\frac{11}{24}$ (B) $\frac{2}{3}$

(C) $\frac{5}{6}$ (D) $\frac{11}{12}$

22. For $f(x) = x^{\sin x}$, $x > 0$, find $f''(\pi/2)/f(\pi/2)$.

(A) $\frac{-\log\left(\left(\frac{\pi}{2}\right)^{\pi^2}\right)-4}{\pi^2}$ (B) $\frac{4-\log\left(\left(\frac{\pi}{2}\right)^{\pi^2}\right)}{\pi^2}$

(C) $-\log(\pi) + \log(2)$ (D) $\log\left(\frac{\pi}{2}\right)$

23. A differentiable nonzero function satisfies $f(x+y) = f(x)f(y)$ for all real x, y, with $f'(0) = 3$. Find $f'(\ln 2)$.

(A) 6 (B) 12

(C) 24 (D) 8

24. A twice differentiable function satisfies $f(x) + 2f(-x) = x^3 + x$ for all real x. Find $f''(1)$.

(A) 0 (B) -6

(C) 6 (D) -3

25. Near $x = -1$, a twice differentiable function satisfies $f(f(x)) = 4x + 3$, $f(-1) = -1$ and $f'(-1) < 0$. Find $f''(-1)$.

(A) 2 (B) 1

(C) -2 (D) 0

26. Let $g(x) = f(x^2)$ with $g'(1) = 8$ and $g''(1) = 20$. For $h(x) = f(1/x)$ near 1, find $h''(1)$.

(A) 11 (B) 3

(C) 7 (D) 14

27. How many real points of nondifferentiability does $f(x) = |x^2 - 1| - 2|x - 1|$ have?

(A) 3 (B) 1

(C) 0 (D) 2

28. The function $f(x) = |x^2 - 1| - 2|x - 1| + a|x + 1|$ is differentiable everywhere. Find $f(0)$.

(A) -1 (B) -5

(C) -3 (D) 1

29. For $f(x) = \arccos((1 - x^2)/(1 + x^2))$, find $f''(-1)$.

(A) 1 (B) -2

(C) -1 (D) 0

30. For $x > 0$, let $f(x) = \arctan((\sqrt{x} - x)/(1 + x^{3/2}))$. Find $f''(1)$.

(A) $-\frac{1}{4}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

31. For $f(x) = (1 + x)/(1 - x + x^2)$, find $f^{(2026)}(0)/2026!$.

(A) 1 (B) 2

(C) -1 (D) -2

32. For $f(x) = e^x \sin x$, find $f^{(2025)}(0)/2^{1012}$.

(A) -1 (B) 0

(C) 1 (D) 2

33. For $x \neq 0$, $f(x) = (e^{ax} - 1 - ax)/x^2$, while $f(0) = b$. It is differentiable at 0 and $f'(0) = 4/3$. Find $a + b$.

(A) 3 (B) 4

(C) 6 (D) 2

34. For real a, define $f(x) = (\sin(ax) - a\sin x)/x^3$ for $x \neq 0$ and $f(0) = -1$. If f is continuous at 0, find $f''(0)$.

(A) $\frac{1}{2}$ (B) 1

(C) $-\frac{1}{2}$ (D) $\frac{1}{4}$

35. For $F(x) = \int_x^{x^2} \ln(1+t^2)\,dt$, find $F''(1)$.

(A) $2\log(2) + 4$ (B) $\log(2) + 3$

(C) $1 + \log(4)$ (D) $\log(4) + 3$

36. A twice differentiable function satisfies $f(x) + \int_0^x f(t)\,dt = e^{2x}$. Find $f''(0)$.

(A) 3 (B) 4

(C) 2 (D) 1

37. For $F(x) = \sum_{k=0}^{10}(k+1)x^k$, find $F''(1)/F'(1)$.

(A) $\frac{27}{4}$ (B) $\frac{9}{2}$

(C) $\frac{29}{4}$ (D) $\frac{27}{2}$

38. Let $P(x) = \prod_{k=1}^{5}(x-k)$. Find $P''(1)/P'(1)$.

(A) $-\frac{25}{12}$ (B) $-\frac{5}{2}$

(C) $-\frac{25}{6}$ (D) $\frac{25}{6}$

39. Let g be the inverse of $f(x) = x|x|$. For how many integers $n \in \{1, 2, \ldots, 8\}$ is $h(x) = g(x^n)$ differentiable at 0?

(A) 7 (B) 6

(C) 4 (D) 5

40. Let g be the real inverse of $f(x) = x^3$. For $h(x) = g(x^3 + x^4)$ near 0, find $h''(0)$.

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$

(C) $\frac{1}{3}$ (D) 0

41–50 / Preserved hard practice

41. Let $f(x) = \begin{cases} a\sin x + b(1-\cos x), & x < 0, \\ x + cx^2, & x \geq 0. \end{cases}$ If f is twice differentiable at 0, find $a + b - 2c$.

(A) 1 (B) 3

(C) 2 (D) 0

42. Let $f(0) = 0$ and $f(x) = x^2 \sin(1/x)$ otherwise. Find $\lim_{h\to 0}[f(h) - f(-h)]/(2h)$.

(A) 0 (B) -1

(C) 2 (D) 1

43. For real p, define $f_p(0) = 0$ and $f_p(x) = |x|^p \sin(1/x)$ for $x \neq 0$. Exactly which condition ensures $f_p\prime(0)$ exists?

(A) $p \geq 1$ (B) $p > 0$

(C) $p \geq 0$ (D) $p > 1$

44. Let $f(0) = a$ and $f(x) = \ln(1 + x)/x$ for nonzero $x > -1$. If f is differentiable at 0, find $a + f\prime(0)$.

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$

(C) $\frac{5}{2}$ (D) $\frac{3}{2}$

45. A differentiable function satisfies $f(x + y) = f(x)f(y)$ for all reals, $f(0) = 1$, and $f\prime(0) = 3$. Find $f\prime(2)/f(2)$.

(A) 5 (B) 4

(C) 2 (D) 3

46. Let $p(x) = x^3 + ax + b$. If the line $y = 3x - 2$ is tangent to its graph at $x = 1$, find $a + b$.

(A) 2 (B) -1

(C) 1 (D) 0

47. Let $A(x) = \begin{pmatrix} e^x & x \\ x & e^{-x} \end{pmatrix}$ and $f(x) = \det(A(x)^2)$. Find $f''(0)$.

(A) -3 (B) -5

(C) -4 (D) -2

48. For $f(x) = (1 + x)^n$, where n is a positive integer, $f''(0) = 30$. Find $f'''(0)$.

(A) 120 (B) 121

(C) 122 (D) 119

49. Let $f(x) = \int_0^{x^2} \cos(t^2)\, dt$. Find $f''(0)$.

(A) 2 (B) 3

(C) 1 (D) 4

50. Let g be the inverse of $f(x) = x + e^x$ near $x = 0$. Find $g''(1)$.

(A) $-\frac{9}{8}$ (B) $-\frac{1}{8}$

(C) $\frac{15}{8}$ (D) $\frac{7}{8}$

Answer key

Continuity & Differentiability

M-CODI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**A**	21	**D**	31	**D**	41	**A**
02	**D**	12	**C**	22	**C**	32	**C**	42	**A**
03	**B**	13	**C**	23	**C**	33	**B**	43	**D**
04	**B**	14	**B**	24	**B**	34	**A**	44	**B**
05	**B**	15	**D**	25	**D**	35	**D**	45	**D**
06	**B**	16	**A**	26	**A**	36	**A**	46	**D**
07	**C**	17	**C**	27	**B**	37	**A**	47	**C**
08	**A**	18	**C**	28	**C**	38	**C**	48	**A**
09	**D**	19	**A**	29	**C**	39	**B**	49	**A**
10	**D**	20	**B**	30	**B**	40	**A**	50	**B**

Chapter 14

Application of Derivatives

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The closest point on $y = x^2$ to $(0, 1)$ with positive x-coordinate has $x^2 =$

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{5}{2}$

02. The maximum of $x(6 - x)^2$ on $[0, 6]$ is

(A) 32 (B) 33

(C) 34 (D) 31

03. For $f(x) = x^4 - 4x^2$, the number of distinct local minima is

(A) 2 (B) 4

(C) 1 (D) 3

04. For $f(x) = x^3 - 3ax$, where $a > 0$, the difference between its local maximum and local minimum values is

(A) $4a^{\frac{3}{2}} + 2$ (B) $4a^{\frac{3}{2}} + 1$

(C) $4a^{\frac{3}{2}}$ (D) $4a^{\frac{3}{2}} - 1$

05. For $f(x) = x^3 - 3x + 1$, the number of real zeros is

(A) 2 (B) 4

(C) 5 (D) 3

06. For $f(x) = x^4 - 2x^2$, the number of inflection points is

(A) 4 (B) 3

(C) 2 (D) 1

07. If $f'(x) = (x-1)^2(x+2)$, at how many distinct real points does f have a local extremum?

(A) 1 (B) 0

(C) 3 (D) 2

08. The smallest real a for which $f(x) = x^3 - 3x^2 + ax$ is nondecreasing on $\mathbb{R}$ is

(A) 4 (B) 3

(C) 5 (D) 2

09. The line $y = mx + 1$ is tangent to $y = x^2$ for how many real values of m?

(A) 2 (B) 1

(C) 0 (D) -1

10. A rectangle lies under $y = 4 - x^2$, above the x-axis, symmetric about the y-axis, with upper vertices on the parabola. Its maximum area is

(A) $-1 + \frac{32\sqrt{3}}{9}$ (B) $1 + \frac{32\sqrt{3}}{9}$

(C) $\frac{32\sqrt{3}}{9}$ (D) $2 + \frac{32\sqrt{3}}{9}$

11–40 / New middle-to-hard practice

11. For how many integers $a \in [-4, 6]$ does $f(x) = x^3 - 3ax^2 + 3(a+2)x$ have a local maximum at a point strictly between 0 and 1?

(A) 3 (B) 2

(C) 5 (D) 4

12. For how many positive integers a do both stationary points of $f(x) = x^3 - 3(a+1)x^2 + 3(a^2+1)x$ lie strictly between 0 and 4?

(A) 1 (B) 3

(C) 2 (D) 0

13. The set of real a for which $x^3 + 3ax^2 + 3(a+2)x$ is nondecreasing on the whole real line is an interval. Find its length.

(A) 4 (B) 3

(C) 1 (D) 2

14. Find the largest real a for which $f(x) = x^3/3 - ax^2/2 + 2x$ is nondecreasing throughout $[1, 3]$.

(A) $\sqrt{2}$ (B) $\frac{8}{3}$

(C) $2\sqrt{2}$ (D) 3

15. How many distinct local extrema does $f(x) = |x^2 - 4x + 3| + x$ have in the open interval $(0, 4)$?

(A) 4 (B) 3

(C) 1 (D) 2

16. Find the greatest value of $x^3 - 3x + |x - 1|$ on $[-2, 2]$.

(A) 3 (B) $1 + \frac{16\sqrt{3}}{9}$

(C) $1 + \frac{8\sqrt{3}}{9}$ (D) $1 + \frac{16\sqrt{3}}{3}$

17. Two distinct tangents to $y = x^3 - 3x$ pass through $(1, -3)$. Find the sum of their slopes.

(A) $\frac{15}{4}$ (B) $\frac{9}{4}$

(C) $-\frac{3}{4}$ (D) $\frac{3}{4}$

18. Find the sum of the slopes of all distinct tangents to $y = x^3$ that pass through $(1, 1)$.

(A) $\frac{15}{4}$ (B) 3

(C) $\frac{27}{4}$ (D) $\frac{9}{4}$

19. Two distinct straight lines are tangent to both $y = x^2$ and $y = -x^2 + 4x - 6$. Find the product of their slopes.

(A) -4 (B) 4

(C) -8 (D) 8

20. A common tangent touches $y = x^2$ at P and $y = (x - 3)^2 + 3$ at Q. Find PQ.

(A) $3\sqrt{2}$ (B) 3

(C) $6\sqrt{2}$ (D) 6

21. A cone of height 12 cm and radius 6 cm is filled at $2\pi\,\text{cm}^3/\text{min}$. Initially water is 2 cm deep. After 7/3 min, at what rate is the wet lateral area increasing, in cm^2/min?

(A) 4π (B) $\sqrt{5}\pi$

(C) $2\sqrt{5}\pi$ (D) $\frac{\sqrt{5}\pi}{2}$

22. A spherical balloon initially has radius 3 m and loses gas at $4\pi\,\text{m}^3/\text{min}$. After 19/3 min, find the positive rate of decrease of its surface area, in m^2/min.

(A) π (B) 4π

(C) 2π (D) 8π

23. A closed cylinder has volume 54π. Material for each circular end costs twice as much per unit area as the curved surface. At minimum total material cost, find the ratio height/radius.

(A) 2 (B) 8

(C) 1 (D) 4

24. A 400 m running track consists of two straight stretches and two semicircles of radius r. Find the greatest area of its rectangular central portion, in m^2.

(A) $\frac{40000}{\pi}$ (B) $\frac{10000}{\pi}$

(C) 20000 (D) $\frac{20000}{\pi}$

25. Find the shortest distance from $(3, 0)$ to the curve $y = x^2$.

(A) $\sqrt{10}$ (B) $\sqrt{5}$

(C) 5 (D) 2

26. For $x, y > 0$ with $xy = 4$, the quantity $9x^2 + 4y^2$ is least at P. Find the square of the x coordinate of P.

(A) $\frac{16}{3}$ (B) $\frac{4}{3}$

(C) $\frac{8}{3}$ (D) $\frac{3}{8}$

27. For how many integers $a \in [-5, 5]$ does $x^3 - 3x = a^2 - 3$ have three distinct real roots?

(A) 1 (B) 2

(C) 3 (D) 4

28. For $a > 0$, the equation $|x^2 - 4| = ax$ has two real roots, and the larger is three times the smaller. Find a.

(A) $4\sqrt{3}$ (B) $\frac{4\sqrt{3}}{3}$

(C) $2\sqrt{3}$ (D) $\frac{2\sqrt{3}}{3}$

29. For $f(x) = x^3 - 3ax + b$, $a > 0$, the local maximum and minimum values are 10 and -22. Find $a + b$.

(A) -2 (B) 2

(C) -10 (D) 4

30. A cubic f has stationary points at $x = 1, 3$ and $f(3) - f(1) = -4$. Find $f\prime(0)$.

(A) 12 (B) 3

(C) 6 (D) 9

31. For $x = t^2 + 1$, $y = t^3 - 3t$, the normal at a point with $t > 0$ has slope $-2/3$. Find its point of contact x coordinate.

(A) 3 (B) $\frac{5}{2} - \frac{\sqrt{5}}{2}$

(C) $\frac{\sqrt{5}}{2} + \frac{3}{2}$ (D) $\frac{\sqrt{5}}{2} + \frac{5}{2}$

32. For $x = t^2 - t$, $y = t^2 + t$, a tangent is perpendicular to $x + 2y = 0$. Find the y intercept of the normal at that point.

(A) $\frac{3}{8}$ (B) $\frac{33}{8}$

(C) $\frac{27}{8}$ (D) $\frac{15}{4}$

33. For $f(x) = x^3$ on $[-1, a]$, $a > 0$, there are exactly two distinct mean-value-theorem points c in the open interval. The permitted a form an interval. Find its length.

(A) 2 (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) 3

34. For $0 < a < 2$, let $f(x) = x(x-a)(x-2)$. The product of its two stationary abscissae is 1. Find their sum.

(A) $\frac{14}{3}$ (B) $\frac{5}{3}$

(C) $\frac{3}{2}$ (D) $\frac{7}{3}$

35. For each integer $a \in \{-2, -1, 0, 1, 2\}$ a function satisfies $f\prime(x) = (x-a)(x^2-1)^2$. Add, over these five values of a, the number of stationary points that are not local extrema.

(A) 6 (B) 7

(C) 8 (D) 10

36. For how many integers $a \in [-3, 3]$ does a function with $f\prime(x) = (x^2 - a)(x^2 - 1)$ have exactly two distinct local extrema?

(A) 5 (B) 6

(C) 3 (D) 4

37. Let $f(x) = x^a e^{-bx}$ for $x > 0$, with $a, b > 0$. It is stationary at $x = 2$ and $f\prime(1)/f(1) = 3$. Find its greatest value.

(A) $\frac{64}{e^3}$ (B) $\frac{32}{e^6}$

(C) $\frac{729}{e^9}$ (D) $\frac{64}{e^6}$

38. Let $f(x) = (x-c)e^{-bx}$ on $x \geq c$, with $b > 0$. Its zero is at $x = 1$ and its maximum occurs at $x = 3$. Find this maximum.

(A) $\frac{3}{e^{\frac{3}{2}}}$ (B) $e^{-\frac{3}{2}}$

(C) $\frac{2}{e^3}$ (D) $\frac{2}{e^{\frac{3}{2}}}$

39. An axis-parallel rectangle centred at the origin has its vertices on $x^2/9 + y^2/4 = 1$. Its horizontal side length must be at least 5. Find its greatest area.

(A) 12 (B) $\frac{10\sqrt{11}}{3}$

(C) $\frac{5\sqrt{11}}{3}$ (D) $\frac{10\sqrt{11}}{9}$

40. A rectangle is inscribed in a circle of radius 5. Its longer side must be at least 8. Find its greatest possible area.

(A) 50 (B) 24

(C) 40 (D) 48

41–50 / Preserved hard practice

41. A line through $(1,2)$ in the first quadrant cuts positive intercepts a, b on the axes. The least possible ab is

(A) 7 (B) 9

(C) 8 (D) 10

42. A rectangular sheet 12 cm by 12 cm has equal squares of side x cut from its four corners and folds into an open box. The largest volume, in cm^3, is

(A) 128 (B) 129

(C) 127 (D) 130

43. For $a > 0$, the function $f(x) = x^3 - 3x + a$ has exactly two distinct real zeros when $a =$

(A) 1 (B) 2

(C) 4 (D) 3

44. The maximum of $f(x) = x^2 e^{-x}$ on $[0, \infty)$ is

(A) $-1 + \frac{4}{e^2}$ (B) $\frac{4}{e^2} + 1$

(C) $\frac{4}{e^2}$ (D) $\frac{4}{e^2} + 2$

45. For $f(x) = x^4 - 4x^3 + 4x^2$, the number of distinct real solutions of $f(x) = 1$ is

(A) 3 (B) 2

(C) 4 (D) 5

46. The maximum area of an axis-parallel rectangle inscribed in $x^2/9 + y^2/4 = 1$ is

(A) 12 (B) 13

(C) 11 (D) 14

47. A right circular cone is inscribed in a sphere of radius 3, with its apex and base circumference on the sphere. Its symmetry axis passes through the sphere centre, and its base plane may vary. At maximum volume, what is its height?

(A) 6 (B) 5

(C) 4 (D) 3

48. If $F(x) = \int_0^x (t^2 - 4t + 3)\,dt$, the local maximum value of F is

(A) $\frac{10}{3}$ (B) $\frac{4}{3}$

(C) $\frac{1}{3}$ (D) $\frac{7}{3}$

49. For $a > 0$, the maximum of $f(x) = \ln x - ax$ over $x > 0$ equals -3. Find a.

(A) $1 + e^2$ (B) $2 + e^2$

(C) $-1 + e^2$ (D) e^2

50. The sum of the x-coordinates of all points on $y = x^2$ whose normals pass through $(0, 3)$ is

(A) 2 (B) -1

(C) 0 (D) 1

Answer key

Application of Derivatives

M-APDE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**A**	21	**B**	31	**D**	41	**C**
02	**A**	12	**A**	22	**B**	32	**B**	42	**A**
03	**A**	13	**B**	23	**D**	33	**C**	43	**B**
04	**C**	14	**C**	24	**D**	34	**D**	44	**C**
05	**D**	15	**B**	25	**B**	35	**C**	45	**A**
06	**C**	16	**B**	26	**C**	36	**D**	46	**A**
07	**A**	17	**D**	27	**B**	37	**D**	47	**C**
08	**B**	18	**A**	28	**B**	38	**D**	48	**B**
09	**C**	19	**A**	29	**A**	39	**B**	49	**D**
10	**C**	20	**A**	30	**D**	40	**D**	50	**C**

Chapter 15

Indefinite Integration

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. An antiderivative of xe^x is

(A) $\frac{x^2e^x}{2}$ (B) $(x-1)\,e^x$

(C) xe^x (D) $(x+1)\,e^x$

02. An antiderivative of $\log(x)$ for $x>0$ is

(A) $x\log(x)+x$ (B) $\frac{\log(x)}{x}$

(C) $x\log(x)-x$ (D) $x\log(x)$

03. An antiderivative of $x\sin(x)$ is

(A) $-x\cos(x)+\sin(x)$ (B) $x\cos(x)-\sin(x)$

(C) $-x\cos(x)$ (D) $x\sin(x)+\cos(x)$

04. An antiderivative of $\frac{1}{x(x+1)}$ for $x>0$ is

(A) $\frac{1}{x+1}$ (B) $\log(x)-\log(x+1)$

(C) $-\log(x)+\log(x+1)$ (D) $\log(x)+\log(x+1)$

05. An antiderivative of $\frac{x^2}{x^2+1}$ is

(A) $-x+\operatorname{atan}(x)$ (B) $\log(x^2+1)$

(C) $x-\operatorname{atan}(x)$ (D) $x+\operatorname{atan}(x)$

06. An antiderivative of $\frac{1}{e^x+e^{-x}}$ is

(A) $\operatorname{atan}(e^{-x})$ (B) $2\operatorname{atan}(e^x)$

(C) $\log(e^x+e^{-x})$ (D) $\operatorname{atan}(e^x)$

07. An antiderivative of $\frac{x}{x^4+1}$ is

(A) $\frac{\log\left(x^4+1\right)}{4}$ (B) $\frac{\operatorname{atan}(x)}{2}$

(C) $\operatorname{atan}\left(x^2\right)$ (D) $\frac{\operatorname{atan}\left(x^2\right)}{2}$

08. An antiderivative of x^2e^x is

(A) x^2e^x (B) $\left(x^2-2x\right)e^x$

(C) $\left(x^2+2x+2\right)e^x$ (D) $\left(x^2-2x+2\right)e^x$

09. An antiderivative of $\sin^3(x)$ is

(A) $-\frac{\cos^3(x)}{3}$ (B) $\frac{\cos^3(x)}{3}-\cos(x)$

(C) $\frac{\sin^4(x)}{4}$ (D) $-\frac{\cos^3(x)}{3}+\cos(x)$

10. If $F\prime(x)=\ln x$ for $x>0$ and $F(1)=0$, find $F(e)$.

(A) 2 (B) 1

(C) 3 (D) 0

11–40 / New middle-to-hard practice

11. For $x>1$, an antiderivative of $\frac{x+3}{x^2+2x+5}$ is

(A) $\log\left(x^2+2x+5\right)+\arctan\left(\frac{x}{2}+\frac{1}{2}\right)$

(B) $\frac{\log\left(x^2+2x+5\right)}{2}+\arctan\left(\frac{x}{2}+\frac{1}{2}\right)$

(C) $\frac{\log\left(x^2+2x+5\right)}{2}-\arctan\left(\frac{x}{2}+\frac{1}{2}\right)$

(D) $-\frac{\log\left(x^2+2x+5\right)}{2}+\arctan\left(\frac{x}{2}+\frac{1}{2}\right)$

12. For $x>1$, an antiderivative of $\frac{2x-4}{x^2-2x+5}$ is

(A) $-\log\left(x^2-2x+5\right)-\arctan\left(\frac{x}{2}-\frac{1}{2}\right)$

(B) $\log\left(x^2-2x+5\right)-\arctan\left(\frac{x}{2}-\frac{1}{2}\right)$

(C) $\log\left(x^2-2x+5\right)-2\arctan\left(\frac{x}{2}-\frac{1}{2}\right)$

(D) $\log\left(x^2-2x+5\right)+\arctan\left(\frac{x}{2}-\frac{1}{2}\right)$

13. For $x>1$, an antiderivative of $\frac{1}{x(x+1)^2}$ is

(A) $\frac{\log\left(\left(\frac{x}{x+1}\right)^{x+1}\right)-1}{x+1}$

(B) $\frac{\log\left(\left(\frac{x}{x+1}\right)^{x+1}\right)+1}{x+1}$

(C) $\frac{\log\left(\left(\frac{x}{x+1}\right)^{x+1}\right)+2}{x+1}$

(D) $\frac{1-\log\left((x(x+1))^{x+1}\right)}{x+1}$

14. For $x>1$, an antiderivative of $\frac{2x+1}{x^2(x+1)}$ is

(A) $\frac{-\log\left((x(x+1))^{x}\right)-1}{x}$

(B) $\frac{\log\left(\left(\frac{x}{x+1}\right)^{x}\right)-2}{x}$

(C) $\frac{\log\left(\left(\frac{x}{x+1}\right)^{x}\right)-1}{x}$

(D) $\frac{\log\left(\left(\frac{x}{x+1}\right)^{x}\right)+1}{x}$

15. For $x>1$, an antiderivative of $\frac{1}{2x^2+2x+5}$ is

(A) $\frac{\arctan\left(\frac{2x}{3}+\frac{1}{3}\right)}{6}$

(B) $\frac{2\arctan\left(\frac{2x}{3}+\frac{1}{3}\right)}{3}$

(C) $\frac{\arctan\left(\frac{2x}{3}+\frac{1}{3}\right)}{3}$

(D) $-\frac{\arctan\left(\frac{2x}{3}+\frac{1}{3}\right)}{3}$

16. For $x>1$, an antiderivative of $\frac{1}{x^2+4x+1}$ is

(A) $\frac{\sqrt{3}\log\left(\frac{x-\sqrt{3}+2}{x+\sqrt{3}+2}\right)}{6}$

(B) $-\frac{\sqrt{3}\log\left(\frac{x-\sqrt{3}+2}{x+\sqrt{3}+2}\right)}{6}$

(C) $\frac{\sqrt{3}\log\left(\frac{x-\sqrt{3}+2}{x+\sqrt{3}+2}\right)}{3}$

(D) $\frac{\sqrt{3}\log\left(\frac{x-\sqrt{3}+2}{x+\sqrt{3}+2}\right)}{12}$

17. For $x>1$, an antiderivative of $\frac{\log(x)^2}{x(\log(x)+1)^2}$ is

(A) $\frac{(-\log(x)-2\log(\log(x)+1))(\log(x)+1)-1}{\log(x)+1}$

(B) $\frac{(\log(x)+1)(\log(x)-2\log(\log(x)+1))-2}{\log(x)+1}$

(C) $\frac{(\log(x)+1)(\log(x)-2\log(\log(x)+1))-1}{\log(x)+1}$

(D) $\frac{(\log(x)+1)(\log(x)-2\log(\log(x)+1))+1}{\log(x)+1}$

18. For $x > 1$, an antiderivative of $\frac{\log(x)+1}{x\left(\log(x)^2+1\right)}$ is

(A) $\frac{\log\left(\log(x)^2+1\right)}{2} + \arctan(\log(x))$

(B) $\frac{\log\left(\log(x)^2+1\right)}{2} - \arctan(\log(x))$

(C) $\log\left(\log(x)^2+1\right) + \arctan(\log(x))$

(D) $-\frac{\log\left(\log(x)^2+1\right)}{2} + \arctan(\log(x))$

19. For $x > 1$, an antiderivative of $\frac{e^x}{(e^x+1)(e^x+2)}$ is

(A) $\log\left(\frac{e^x+1}{(e^x+2)^2}\right)$

(B) $\log\left((e^x+1)(e^x+2)\right)$

(C) $\log\left(\frac{e^x+1}{e^x+2}\right)$

(D) $-\log\left((e^x+1)(e^x+2)\right)$

20. For $x > 1$, an antiderivative of $\frac{e^{2x}}{(e^x+1)(e^x+2)}$ is

(A) $\log\left(\frac{(e^x+2)^2}{e^x+1}\right)$

(B) $\log\left((e^x+1)(e^x+2)^2\right)$

(C) $-2\log(e^x+1) + 2\log(e^x+2)$

(D) $-\log\left((e^x+1)(e^x+2)^2\right)$

21. For $0 < x < \pi/4$, an antiderivative of $\frac{\tan^2(x)+1}{\tan^2(x)+3\tan(x)+2}$ is

(A) $\frac{\log\left(\frac{\tan(x)+1}{\tan(x)+2}\right)}{2}$

(B) $2\log\left(\frac{\tan(x)+1}{\tan(x)+2}\right)$

(C) $\log\left(\frac{\tan(x)+1}{\tan(x)+2}\right)$

(D) $-\log\left(\frac{\tan(x)+1}{\tan(x)+2}\right)$

22. For $0 < x < \pi/4$, an antiderivative of $\frac{(\tan(x)+3)\left(\tan^2(x)+1\right)}{\tan^2(x)+2\tan(x)+5}$ is

(A) $-\frac{\log\left(\tan^2(x)+2\tan(x)+5\right)}{2} + \arctan\left(\frac{\tan(x)}{2}+\frac{1}{2}\right)$

(B) $\frac{\log\left(\tan^2(x)+2\tan(x)+5\right)}{2} + \arctan\left(\frac{\tan(x)}{2}+\frac{1}{2}\right)$

(C) $\frac{\log\left(\tan^2(x)+2\tan(x)+5\right)}{2} - \arctan\left(\frac{\tan(x)}{2}+\frac{1}{2}\right)$

(D) $\log\left(\tan^2(x)+2\tan(x)+5\right) + \arctan\left(\frac{\tan(x)}{2}+\frac{1}{2}\right)$

23. For $x > 1$, an antiderivative of x^3e^{2x} is

(A) $\frac{\left(4x^3-6x^2+6x-3\right)e^{2x}}{8}$

(B) $\frac{\left(4x^3-6x^2+6x+3\right)e^{2x}}{8}$

(C) $\frac{\left(4x^3-6x^2-6x-3\right)e^{2x}}{8}$

(D) $\frac{\left(2x^3-3x^2+3x-3\right)e^{2x}}{4}$

24. For $x > 1$, an antiderivative of $x^2 \cos(2x)$ is

(A) $\frac{x^2 \sin(2x)}{2} + \frac{x \cos(2x)}{2} - \frac{\sin(2x)}{4}$

(B) $\frac{x^2 \sin(2x)}{2} + \frac{x \cos(2x)}{2} - \frac{\sin(2x)}{2}$

(C) $\frac{x^2 \sin(2x)}{2} + \frac{x \cos(2x)}{2} + \frac{\sin(2x)}{4}$

(D) $-\frac{x^2 \sin(2x)}{2} + \frac{x \cos(2x)}{2} - \frac{\sin(2x)}{4}$

25. For $x > 1$, an antiderivative of $x \log(x)^2$ is

(A) $\frac{x^2\left(2\log(x)^2 - 2\log(x) + 1\right)}{4}$

(B) $\frac{x^2\left(\log(x)^2 - \log(x) + 1\right)}{2}$

(C) $\frac{x^2\left(2\log(x)^2 + 2\log(x) + 1\right)}{4}$

(D) $\frac{x^2\left(2\log(x)^2 - 2\log(x) - 1\right)}{4}$

26. For $x > 1$, an antiderivative of $\frac{\log(x)}{x^3}$ is

(A) $\frac{2\log(x) - 1}{4x^2}$

(B) $\frac{-\log(x) - 1}{2x^2}$

(C) $\frac{1 - 2\log(x)}{4x^2}$

(D) $\frac{-2\log(x) - 1}{4x^2}$

27. For $x > 1$, an antiderivative of $\frac{\sqrt{x^2+1}}{x^2}$ is

(A) $\log\left(x + \sqrt{x^2+1}\right) - \frac{2\sqrt{x^2+1}}{x}$

(B) $\log\left(x + \sqrt{x^2+1}\right) + \frac{\sqrt{x^2+1}}{x}$

(C) $\log\left(x + \sqrt{x^2+1}\right) - \frac{\sqrt{x^2+1}}{x}$

(D) $-\log\left(x + \sqrt{x^2+1}\right) - \frac{\sqrt{x^2+1}}{x}$

28. For $x > 1$, an antiderivative of $\frac{\sqrt{x^2-1}}{x}$ is

(A) $2\sqrt{x^2-1} - \arccos\left(\frac{1}{x}\right)$

(B) $\sqrt{x^2-1} + \arccos\left(\frac{1}{x}\right)$

(C) $\sqrt{x^2-1} - \arccos\left(\frac{1}{x}\right)$

(D) $-\sqrt{x^2-1} - \arccos\left(\frac{1}{x}\right)$

29. For $x > 1$, an antiderivative of $\frac{1}{(x-1)^{\frac{2}{5}}(x+3)^{\frac{8}{5}}}$ is

(A) $\frac{5(x-1)^{\frac{3}{5}}}{6(x+3)^{\frac{3}{5}}}$

(B) $-\frac{5(x-1)^{\frac{3}{5}}}{12(x+3)^{\frac{3}{5}}}$

(C) $\frac{5(x-1)^{\frac{3}{5}}}{12(x+3)^{\frac{3}{5}}}$

(D) $\frac{5(x-1)^{\frac{3}{5}}}{24(x+3)^{\frac{3}{5}}}$

30. For $x > 1$, an antiderivative of $\frac{1}{(x-1)^{\frac{3}{2}}\sqrt{x+2}}$ is

(A) $-\frac{\sqrt{x+2}\sqrt{\frac{1}{x-1}}}{3}$

(B) $-\frac{4\sqrt{x+2}\sqrt{\frac{1}{x-1}}}{3}$

(C) $\frac{2\sqrt{x+2}\sqrt{\frac{1}{x-1}}}{3}$

(D) $-\frac{2\sqrt{x+2}\sqrt{\frac{1}{x-1}}}{3}$

31. For $x > 1$, an antiderivative of $\frac{4x^{11}\left(x^4+2x^2+3\right)}{\left(x^4+x^2+1\right)^3}$ is

(A) $-\frac{x^{12}}{\left(x^4+x^2+1\right)^2}$

(B) $\frac{x^{12}}{\left(x^4+x^2+1\right)^2}$

(C) $\frac{2x^{12}}{\left(x^4+x^2+1\right)^2}$

(D) $\frac{x^{12}}{2\left(x^4+x^2+1\right)^2}$

32. For $x > 1$, an antiderivative of $\frac{4x^{11}\left(x^2+3\right)}{\left(x^2+1\right)^5}$ is

(A) $\frac{x^{12}}{\left(x^2+1\right)^4}$

(B) $\frac{2x^{12}}{\left(x^2+1\right)^4}$

(C) $-\frac{x^{12}}{\left(x^2+1\right)^4}$

(D) $\frac{x^{12}}{2\left(x^2+1\right)^4}$

33. For $x > 1$, an antiderivative of $x \arctan(x)$ is

(A) $\frac{x^2 \arctan(x)}{2} - \frac{x}{2} - \frac{\arctan(x)}{2}$

(B) $-\frac{x}{2} + \frac{\left(x^2+1\right)\arctan(x)}{2} + \frac{\arctan(x)}{2}$

(C) $-\frac{x^2 \arctan(x)}{2} - \frac{x}{2} + \frac{\arctan(x)}{2}$

(D) $-\frac{x}{2} + \frac{\left(x^2+1\right)\arctan(x)}{2}$

34. For $0 < x < 1$, an antiderivative of $\frac{\arcsin(x)}{x^2}$ is

(A) $\log\left(\frac{x}{\sqrt{1-x^2}+1}\right) - \frac{2\arcsin(x)}{x}$

(B) $\log\left(\frac{x}{\sqrt{1-x^2}+1}\right) + \frac{\arcsin(x)}{x}$

(C) $\log\left(\frac{x}{\sqrt{1-x^2}+1}\right) - \frac{\arcsin(x)}{x}$

(D) $-\log\left(\frac{x}{\sqrt{1-x^2}+1}\right) - \frac{\arcsin(x)}{x}$

35. For $0 < x < \pi/4$, an antiderivative of $\sec^3(x)$ is

(A) $-\frac{\log(\tan(x)+\sec(x))}{2} + \frac{\tan(x)\sec(x)}{2}$

(B) $\frac{\log(\tan(x)+\sec(x))}{2} - \frac{\tan(x)\sec(x)}{2}$

(C) $\log(\tan(x)+\sec(x)) + \frac{\tan(x)\sec(x)}{2}$

(D) $\frac{\log(\tan(x)+\sec(x))}{2} + \frac{\tan(x)\sec(x)}{2}$

36. For $0 < x < \pi/4$, an antiderivative of $\tan^3(x)$ is

(A) $\log(\cos(x)) - \frac{\tan^2(x)}{2}$

(B) $-\log(\cos(x)) + \frac{\tan^2(x)}{2}$

(C) $\log(\cos(x)) + \tan^2(x)$

(D) $\log(\cos(x)) + \frac{\tan^2(x)}{2}$

37. For $0 < x < \pi/4$, an antiderivative of $\sin^5(x)\cos^3(x)$ is

(A) $-\frac{\sin^8(x)}{8} - \frac{\sin^6(x)}{6}$

(B) $-\frac{\sin^8(x)}{4} + \frac{\sin^6(x)}{6}$

(C) $-\frac{\sin^8(x)}{8} + \frac{\sin^6(x)}{6}$

(D) $\frac{\sin^8(x)}{8} + \frac{\sin^6(x)}{6}$

38. For $0 < x < \pi/4$, an antiderivative of $\sin^4(x)\cos^4(x)$ is

(A) $-\frac{3x}{128} - \frac{\sin(4x)}{128} + \frac{\sin(8x)}{1024}$

(B) $\frac{3x}{128} - \frac{\sin(4x)}{64} + \frac{\sin(8x)}{1024}$

(C) $\frac{3x}{128} + \frac{\sin(4x)}{128} + \frac{\sin(8x)}{1024}$

(D) $\frac{3x}{128} - \frac{\sin(4x)}{128} + \frac{\sin(8x)}{1024}$

39. For $x > 1$, an antiderivative of $\frac{x^2-1}{x^4+3x^2+1}$ is

(A) $\arctan\left(x + \frac{1}{x}\right)$

(B) $2\arctan\left(x + \frac{1}{x}\right)$

(C) $-\arctan\left(x + \frac{1}{x}\right)$

(D) $\frac{\arctan\left(x+\frac{1}{x}\right)}{2}$

40. For $x > 1$, an antiderivative of $\frac{x^2+1}{x^4-x^2+1}$ is

(A) $2\arctan\left(x - \frac{1}{x}\right)$

(B) $\arctan\left(x - \frac{1}{x}\right)$

(C) $-\arctan\left(x - \frac{1}{x}\right)$

(D) $\frac{\arctan\left(x-\frac{1}{x}\right)}{2}$

41–50 / Preserved hard practice

41. An antiderivative of $\frac{1}{x^2+2x+5}$ is

(A) $\frac{\log\left(x^2+2x+5\right)}{2}$

(B) $\frac{\operatorname{atan}(x+1)}{2}$

(C) $\operatorname{atan}\left(\frac{x}{2} + \frac{1}{2}\right)$

(D) $\frac{\operatorname{atan}\left(\frac{x}{2}+\frac{1}{2}\right)}{2}$

42. An antiderivative of $\frac{x^3}{(x^2+1)^2}$ is

(A) $\frac{\log\left(x^2+1\right)}{2}$ (B) $\frac{1}{2x^2+2}$

(C) $\frac{\log\left(x^2+1\right)}{2} - \frac{1}{2(x^2+1)}$ (D) $\frac{\log\left(x^2+1\right)}{2} + \frac{1}{2(x^2+1)}$

43. An antiderivative of $(\sin(x) + \cos(x))\, e^x$ is

(A) $e^x \sin(x)$ (B) $(\sin(x) + \cos(x))\, e^x$

(C) $\frac{(\sin(x) - \cos(x))e^x}{2}$ (D) $e^x \cos(x)$

44. An antiderivative of $\frac{1}{x\left(\log(x)^2+1\right)}$ for $x > 0$ is

(A) $\log\left(\log(x)^2 + 1\right)$ (B) $\operatorname{atan}(x)$

(C) $\frac{\operatorname{atan}(\log(x))}{2}$ (D) $\operatorname{atan}(\log(x))$

45. An antiderivative of $\frac{x^2+1}{x^4-x^2+1}$ on an interval not containing 0 is

(A) $\log\left(x^4 - x^2 + 1\right)$ (B) $\operatorname{atan}\left(x - \frac{1}{x}\right)$

(C) $\operatorname{atan}\left(x^2\right)$ (D) $\operatorname{atan}\left(x + \frac{1}{x}\right)$

46. Let $F\prime(x) = (1+x)^7$ and $F(0) = 0$. The sum of all coefficients of $F(x)$ is

(A) $\frac{255}{8}$ (B) $\frac{271}{8}$

(C) $\frac{263}{8}$ (D) $\frac{247}{8}$

47. Let $F\prime(x) = x + 1/x$ for $x > 0$ and $F(1) = 0$. Find $e^{F(2)-3/2}$.

(A) 2 (B) 3

(C) 1 (D) 4

48. A monic cubic P satisfies $P\prime(x) = 3(x-1)(x-3)$ and $P(1) = 0$. Find $P(3)$.

(A) -2 (B) -4

(C) -5 (D) -3

49. If $F\prime(x) = 1/(1+x^2)$ and $F(1) = \pi/4$, find $\tan(F(\sqrt{3}) - F(0))$.

(A) $1 + \sqrt{3}$ (B) $\sqrt{3}$

(C) $\sqrt{3} + 2$ (D) $-1 + \sqrt{3}$

50. Let $F\prime(x) = x^2$ and $F(0) = 0$. If $a, b > 0$ satisfy $a + b = 2$, the minimum of $F(a) + F(b)$ is

(A) $\frac{2}{3}$ (B) $\frac{8}{3}$

(C) $\frac{5}{3}$ (D) $-\frac{1}{3}$

Answer key

Indefinite Integration

M-ININ / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**C**	31	**B**	41	**D**
02	**C**	12	**B**	22	**B**	32	**A**	42	**D**
03	**A**	13	**B**	23	**A**	33	**D**	43	**A**
04	**B**	14	**C**	24	**A**	34	**C**	44	**D**
05	**C**	15	**C**	25	**A**	35	**D**	45	**B**
06	**D**	16	**A**	26	**D**	36	**D**	46	**A**
07	**D**	17	**C**	27	**C**	37	**C**	47	**A**
08	**D**	18	**A**	28	**C**	38	**D**	48	**B**
09	**B**	19	**C**	29	**C**	39	**A**	49	**B**
10	**B**	20	**A**	30	**D**	40	**B**	50	**A**

Chapter 16

Definite Integration

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Evaluate $\int_0^1 xe^x\,dx$.

(A) 1
(B) 2
(C) 3
(D) 0

02. Evaluate $\int_0^{\pi} x\sin(x)\,dx$.

(A) $2+\pi$
(B) $1+\pi$
(C) $-1+\pi$
(D) π

03. Evaluate $\int_0^1 \frac{x^2}{x^2+1}\,dx$.

(A) $2-\frac{\pi}{4}$
(B) $1-\frac{\pi}{4}$
(C) $3-\frac{\pi}{4}$
(D) $-\frac{\pi}{4}$

04. Evaluate $\int_0^{\frac{\pi}{2}} \sin^3(x)\,dx$.

(A) $\frac{2}{3}$
(B) $\frac{5}{3}$
(C) $\frac{8}{3}$
(D) $-\frac{1}{3}$

05. Evaluate $\int_{-1}^1 \frac{1}{e^x+1}\,dx$.

(A) 0
(B) 2
(C) 3
(D) 1

06. Evaluate $\displaystyle\int_0^{\log(3)} \frac{e^x}{e^x+1}\,dx$.

(A) $-1+\log(2)$ (B) $\log(2)$

(C) $\log(2)+1$ (D) $\log(2)+2$

07. A continuous function satisfies $f(x)+f(1-x)=x^2+(1-x)^2$ on $[0,1]$. Find $\int_0^1 f(x)\,dx$.

(A) $\frac{7}{3}$ (B) $\frac{4}{3}$

(C) $-\frac{2}{3}$ (D) $\frac{1}{3}$

08. If $F(x)=\int_0^{x^2}(1+t^2)\,dt$, find $F\prime(1)$.

(A) 3 (B) 6

(C) 5 (D) 4

09. If $f\prime(x)=2x+1$ and $f(0)=3$, find $\int_0^1 f(x)\,dx$.

(A) $\frac{35}{6}$ (B) $\frac{23}{6}$

(C) $\frac{29}{6}$ (D) $\frac{17}{6}$

10. The average value of $f(x)=x^2$ on the interval joining the real roots of $x^2-4=0$ is

(A) $\frac{1}{3}$ (B) $\frac{4}{3}$

(C) $\frac{10}{3}$ (D) $\frac{7}{3}$

11–40 / New middle-to-hard practice

11. Evaluate $\int_{-1}^{1}(x^2+2)/(1+e^x)\,dx$.

(A) $\frac{7}{3}$ (B) $\frac{5}{3}$

(C) $\frac{7}{6}$ (D) $\frac{14}{3}$

12. Evaluate $\int_0^{\pi} x\sin x/(1+\cos^2 x)\,dx$.

(A) $\frac{\pi}{4}$ (B) $\frac{\pi^2}{2}$

(C) $\frac{\pi^2}{4}$ (D) $\frac{\pi^2}{8}$

13. Evaluate $\int_0^{\infty} dx/[(1+x)(1+x^2)]$.

(A) $\frac{\pi}{8}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$ (D) $\log(2)$

14. Evaluate the convergent improper integral $\int_0^{\infty} \ln x/(1+x^2)\,dx$.

(A) $\frac{\pi}{4}$ (B) 0

(C) $-\pi\log(2)$ (D) $\pi\log(2)$

15. Evaluate $\int_0^{\pi/2}[x^2+(\pi/2-x)^2]\,\sqrt{\sin x}/(\sqrt{\sin x}+\sqrt{\cos x})\,dx$.

(A) $\frac{\pi^3}{24}$ (B) $\frac{\pi^3}{48}$

(C) $\frac{\pi^3}{12}$ (D) $\frac{\pi^2}{24}$

16. Evaluate $\int_0^{\pi/2} x/[\tan x+\cot x+\csc x+\sec x]\,dx$ as an improper endpoint integral.

(A) $\frac{\pi(4-\pi)}{16}$ (B) $1-\frac{\pi}{4}$

(C) $\frac{\pi(\pi+4)}{16}$ (D) $\frac{\pi(4-\pi)}{8}$

17. Evaluate $\int_0^{\pi} x\ln(\sin x)\,dx$.

(A) $-\pi^2\log(2)$ (B) $-\frac{\pi^2\log(2)}{2}$

(C) $-\pi\log(2)$ (D) $\frac{\pi^2\log(2)}{2}$

18. Evaluate $\int_0^{\pi/4}[\ln(\sin x)+\ln(\cos x)]\,dx$.

(A) $\frac{\pi\log(2)}{2}$ (B) $-\frac{\pi\log(2)}{2}$

(C) $-\pi\log(2)$ (D) $-\frac{\pi\log(2)}{4}$

19. Evaluate $\int_0^{\infty}(x^2+3)/[(x^2+1)(x^2+4)]\,dx$.

(A) $\frac{5\pi}{24}$ (B) $\frac{\pi}{12}$

(C) $\frac{7\pi}{12}$ (D) $\frac{5\pi}{12}$

20. Evaluate $\int_0^{\infty} x^2/[(x^2+1)(x^2+4)(x^2+9)]\,dx$.

(A) $\frac{\pi}{60}$ (B) $\frac{\pi}{24}$

(C) $\frac{\pi}{120}$ (D) $\frac{\pi}{240}$

21. For $a>0$, $\int_0^{\infty} dx/(x^2+a^2)=\pi/6$. Find $\int_0^{\infty} dx/(x^2+a^2)^2$.

(A) $\frac{\pi}{108}$ (B) $\frac{\pi}{216}$

(C) $\frac{\pi}{36}$ (D) $\frac{\pi}{54}$

22. For $b>0$, $\int_0^1 dx/(1+bx)=(\ln 3)/2$. Find $\int_0^1 x/(1+bx)^2\,dx$.

(A) $-\frac{1}{6}+\frac{\log(3)}{4}$ (B) $-\frac{1}{12}+\frac{\log(3)}{8}$

(C) $-\frac{1}{3}+\frac{\log(3)}{2}$ (D) $\frac{1}{6}+\frac{\log(3)}{4}$

23. A continuous f on $[0,1]$ satisfies $f(x)+2f(1-x)=x^2$. Find $\int_0^1 xf(x)\,dx$.

(A) $-\frac{1}{36}$ (B) $\frac{1}{12}$

(C) $-\frac{1}{12}$ (D) $\frac{1}{36}$

24. For continuous f on $[0,1]$, $f(x)+f(1-x)=1$ and $\int_0^1 xf(x)\,dx = 1/6$. Find $\int_0^1 x^2 f(x)\,dx$.

(A) $\frac{1}{24}$ (B) $\frac{1}{3}$
(C) $\frac{1}{6}$ (D) $\frac{1}{12}$

25. A quadratic polynomial f satisfies $d[\int_x^{2x} f(t)\,dt]/dx = 3x^2$ for all real x. Find $\int_0^1 f(t)\,dt$.

(A) $\frac{1}{7}$ (B) $\frac{3}{7}$
(C) $\frac{1}{3}$ (D) $\frac{1}{21}$

26. A continuous f satisfies $d[\int_0^{x^2} f(t)\,dt]/dx = 4x^3+2x$ for all $x>0$. Find $\int_1^3 [f(t)]^2\,dt$.

(A) $\frac{158}{3}$ (B) $\frac{79}{3}$
(C) $\frac{152}{3}$ (D) $\frac{160}{3}$

27. Evaluate $\lim_{n\to\infty}(1/n)\sum_{k=1}^{n}(k/n)\ln(1+k/n)$.

(A) $-\frac{1}{2}+\log(2)$ (B) $\frac{\log(2)}{2}$
(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

28. Evaluate $\lim_{n\to\infty}(1/n)\sum_{k=1}^{n}(k/n)^3/[1+(k/n)^2]$.

(A) $1-\log(2)$ (B) $\frac{\log(2)}{2}+\frac{1}{2}$
(C) $1-\frac{\pi}{4}$ (D) $\frac{1}{2}-\frac{\log(2)}{2}$

29. Evaluate $\int_0^1 x^2\ln(1+x)\,dx$.

(A) $-\frac{5}{9}+\frac{2\log(2)}{3}$ (B) $-\frac{5}{18}+\frac{\log(2)}{3}$
(C) $-\frac{5}{18}+\frac{2\log(2)}{3}$ (D) $\frac{5}{18}+\frac{2\log(2)}{3}$

30. Evaluate $\int_0^{\pi/2} x^2\cos x\,dx$.

(A) $\frac{\pi^2}{4}$ (B) $-2+\frac{\pi^2}{4}$
(C) $-1+\frac{\pi^2}{8}$ (D) $2+\frac{\pi^2}{4}$

31. Evaluate $\int_0^3 x|x^2-1|/(x+1)\,dx$.

(A) $\frac{9}{2}$ (B) $\frac{31}{6}$
(C) $\frac{29}{6}$ (D) $\frac{14}{3}$

32. Evaluate $\int_0^2 \lfloor 2x\rfloor/(1+x)\,dx$.

(A) $\log\left(\frac{36}{5}\right)$ (B) $\log\left(\frac{18}{25}\right)$
(C) $\log\left(\frac{9}{5}\right)$ (D) $\log\left(\frac{18}{5}\right)$

33. Let g be the inverse of $f(x) = x^3 + x$ on $[0, 1]$. Find $\int_0^2 [g(y)]^2\, dy$.

(A) $\frac{16}{15}$ (B) $\frac{2}{3}$

(C) $\frac{7}{15}$ (D) $\frac{14}{15}$

34. Let g be the inverse of $f(x) = x^2 + 2x$ on $[0, 1]$. Evaluate $\int_0^3 yg(y)\, dy$.

(A) $\frac{37}{15}$ (B) $\frac{67}{30}$

(C) $\frac{97}{30}$ (D) $\frac{97}{60}$

35. A continuous f on $[0, 1]$ satisfies $f(x) + f((1 - x)/(1 + x)) = \ln 2$. Find $\int_0^1 f(x)/(1 + x^2)\, dx$.

(A) $\frac{\pi \log(2)}{8}$ (B) $\frac{\pi \log(2)}{16}$

(C) $\frac{\log(2)}{2}$ (D) $\frac{\pi \log(2)}{4}$

36. Evaluate $\int_0^1 \arctan((1 - x)/(1 + x))/(1 + x^2)\, dx$.

(A) $\frac{\pi}{8}$ (B) $\frac{\pi^2}{16}$

(C) $\frac{\pi^2}{32}$ (D) $\frac{\pi^2}{64}$

37. Evaluate $\int_0^1 \sqrt{(1 - x)/(1 + x)}\, dx$.

(A) $-1 + \frac{\pi}{4}$ (B) $-2 + \pi$

(C) $1 + \frac{\pi}{2}$ (D) $-1 + \frac{\pi}{2}$

38. Evaluate the improper integral $\int_0^1 x\sqrt{(1 + x)/(1 - x)}\, dx$.

(A) $\frac{\pi}{4} + 1$ (B) $1 + \frac{\pi}{2}$

(C) $-1 + \frac{\pi}{4}$ (D) $\frac{1}{2} + \frac{\pi}{4}$

39. A continuous real f on $[0, 1]$ satisfies $\int_0^1 f(x)\, dx = 1$ and $\int_0^1 f(x)^2\, dx = 1$. Find $\int_0^1 x^2 f(x)\, dx$.

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) 1

40. A continuous real f on $[0, 1]$ satisfies $\int_0^1 f(x)^2\, dx = 1/3$ and $\int_0^1 xf(x)\, dx = 1/3$. Find $\int_0^1 f(x)^3\, dx$.

(A) $\frac{1}{6}$ (B) $\frac{1}{9}$

(C) $\frac{1}{4}$ (D) $\frac{1}{3}$

41–50 / Preserved hard practice

41. For $a > 0$, $\int_0^{\pi/2} \frac{\sin^2 x}{a^2\cos^2 x + \sin^2 x} dx = \pi/6$. Find a.

(A) 3 (B) 4

(C) 2 (D) 1

42. Evaluate $\displaystyle\int_0^{\pi/2} \frac{\sin^4 x}{\sin^4 x + \cos^4 x}\, dx$.

(A) $-1 + \frac{\pi}{4}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{4} + 1$ (D) $\frac{\pi}{4} + 2$

43. Evaluate $\displaystyle\int_0^1 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{1-x}}\, dx$.

(A) $-\frac{1}{2}$ (B) $\frac{3}{2}$

(C) $\frac{5}{2}$ (D) $\frac{1}{2}$

44. Evaluate $\displaystyle\int_0^{\pi} x \sin^5 x\, dx$.

(A) $-1 + \frac{8\pi}{15}$ (B) $1 + \frac{8\pi}{15}$

(C) $\frac{8\pi}{15}$ (D) $\frac{8\pi}{15} + 2$

45. Evaluate $\displaystyle\int_0^1 \frac{x^3}{(x^2+1)^2}\, dx$.

(A) $\frac{\log(2)}{2} + \frac{3}{4}$ (B) $-\frac{1}{4} + \frac{\log(2)}{2}$

(C) $\frac{\log(2)}{2} + \frac{7}{4}$ (D) $-\frac{5}{4} + \frac{\log(2)}{2}$

46. A continuous function satisfies $f(x) + f(1-x) = 6x(1-x)$ on $[0,1]$. Find $\int_0^1 [f(x) + x^2] dx$.

(A) $\frac{17}{6}$ (B) $-\frac{1}{6}$

(C) $\frac{5}{6}$ (D) $\frac{11}{6}$

47. Let $F(x) = \int_0^x (t^2 - 2t) dt$. The minimum of F on $[0,3]$ is

(A) $-\frac{7}{3}$ (B) $-\frac{4}{3}$

(C) $\frac{2}{3}$ (D) $-\frac{1}{3}$

48. For $n \geq 1$, let $S_n = \sum_{k=1}^{n} \int_0^1 x^{k-1}(1-x) dx$. Find $\lim_{n\to\infty} S_n$.

(A) 3 (B) 2

(C) 1 (D) 0

49. Let $p(x) = ax + b$ satisfy $\int_0^1 p(x)dx = 1$ and $\int_0^1 xp(x)dx = 2/3$. Find $a + b$.

(A) 3 (B) 2

(C) 4 (D) 1

50. A coin has head probability $p = \int_0^1 2x(1-x)dx$. In three independent tosses, what is the probability of at least two heads?

(A) $-\frac{20}{27}$ (B) $\frac{34}{27}$

(C) $\frac{61}{27}$ (D) $\frac{7}{27}$

Answer key

Definite Integration

M-DEIN / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**A**	21	**A**	31	**C**	41	**C**
02	**D**	12	**C**	22	**A**	32	**D**	42	**B**
03	**B**	13	**B**	23	**A**	33	**D**	43	**D**
04	**A**	14	**B**	24	**D**	34	**C**	44	**C**
05	**D**	15	**A**	25	**A**	35	**A**	45	**B**
06	**B**	16	**A**	26	**A**	36	**C**	46	**C**
07	**D**	17	**B**	27	**C**	37	**D**	47	**B**
08	**D**	18	**B**	28	**D**	38	**A**	48	**C**
09	**B**	19	**D**	29	**C**	39	**A**	49	**B**
10	**B**	20	**C**	30	**B**	40	**C**	50	**D**

Chapter 17

Area under Curves

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Find the total area between $y = x^2 - 1$ and the x-axis on $[-2, 2]$.

(A) 6 (B) 5

(C) 4 (D) 3

02. Find the area enclosed by $y = x^3$ and $y = x$.

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$

(C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

03. Find the area enclosed by $y = |x|$ and $y = x^2$.

(A) $\frac{7}{3}$ (B) $\frac{1}{3}$

(C) $-\frac{2}{3}$ (D) $\frac{4}{3}$

04. Find the area between $y = \sin x$ and $y = \cos x$ on $[0, \pi/2]$.

(A) $-1 + 2\sqrt{2}$ (B) $-3 + 2\sqrt{2}$

(C) $-2 + 2\sqrt{2}$ (D) $2\sqrt{2}$

05. Find the area enclosed by $y = x^2$, $y = 2 - x$, and the y-axis in the first quadrant.

(A) $\frac{13}{6}$ (B) $\frac{1}{6}$

(C) $\frac{7}{6}$ (D) $\frac{19}{6}$

06. For $a > 0$, the area enclosed by $y = ax - x^2$ and the x-axis is $9/2$. Find a.

(A) 2 (B) 5

(C) 3 (D) 4

07. The area between $y = x^2$ and $y = k$ for $k > 0$ is $32/3$. Find k.

(A) 4 (B) 3

(C) 5 (D) 6

08. Find the area between $y = x(1 - x)$ and the x-axis on $[-1, 2]$.

(A) $\frac{23}{6}$ (B) $\frac{5}{6}$

(C) $\frac{11}{6}$ (D) $\frac{17}{6}$

09. Find the area enclosed by $y = x^2 - 2x$ and $y = 2x - x^2$.

(A) $\frac{11}{3}$ (B) $\frac{5}{3}$

(C) $\frac{14}{3}$ (D) $\frac{8}{3}$

10. Find the area bounded by $y = e^x$, $y = e^{-x}$, and $x = 1$.

(A) $-1 + e + e^{-1}$ (B) $-3 + e + e^{-1}$

(C) $-2 + e + e^{-1}$ (D) $e + e^{-1}$

11–40 / New middle-to-hard practice

11. For $a > 0$, the area enclosed by $y = ax - x^2$ and the x axis is 36. Find the area enclosed by this parabola and $y = 5$.

(A) $\frac{32}{3}$ (B) 12

(C) 18 (D) $\frac{64}{3}$

12. The area enclosed by $y = x^2$ and $y = k$, $k > 0$, is 36. Find the area of the part of this enclosed region with $x \geq 2$.

(A) $\frac{8}{3}$ (B) $\frac{4}{3}$

(C) $\frac{16}{3}$ (D) 6

13. Find the total area between $y = x(x - 1)(x - 3)$ and the x axis for $0 \leq x \leq 4$.

(A) $\frac{8}{3}$ (B) 8

(C) $\frac{10}{3}$ (D) 16

14. Find the total area between $y = (x^2 - 1)(x^2 - 4)$ and the x axis on $0 \leq x \leq 2$.

(A) $\frac{38}{15}$ (B) 4

(C) $\frac{22}{15}$ (D) $\frac{16}{15}$

15. The parabola $y = x^2$, its tangent at $(2, 4)$ and the x axis enclose a curvilinear region. Find its area.

(A) $\frac{4}{3}$ (B) $\frac{1}{3}$

(C) 2 (D) $\frac{2}{3}$

16. The parabola $x = y^2$, its tangent at (a^2, a) with $a > 0$, and the y axis enclose area 18. Find the x coordinate of the contact point.

(A) 6 (B) 36

(C) 72 (D) 18

17. Find the area of the bounded loop of $y^2 = x^2(4 - x)$.

(A) $\frac{64}{15}$ (B) $\frac{512}{15}$

(C) $\frac{128}{15}$ (D) $\frac{256}{15}$

18. Find the area of the bounded loop of $y^2 = x(1 - x)^2$.

(A) $\frac{16}{15}$ (B) $\frac{2}{3}$

(C) $\frac{4}{15}$ (D) $\frac{8}{15}$

19. Find the total area between $y = |\sin x + \cos x|$ and $y = |\sin x - \cos x|$ on $[0, \pi]$.

(A) $4 - 2\sqrt{2}$ (B) $8 - 4\sqrt{2}$

(C) $8 - 2\sqrt{2}$ (D) $4\sqrt{2}$

20. Find the total area between $y = |\sin 2x|$ and $y = \sin x$ on $[0, \pi]$.

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) 2 (D) 1

21. The disk $x^2 + (y - 1)^2 \leq 9$ is divided by $y = 1 + x^3 - x$. Find the area of the part above the curve.

(A) $\frac{9\pi}{2}$ (B) 9π

(C) 3π (D) $\frac{9\pi}{4}$

22. Find the area inside $x^2/9 + (y - 2)^2/4 \leq 1$ and above $y = 2 + x^3 - x$.

(A) $\frac{3\pi}{2}$ (B) 4π

(C) 6π (D) 3π

23. In the region $0 \leq x \leq 1$, $0 \leq y \leq 1 - x^2$, the part above $y = mx$, $m > 0$, has 13/32 of the total area. Find m.

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) 2 (D) $\frac{5}{2}$

24. In $0 \leq x \leq 2$, $0 \leq y \leq x^2$, the area above $y = mx$ is 5/6, where $0 < m < 2$. Find m.

(A) 1 (B) $\frac{1}{2}$

(C) 2 (D) $\frac{3}{2}$

25. The area enclosed by $y = x^2$ and $y = 2x + k$ is $32/3$. Find the area of the part of this region with $x \geq 0$.

(A) 12 (B) $\frac{16}{3}$

(C) $\frac{32}{3}$ (D) 9

26. For $a > 0$, the area enclosed by $y = x^2$ and $y = ax + 1$ is $8\sqrt{2}/3$. Find the area under $y = ax + 1$ above the x axis on $[0, 2]$.

(A) 4 (B) 8

(C) 6 (D) 3

27. Find the area between $y = e^x$ and its tangent at $x = 0$, for $0 \leq x \leq \ln 2$.

(A) $\frac{\log(2)^2}{2}$ (B) $-\log(2) + \frac{\log(2)^2}{2} + 1$

(C) $-\log(2) - \frac{\log(2)^2}{2} + 1$ (D) $1 - \log(2)$

28. Find the area enclosed by $y = \ln x$ and the line segment joining $(1, 0)$ to $(e, 1)$.

(A) $\frac{3}{2} - \frac{e}{2}$ (B) $-\frac{1}{2} + \frac{e}{2}$

(C) $3 - e$ (D) $-1 + \frac{e}{2}$

29. Let g be the inverse of $f(x) = x^3 + x$. Find the area between their graphs on $0 \leq x \leq 2$.

(A) $\frac{19}{4}$ (B) $\frac{21}{4}$

(C) $\frac{17}{4}$ (D) $\frac{5}{4}$

30. Let g be the inverse, on $[0, \infty)$, of $f(x) = x^2 + x$. Find the area between their graphs for $0 \leq x \leq 2$.

(A) $\frac{14}{3}$ (B) $\frac{7}{2}$

(C) $\frac{17}{6}$ (D) $\frac{7}{6}$

31. Find the area enclosed by the parabola $x = y^2$ and the line $x = y + 2$.

(A) 9 (B) $\frac{9}{2}$

(C) $\frac{3}{2}$ (D) 3

32. Find the area inside $x^2 + y^2 \leq 2$ and above $y = x^2$.

(A) $-\frac{1}{3} + \frac{\pi}{2}$ (B) $\frac{1}{3} + \frac{\pi}{4}$

(C) $\frac{1}{3} + \frac{\pi}{2}$ (D) $\frac{1}{3} + \pi$

33. For $0 \leq x \leq 2$, find the area above the x axis and below all three graphs $y = x/2$, $y = 1/2$ and $y = (x - 1)^2$.

(A) $\frac{\sqrt{2}}{6} + \frac{29}{48}$ (B) $\frac{5}{6}$

(C) $\frac{29}{48} - \frac{\sqrt{2}}{12}$ (D) $\frac{29}{48} - \frac{\sqrt{2}}{6}$

34. For $0 \le x \le 2$, consider the two regions between the x axis and respectively $y = x^2$ and $y = 2x - x^2$. Find the area lying in exactly one of these regions.

(A) 1 (B) 2

(C) 3 (D) $\frac{4}{3}$

35. A point (X, Y) is uniform in $[0, 2]^2$. Given $Y \ge X^2$, find the probability that also $Y \le 2 - X$.

(A) $\frac{7\sqrt{2}}{8}$ (B) $\frac{7\sqrt{2}}{16}$

(C) $\frac{7\sqrt{2}}{32}$ (D) $\frac{7}{24}$

36. A point (X, Y) is uniform in the unit square. Given $Y \le X/2$, find the probability that $Y \ge X^2$.

(A) $\frac{1}{48}$ (B) $\frac{1}{12}$

(C) $\frac{1}{3}$ (D) $\frac{1}{6}$

37. The area between $y = x^2 - 2x$ and the x axis on $[0, c]$ is one quarter of its total area on $[0, 3]$. Find c.

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) 2

38. The area between $y = x^2 - 1$ and the x axis on $[0, c]$ is $23/48$ of its total area on $[0, 2]$. Find c.

(A) $\frac{1}{2}$ (B) $\frac{7}{4}$

(C) $\frac{5}{4}$ (D) $\frac{3}{2}$

39. Let $f\prime(x) = 3x^2 - 2ax$, $f(0) = 0$ and $f(2) = 0$. Find the area enclosed by $y = f(x)$ and the x axis between 0 and 2.

(A) $-\frac{4}{3}$ (B) 2

(C) $\frac{8}{3}$ (D) $\frac{4}{3}$

40. A twice differentiable function satisfies $f\prime\prime(x) = 6x$, $f(0) = 1$ and $f(1) = 1$. Find the total area enclosed by $y = f(x)$ and $y = 1$.

(A) $\frac{1}{4}$ (B) 1

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

41–50 / Preserved hard practice

41. Find the total area enclosed between $y = x^4$ and $y = x^2$.

(A) $\frac{4}{15}$ (B) $-\frac{11}{15}$

(C) $\frac{19}{15}$ (D) $\frac{34}{15}$

42. Find the total area between $y = x^3 - 3x$ and the x-axis on $[-2, 2]$.

(A) 7 (B) 4

(C) 5 (D) 6

43. Find the area bounded by $y = 1/x$, $y = 1/x^2$, and $x = 2$.

(A) $\frac{1}{2} + \log(2)$ (B) $-\frac{3}{2} + \log(2)$

(C) $\log(2) + \frac{3}{2}$ (D) $-\frac{1}{2} + \log(2)$

44. Find the area enclosed by $y = x^2$ and $y = 2|x|$.

(A) $\frac{8}{3}$ (B) $\frac{14}{3}$

(C) $\frac{5}{3}$ (D) $\frac{11}{3}$

45. For $a > 0$, the region between $y = x^2$ and $y = 2ax$ has area 36. Find a.

(A) 4 (B) 3

(C) 2 (D) 5

46. The tangent to $y = x^2$ at $x = 2$ and the coordinate axes enclose a triangle. Find its area.

(A) 1 (B) 3

(C) 2 (D) 4

47. A circle of radius 2 centred at the origin and the line $x = 1$ bound a minor circular segment. Find its area.

(A) $-\sqrt{3} + \frac{4\pi}{3}$ (B) $-\sqrt{3} - 1 + \frac{4\pi}{3}$

(C) $-\sqrt{3} + 1 + \frac{4\pi}{3}$ (D) $-\sqrt{3} + 2 + \frac{4\pi}{3}$

48. Find the area enclosed by $x = y^2$ and $x = 2 - y^2$.

(A) $\frac{8}{3}$ (B) $\frac{5}{3}$

(C) $\frac{11}{3}$ (D) $\frac{14}{3}$

49. The region under $y = \sin x$ on $[0, \pi]$ is cut into two equal areas by $x = a$, with $0 < a < \pi$. Find a.

(A) $\frac{\pi}{2}$ (B) $1 + \frac{\pi}{2}$

(C) $-1 + \frac{\pi}{2}$ (D) $\frac{\pi}{2} + 2$

50. For $a > 0$, let $A(a)$ be the area enclosed by $y = x^2$, its tangent at $x = a$, and the y-axis. If $A(a) = 8/3$, find a.

(A) 4 (B) 2

(C) 1 (D) 3

Answer key

Area under Curves

M-ARCU / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**A**	21	**A**	31	**B**	41	**A**
02	**C**	12	**A**	22	**D**	32	**C**	42	**C**
03	**B**	13	**B**	23	**B**	33	**D**	43	**D**
04	**C**	14	**B**	24	**A**	34	**B**	44	**A**
05	**C**	15	**D**	25	**D**	35	**B**	45	**B**
06	**C**	16	**B**	26	**C**	36	**B**	46	**C**
07	**A**	17	**D**	27	**C**	37	**A**	47	**A**
08	**C**	18	**D**	28	**A**	38	**D**	48	**A**
09	**D**	19	**B**	29	**A**	39	**D**	49	**A**
10	**C**	20	**D**	30	**B**	40	**D**	50	**B**

Chapter 18

Differential Equations

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For $x > 0$, let $xy' + y = x^2$ and $y(1) = 2$. Find $y(2)$.

(A) $\frac{25}{6}$ (B) $\frac{7}{6}$

(C) $\frac{19}{6}$ (D) $\frac{13}{6}$

02. If $y' + 2y = e^x$ and $y(0) = 1$, find $y(\ln 2)$.

(A) $-\frac{1}{6}$ (B) $\frac{11}{6}$

(C) $\frac{17}{6}$ (D) $\frac{5}{6}$

03. If $y' - y = x$ and $y(0) = 0$, find $y(1)$.

(A) $-2 + e$ (B) e

(C) $-3 + e$ (D) $-1 + e$

04. If $y' = xy^2$ and $y(0) = 1$, find $y(1)$ on the branch through 0.

(A) 2 (B) 4

(C) 1 (D) 3

05. If $y' = y \ln y$, $y > 0$, and $y(0) = e$, find $\ln y(\ln 2)$.

(A) 2 (B) 1

(C) 4 (D) 3

06. If $y' + y \tan x = \sec x$ and $y(0) = 0$, find $y(\pi/4)$.

(A) $\frac{\sqrt{2}}{2} + 2$ (B) $\frac{\sqrt{2}}{2} + 1$

(C) $-1 + \frac{\sqrt{2}}{2}$ (D) $\frac{\sqrt{2}}{2}$

07. If $y' = 2\sqrt{1-y^2}$ and $y(0) = 0$, find $y(\pi/8)$ before the first point where $y = 1$.

(A) $\frac{\sqrt{2}}{2} + 2$ (B) $\frac{\sqrt{2}}{2}$

(C) $-1 + \frac{\sqrt{2}}{2}$ (D) $\frac{\sqrt{2}}{2} + 1$

08. If $y' = x(1+y^2)$ and $y(0) = 1$, find $y(\sqrt{\pi/6})$ on its branch through 0.

(A) $1 + \sqrt{3}$ (B) $\sqrt{3} + 2$

(C) $\sqrt{3}$ (D) $-1 + \sqrt{3}$

09. If $y' + 2xy = 2x$ and $y(0) = 3$, find $y(1)$.

(A) $\frac{2}{e} + 3$ (B) $\frac{2}{e} + 1$

(C) $\frac{2}{e}$ (D) $\frac{2}{e} + 2$

10. If $y' = x/y$, $y(0) = -2$, find $y(1)$.

(A) $-\sqrt{5}$ (B) $-\sqrt{5} - 1$

(C) $1 - \sqrt{5}$ (D) $2 - \sqrt{5}$

11–40 / New middle-to-hard practice

11. For $x > -1$, $y' + 2y/(x+1) = (x+1)^2$, $y(0) = 1$. Find $y(1)$.

(A) $\frac{9}{10}$ (B) $\frac{13}{5}$

(C) $\frac{9}{5}$ (D) $\frac{8}{5}$

12. If $y' - 2y = e^{2x}(x+1)$ and $y(0) = 1$, find $y(1)$.

(A) $\frac{5e^2}{2}$ (B) $\frac{3e^2}{2}$

(C) e^2 (D) $5e^2$

13. If $y' + y = xy^3$ and $y(0) = -1$, find $y(1)$ on this solution branch.

(A) $\frac{\sqrt{2}}{\sqrt{3+e^2}}$ (B) $-\frac{\sqrt{2}}{\sqrt{1+e^2}}$

(C) $-\frac{\sqrt{2}}{\sqrt{3+e^2}}$ (D) $-\frac{1}{\sqrt{1+e^2}}$

14. For $x > 0$, $y' - y/x = xy^2$ and $y(1) = -1$. Find $y(2)$.

(A) $\frac{3}{5}$ (B) $-\frac{1}{5}$

(C) $-\frac{6}{5}$ (D) $-\frac{3}{5}$

15. On the branch $|y| < \pi/2$, solve $y' + \tan y = x \sec y$ with $y(0) = 0$. Find $y(1)$.

(A) $-\arcsin\left(e^{-1}\right)$ (B) $\arctan\left(e^{-1}\right)$

(C) $\arcsin\left(1 - e^{-1}\right)$ (D) $\arcsin\left(e^{-1}\right)$

16. For $x > 0$, $xy' \cos y + 2 \sin y = 3x^2$, $y(1) = 0$. On $|y| < \pi/2$, find $y(\sqrt{3/2})$.

(A) $\arcsin\left(\frac{7}{8}\right)$ (B) $\arcsin\left(\frac{5}{8}\right)$

(C) $-\arcsin\left(\frac{5}{8}\right)$ (D) $\arcsin\left(\frac{3}{4}\right)$

17. On $|y| < \pi/2$, $y' + x \sin 2y = x^3 \cos^2 y$, $y(0) = \pi/4$. Find $y(1)$.

(A) $-\arctan\left(\frac{3}{2e}\right)$ (B) $\arctan\left(\frac{3}{e}\right)$

(C) $\arctan\left(\frac{3}{2e}\right)$ (D) $\arctan\left(e^{-1}\right)$

18. On $|y| < \pi/2$, $y' - \sin y \cos y = e^x \cos^2 y$, $y(0) = 0$. Find $y(\ln 2)$.

(A) $-\arctan(\log(4))$ (B) $\arctan(\log(2))$

(C) $\arctan(\log(4))$ (D) $\arctan(\log(16))$

19. For $x > 0$, $y' = y/x + x/y$, $y(1) = -1$. Find $y(\sqrt{e})$ on the negative solution branch.

(A) $-e^{\frac{1}{2}}$ (B) $-2e^{\frac{1}{2}}$

(C) $-\sqrt{2}e^{\frac{1}{2}}$ (D) $\sqrt{2}e^{\frac{1}{2}}$

20. For $x > 0$, $y' = y/x + (y/x)^2$, $y(1) = 1$. Find the finite upper endpoint of the maximal solution interval containing 1.

(A) e^2 (B) $e^{\frac{1}{2}}$

(C) $2e$ (D) e

21. If $y' = (x + y + 1)/(x + y - 1)$ and $y(0) = 2$, find the first x for which $x + y = 4$.

(A) $1 - \frac{\log(2)}{2}$ (B) $\frac{\log(2)}{2} + 1$

(C) $2 - \log(2)$ (D) $\frac{\log(2)}{2}$

22. If $y' = (x + y)^2 - 1$ and $y(0) = 1$, find $y(1/2)$.

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) 2 (D) $\frac{5}{2}$

23. If $y' = (1 + y^2)/(1 + x^2)$ and $y(1) = 0$, find $\int_0^1 y(x)\,dx$.

(A) $-1 + \log(4)$ (B) $1 - \log(2)$

(C) $-\log(4)$ (D) $1 - \log(4)$

24. If $y' = 2xy^2$ and $y(0) = -1$, find the geometric area between this solution and the x axis on $[0, 1]$.

(A) $-\frac{\pi}{4}$ (B) 1

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

25. A real solution of $y' = y(y-2)$ on $[0,1]$ satisfies $y(0) = y(1)$ and $\int_0^1 y^2\,dx = 4$. Find $y(0)$.

(A) -2 (B) 0

(C) 4 (D) 2

26. A real solution of $y' = (y-1)(y-2)(y-4)$ on $[0,2]$ satisfies $y(0) = y(2)$ and $\int_0^2 y^2\,dx = 8$. Find $y(0) + y(1)$.

(A) 8 (B) 2

(C) 4 (D) 6

27. If $(1+x^2)y' + 2xy = 1/(1+x^2)$ and $y(0) = 1$, find $y(1)$.

(A) $\frac{\pi}{4} + 1$ (B) $\frac{\pi}{8} + \frac{1}{2}$

(C) $\frac{\pi}{8}$ (D) $\frac{1}{4} + \frac{\pi}{8}$

28. On $(-\pi/2, \pi/2)$, $\cos x\, y' - \sin x\, y = \sec^2 x$ and $y(0) = 2$. Find $y(\pi/4)$.

(A) $\sqrt{2}$ (B) $\frac{3\sqrt{2}}{2}$

(C) $3\sqrt{2}$ (D) $2\sqrt{2}$

29. The solution of $y' = y/(x+y^2)$ passes through $(0,1)$. Find $y(2)$ on this branch.

(A) 1 (B) 2

(C) -1 (D) 3

30. The solution of $y' = 1/(x+e^y)$ has $y(0) = 0$. Find the x coordinate where $y = \ln 2$ on this branch.

(A) $\log(16)$ (B) $\log(4)$

(C) $\frac{2}{\log(2)}$ (D) $\log(2)$

31. A curve orthogonal to every member of $y = Cx^2$ passes through $(2,1)$. Find its positive ordinate at $x = 1$.

(A) $\frac{\sqrt{6}}{2}$ (B) $\frac{\sqrt{10}}{2}$

(C) $\sqrt{5}$ (D) $-\frac{\sqrt{10}}{2}$

32. A curve orthogonal to the family $y = C/x^2$ passes through $(2,-1)$. Find its ordinate at $x = 3$ on the same branch.

(A) $\frac{\sqrt{14}}{2}$ (B) $-\sqrt{7}$

(C) $-\frac{\sqrt{14}}{2}$ (D) $-\frac{\sqrt{10}}{2}$

33. A continuous solution has $y(0) = 0$, $y' + y = 1$ for $0 < x < \ln 2$, and $y' + y = 3$ for $x > \ln 2$. Find $y(\ln 4)$.

(A) $\frac{11}{4}$ (B) $\frac{7}{4}$

(C) $\frac{5}{4}$ (D) $\frac{3}{4}$

34. A continuous function is differentiable away from $x = 1$, has $y(0) = 2$, and satisfies $y\prime = -y$ for $x < 1$, $y\prime = -2y$ for $x > 1$. Find $y(2)$.

(A) $\frac{2}{e^4}$ (B) $\frac{2}{e^3}$

(C) $\frac{2}{e^2}$ (D) e^{-3}

35. For real constants a, b, $y\prime = ay + b$, $y(0) = 0$, $y(\ln 2) = 1$ and $y(\ln 4) = 3$. Find $y\prime(\ln 8)$.

(A) 8 (B) 4

(C) 9 (D) 7

36. For constant a, b, $y\prime = ay + b$ with $y(0) = 1$, $y(1) = 3$ and $y(2) = 7$. Find $y(3)$.

(A) 14 (B) 11

(C) 16 (D) 15

37. A temperature obeys $T\prime = -k(T - A)$ with unknown constants $k > 0, A$. Its values at times 0,1,2 are 100,70,50. Find $T(3)$.

(A) $\frac{110}{3}$ (B) $\frac{100}{3}$

(C) 40 (D) 30

38. A population model obeys $N\prime = kN(1 - N/K)$, $k, K > 0$. Given $N(0) = K/5$ and $N(2) = K/2$, find $N(3)/K$.

(A) $\frac{2}{3}$ (B) $\frac{4}{5}$

(C) $\frac{1}{3}$ (D) $\frac{3}{4}$

39. For $\sqrt{1 + (y\prime\prime)^2} = y\prime + 1$, which gives the order, degree after legitimate radical removal, and the full admissible range of $y\prime$?

(A) Order 2, degree 2; $y\prime \in \mathbb{R}$ (B) Order 2, degree 1; $y\prime \geq -1$

(C) Order 2, degree 2; $y\prime \geq 0$ (D) Order 1, degree 2; $y\prime \leq -2$

40. Interpreting the real power $(y\prime\prime)^{2/3}$ as the square of a real cube root, consider $(y\prime\prime)^{2/3} + (y\prime)^2 = 1$. Which gives its polynomial degree after radical removal and the allowed slopes?

(A) Degree 2; $-1 \leq y\prime \leq 1$ (B) Degree 2; every real $y\prime$

(C) Degree 6; every real $y\prime$ (D) Degree 3; $-1 \leq y\prime \leq 1$

41–50 / Preserved hard practice

41. If $y' = (x + y + 1)/(x + 1)$ for $x > -1$ and $y(0) = 1$, find $y(1)$.

(A) $2\log(2) + 4$ (B) $2\log(2) + 2$

(C) $\log(4) + 3$ (D) $1 + \log(4)$

42. For $x > 0$, let $xy' - y = x^2 \ln x$ and $y(1) = 0$. Find $y(e)$.

(A) $2 + e$ (B) e

(C) $-1 + e$ (D) $1 + e$

43. Let $y > 0$ satisfy $y' = (1+y^2)/(1+x^2)$ and $y(0) = 1$. Find $y(1/2)$ before the first vertical asymptote.

(A) 3 (B) 4

(C) 2 (D) 5

44. If $y' + 2y = \sin x$ and $y(0) = 0$, find $5y(\pi)$.

(A) $e^{-2\pi} + 3$ (B) $e^{-2\pi} + 2$

(C) $e^{-2\pi}$ (D) $e^{-2\pi} + 1$

45. For $x > 0$, let $y' + 2y/x = x^2$ and $y(1) = 1$. Find $y(2)$.

(A) $\frac{9}{5}$ (B) $\frac{19}{5}$

(C) $\frac{4}{5}$ (D) $\frac{14}{5}$

46. A curve satisfies $y' = 2x$ and has a tangent through the origin at $x = 2$. Find $y(0)$.

(A) 3 (B) 6

(C) 5 (D) 4

47. Let $y' = 2xy$, $y(0) > 0$, and $\int_0^1 2xy(x)dx = e - 1$. Find $y(1)$.

(A) $-1 + e$ (B) $2 + e$

(C) e (D) $1 + e$

48. A positive quantity N satisfies $N' = -kN$, $k > 0$. It drops from 80 to 20 in 4 hours. How many hours from the start are needed to reach 5?

(A) 8 (B) 9

(C) 7 (D) 10

49. Let $y' = 3x^2 - 6x$ and $y(0) = a$. If the curve touches the x-axis at its local minimum, find a.

(A) 5 (B) 6

(C) 3 (D) 4

50. For $x > 0$, $xy\prime + y = x^2$ and $y(1) = 1$. Find the area under the solution curve on $[1, 2]$.

(A) $\frac{2\log(2)}{3} + \frac{16}{9}$

(B) $\frac{2\log(2)}{3} + \frac{7}{9}$

(C) $\frac{2\log(2)}{3} + \frac{25}{9}$

(D) $-\frac{2}{9} + \frac{2\log(2)}{3}$

Answer key

Differential Equations

M-DIEQ / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**A**	31	**B**	41	**B**
02	**D**	12	**A**	22	**B**	32	**C**	42	**B**
03	**A**	13	**C**	23	**D**	33	**B**	43	**A**
04	**A**	14	**D**	24	**C**	34	**B**	44	**D**
05	**A**	15	**D**	25	**D**	35	**A**	45	**A**
06	**D**	16	**B**	26	**C**	36	**D**	46	**D**
07	**B**	17	**C**	27	**B**	37	**A**	47	**C**
08	**C**	18	**C**	28	**C**	38	**A**	48	**A**
09	**B**	19	**C**	29	**B**	39	**C**	49	**D**
10	**A**	20	**D**	30	**B**	40	**A**	50	**B**

Chapter 19

Straight Lines

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The foot of the perpendicular from $(3,4)$ to $y=x$ has x coordinate

(A) $\frac{9}{2}$ (B) $\frac{7}{2}$

(C) $\frac{11}{2}$ (D) $\frac{5}{2}$

02. The line through origin and intersection of $x+2y=5$ and $2x-y=0$ has slope

(A) 2 (B) 4

(C) 3 (D) 1

03. A line passes through $(1,2)$ and has equal positive intercepts on both axes. Its distance from the origin is

(A) $1+\frac{3\sqrt{2}}{2}$ (B) $\frac{3\sqrt{2}}{2}$

(C) $-1+\frac{3\sqrt{2}}{2}$ (D) $2+\frac{3\sqrt{2}}{2}$

04. The point $(2,1)$ is reflected in $x+y=0$. Sum of coordinates of its image is

(A) -1 (B) -3

(C) -4 (D) -2

05. The distance from origin to any line through $(3,4)$ is at most

(A) 6 (B) 5

(C) 4 (D) 7

06. A line $y=mx+1$ is tangent to the circle $x^2+y^2=1$. Find m^2.

(A) 0 (B) -1

(C) 2 (D) 1

07. A line cuts parabola $y^2 = 4x$ at points with y coordinates 1 and 3. Its slope is

(A) 0 (B) 2

(C) 3 (D) 1

08. How many lines through $(0,0)$ and one of $(1,1), (2,2), (1,2), (2,1)$ are distinct?

(A) 5 (B) 2

(C) 3 (D) 4

09. A line has direction vector $(3,4)$. Unit normal with positive first component is (a,b). Find a^2.

(A) $\frac{16}{25}$ (B) $-\frac{9}{25}$

(C) $\frac{41}{25}$ (D) $\frac{66}{25}$

10. A line has positive intercepts a, b and passes through $(1,1)$. Minimum triangle area $ab/2$ is

(A) 1 (B) 3

(C) 4 (D) 2

11–40 / New middle-to-hard practice

11. The ray $r = (1,2) + t(2,1)$, $t \geq 0$, meets $3x - 4y + c = 0$ at distance $3\sqrt{5}$ from its starting point. Find c.

(A) 11 (B) -1

(C) 1 (D) -11

12. The ray $r = (2,-1) + t(1,1)$, $t \geq 0$, meets $2x + y + k = 0$ at distance $2\sqrt{2}$ from its starting point. Find k.

(A) -3 (B) 9

(C) -9 (D) 3

13. Reflect $P = (3,2)$ in $x - 2y = 1$, obtaining R. Find the y intercept of the line joining R to $(1,1)$.

(A) $\frac{17}{7}$ (B) $\frac{3}{14}$

(C) $\frac{17}{14}$ (D) $\frac{11}{14}$

14. The intersection of $x + y = 4$ and $x - y = 2$ is reflected in $2x + y = 2$, obtaining R. Find the distance from R to $(2,3)$.

(A) 10 (B) $\sqrt{13}$

(C) $\sqrt{17}$ (D) 5

15. The foot of the perpendicular from $(4,5)$ to a line L is $(1,1)$. Find the distance of L from the origin.

(A) $\frac{7}{5}$ (B) $\frac{1}{5}$

(C) $\frac{14}{5}$ (D) $\frac{7}{25}$

16. The foot of the perpendicular from the origin to L is $(1, 2)$. If H is the foot from $(3, 4)$ to L, find $x_H + y_H$.

(A) $\frac{19}{5}$ (B) $\frac{11}{5}$

(C) $\frac{7}{5}$ (D) $\frac{17}{5}$

17. P is $(t, 0)$, $t > 0$, equidistant from $3x + 4y = 12$ and $4x + 3y = 8$. Find this common distance.

(A) $\frac{24}{7}$ (B) $\frac{8}{7}$

(C) $\frac{24}{35}$ (D) $\frac{12}{35}$

18. The angle bisector with negative slope of $3x + 4y = 0$ and $4x + 3y = 0$ meets $x + 2y = 6$ at P. Find OP.

(A) $2\sqrt{2}$ (B) $6\sqrt{2}$

(C) 6 (D) $3\sqrt{2}$

19. For $a > 0$, the line $x + ay + a^2 = 0$ cuts a triangle of area 4 with the coordinate axes. Find its distance from the origin.

(A) $2\sqrt{5}$ (B) $\frac{4\sqrt{3}}{3}$

(C) $\frac{2\sqrt{5}}{5}$ (D) $\frac{4\sqrt{5}}{5}$

20. Positive a, c, b are in that order in GP, $a + b = 5$, $c = 2$, and $a < b$. Find the distance of $ax + by + c = 0$ from the origin.

(A) $\frac{2\sqrt{5}}{5}$ (B) $\frac{\sqrt{17}}{17}$

(C) $\frac{2\sqrt{17}}{17}$ (D) $\frac{4\sqrt{17}}{17}$

21. The equation $2xy - 4x - 6y + c = 0$ represents two straight lines. These and the coordinate axes form a rectangle. Find its diagonal length.

(A) $\sqrt{5}$ (B) $\sqrt{13}$

(C) 5 (D) 13

22. The equation $4xy + 6x - 8y + c = 0$ represents two straight lines. Find the area of the rectangle that these lines form with the coordinate axes.

(A) 6 (B) $\frac{3}{2}$

(C) 3 (D) 12

23. The four lines $x \pm 2y \pm c = 0$, $c > 0$, enclose a rhombus whose incircle radius is 2. Find its area.

(A) 10 (B) 20

(C) $8\sqrt{5}$ (D) 40

24. A rhombus has diagonals in ratio $3 : 4$ and area 24. Find its incircle radius.

(A) $\frac{6}{5}$ (B) $\frac{5}{2}$

(C) $\frac{12}{5}$ (D) $\frac{24}{5}$

25. The lines $x+2y=5$, $2x-y=0$ and $(k+1)x+(k-1)y=8$ are concurrent. Find the triangle area cut off by the third line on the coordinate axes.

(A) 4 (B) 8

(C) 2 (D) 16

26. The lines $x+y=1$, $x-2y=4$ and $ax+(a+1)y=3$ are concurrent. Find the area cut off by the third line on the coordinate axes.

(A) $\frac{3}{20}$ (B) $\frac{9}{10}$

(C) $\frac{9}{20}$ (D) $\frac{9}{40}$

27. As real k varies, find the greatest distance from the origin to $(k+1)x+(2k-1)y+3-k=0$.

(A) $\frac{5}{3}$ (B) $\frac{\sqrt{41}}{3}$

(C) $\sqrt{41}$ (D) $\frac{4}{3}$

28. As real k varies, find the largest distance from $(1,4)$ to $(k-1)x+(k+2)y-3k=0$.

(A) $\sqrt{5}$ (B) 3

(C) $2\sqrt{10}$ (D) $\sqrt{10}$

29. A line has positive intercepts and passes through $(2,3)$. Among such lines it cuts off the least possible triangle area with the axes. Find its distance from the origin.

(A) $6\sqrt{13}$ (B) $\frac{12\sqrt{5}}{5}$

(C) $\frac{12\sqrt{13}}{13}$ (D) $\frac{6\sqrt{13}}{13}$

30. A line with positive axis intercepts passes through $(1,4)$ and has the least possible sum of its intercepts. Find the area of its intercept triangle.

(A) 8 (B) $\frac{9}{2}$

(C) 9 (D) 12

31. For $A=(1,2)$, $B=(5,4)$ and $P=(t,0)$, triangle ABP has area 6 and $t>0$. Find t.

(A) 6 (B) -9

(C) 9 (D) 3

32. Triangle with vertices $(0,1)$, $(4,3)$ and $P=(t,t)$ has area 5. P is in the first quadrant. Find OP.

(A) $3\sqrt{2}$ (B) $5\sqrt{2}$

(C) 7 (D) $7\sqrt{2}$

33. The line parallel and equidistant from $3x+4y=5$ and $3x+4y=20$ meets $x-y=1$ at P. Find x_P+y_P.

(A) $\frac{26}{7}$ (B) $\frac{13}{7}$

(C) $\frac{19}{7}$ (D) $\frac{33}{7}$

34. For $a, b > 0$, the distance between $ax + by = 1$ and $2ax + 2by + 6 = 0$ is 2, and $a + b = \sqrt{6}$. Find the acute angle between $ax + by = 1$ and $bx + ay = 1$.

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) 0 (D) $\frac{\pi}{6}$

35. The acute angle between $y = mx$ and $y = (m + 1)x$ is $\arctan(1/7)$, with $m > 0$. The first line meets $x + y = 6$ at P. Find OP.

(A) $2\sqrt{2}$ (B) $3\sqrt{5}$

(C) $\sqrt{5}$ (D) $2\sqrt{5}$

36. Lines $y = mx + 1$ and $y = (2m - 3)x - 1$ are perpendicular, with $0 < m < 1$. Find the sum of coordinates of their intersection.

(A) -1 (B) $\frac{1}{5}$

(C) $\frac{7}{5}$ (D) $-\frac{1}{5}$

37. New coordinates (X, Y) have origin $(1, 2)$, positive X direction $(1, 1)$ and positive Y direction $(-1, 1)$. The line is $X + 2Y = 3$. Find the sum of its old x and y intercepts.

(A) $-\frac{10}{3} - 2\sqrt{2}$ (B) $-5 - 3\sqrt{2}$

(C) $2\sqrt{2} + \frac{10}{3}$ (D) $-\frac{10}{3} + 2\sqrt{2}$

38. New coordinates have origin $(2, -1)$, positive X direction along the old positive y axis and positive Y direction along the old negative x axis. Find the area cut off on the old axes by $X - 2Y = 4$.

(A) $\frac{49}{4}$ (B) $\frac{9}{4}$

(C) $\frac{25}{4}$ (D) $\frac{49}{2}$

39. L varies over all lines through $(2, 1)$. H is the foot of the perpendicular from the origin to L. Find the greatest possible x coordinate of H.

(A) $1 + \frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{5}}{2}$

(C) $1 + \sqrt{5}$ (D) 2

40. Reflect $A = (1, 0)$ in a variable line through $P = (3, 2)$ to obtain Q. Find the greatest possible OQ.

(A) $-2\sqrt{2} + \sqrt{13}$ (B) $\sqrt{21}$

(C) $2\sqrt{2} + \sqrt{13}$ (D) $\sqrt{2} + \sqrt{13}$

41–50 / Preserved hard practice

41. For each real t, let $L_t : (t+1)x + (2t-1)y = 5t+1$. The member perpendicular to $3x+4y=7$ reflects the origin to P. Find $x_P + y_P$.

(A) $\frac{32}{75}$ (B) $\frac{107}{75}$

(C) $\frac{182}{75}$ (D) $-\frac{43}{75}$

42. A line through $(1,2)$ cuts finite positive intercepts a, b on the coordinate axes. Its distance from the origin is 2. Find ab.

(A) $\frac{31}{3}$ (B) $\frac{25}{3}$

(C) $\frac{22}{3}$ (D) $\frac{28}{3}$

43. Lines through $(3,4)$ have perpendicular feet F from the origin. For those with $x_F + y_F = 1$, find the product of their squared distances from the origin.

(A) $\frac{29}{2}$ (B) $\frac{23}{2}$

(C) $\frac{25}{2}$ (D) $\frac{27}{2}$

44. Point M varies on $x+y=0$. Find the minimum of $PM + MQ$, where $P=(1,3)$ and $Q=(4,0)$.

(A) $2+5\sqrt{2}$ (B) $5\sqrt{2}$

(C) $-1+5\sqrt{2}$ (D) $1+5\sqrt{2}$

45. Two distinct members of $y = tx + t^2 + 1$ pass through $(2,k)$ and make an acute angle of 45°. Find the product of all possible real values of k.

(A) 4 (B) 3

(C) 5 (D) 6

46. A line through $(2,3)$ cuts positive intercepts a, b. Find the minimum value of a^2+b^2.

(A) $2+\left(2^{\frac{2}{3}}+3^{\frac{2}{3}}\right)^3$ (B) $1+\left(2^{\frac{2}{3}}+3^{\frac{2}{3}}\right)^3$

(C) $-1+\left(2^{\frac{2}{3}}+3^{\frac{2}{3}}\right)^3$ (D) $\left(2^{\frac{2}{3}}+3^{\frac{2}{3}}\right)^3$

47. Line $y = mx + c$, with $m, c > 0$, is tangent to $y^2 = 8x$ and has distance 1 from the origin. Find m^2.

(A) $-\frac{1}{2}+\frac{\sqrt{17}}{2}$ (B) $-\frac{3}{2}+\frac{\sqrt{17}}{2}$

(C) $\frac{1}{2}+\frac{\sqrt{17}}{2}$ (D) $\frac{3}{2}+\frac{\sqrt{17}}{2}$

48. Circles $x^2+y^2=4$ and $(x-4)^2+y^2=1$ have external and internal common tangents. If u is the squared slope of an external common tangent and v that of an internal common tangent, find uv.

(A) $\frac{73}{35}$ (B) $\frac{38}{35}$

(C) $\frac{3}{35}$ (D) $-\frac{32}{35}$

49. How many lines through $(2,3)$ have positive integer intercepts on both coordinate axes?

(A) 6 (B) 4

(C) 3 (D) 5

50. An integer point (x,y) is chosen uniformly from $x,y \geq 0$, $x+y \leq 6$. Find the probability its distance from $2x-y-1=0$ is at most $1/\sqrt{5}$.

(A) $\frac{1}{7}$ (B) $\frac{1}{4}$

(C) $\frac{2}{7}$ (D) $\frac{3}{14}$

Answer key

Straight Lines

M-STLI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**B**	31	**D**	41	**A**
02	**A**	12	**C**	22	**C**	32	**D**	42	**B**
03	**B**	13	**C**	23	**B**	33	**A**	43	**C**
04	**B**	14	**D**	24	**C**	34	**A**	44	**B**
05	**B**	15	**A**	25	**A**	35	**D**	45	**A**
06	**A**	16	**D**	26	**D**	36	**D**	46	**D**
07	**D**	17	**C**	27	**B**	37	**A**	47	**A**
08	**C**	18	**B**	28	**D**	38	**A**	48	**C**
09	**A**	19	**D**	29	**C**	39	**A**	49	**B**
10	**D**	20	**C**	30	**C**	40	**C**	50	**B**

Chapter 20

Circle

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The radical axis of $x^2+y^2=25$ and $x^2+y^2-6x=7$ has x coordinate

(A) 5 (B) 3

(C) 2 (D) 4

02. The chord of contact from $(5,0)$ to $x^2+y^2=9$ is a vertical line at $x=$

(A) $\frac{19}{5}$ (B) $\frac{14}{5}$

(C) $\frac{4}{5}$ (D) $\frac{9}{5}$

03. The midpoint of chord $x+y=2$ of $x^2+y^2=4$ has x coordinate

(A) 2 (B) 3

(C) 1 (D) 0

04. A circle tangent to both positive coordinate axes has centre (r,r). If it passes through $(2,2)$, the product of its two possible radii is

(A) 8 (B) 10

(C) 7 (D) 9

05. A chord of $x^2+y^2=25$ has midpoint $(1,2)$. Its length is

(A) $2+4\sqrt{5}$ (B) $-1+4\sqrt{5}$

(C) $4\sqrt{5}$ (D) $1+4\sqrt{5}$

06. A circle centre lies on $y=x$ and passes through $(0,0),(4,0)$. Radius squared is

(A) 10 (B) 7

(C) 9 (D) 8

07. The two tangents from a point P to a unit circle meet at angle $60°$. Distance from P to centre is

(A) 1 (B) 2

(C) 3 (D) 4

08. How many integer points lie on $x^2 + y^2 = 5$?

(A) 10 (B) 8

(C) 7 (D) 9

09. An integer point is chosen uniformly from the disc $x^2+y^2 \le 9$. Find the probability that $x^2+y^2 \le 1$.

(A) $\frac{5}{29}$ (B) $\frac{5}{21}$

(C) $\frac{1}{9}$ (D) $\frac{1}{5}$

10. The line $y = x + b$ is tangent to $x^2 + y^2 = 8$. Find b^2.

(A) 18 (B) 16

(C) 17 (D) 15

11–40 / New middle-to-hard practice

11. A circle is tangent to the x axis and passes through $(0, 1)$ and $(4, 3)$. Its centre has positive x coordinate. Find its radius.

(A) $5 - \sqrt{15}$ (B) $2\sqrt{15}$

(C) $10 - 2\sqrt{15}$ (D) $2\sqrt{15} + 10$

12. A circle lies in the first quadrant, is tangent to both coordinate axes, and passes through $(2, 4)$. Its radius is less than 5. Find its area.

(A) 16π (B) 4π

(C) 2π (D) 100π

13. Find the length of the common chord of $x^2 + y^2 = 25$ and $x^2 + y^2 - 8x - 6y + 5 = 0$.

(A) 6 (B) 4

(C) 10 (D) 8

14. Find the common chord length of $x^2 + y^2 - 2x + 2y - 23 = 0$ and $x^2 + y^2 - 8x - 6y = 0$.

(A) $\frac{5\sqrt{3}}{2}$ (B) $5\sqrt{3}$

(C) 10 (D) $5\sqrt{2}$

15. P runs along $x + y = 5$. Tangents from P touch $x^2 + y^2 = 9$, and M is the midpoint of their contact chord. The locus of M lies on a circle. Find that circle radius.

(A) $\frac{9}{10}$ (B) $\frac{3\sqrt{2}}{5}$

(C) $\frac{9\sqrt{2}}{10}$ (D) $\frac{9\sqrt{2}}{5}$

16. P runs along $x + y = 8$. Tangents from P touch $(x-1)^2 + (y-1)^2 = 4$. Their contact chord midpoint lies on a circle. Find the distance of that locus-circle centre from the origin.

(A) $\frac{2\sqrt{2}}{3}$ (B) $4\sqrt{2}$

(C) $\frac{\sqrt{2}}{3}$ (D) $\frac{4\sqrt{2}}{3}$

17. From $P = (7, 10)$, tangents PA, PB to a circle centred at $(1, 2)$ make an angle $60°$. Find AB.

(A) $10\sqrt{3}$ (B) 10

(C) $5\sqrt{3}$ (D) 5

18. Tangents PA, PB to $x^2 + y^2 = 9$ are perpendicular. Find the area of triangle PAB.

(A) $\frac{9}{4}$ (B) 9

(C) 18 (D) $\frac{9}{2}$

19. A circle passes through $(2, 0)$ and $(0, 4)$ and intersects $x^2 + y^2 = 4$ orthogonally. Find its radius.

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$

(C) $\sqrt{5}$ (D) $\frac{25}{4}$

20. A circle with centre on the x axis passes through the origin and intersects $(x-1)^2 + y^2 = 4$ orthogonally. Find the x coordinate of its other x-axis intersection.

(A) -6 (B) -3

(C) 3 (D) $-\frac{3}{2}$

21. Two circles have radii with sum 9 and absolute difference 3. A common external tangent segment between contact points has length 8. How many common tangent lines do the circles have?

(A) 2 (B) 4

(C) 1 (D) 0

22. Two disjoint circles have centre distance 15. Their common external and internal tangent segments have lengths 12 and 9, respectively. Find the product of their radii.

(A) $\frac{81}{4}$ (B) $\frac{63}{2}$

(C) 36 (D) $\frac{63}{4}$

23. A circle passes through $(0, 0)$ and $(6, 0)$. A chord parallel to $(1, 2)$ has midpoint $(2, 1)$. Find this chord length.

(A) $4\sqrt{2}$ (B) 4

(C) $2\sqrt{37}$ (D) $2\sqrt{2}$

24. A circle has centre on $y = 2x$. A chord parallel to $(1, 1)$ has midpoint $(3, 1)$ and length 4. Find the squared radius.

(A) $\frac{68}{9}$ (B) $\frac{50}{9}$

(C) $\frac{122}{9}$ (D) $\frac{86}{9}$

25. An integer point is chosen uniformly from $x > 0$, $y > 0$, $x^2 + y^2 \leq 25$. Find the probability that $x + y > 5$.

(A) $\frac{1}{3}$ (B) $\frac{2}{5}$

(C) $\frac{1}{5}$ (D) $\frac{5}{16}$

26. Choose a uniformly from $[-2, 2]$. Find the probability that $(2a, a + 1)$ lies strictly inside the larger segment of $x^2 + y^2 - 2x - 2y - 8 = 0$ cut by $x - y + 1 = 0$.

(A) $\frac{7}{10}$ (B) $\frac{1}{5}$

(C) $\frac{9}{10}$ (D) $\frac{9}{20}$

27. A rectangle of area 48 is inscribed in a circle of radius 5. Find the ratio of its longer side to its shorter side.

(A) $\frac{4}{3}$ (B) $\frac{3}{4}$

(C) 2 (D) $\frac{5}{3}$

28. A rectangle has perimeter 20 and area 24. Find the area of its circumcircle.

(A) 25π (B) 12π

(C) 26π (D) 13π

29. A circle has centre $(h, 0)$, with $1 < h < 4$. It touches $x + y = 4$ and touches $x^2 + y^2 = 1$ externally. Find its radius.

(A) $-3 + 3\sqrt{2}$ (B) $-1 + \sqrt{2}$

(C) $3 + 3\sqrt{2}$ (D) $-1 + 2\sqrt{2}$

30. A circle with centre $(h, 0)$, $0 < h < 4$, contains the unit disk, is internally tangent to its boundary, and passes through $(4, 0)$. Find its radius.

(A) 3 (B) $\frac{5}{2}$

(C) $\frac{3}{2}$ (D) 4

31. Find the common power of the radical centre of $x^2 + y^2 = 25$, $x^2 + y^2 - 6x - 7 = 0$ and $x^2 + y^2 - 8y - 9 = 0$.

(A) -25 (B) -13

(C) -12 (D) 12

32. The radical centre P of $x^2+y^2-2x-4y-4 = 0$, $x^2+y^2-6x+2y-2 = 0$ and $x^2+y^2+2x-2y-14 = 0$ is formed. Find its distance from the first circle centre.

(A) $\sqrt{2}$ (B) $\sqrt{5}$

(C) 2 (D) 1

33. A circle touches the lines PA and PB at $A = (1, 0)$ and $B = (0, 1)$, where $P = (2, 2)$. Find its radius.

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{\sqrt{5}}{3}$

(C) $\frac{5}{9}$ (D) $\sqrt{5}$

34. Tangents to a circle at $A = (1, 1)$ and $B = (3, 1)$ meet at $P = (2, 5)$. Find its radius.

(A) $\frac{\sqrt{17}}{4}$ (B) $\frac{17}{16}$

(C) $\frac{\sqrt{17}}{2}$ (D) $\frac{5}{4}$

35. From an external point P, a secant enters a circle at A and exits at B, with $PA = 3$, $AB = 5$. The nearest circle point is 2 units from P. Find the circle radius.

(A) 12 (B) 5

(C) 6 (D) 3

36. Chords AB and CD intersect at P inside a circle. Given $PA = 2$, $PB = 6$, $PC = 3$ and $AB \perp CD$, find the area of quadrilateral ACBD.

(A) 56 (B) 28

(C) 24 (D) 14

37. The circle $x^2 + y^2 - 2ax - 4y + a^2 - 5 = 0$ touches $x + y = 1$, with $a > 0$. Find the squared distance of its centre from the origin.

(A) $23 - 6\sqrt{2}$ (B) $6\sqrt{2} + 23$

(C) $19 - 6\sqrt{2}$ (D) 9

38. The circle $x^2 + y^2 - 2kx - 2y + k^2 - 3 = 0$ touches $y = x + 1$, where $k < 0$. Find its larger x intercept.

(A) $-2\sqrt{2} + \sqrt{3}$ (B) $2 - 2\sqrt{2}$

(C) $\sqrt{3} + 2\sqrt{2}$ (D) $-2\sqrt{2} - \sqrt{3}$

39. A right triangle has perimeter 12 and altitude 12/5 from the right-angle vertex to the hypotenuse. Find its circumradius.

(A) $\frac{12}{5}$ (B) 5

(C) $\frac{5}{2}$ (D) 3

40. AB is a chord of length 8 in a circle of radius 5. As C varies over the circle, find the greatest possible area of triangle ABC.

(A) 32 (B) 20

(C) 24 (D) 40

41–50 / Preserved hard practice

41. Two circles pass through $(1,0),(3,0)$ and touch $y=x$. Find the product of their areas divided by π^2.

(A) 27 (B) 24

(C) 25 (D) 26

42. A circle passes through $(1,0),(-1,0)$. Tangents from $(0,3)$ make an angle of 60° with each other. Find the sum of the squared radii of all such circles.

(A) $\frac{34}{3}$ (B) $\frac{28}{3}$

(C) $\frac{25}{3}$ (D) $\frac{31}{3}$

43. A circle passes through the two intersections of $x^2+y^2=25$ and $x^2+y^2-8x=9$, and through the origin. Find its squared radius.

(A) $\frac{657}{16}$ (B) $\frac{641}{16}$

(C) $\frac{609}{16}$ (D) $\frac{625}{16}$

44. Circles tangent to both positive coordinate axes have centres in the first quadrant and also touch $3x+4y=12$. Find the sum of their squared radii.

(A) 38 (B) 36

(C) 39 (D) 37

45. A point P is 5 units from the centre of a circle. The chord joining its two tangency points has length 4. Two radii are possible. Find their product.

(A) 11 (B) 12

(C) 9 (D) 10

46. A circle passes through the intersections of $x^2+y^2=1$ and $x+y=1$. It is orthogonal to $(x-2)^2+y^2=1$. Find its squared radius.

(A) 2 (B) 0

(C) 1 (D) 3

47. A circle centred at $(-2,0)$ intersects $y^2=4x$ orthogonally. Find the squared radius of the circle.

(A) 25 (B) 23

(C) 24 (D) 26

48. Two distinct vertices of a regular 12-gon inscribed in a circle of radius 2 are chosen uniformly. Find the probability that their joining chord is at distance at least $\sqrt{2}$ from the centre.

(A) $\frac{7}{11}$ (B) $\frac{5}{11}$

(C) $\frac{6}{11}$ (D) $\frac{4}{11}$

49. Point $P = (x, y)$ lies on $x^2 + y^2 = 1$. For $A = (2, 0)$ and $B = (0, 2)$, find the minimum of $PA^2\, PB^2$.

(A) $\frac{7}{2}$ (B) $\frac{11}{2}$

(C) $\frac{9}{2}$ (D) $\frac{13}{2}$

50. In the complex plane, $|z - 1| = 2|z + 1|$. Find the minimum of $|z - 2i|^2$.

(A) $\frac{77}{9} - \frac{8\sqrt{61}}{9}$ (B) $\frac{95}{9} - \frac{8\sqrt{61}}{9}$

(C) $\frac{86}{9} - \frac{8\sqrt{61}}{9}$ (D) $\frac{68}{9} - \frac{8\sqrt{61}}{9}$

Answer key

Circle

M-CIRC / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**C**	21	**A**	31	**C**	41	**C**
02	**D**	12	**B**	22	**D**	32	**A**	42	**B**
03	**C**	13	**D**	23	**A**	33	**B**	43	**D**
04	**A**	14	**B**	24	**D**	34	**A**	44	**D**
05	**C**	15	**C**	25	**A**	35	**B**	45	**D**
06	**D**	16	**D**	26	**D**	36	**B**	46	**C**
07	**B**	17	**C**	27	**A**	37	**A**	47	**C**
08	**B**	18	**D**	28	**D**	38	**A**	48	**C**
09	**A**	19	**B**	29	**A**	39	**C**	49	**C**
10	**B**	20	**B**	30	**B**	40	**A**	50	**A**

Chapter 21

Parabola

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For $y^2 = 4x$, the focal chord has one endpoint parameter 2. The other parameter is

(A) $-\frac{3}{2}$ (B) $\frac{3}{2}$

(C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

02. The chord of $y^2 = 4x$ joining parameters 1 and 3 has slope

(A) $-\frac{1}{2}$ (B) $\frac{3}{2}$

(C) $\frac{1}{2}$ (D) $\frac{5}{2}$

03. For $y^2 = 8x$, point (x, y) on it has focal distance 10. Find x.

(A) 9 (B) 8

(C) 10 (D) 7

04. The area of triangle formed by the two latus rectum endpoints and the intersection of their tangents, for $y^2 = 4x$, is

(A) 6 (B) 3

(C) 4 (D) 5

05. The normal at $(1, 2)$ on $y^2 = 4x$ cuts x-axis at abscissa

(A) 3 (B) 5

(C) 4 (D) 2

06. The chord joining parameters $t_1 = 1, t_2 = 2$ on $y^2 = 4x$ passes through $(0, b)$. Find b.

(A) $\frac{4}{3}$ (B) $\frac{7}{3}$

(C) $\frac{10}{3}$ (D) $\frac{1}{3}$

07. For $y^2 = 4x$, the tangent at parameter t passes through $(3, 4)$. Product of its two possible parameters is

(A) 3 (B) 4

(C) 5 (D) 2

08. The number of distinct real tangents from $(-1, 0)$ to $y^2 = 4x$ is

(A) 4 (B) 3

(C) 1 (D) 2

09. How many real points satisfy $y^2 = 4x$ and $x^2 + y^2 = 5$?

(A) 2 (B) 4

(C) 1 (D) 3

10. The area bounded by $y^2 = 4x$ and $x = 1$ is

(A) $\frac{8}{3}$ (B) $\frac{11}{3}$

(C) $\frac{14}{3}$ (D) $\frac{5}{3}$

11–40 / New middle-to-hard practice

11. For $(y-2)^2 = 4a(x-h)$, $a > 0$, the directrix is $x = -1$ and $(5, 6)$ lies on the curve. The vertex has x coordinate greater than 1. Find the focus x coordinate.

(A) $5 - 2\sqrt{5}$ (B) $2\sqrt{5} + 5$

(C) $3 + 2\sqrt{5}$ (D) $2 + \sqrt{5}$

12. The parabola $(y-k)^2 = 4a(x-1)$ has directrix $x = -1$, passes through $(3, 0)$, and has its axis below the x axis. Find the distance of its focus from the origin.

(A) 5 (B) 3

(C) 4 (D) $\sqrt{17}$

13. On $y^2 = 8x$, a focal chord has endpoint parameters p, q with $p + q = 3$. Find the distance from the intersection of the endpoint tangents to the focus.

(A) $2\sqrt{13}$ (B) $\sqrt{13}$

(C) $2\sqrt{5}$ (D) $2\sqrt{10}$

14. Two points on $y^2 = 4x$ have ordinates summing to 12 and abscissae summing to 20. Find the distance from the intersection of their tangents to the focus.

(A) $\sqrt{89}$ (B) $\sqrt{65}$

(C) $\sqrt{85}$ (D) 10

15. Tangents from $(-3, 4)$ touch $y^2 = 4x$ at A, B. Find AB.

(A) $8\sqrt{35}$ (B) $4\sqrt{35}$

(C) $4\sqrt{7}$ (D) $2\sqrt{35}$

16. Tangents from $P = (0, 4)$ touch $y^2 = 4x$ at A, B. Find the area of triangle PAB.

(A) 16 (B) 32

(C) 8 (D) 64

17. Three distinct normals from $(6, 0)$ meet $y^2 = 4x$ at their contact points. Find the product of the three contact-point focal distances.

(A) 11 (B) 25

(C) 125 (D) 10

18. Three distinct normals from $(7, 2)$ touch $y^2 = 4x$. Find the sum of the distances of their contact points from the focus.

(A) 8 (B) 10

(C) 16 (D) 13

19. A focal chord of $y^2 = 4x$ has length $25/4$. Find the larger endpoint abscissa.

(A) $\frac{25}{4}$ (B) $\frac{17}{4}$

(C) $\frac{1}{4}$ (D) 4

20. A focal chord of $y^2 = 8x$ has length 18. Find the sum of its endpoint abscissae.

(A) 7 (B) 16

(C) 18 (D) 14

21. Two distinct common tangents to $x^2 + y^2 = 2$ and $y^2 = 4x$ have finite slopes m_1, m_2. Find $m_1 m_2$.

(A) $\frac{1}{2} + \frac{\sqrt{3}}{2}$ (B) $1 - \sqrt{3}$

(C) $\frac{1}{2} - \frac{\sqrt{3}}{2}$ (D) $-\frac{1}{2} + \frac{\sqrt{3}}{2}$

22. How many real lines are tangent to both $(x - 1)^2 + y^2 = 1$ and $y^2 = 4x$?

(A) 1 (B) 0

(C) 3 (D) 2

23. Two perpendicular tangents to $y^2 = 8x$ meet at a point of ordinate 4. Find the area of the triangle formed by the two contact points and their tangent intersection.

(A) $64\sqrt{2}$ (B) $32\sqrt{2}$

(C) $16\sqrt{2}$ (D) 32

24. Tangents from $(h, 3)$ to $y^2 = 4x$ make an acute angle $\pi/3$, where $h > 0$. Find h.

(A) $-\frac{\sqrt{43}}{3} - \frac{5}{3}$ (B) $\frac{5}{3} + \frac{\sqrt{43}}{3}$

(C) $-\frac{5}{3} + \frac{\sqrt{43}}{3}$ (D) $-\frac{2}{3} + \frac{\sqrt{19}}{3}$

25. A chord of $y^2 = 4x$ has midpoint $(5, 2)$. Find its length.

(A) 16 (B) 8

(C) $4\sqrt{2}$ (D) $8\sqrt{2}$

26. A chord of $y^2 = 4x$ is parallel to $x - 2y = 0$, and its midpoint abscissa is 5. Find its length.

(A) $8\sqrt{5}$ (B) $4\sqrt{30}$

(C) $4\sqrt{5}$ (D) $4\sqrt{6}$

27. On $y^2 = 4ax$, $a > 0$, the tangent and normal at a point with positive parameter have y intercepts c and b. Given $b/c = 6$ and $b = 24$, find the focal distance of the contact point.

(A) 12 (B) 20

(C) 8 (D) 10

28. The tangent and normal at the same nonvertex point of $y^2 = 4ax$, $a > 0$, cut the x axis at -3 and 7, respectively. Find the latus rectum length.

(A) 16 (B) 5

(C) 4 (D) 8

29. Choose k uniformly from $[0, 12]$. Find the probability that $(x - 3)^2 + y^2 = k$ and $y^2 = 4x$ have exactly four distinct real common points.

(A) $\frac{1}{12}$ (B) $\frac{1}{6}$

(C) $\frac{1}{3}$ (D) $\frac{1}{4}$

30. The smallest-radius circle centred at $(5, 0)$ that meets $y^2 = 4x$ touches it at two points. Find the distance between those points.

(A) 4 (B) $4\sqrt{3}$

(C) $2\sqrt{3}$ (D) $8\sqrt{3}$

31. On $y^2 = 4x$, points have parameters $-1, 1, t$, where $t > 0$. Their triangle area is 12. Find the focal distance of the third point.

(A) 14 (B) 12

(C) 8 (D) 7

32. On $y^2 = 8x$, three points have parameters $0, 1, t$, with $t > 1$. Their triangle area is 24. Find the third point abscissa.

(A) 24 (B) 18

(C) 12 (D) 9

33. A point with positive ordinate on $y^2 = 4ax$, $a > 0$, has focal distance $5a$. Its normal passes through $(3, 18)$. Find the latus rectum length.

(A) 8 (B) 16

(C) 10 (D) 4

34. For $y^2 = 4ax$, $a > 0$, a positive-ordinate point P has its focal radius directed at angle $\pi/3$ to the positive x axis. Its tangent is at distance 3 from the origin. Find SP, where S is the focus.

(A) 4 (B) 6

(C) 12 (D) 8

35. The focus of $y^2 = 4x$ is reflected in its tangent at $(4, 4)$ to F. Find the distance from F to $(4, 4)$.

(A) 3 (B) 5

(C) $\sqrt{17}$ (D) 4

36. On $y^2 = 8x$, the tangent at a positive-ordinate point has a perpendicular from the focus whose foot is 6 units from the vertex. Find the x intercept of the normal at that point.

(A) 26 (B) 22

(C) 20 (D) 18

37. Find the tangent of the acute angle between $y^2 = 4x$ and $x^2 = 4y$ at their nonzero common point.

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{4}{3}$ (D) $\frac{3}{2}$

38. Find the tangent of the acute angle between $y^2 = 4x$ and $x^2 = 32y$ at their nonzero common point.

(A) $\frac{3}{5}$ (B) $\frac{5}{3}$

(C) $\frac{1}{5}$ (D) $\frac{3}{4}$

39. A parabola has focus $(2, 0)$ and directrix $x + y = 0$. Find the sum of its vertex coordinates.

(A) 2 (B) 1

(C) 0 (D) 3

40. A parabola has focus at the origin and directrix $x - 2y = c$, $c > 0$. It passes through $(1, 1)$. Find its latus rectum length.

(A) $\frac{2\sqrt{5}}{5} + 2\sqrt{2}$ (B) $-\frac{\sqrt{5}}{5} + \sqrt{2}$

(C) $-\frac{2\sqrt{5}}{5} + 2\sqrt{2}$ (D) $-2 + 2\sqrt{10}$

41–50 / Preserved hard practice

41. A focal chord of $y^2 = 4x$ has endpoint parameters t, u. The normals at its endpoints meet at $N = (12, k)$. Find k^2.

(A) 11 (B) 9

(C) 8 (D) 10

42. The distinct real normals to $y^2 = 4x$ through $(6, 2)$ touch at parameters t_1, t_2, t_3. Find $t_1^2 + t_2^2 + t_3^2$.

(A) 9 (B) 10

(C) 7 (D) 8

43. Two perpendicular normals to $y^2 = 4x$ meet at a point with abscissa 4. Considering both possible intersection points, find the sum of squares of ordinates of all four contact points.

(A) 25 (B) 23

(C) 26 (D) 24

44. For $y^2 = 4x$, normals at the two real parameters satisfying $t^2 - 3t + 1 = 0$ meet at N. Find the squared distance of N from the focus.

(A) 90 (B) 92

(C) 91 (D) 89

45. Point P varies on $y^2 = 4x$, with focus $S = (1, 0)$. For $Q = (4, 2)$, find the minimum of $PS + PQ$.

(A) 4 (B) 6

(C) 7 (D) 5

46. A rectangle has one side on $x = 2$ and its other two vertices on $y^2 = 4x$, symmetric about the x-axis. The opposite vertical side is $x = u$, $0 \le u \le 2$. Maximum rectangle area is

(A) $-1 + \frac{16\sqrt{6}}{9}$ (B) $2 + \frac{16\sqrt{6}}{9}$

(C) $\frac{16\sqrt{6}}{9}$ (D) $1 + \frac{16\sqrt{6}}{9}$

47. A circle passes through $(0, 0), (1, 2), (4, 4)$. It meets $y^2 = 4x$ at one further point. Find the ordinate of that point.

(A) -7 (B) -4

(C) -6 (D) -5

48. Let $z \neq 0$ satisfy $|z - 2| = \operatorname{Re} z + 2$ and $\arg z = \pi/4$. Find $|z|^2$.

(A) 127 (B) 130

(C) 129 (D) 128

49. Tangents to $y^2 = 4x$ at the three parameters solving $t^3 - 3t + 1 = 0$ form a triangle. Find its area.

(A) $\frac{11}{2}$ (B) $\frac{9}{2}$

(C) $\frac{7}{2}$ (D) $\frac{13}{2}$

50. Two distinct parameters are chosen uniformly from $\{-3, -2, -1, 0, 1, 2, 3\}$ to select points on $y^2 = 4x$. Find the probability their joining line meets the x-axis at a point with $0 < x \leq 2$.

(A) $\frac{1}{3}$ (B) $\frac{1}{21}$

(C) $\frac{1}{7}$ (D) $\frac{2}{7}$

Answer key

Parabola

M-PARA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**C**	31	**C**	41	**B**
02	**C**	12	**A**	22	**A**	32	**B**	42	**D**
03	**B**	13	**A**	23	**B**	33	**A**	43	**D**
04	**C**	14	**C**	24	**C**	34	**D**	44	**A**
05	**A**	15	**B**	25	**D**	35	**B**	45	**D**
06	**A**	16	**B**	26	**C**	36	**B**	46	**C**
07	**A**	17	**B**	27	**D**	37	**A**	47	**C**
08	**D**	18	**D**	28	**D**	38	**A**	48	**D**
09	**A**	19	**D**	29	**A**	39	**B**	49	**B**
10	**A**	20	**D**	30	**B**	40	**C**	50	**C**

Chapter 22

Ellipse

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The tangent to $x^2/25 + y^2/16 = 1$ at $(3, 16/5)$ has y intercept

(A) 6 (B) 4

(C) 7 (D) 5

02. An ellipse has major-axis length 10 and focal separation 6. Its area divided by π is

(A) 21 (B) 19

(C) 22 (D) 20

03. For $x^2/25 + y^2/9 = 1$, the chord through centre along $y = x$ has length

(A) $2 + \frac{30\sqrt{17}}{17}$ (B) $\frac{30\sqrt{17}}{17}$

(C) $-1 + \frac{30\sqrt{17}}{17}$ (D) $1 + \frac{30\sqrt{17}}{17}$

04. For ellipse $x^2/25 + y^2/16 = 1$, the tangent with slope 1 and positive intercept has intercept

(A) $-1 + \sqrt{41}$ (B) $1 + \sqrt{41}$

(C) $\sqrt{41}$ (D) $2 + \sqrt{41}$

05. For ellipse $x^2/25 + y^2/16 = 1$, a tangent $y = mx + 5$ has m^2 equal to

(A) $-\frac{16}{25}$ (B) $\frac{9}{25}$

(C) $\frac{59}{25}$ (D) $\frac{34}{25}$

06. For $x^2/25 + y^2/16 = 1$, parameter satisfies $\cos t = 3/5$, $\sin t = 4/5$. Distance to right focus is

(A) $\frac{11}{5}$ (B) $\frac{16}{5}$

(C) $\frac{21}{5}$ (D) $\frac{26}{5}$

07. An ellipse has eccentricity $3/5$ and latus rectum length $32/5$. Find semi-major axis.

(A) 4 (B) 7

(C) 6 (D) 5

08. A point $(5\cos t, 3\sin t)$ is selected by choosing t uniformly from $\{0, \pi/2, \pi, 3\pi/2\}$. Find the expected value of x^2.

(A) $\frac{23}{2}$ (B) $\frac{25}{2}$

(C) $\frac{29}{2}$ (D) $\frac{27}{2}$

09. How many points satisfy both $x^2/25 + y^2/9 = 1$ and $x^2 + y^2 = 9$?

(A) 2 (B) 4

(C) 1 (D) 3

10. For $x^2/25 + y^2/9 = 1$, slope of tangent at (x, y) satisfies $y \neq 0$. At point $(4, 9/5)$, slope is

(A) $\frac{6}{5}$ (B) $\frac{1}{5}$

(C) $-\frac{4}{5}$ (D) $-\frac{9}{5}$

11–40 / New middle-to-hard practice

11. An ellipse has eccentricity $\sqrt{5}/3$, area 12π, and major axis along the x axis. Find its latus rectum length.

(A) $\frac{8}{3}$ (B) $\frac{8\sqrt{2}}{3}$

(C) $\frac{16\sqrt{2}}{3}$ (D) $\frac{4\sqrt{2}}{3}$

12. An ellipse centred at the origin has foci $(\pm 3, 0)$ and passes through $(4, 12/5)$. Find its area divided by π.

(A) 25 (B) 16

(C) 12 (D) 20

13. The line $2x + 3y = c$, $c > 0$, touches $x^2/25 + y^2/9 = 1$ at P. Find $x_P + y_P$.

(A) $\frac{23\sqrt{181}}{181}$ (B) $\sqrt{181}$

(C) $\frac{77\sqrt{109}}{109}$ (D) $\frac{77\sqrt{181}}{181}$

14. The ellipse $x^2/A + y^2/B = 1$, $A > B > 0$, has focal separation 6 and is tangent to $3x + 4y = 12$. Find its eccentricity squared.

(A) $\frac{25}{32}$ (B) $\frac{9}{32}$

(C) $\frac{7}{32}$ (D) $\frac{25}{64}$

15. Tangents at the endpoints of chord $x + 2y = 4$ of $x^2/16 + y^2/9 = 1$ meet at P. Find OP.

(A) $\sqrt{145}$ (B) $\frac{\sqrt{97}}{2}$

(C) $\frac{\sqrt{145}}{2}$ (D) $\frac{17}{2}$

16. Tangents from $(10, 8)$ touch $x^2/25 + y^2/16 = 1$ at A, B. Find the area cut off by the line AB on the coordinate axes.

(A) 10 (B) $\frac{5}{2}$

(C) $\frac{5}{4}$ (D) 5

17. At a first-quadrant point P of $x^2/25 + y^2/16 = 1$, the normal has y intercept $-9/5$. Find the distance from P to the right focus.

(A) $\frac{34}{5}$ (B) $\frac{12}{5}$

(C) $\frac{16}{5}$ (D) $\frac{9}{5}$

18. The normal at a first-quadrant point P of $x^2/25 + y^2/9 = 1$ is parallel to $5x - 4y = 0$. Find OP.

(A) $\frac{\sqrt{481}}{5}$ (B) $\frac{\sqrt{97}}{5}$

(C) $\frac{29}{5}$ (D) $\frac{\sqrt{481}}{10}$

19. An ellipse centred at the origin has eccentricity $3/5$ and major axis along the x axis. Two perpendicular tangents meet at $(\sqrt{41}, 0)$. Find its area divided by π.

(A) 15 (B) 25

(C) 20 (D) 16

20. Two perpendicular tangents to $x^2/25 + y^2/9 = 1$ meet at (t, t), $t > 0$. Find the sum of the axis intercepts of their contact chord.

(A) $\sqrt{17}$ (B) $2\sqrt{17}$

(C) 34 (D) $4\sqrt{17}$

21. A chord of $x^2/16 + y^2/9 = 1$ has midpoint $(2, 1)$. Find its squared length.

(A) $\frac{3335}{468}$ (B) $\frac{145}{13}$

(C) $\frac{3335}{39}$ (D) $\frac{3335}{117}$

22. A chord of $x^2/25 + y^2/9 = 1$ is parallel to $x + 2y = 0$, and its midpoint has x coordinate 1. Find the distance of the chord line from the origin.

(A) $\frac{\sqrt{5}}{25}$ (B) $\frac{61\sqrt{5}}{125}$

(C) $\frac{61}{25}$ (D) $\frac{61\sqrt{5}}{250}$

23. An ellipse has eccentricity $\sqrt{3}/2$ and major axis along the x axis. Its central chord on $y = x$ has length $4\sqrt{2/5}$. Find its area divided by π.

(A) 4 (B) 2

(C) $\sqrt{3}$ (D) 1

24. A line $y = mx$, $m > 0$, cuts a central chord of length $3\sqrt{5}$ from $x^2/25 + y^2/9 = 1$. Find m^2.

(A) $\frac{11}{5}$ (B) $\frac{5}{11}$

(C) $\frac{9}{5}$ (D) $\frac{11}{9}$

25. For a point P of $x^2/25 + y^2/16 = 1$ above the x axis, its distances from the right and left foci are in ratio $2 : 3$. Find its y coordinate.

(A) $-\frac{8\sqrt{2}}{3}$ (B) $\frac{8}{3}$

(C) $\frac{8\sqrt{2}}{3}$ (D) $\frac{4\sqrt{2}}{3}$

26. An ellipse has foci $(\pm 4, 0)$. A point on it is 5 units from the right focus and 11 from the left. Find its area divided by π.

(A) $48\sqrt{3}$ (B) $16\sqrt{3}$

(C) $32\sqrt{3}$ (D) 32

27. On $x^2/9 + y^2/4 = 1$ with $x \le 1$, find the greatest value of $2x + y$.

(A) $\frac{4\sqrt{2}}{3} + 2$ (B) $2\sqrt{10}$

(C) $2 + 4\sqrt{2}$ (D) $\frac{2\sqrt{2}}{3} + 2$

28. An ellipse $x^2/A + y^2/B = 1$, $A > B > 0$, has focal separation 6. The greatest value of $2x + y$ on it is $2\sqrt{29}$. Find its latus rectum length.

(A) $\frac{32}{5}$ (B) $\frac{32}{3}$

(C) $\frac{16}{5}$ (D) $\frac{64}{5}$

29. Choose k uniformly from $[0, 30]$. Find the probability that $x^2 + y^2 = k$ and $x^2/25 + y^2/9 = 1$ have four distinct real intersections.

(A) $\frac{8}{15}$ (B) $\frac{4}{15}$

(C) $\frac{3}{5}$ (D) $\frac{5}{6}$

30. The four intersections of $x^2 + y^2 = 20$ and $x^2/25 + y^2/16 = 1$ form a rectangle. Find its area.

(A) $\frac{80\sqrt{5}}{9}$ (B) $\frac{40\sqrt{5}}{9}$

(C) $\frac{160\sqrt{5}}{3}$ (D) $\frac{160\sqrt{5}}{9}$

31. A line of slope 1 touches $x^2/25 + y^2/9 = 1$ and also a circle centred at the origin. Find the distance between the two contact points on that line.

(A) $\frac{4\sqrt{17}}{17}$ (B) $\frac{8\sqrt{17}}{17}$

(C) $\frac{4\sqrt{34}}{17}$ (D) $4\sqrt{17}$

32. The line $2x + y = c$, $c > 0$, touches $x^2/16 + y^2/9 = 1$ and a circle centred at $(1, 0)$. Find that circle radius.

(A) $\frac{\sqrt{365}}{5}$ (B) $-\frac{\sqrt{5}}{5} + \frac{\sqrt{365}}{5}$

(C) $\frac{\sqrt{5}(2+\sqrt{73})}{5}$ (D) $\frac{\sqrt{5}(-2+\sqrt{73})}{5}$

33. A first-quadrant tangent to $x^2/25+y^2/16=1$ cuts the least possible triangle area from the positive axes. Find the ratio x/y of its contact point.

(A) $\frac{25}{16}$ (B) $\frac{5}{2}$

(C) $\frac{5}{4}$ (D) $\frac{4}{5}$

34. A tangent to $x^2/25+y^2/16=1$ has positive axis intercepts u, v, with $u>v$, and cuts a triangle of area 25 from the axes. Find the contact point x coordinate.

(A) $\sqrt{5}$ (B) $\frac{5\sqrt{2}}{2}$

(C) $2\sqrt{5}$ (D) $\frac{\sqrt{5}}{2}$

35. An axis-parallel rectangle centred at the origin has vertices on $x^2/25+y^2/9=1$ and area 24. Its horizontal side is longer than its vertical side. Find the square of its positive vertex x coordinate.

(A) 10 (B) 5

(C) $\frac{9}{5}$ (D) 20

36. An ellipse centred at the origin has eccentricity $\sqrt{5}/3$ and major axis along x. It passes through the vertex $(3,2)$ of an axis-parallel rectangle. Find the greatest area of any such centred rectangle inscribed in this ellipse.

(A) 18 (B) 24

(C) 12 (D) 36

37. An ellipse with one focus at the polar origin obeys $r=\ell/(1+e\cos\theta)$, $0<e<1$. Its radii at $\theta=0,\pi$ are 2 and 8. Find its major-axis length.

(A) 8 (B) 10

(C) 5 (D) 6

38. For the focus-polar ellipse $r=\ell/(1+e\cos\theta)$, the radii at $\pi/2$ and $\pi/3$ are 3 and 12/5. Find its area divided by π.

(A) $8\sqrt{3}$ (B) 12

(C) $4\sqrt{3}$ (D) $16\sqrt{3}$

39. Choose t uniformly from $[0,2\pi)$ and let $P=(5\cos t,3\sin t)$. Given $x_P>0$, find the probability that P is closer to the right focus than to the origin.

(A) $\frac{\arccos\left(\frac{2}{5}\right)}{\pi}$ (B) $\frac{\pi-2\arccos\left(\frac{2}{5}\right)}{\pi}$

(C) $\frac{2\arccos\left(\frac{2}{5}\right)}{\pi}$ (D) $\frac{2\arcsin\left(\frac{2}{5}\right)}{\pi}$

40. Choose t uniformly from $(0,\pi/2)$ on $P=(5\cos t,4\sin t)$. Find the probability that the tangent at P cuts off a triangle of area at most 40 from the positive axes.

(A) $\frac{5}{6}$ (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) $\frac{1}{2}$

41–50 / Preserved hard practice

41. How many points on $x^2/25 + y^2/9 = 1$ have a normal passing through $(3, 0)$? Count contact points, even if two normals are the same line.

(A) 4 (B) 5

(C) 6 (D) 3

42. For $x^2/25 + y^2/9 = 1$, find the maximum of $x^2 + 4xy + y^2$.

(A) $18 + 2\sqrt{241}$ (B) $16 + 2\sqrt{241}$

(C) $19 + 2\sqrt{241}$ (D) $17 + 2\sqrt{241}$

43. The tangent at a first-quadrant point P of $x^2/25 + y^2/9 = 1$ cuts the axes at A, B. When AB is minimum, find the distance from P to the right focus.

(A) $7 - \sqrt{10}$ (B) $6 - \sqrt{10}$

(C) $5 - \sqrt{10}$ (D) $4 - \sqrt{10}$

44. The normal at a non-axial point P of $x^2/25 + y^2/9 = 1$ meets the major axis at N. If $PN^2 = 5$, find the product of the two focal distances of P.

(A) $\frac{125}{9}$ (B) $\frac{116}{9}$

(C) $\frac{143}{9}$ (D) $\frac{134}{9}$

45. Tangents from T touch $x^2/25 + y^2/9 = 1$ at P, Q. If $x_T^2/25 + y_T^2/9 = 4$, find the area of triangle TPQ.

(A) $-1 + \frac{45\sqrt{3}}{4}$ (B) $1 + \frac{45\sqrt{3}}{4}$

(C) $2 + \frac{45\sqrt{3}}{4}$ (D) $\frac{45\sqrt{3}}{4}$

46. A rectangle has left side on $x = 1$ and right side on $x = u$, where $1 \le u \le 5$. Its upper and lower right vertices lie on $x^2/25 + y^2/9 = 1$, and its sides are parallel to the axes. At maximum area, find u.

(A) $\frac{1}{4} + \frac{\sqrt{201}}{4}$ (B) $\frac{5}{4} + \frac{\sqrt{201}}{4}$

(C) $-\frac{3}{4} + \frac{\sqrt{201}}{4}$ (D) $\frac{9}{4} + \frac{\sqrt{201}}{4}$

47. Find the area inside $x^2/25 + y^2/9 \le 1$ and outside the strip $|x| \le 5/2$.

(A) $-\frac{15\sqrt{3}}{2} - 1 + 10\pi$ (B) $-\frac{15\sqrt{3}}{2} + 1 + 10\pi$

(C) $-\frac{15\sqrt{3}}{2} + 10\pi$ (D) $-\frac{15\sqrt{3}}{2} + 2 + 10\pi$

48. Two tangents from $(0, 5)$ touch $x^2/25 + y^2/9 = 1$ at P, Q. Find the area of triangle OPQ, where O is the origin.

(A) $\frac{46}{5}$ (B) $\frac{41}{5}$

(C) $\frac{36}{5}$ (D) $\frac{31}{5}$

49. A complex number satisfies $|z-4|+|z+4|=10$ and $|z+4|=3|z-4|$. Find $|z|^2$.

(A) $\frac{65}{4}$ (B) $\frac{69}{4}$

(C) $\frac{57}{4}$ (D) $\frac{61}{4}$

50. Two distinct points are chosen uniformly from the eight points $(5\cos(k\pi/4), 3\sin(k\pi/4))$, $k = 0, \ldots, 7$. Find the probability that their tangents to the ellipse are perpendicular.

(A) $\frac{1}{7}$ (B) $\frac{2}{7}$

(C) $\frac{3}{7}$ (D) $\frac{1}{14}$

Answer key

Ellipse

M-ELLI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**B**	21	**D**	31	**B**	41	**A**
02	**D**	12	**D**	22	**B**	32	**D**	42	**D**
03	**B**	13	**D**	23	**B**	33	**C**	43	**C**
04	**C**	14	**A**	24	**A**	34	**A**	44	**A**
05	**B**	15	**C**	25	**C**	35	**D**	45	**D**
06	**B**	16	**B**	26	**C**	36	**B**	46	**A**
07	**D**	17	**C**	27	**A**	37	**B**	47	**C**
08	**B**	18	**A**	28	**A**	38	**A**	48	**C**
09	**A**	19	**C**	29	**A**	39	**C**	49	**D**
10	**C**	20	**B**	30	**D**	40	**B**	50	**A**

Chapter 23

Hyperbola

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For $x^2/9 - y^2/16 = 1$, the tangent at $(5, 16/3)$ has x intercept

(A) $\frac{4}{5}$ (B) $\frac{19}{5}$

(C) $\frac{9}{5}$ (D) $\frac{14}{5}$

02. For $x^2/9 - y^2/16 = 1$, a tangent of slope 2 with positive intercept has intercept

(A) $2\sqrt{5}$ (B) $-1 + 2\sqrt{5}$

(C) $1 + 2\sqrt{5}$ (D) $2 + 2\sqrt{5}$

03. For $x^2/9 - y^2/16 = 1$, a tangent $y = mx + 2$ has m^2 equal to

(A) $\frac{11}{9}$ (B) $\frac{20}{9}$

(C) $\frac{29}{9}$ (D) $\frac{38}{9}$

04. A hyperbola has semi-transverse axis 3 and eccentricity $5/3$. Its b^2 is

(A) 18 (B) 15

(C) 17 (D) 16

05. For $x^2/9 - y^2/16 = 1$, the product of distances from right vertex to the two foci is

(A) 15 (B) 18

(C) 16 (D) 17

06. For $xy = 4$, the area of triangle between the tangent at any first-quadrant point and positive axes is

(A) 10 (B) 9

(C) 8 (D) 7

07. An asymptote of $x^2/9 - y^2/16 = 1$ makes angle θ with positive x axis in first quadrant. Find $\sin\theta$.

(A) $\frac{9}{5}$ (B) $-\frac{1}{5}$

(C) $\frac{14}{5}$ (D) $\frac{4}{5}$

08. For $x^2/9 - y^2/16 = 1$, point on right branch has distances 10 and d to left and right foci. Find d.

(A) 4 (B) 3

(C) 5 (D) 6

09. How many points satisfy both $x^2 - y^2 = 1$ and $x^2 + y^2 = 5$?

(A) 4 (B) 6

(C) 5 (D) 3

10. For $x^2/9 - y^2/16 = 1$, line $y = mx$ meets the hyperbola in two real points when $m = 1$. Positive intersection abscissa is

(A) $-1 + \frac{12\sqrt{7}}{7}$ (B) $\frac{12\sqrt{7}}{7}$

(C) $1 + \frac{12\sqrt{7}}{7}$ (D) $2 + \frac{12\sqrt{7}}{7}$

11–40 / New middle-to-hard practice

11. A hyperbola centred at the origin has foci $(\pm 5, 0)$ and asymptotes $y = \pm 4x/3$. Find its latus rectum length.

(A) $\frac{32}{3}$ (B) $\frac{64}{3}$

(C) $\frac{25}{3}$ (D) $\frac{16}{3}$

12. A hyperbola has equation $x^2/A - y^2/B = 1$, asymptotes $y = \pm\sqrt{3}x$, and passes through $(4, 6)$. Find its latus rectum length.

(A) 8 (B) 6

(C) 12 (D) 24

13. The line $2x + y = c$, $c > 0$, touches $x^2/9 - y^2/4 = 1$ at P. Find $x_P + y_P$.

(A) $\frac{7\sqrt{2}}{2}$ (B) $\frac{7\sqrt{2}}{4}$

(C) $\sqrt{2}$ (D) $\frac{11\sqrt{2}}{4}$

14. A hyperbola $x^2/A - y^2/B = 1$ has asymptotes $y = \pm x/2$ and is tangent to $2x + 3y = 6$. Find its latus rectum length.

(A) $\frac{6\sqrt{7}}{7}$ (B) $\frac{12\sqrt{7}}{7}$

(C) $\frac{3\sqrt{7}}{7}$ (D) $2\sqrt{3}$

15. The chord of contact of tangents from P to $x^2/9 - y^2/16 = 1$ is $x = 6$. Find the distance of P from the right focus.

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$

(C) $\frac{13}{2}$ (D) $\frac{7}{2}$

16. Tangents from $(1, 2)$ touch $x^2/16 - y^2/9 = 1$. Find the area cut off on the axes by the line joining the contact points.

(A) 9 (B) 18

(C) 72 (D) 36

17. The normal at a first-quadrant point of $x^2/9 - y^2/16 = 1$ is parallel to $y = -x/2$. Find its y intercept.

(A) $\frac{5\sqrt{5}}{2}$ (B) $\frac{\sqrt{5}}{2}$

(C) $\frac{3\sqrt{5}}{2}$ (D) $5\sqrt{5}$

18. The normal at a first-quadrant point P of $x^2/16 - y^2/9 = 1$ has y intercept $25/4$. Find the distance from P to the right focus.

(A) $\frac{9}{2}$ (B) $\frac{9}{4}$

(C) $\frac{41}{4}$ (D) $\frac{5}{4}$

19. A first-quadrant point P of $x^2/9 - y^2/16 = 1$ is 13 units from the left focus. Find its y coordinate.

(A) $4\sqrt{3}$ (B) $4\sqrt{2}$

(C) $-4\sqrt{3}$ (D) $8\sqrt{3}$

20. A hyperbola has foci $(\pm 5, 0)$. A right-branch point is 14 units from the left focus and 6 from the right. Find the latus rectum length.

(A) $\frac{25}{2}$ (B) $\frac{9}{4}$

(C) 8 (D) $\frac{9}{2}$

21. A hyperbola has eccentricity $\sqrt{5}/2$ and focal separation 10. Find the latus rectum length of its conjugate hyperbola.

(A) $\frac{8\sqrt{5}}{5}$ (B) $4\sqrt{5}$

(C) $8\sqrt{5}$ (D) $2\sqrt{5}$

22. A hyperbola has latus rectum length 12, and its conjugate has eccentricity $\sqrt{3/2}$. Find the transverse-axis length of the conjugate.

(A) $3\sqrt{2}$ (B) $12\sqrt{2}$

(C) $6\sqrt{2}$ (D) 6

23. Two perpendicular tangents to $x^2/16 - y^2/9 = 1$ meet at $(t, 2)$, $t > 0$. Find the x intercept of their contact chord.

(A) 16 (B) $\frac{8\sqrt{3}}{3}$

(C) $\frac{16\sqrt{3}}{3}$ (D) $\frac{16\sqrt{7}}{7}$

24. A hyperbola with foci $(\pm 5, 0)$ has a pair of perpendicular tangents meeting at distance 3 from the origin. Find its latus rectum length.

(A) $2\sqrt{17}$ (B) $\frac{8\sqrt{17}}{17}$

(C) $4\sqrt{2}$ (D) $\frac{16\sqrt{17}}{17}$

25. A first-quadrant tangent to $xy = 4$ has positive x and y intercepts u, v, with $u - v = 6$. Find the distance from its contact point to the origin.

(A) 5 (B) $\sqrt{17}$

(C) $2\sqrt{17}$ (D) $\sqrt{13}$

26. A first-quadrant tangent to $xy = k$, $k > 0$, cuts a triangle of area 18 from the axes. Its positive intercepts sum to 20. Find the distance of its contact point from the origin.

(A) $\sqrt{118}$ (B) $2\sqrt{41}$

(C) $\sqrt{82}$ (D) $2\sqrt{82}$

27. A chord of $x^2/9 - y^2/4 = 1$ has midpoint $(6, 2)$. Find its squared length.

(A) $\frac{400}{3}$ (B) $\frac{200}{3}$

(C) $\frac{50}{3}$ (D) $\frac{100}{3}$

28. A chord of $x^2/16 - y^2/9 = 1$ is parallel to $x - 2y = 0$ and its midpoint has ordinate 3. Find the distance of its line from the origin.

(A) $\frac{10}{3}$ (B) $\frac{\sqrt{5}}{3}$

(C) $\frac{2\sqrt{5}}{3}$ (D) $2\sqrt{5}$

29. Choose m uniformly from $\{-2, -1, 0, 1, 2\}$, conditional on $y = mx$ meeting $x^2/9 - y^2/16 = 1$ at two real points. Find the probability that its chord length exceeds 10.

(A) $\frac{3}{5}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{2}{5}$

30. The central chord of $x^2/16 - y^2/9 = 1$ on $y = mx$, $m > 0$, has length 10. Find m^2.

(A) $\frac{81}{544}$ (B) $\frac{81}{272}$

(C) $\frac{544}{81}$ (D) $\frac{9}{136}$

31. A hyperbola $x^2/a^2 - y^2/b^2 = 1$ has eccentricity $5/3$. The triangle bounded by either tangent and the two asymptotes has area 12. Find the focal separation.

(A) 6 (B) 5

(C) 10 (D) 8

32. A hyperbola has asymptotes $y = \pm 2x$. The triangle bounded by any tangent and its two asymptotes has area 18. Find its latus rectum length.

(A) 12 (B) 48

(C) 18 (D) 24

33. On the right branch of $x^2/9 - y^2/16 = 1$, find the least value of $5x - 3y$.

(A) 15 (B) 6

(C) 3 (D) 9

34. On $x^2/16 - y^2/9 = 1$ with $4 \le x \le 5$, find the least value of $x + y$.

(A) $\frac{29}{4}$ (B) $\frac{9}{4}$

(C) $\sqrt{7}$ (D) $\frac{11}{4}$

35. The four intersections of $x^2/9 - y^2/16 = 1$ and $x^2 + y^2 = 25$ form a rectangle. Find its area.

(A) $\frac{48\sqrt{41}}{25}$ (B) $\frac{192\sqrt{41}}{25}$

(C) $\frac{192\sqrt{41}}{5}$ (D) $\frac{96\sqrt{41}}{25}$

36. Choose k uniformly from $[0, 25]$. Find the probability that $x^2 + y^2 = k$ and $x^2/9 - y^2/16 = 1$ have four distinct real intersections.

(A) $\frac{4}{5}$ (B) $\frac{8}{25}$

(C) $\frac{9}{25}$ (D) $\frac{16}{25}$

37. A chord of $x^2 + y^2 = 25$ is tangent to $x^2/9 - y^2/4 = 1$. Its midpoint is $(h, 1)$ with $h > 2$. Find h^2.

(A) $\sqrt{29} + 7$ (B) $\frac{\sqrt{29}}{2} + \frac{9}{2}$

(C) $\frac{7}{2} - \frac{\sqrt{29}}{2}$ (D) $\frac{\sqrt{29}}{2} + \frac{7}{2}$

38. A chord of $x^2 + y^2 = 16$ is tangent to $x^2/16 - y^2/9 = 1$. Its midpoint is $(3, k)$. Find k^2.

(A) $-\frac{27}{2} + \frac{3\sqrt{109}}{2}$ (B) $\frac{27}{2} + \frac{3\sqrt{109}}{2}$

(C) $-27 + 3\sqrt{109}$ (D) $-\frac{9}{2} + \frac{\sqrt{109}}{2}$

39. For $z = x + iy$, the locus $\mathrm{Re}((z - (1 + i))^2) = 8$ is a hyperbola. Find the distance from the origin to its right focus.

(A) $\sqrt{10}$ (B) 4

(C) $3\sqrt{2}$ (D) $\sqrt{26}$

40. For $z = x + iy$, the hyperbola $\mathrm{Im}((z-(1-2i))^2) = 18$ has foci F, G. Find $OF^2 + OG^2$.

(A) 82 (B) 154

(C) 46 (D) 41

41–50 / Preserved hard practice

41. The normals at the two non-axial points of $x^2/9 - y^2/16 = 1$ passing through $(10, 0)$ have contact ordinates y_1, y_2. Find $y_1^2 + y_2^2$.

(A) $\frac{327}{25}$ (B) $\frac{402}{25}$

(C) $\frac{352}{25}$ (D) $\frac{377}{25}$

42. Tangents from $(0, 5)$ touch $x^2/9 - y^2/16 = 1$ at P, Q. Find PQ.

(A) $\frac{6\sqrt{41}}{5}$ (B) $2 + \frac{6\sqrt{41}}{5}$

(C) $-1 + \frac{6\sqrt{41}}{5}$ (D) $1 + \frac{6\sqrt{41}}{5}$

43. How many distinct normals to $xy = 4$ pass through $(5, 5)$ and touch the first-quadrant branch?

(A) 2 (B) 3

(C) 5 (D) 4

44. The tangent at any point of $x^2/9 - y^2/16 = 1$ meets the two asymptotes at A, B. Find the area of triangle OAB, where $O = (0, 0)$.

(A) 11 (B) 12

(C) 14 (D) 13

45. A point P on the right branch of $x^2/9 - y^2/16 = 1$ has its distance to the left focus twice its distance to the right focus. Find OP^2.

(A) 66 (B) 65

(C) 64 (D) 67

46. A chord of $xy = 4$ has midpoint (h, k) in the first quadrant, with $h + k = 6$. Find the maximum possible squared chord length.

(A) $175 - 96\sqrt{2}$ (B) $176 - 96\sqrt{2}$

(C) $177 - 96\sqrt{2}$ (D) $178 - 96\sqrt{2}$

47. For two real nonzero values of k, the curves $y^2 = 4x$ and $x^2 - y^2 = k$ intersect at an acute angle of $45°$. Find the product of these two values of k.

(A) -129 (B) -128

(C) -126 (D) -127

48. Two first-quadrant tangents to $xy = 4$ each cut positive coordinate intercepts whose sum is 10. Their normals at the respective contact points meet at N. Find ON^2.

(A) 50 (B) 49

(C) 51 (D) 52

49. Two distinct numbers p, q are chosen uniformly from $\{1, 2, 3, 4, 6, 12\}$. Tangents to $xy = 12$ at abscissae p, q meet at T. Find the probability that $x_T + y_T < 8$.

(A) $\frac{13}{15}$ (B) $\frac{3}{5}$

(C) $\frac{4}{5}$ (D) $\frac{11}{15}$

50. How many complex numbers $z = x + iy$ with integer x, y satisfy $\mathrm{Re}(z^2) = 45$?

(A) 13 (B) 12

(C) 14 (D) 11

Answer key

Hyperbola

M-HYPE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**A**	21	**C**	31	**C**	41	**C**
02	**A**	12	**C**	22	**C**	32	**D**	42	**A**
03	**B**	13	**B**	23	**C**	33	**D**	43	**B**
04	**D**	14	**A**	24	**D**	34	**D**	44	**B**
05	**C**	15	**D**	25	**B**	35	**B**	45	**B**
06	**C**	16	**D**	26	**C**	36	**D**	46	**B**
07	**D**	17	**A**	27	**B**	37	**D**	47	**B**
08	**A**	18	**B**	28	**C**	38	**A**	48	**A**
09	**A**	19	**A**	29	**C**	39	**D**	49	**A**
10	**B**	20	**D**	30	**A**	40	**A**	50	**B**

Chapter 24

Vectors

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. If $|\mathbf{a}+\mathbf{b}|=5$ and $|\mathbf{a}-\mathbf{b}|=3$, then $\mathbf{a}\cdot\mathbf{b}$ is

(A) 5 (B) 6

(C) 3 (D) 4

02. For $\mathbf{a}=(1,0,1), \mathbf{b}=(0,1,1), \mathbf{c}=(1,1,0)$, scalar triple product $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$ is

(A) 0 (B) -1

(C) -3 (D) -2

03. For $\mathbf{a}=(1,2,0)$ and $\mathbf{b}=(0,1,2)$, magnitude of the cross product is

(A) $2+\sqrt{21}$ (B) $\sqrt{21}$

(C) $-1+\sqrt{21}$ (D) $1+\sqrt{21}$

04. Unit vectors $\mathbf{a},\mathbf{b}$ have $|\mathbf{a}-\mathbf{b}|=1$. Magnitude of cross product is

(A) $-1+\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{2}+1$ (D) $\frac{\sqrt{3}}{2}+2$

05. Vectors $\mathbf{a},\mathbf{b}$ are unit and perpendicular. Then $|(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{b})|$ is

(A) 4 (B) 3

(C) 1 (D) 2

06. For $\mathbf{a}=\hat{i}, \mathbf{b}=\hat{j}, \mathbf{c}=\hat{i}$, the $\hat{j}$ component of $(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$ is

(A) 1 (B) 0

(C) 3 (D) 2

07. The plane $x + 2y + 2z = 3$ has a unit normal whose positive x component is

(A) $\frac{4}{3}$ (B) $\frac{7}{3}$

(C) $\frac{1}{3}$ (D) $-\frac{2}{3}$

08. The determinant with rows $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is 6. The volume of tetrahedron with these edges from one vertex is

(A) 0 (B) 3

(C) 2 (D) 1

09. A unit vector $(\cos(k\pi/3), \sin(k\pi/3))$ is selected by choosing k uniformly from $\{0, 1, 2, 3, 4, 5\}$. Find the probability that its dot product with $\hat{i}$ is positive.

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$

(C) $\frac{1}{6}$ (D) $\frac{1}{3}$

10. For real t, squared norm of $(t, 1 - t, 0)$ is minimized at $t =$

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$

(C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

11–40 / New middle-to-hard practice

11. Vectors a, b satisfy $|a|^2 = 5$, $|a + 2b|^2 = 41$ and $|2a - b|^2 = 29$. Find $|a \times b|^2$.

(A) 50 (B) 36

(C) 45 (D) 41

12. Vectors a, b satisfy $|a + b|^2 = 25$, $|a - b|^2 = 9$, and $|a|^2 - |b|^2 = 6$. Find $|a \times b|^2$.

(A) $\frac{125}{4}$ (B) $\frac{253}{4}$

(C) $\frac{189}{4}$ (D) $\frac{189}{2}$

13. Noncollinear unit vectors a, b make an obtuse angle and satisfy $|(a+b)/3 + a \times b| = 1$. Find $|a \times b|^2$.

(A) $\frac{2\sqrt{19}}{81} + \frac{61}{81}$ (B) $\frac{61}{81} - \frac{2\sqrt{19}}{81}$

(C) $\frac{3}{4}$ (D) $\frac{8}{9}$

14. Unit vectors a, b satisfy $|a + b + a \times b|^2 = 3$. Find the cosine of the angle between $a + b$ and $2a - b$.

(A) $\frac{1}{2}$ (B) $-\frac{\sqrt{10}}{10}$

(C) $\frac{\sqrt{10}}{10}$ (D) $\frac{\sqrt{5}}{5}$

15. Let $a = (1, 1, 0)$, $b = (0, 1, 2)$. A unit vector c lies in their span, is perpendicular to a and has positive z component. Find $c \cdot b$.

(A) $\sqrt{5}$ (B) $-\frac{3\sqrt{2}}{2}$

(C) $\frac{3\sqrt{2}}{2}$ (D) $\frac{9}{2}$

16. Let $a = (1, 2, 0)$, $b = (1, 0, 2)$. Vector c lies in their span and satisfies $c \cdot b = 0$, $c \cdot a = 3$. Find $|c|^2$.

(A) $\frac{24}{5}$ (B) $\frac{3}{2}$

(C) $\frac{15}{8}$ (D) $\frac{15}{4}$

17. Write $[u, v, w] = u \cdot (v \times w)$. Vectors satisfy $[a, b, c] = 3$ and $[a + \lambda b, b + \lambda c, c + \lambda a] = 27$, with $\lambda > 0$. Find $[a + \lambda b, b - c, \lambda a + c]$.

(A) -9 (B) -3

(C) 27 (D) 9

18. Write $[u, v, w] = u \cdot (v \times w)$. Suppose $[a, b, c] = -2$ and vectors $2a + b$, $a + 2b + c$, $b + tc$ are linearly dependent. Find $[a + tb, b + tc, c + ta]$.

(A) $-\frac{70}{27}$ (B) $-\frac{16}{27}$

(C) $-\frac{35}{27}$ (D) $\frac{70}{27}$

19. Vectors a, b have lengths 3,4 and $|a \times b| = 6$. Their angle is obtuse. Find $|2a + b|^2$.

(A) 16 (B) $24\sqrt{3} + 52$

(C) $52 - 24\sqrt{3}$ (D) $52 - 12\sqrt{3}$

20. Vectors a, b satisfy $|a|^2 = 5$, $|b|^2 = 10$ and $a \cdot b = 1$. Find the area of a triangle whose two sides from one vertex are $a + b$ and $2a - b$.

(A) $\frac{7}{2}$ (B) 7

(C) 21 (D) $\frac{21}{2}$

21. Vectors a, b satisfy $|a| = 2$, $|b| = 3$ and $a \cdot b = 3$. Let u be the unit internal angle bisector in the direction $a/2 + b/3$. Find the signed scalar projection of $2a - b$ onto u.

(A) $\sqrt{3}$ (B) $\frac{3}{2}$

(C) $-\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2}$

22. Vectors a, b satisfy $|a| = 3$, $|b| = 4$ and $a \cdot b = 6$. Let u be the unit vector in direction $a/3 - b/4$. Find the signed scalar projection of $a + b$ onto u.

(A) -1 (B) $-\frac{1}{2}$

(C) 1 (D) $\frac{1}{2}$

23. Unit vectors a, b satisfy $|a + 2b|^2 = 7$. Find $((a \times b) \times (2a - b)) \cdot (a + 3b)$.

(A) $\frac{21}{4}$ (B) $\frac{7}{4}$

(C) $\frac{21}{2}$ (D) $-\frac{21}{4}$

24. Unit vectors a, b make an obtuse angle and satisfy $|a \times b|^2 = 3/4$. Find $|a \times (b \times a) - b \times (a \times b)|^2$.

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$

(C) $\frac{9}{4}$ (D) $\frac{3}{2}$

25. Let $A = (1, 0, 0)$, $B = (0, 2, 0)$, $C = (0, 0, 3)$. P divides BC internally with $BP : PC = 2 : 1$, and Q is the midpoint of AC. Find $\overrightarrow{AP} \cdot \overrightarrow{BQ}$.

(A) $\frac{5}{6}$ (B) $\frac{7}{6}$
(C) $\frac{7}{3}$ (D) $-\frac{7}{6}$

26. A tetrahedron has vertices $A = (0, 0, 0)$, $B = (2, 0, 0)$, $C = (0, 3, 0)$, $D = (0, 0, 4)$. P is the centroid of BCD and Q divides AP with $AQ : QP = 3 : 1$. Find $|BQ|^2$.

(A) $\frac{45}{16}$ (B) $\frac{49}{16}$
(C) $\frac{61}{4}$ (D) $\frac{61}{16}$

27. For real t, vectors $a = (1, t, 0)$, $b = (0, 1, t)$ and $c = (-t, 0, 1)$ are coplanar. Find $|a \times b|$.

(A) $\sqrt{2}$ (B) $\sqrt{3}$
(C) 2 (D) 3

28. For real t, vectors $a = (t, 1, 1)$, $b = (1, t, 1)$, $c = (1, 1, t)$ are coplanar, but a and b are not parallel. Find $a \cdot b$.

(A) -6 (B) -3
(C) 0 (D) 3

29. A unit vector n is perpendicular to $(2, -1, 0)$ and $(2, 0, -1)$ and has positive z component. Let $v = (3, 4, 0)$ and $w = v - 2(v \cdot n)n$. Find the y component of w.

(A) $\frac{14}{9}$ (B) $\frac{8}{9}$
(C) $-\frac{44}{9}$ (D) $-\frac{8}{9}$

30. A vector $v = (1, -2, 2)$ is reflected to $w = (-1, 2, 2)$ across a plane through the origin. Let n be a unit normal with positive x component. Find the cosine of the angle between v and n.

(A) $-\frac{\sqrt{5}}{3}$ (B) $\frac{\sqrt{5}}{3}$
(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

31. A unit vector u is perpendicular to $(1, 1, 1)$. Find the largest value of $(1, 2, 3) \cdot u$.

(A) $\sqrt{6}$ (B) $\sqrt{14}$
(C) 2 (D) $\sqrt{2}$

32. A unit vector u has z component $1/2$. Find the largest possible value of $(3, 4, 2) \cdot u$.

(A) $1 + \frac{5\sqrt{3}}{2}$ (B) $\frac{5\sqrt{3}}{2}$
(C) $1 + 5\sqrt{3}$ (D) $1 + \frac{3\sqrt{3}}{2}$

33. Let $a = (1, 1, 0)$ and $b = (0, 1, 1)$. Find the least value of $|ua + vb - (1, 0, 1)|^2$ over real u, v.

(A) $\frac{4}{3}$ (B) 2
(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

34. Let $a = (1,1,0)$, $b = (0,1,1)$ and $p = (t,0,1)$ with $t > 0$, $|p|^2 = 5$. Find the least value of $|ua + vb - p|^2$ over real u, v.

(A) $\frac{4}{3}$ (B) 5

(C) $\frac{1}{3}$ (D) 3

35. A vector r satisfies $r \times (1,1,0) = (0,0,2)$ and $r \cdot (1,1,0) = 4$. Find $|r|^2$.

(A) 12 (B) 10

(C) 8 (D) 20

36. A vector r satisfies $r \times (1,0,1) = (1,2,-1)$ and $r \cdot (1,0,1) = 3$. Find $|r|^2$.

(A) $\frac{15}{2}$ (B) $\frac{13}{2}$

(C) 15 (D) $\frac{17}{2}$

37. Find the shortest distance between lines $r = (1,0,0) + s(1,1,0)$ and $r = (0,1,1) + t(0,1,1)$, with real s, t.

(A) $\frac{\sqrt{3}}{3}$ (B) $\sqrt{3}$

(C) $\frac{1}{3}$ (D) $\frac{2\sqrt{3}}{3}$

38. Two parallel planes are each parallel to directions $(1,1,0)$ and $(1,0,1)$. They pass through $(1,2,3)$ and $(3,1,4)$, respectively. Find their distance.

(A) $2\sqrt{3}$ (B) $\frac{2\sqrt{3}}{3}$

(C) $\sqrt{6}$ (D) $\frac{2}{3}$

39. Unit vectors a, b, c satisfy $a \cdot b = a \cdot c = 1/2$. Find the least possible value of $b \cdot c$.

(A) $-\frac{1}{2}$ (B) $-\frac{3}{4}$

(C) $\frac{1}{2}$ (D) -1

40. Unit vectors a, b, c satisfy $a \cdot b = 3/5$, $a \cdot c = 4/5$. Find the largest possible value of $|b + c|^2$.

(A) $\frac{98}{25}$ (B) $\frac{74}{25}$

(C) 4 (D) $\frac{48}{25}$

41–50 / Preserved hard practice

41. Unit vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ satisfy $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$. Find $|\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}|^2$.

(A) $\frac{35}{4}$ (B) $\frac{23}{4}$

(C) $\frac{27}{4}$ (D) $\frac{31}{4}$

42. Vectors satisfy $|\mathbf{a}| = 2$, $|\mathbf{b}| = 3$, $|\mathbf{a} + \mathbf{b}| = \sqrt{19}$. Find $|\mathbf{a} \times (\mathbf{a} \times \mathbf{b}) + \mathbf{b} \times (\mathbf{b} \times \mathbf{a})|^2$.

(A) 188 (B) 189

(C) 190 (D) 191

43. Unit vectors $\mathbf{a}, \mathbf{b}$ satisfy $\mathbf{a}\cdot\mathbf{b} = 1/2$. A vector $\mathbf{x}$ satisfies $\mathbf{x}\cdot\mathbf{a} = 1$, $\mathbf{x}\cdot\mathbf{b} = 2$, and $\mathbf{x}\cdot(\mathbf{a}\times\mathbf{b}) = 3$. Find $|\mathbf{x}|^2$.

(A) 18 (B) 16

(C) 17 (D) 15

44. Three unit vectors satisfy $\mathbf{a}\cdot\mathbf{b} = \mathbf{a}\cdot\mathbf{c} = 1/2$ and $|\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})| = 3/4$. Find $\mathbf{b}\cdot\mathbf{c}$.

(A) $\frac{1}{4}$ (B) $\frac{9}{4}$

(C) $\frac{5}{4}$ (D) $-\frac{3}{4}$

45. Let $\mathbf{a} = (1,2,2)$ and $\mathbf{b} = (2,-1,k)$. A vector $\mathbf{x}$ satisfies $\mathbf{x}\times\mathbf{a} = \mathbf{b}$ and $\mathbf{x}\cdot\mathbf{a} = 6$. Find $|\mathbf{x}|^2$.

(A) $\frac{59}{9}$ (B) $\frac{50}{9}$

(C) $\frac{32}{9}$ (D) $\frac{41}{9}$

46. A sphere centred at $(1,1,1)$ is tangent to the line of intersection of $x+y=1$ and $y+z=2$. Find its squared radius.

(A) $\frac{8}{3}$ (B) $\frac{2}{3}$

(C) $-\frac{1}{3}$ (D) $\frac{5}{3}$

47. For each real t such that $(t,1,0)$ and $(1,t,0)$ make an angle of 60°, form $1/(t^2+1)$. Find the sum of these values.

(A) 2 (B) 3

(C) 1 (D) 0

48. For $t \in [0, 2\pi)$, let $\mathbf{a} = (\cos t, \sin t, 0)$, $\mathbf{b} = (\cos 2t, \sin 2t, 1)$. How many values of t satisfy $|\mathbf{a}\times\mathbf{b}| = \sqrt{7}/2$ and $(\mathbf{a}\times\mathbf{b})\cdot\hat{k} > 0$?

(A) 4 (B) 1

(C) 2 (D) 3

49. A planar position vector $\mathbf{r} = (x,y)$ satisfies $|\mathbf{r}-(1,0)|^2 + |\mathbf{r}+(1,0)|^2 = 10$ and $\mathbf{r}\cdot(1,1) = 1$. There are two possible positions. Find the product of their squared y coordinates.

(A) $\frac{17}{4}$ (B) $\frac{13}{4}$

(C) $\frac{9}{4}$ (D) $\frac{5}{4}$

50. The determinant whose rows are $\mathbf{a}+\mathbf{b}$, $\mathbf{b}+\mathbf{c}$, $\mathbf{c}+\mathbf{a}$ is 12. Find the volume of the tetrahedron with edges $\mathbf{a}+2\mathbf{b}$, $\mathbf{b}+2\mathbf{c}$, $\mathbf{c}+2\mathbf{a}$ from one vertex.

(A) 10 (B) 8

(C) 9 (D) 11

Answer key

Vectors

M-VECT / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**D**	31	**D**	41	**C**
02	**D**	12	**C**	22	**B**	32	**A**	42	**B**
03	**B**	13	**A**	23	**A**	33	**A**	43	**B**
04	**B**	14	**C**	24	**B**	34	**D**	44	**A**
05	**D**	15	**C**	25	**B**	35	**B**	45	**D**
06	**A**	16	**C**	26	**D**	36	**A**	46	**B**
07	**C**	17	**A**	27	**B**	37	**A**	47	**C**
08	**D**	18	**A**	28	**B**	38	**B**	48	**C**
09	**A**	19	**C**	29	**D**	39	**A**	49	**C**
10	**D**	20	**D**	30	**B**	40	**A**	50	**C**

Chapter 25

3D Geometry

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The line of intersection of $x+y=1$ and $y+z=1$ has direction ratios $(a,1,b)$. Find $a+b$.

(A) -1 (B) -2

(C) 0 (D) -3

02. The intersection of sphere $x^2+y^2+z^2=25$ and plane $z=3$ is a circle of radius

(A) 5 (B) 6

(C) 3 (D) 4

03. Point $(1,2,k)$ lies on $x^2+y^2+z^2=9$, with $k>0$. Find k.

(A) 1 (B) 2

(C) 3 (D) 4

04. The distance from origin to line $\mathbf{r}=(1,2,3)+t(1,1,1)$ is

(A) $-1+\sqrt{2}$ (B) $\sqrt{2}+2$

(C) $\sqrt{2}$ (D) $1+\sqrt{2}$

05. The distance between parallel lines $(t,0,0)$ and $(s,3,4)$ is

(A) 4 (B) 5

(C) 7 (D) 6

06. The sum of positive intercepts of plane $2x+3y+6z=12$ is

(A) 12 (B) 11

(C) 14 (D) 13

07. Two lines have direction vectors $(1,2,2)$ and $(2,1,2)$. Cosine of their acute angle is

(A) $\frac{26}{9}$ (B) $\frac{17}{9}$

(C) $-\frac{1}{9}$ (D) $\frac{8}{9}$

08. The tetrahedron with vertices $(0,0,0),(2,0,0),(0,3,0),(0,0,4)$ has volume

(A) 6 (B) 4

(C) 3 (D) 5

09. How many points of line (t,t,t) lie on sphere $x^2+y^2+z^2=3$?

(A) 1 (B) 2

(C) 3 (D) 4

10. Projection of plane $x+y+z=6$ onto the xy-plane along the z-axis contains the line where $z=0$. That line cuts coordinate axes with triangle area

(A) 20 (B) 19

(C) 18 (D) 17

11–40 / New middle-to-hard practice

11. The line $r=(1,0,1)+t(1,2,-1)$ meets $2x-y+z=k$ at Q. Its distance from $P=(1,0,1)$ is $2\sqrt{6}$, and Q has negative z coordinate. Find k.

(A) 1 (B) 5

(C) 3 (D) -1

12. The line $r=(0,1,0)+t(2,-1,2)$ meets $x+2y+2z=d$ at Q, at distance 3 from $(0,1,0)$. Q has negative x coordinate. Find the distance of this plane from the origin.

(A) $\frac{1}{3}$ (B) $\frac{4}{3}$

(C) 2 (D) $\frac{2}{3}$

13. The acute angle between planes $tx+y=2$ and $x=0$ is $\pi/6$, with $t<0$. Find the distance of $(1,1,0)$ from the first plane.

(A) $-\frac{1}{2}+\frac{\sqrt{3}}{2}$ (B) $1+\sqrt{3}$

(C) $\frac{1}{2}+\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

14. The acute angle between planes $tx+y+z=3$ and $x=0$ has cosine $\sqrt{2/3}$, with $t>0$. Find the distance of $(1,2,3)$ from the first plane.

(A) 2 (B) $\frac{4\sqrt{6}}{3}$

(C) $\frac{2\sqrt{6}}{3}$ (D) $\frac{\sqrt{6}}{3}$

15. At the closest pair of points on lines $r = (1,0,0) + u(1,1,0)$ and $r = (0,1,1) + v(0,1,1)$, find $u + v$.

(A) 2 (B) $-\frac{2}{3}$

(C) $-\frac{4}{3}$ (D) -2

16. P on $r = (1,2,0) + u(1,0,1)$ is closest to the line $r = (0,0,2) + v(0,1,1)$. Find the sum of the coordinates of P.

(A) $\frac{13}{3}$ (B) 5

(C) $\frac{11}{3}$ (D) $\frac{7}{3}$

17. P is $(2,1,3)$. Reflect P first in $x + y = 2$, then reflect the resulting point in $y + z = 1$ to obtain Q. Find PQ^2.

(A) 6 (B) 18

(C) 10 (D) 14

18. Reflect $(2,0,1)$ first in $x = y$ and then in $x + 2z = 1$, obtaining Q. Find the squared distance of Q from the origin.

(A) 5 (B) $\frac{52}{5}$

(C) $\frac{21}{5}$ (D) $\frac{26}{5}$

19. A sphere of radius 3 has centre $(t,1,2)$, with $t > 1$. Its section by $x + y + z = 4$ has radius 2. Find the distance of its centre from the plane $x = 1$.

(A) 15 (B) $\sqrt{5}$

(C) $3\sqrt{5}$ (D) $\sqrt{15}$

20. A sphere has centre $(t,0,0)$ and passes through $(0,3,0)$. Its section by $x = 2$ is a circle of area 13π. Find the squared sphere radius.

(A) 25 (B) 17

(C) 9 (D) 13

21. The line $r = (0,0,k) + t(1,2,2)$ is tangent to $x^2 + y^2 + z^2 = 9$, where $k > 0$. Find the z coordinate of its point of contact.

(A) $\sqrt{5}$ (B) $\frac{9\sqrt{5}}{5}$

(C) $\frac{3\sqrt{5}}{5}$ (D) $-\sqrt{5}$

22. The line $(x,y,z) = (0,3,0) + u(1,k,0)$ is tangent to $(x-1)^2 + y^2 + z^2 = 4$, with $k > 0$. Find k.

(A) $1 + \frac{2\sqrt{6}}{3}$ (B) $1 + \sqrt{6}$

(C) $\frac{2\sqrt{6}}{3}$ (D) $1 - \frac{2\sqrt{6}}{3}$

23. A plane contains the entire line $r = (1,0,1) + t(1,1,0)$ and the point $(0,1,2)$. Find its distance from the origin.

(A) $\sqrt{6}$ (B) $\frac{3}{2}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{6}}{2}$

24. A plane contains $r = (0,1,1) + t(1,0,1)$ and $(1,2,0)$. Find the distance of $(1,1,1)$ from this plane.

(A) $\frac{\sqrt{6}}{3}$ (B) $\frac{\sqrt{6}}{6}$

(C) $\frac{1}{6}$ (D) $\sqrt{6}$

25. A line has direction $(1,t,1)$ and is perpendicular to a line of direction $(1,1,-3)$. Find its acute angle, in radians, with the plane $x+y=1$.

(A) 0 (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

26. A line with direction $(1,a,b)$ is parallel to $x+y+z=0$ and perpendicular to direction $(1,2,-1)$. Find the square of the scalar projection of $(2,1,0)$ onto this line.

(A) $\frac{8}{7}$ (B) $\frac{16}{7}$

(C) $\frac{9}{7}$ (D) $\frac{4}{7}$

27. Find the least value of $x^2+y^2+z^2$ subject to $x+y+z=3$ and $x-y=1$.

(A) 3 (B) $\frac{7}{2}$

(C) $\frac{9}{2}$ (D) 5

28. Find the squared distance from the origin to the line of intersection of $x+2y+z=1$ and $x-y+z=4$.

(A) $\frac{13}{2}$ (B) $\frac{11}{2}$

(C) $\frac{9}{2}$ (D) 5

29. A sphere has centre $(t,t,1)$ and passes through $(1,0,0)$ and $(0,2,1)$. Find its radius.

(A) 2 (B) 1

(C) $\sqrt{2}$ (D) $\sqrt{3}$

30. A sphere has centre $(t,1,0)$, passes through the origin and is tangent to $x=3$. Find its radius.

(A) 3 (B) $\frac{4}{3}$

(C) $\frac{25}{9}$ (D) $\frac{5}{3}$

31. The lines $r=(s,s,0)$ and $r=(1+u,2u,k+u)$ intersect at P. Find $k+x_P$.

(A) 2 (B) 1

(C) -1 (D) 3

32. The lines $r=(1+s,2-s,s)$ and $r=(u,1+2u,k-u)$ intersect at P. Find $|OP|^2$.

(A) $\frac{17}{3}$ (B) 6

(C) $\frac{50}{9}$ (D) $\frac{19}{3}$

33. P is $(t, 0, 0)$ with $t > 2$. It is equidistant from planes $x + y - 2z = 1$ and $2x - y + z = 4$. Find this common distance.

(A) $\frac{\sqrt{6}}{3}$ (B) $\frac{\sqrt{6}}{9}$

(C) $\sqrt{6}$ (D) $\frac{2}{3}$

34. P is $(t, 0, 0)$ with $0 < t < 3$. It is equidistant from $x + y + z = 0$ and $2x - y + 2z = 3$. Find OP.

(A) $6 - 3\sqrt{3}$ (B) $2\sqrt{3}$

(C) $3 - \sqrt{3}$ (D) $3\sqrt{3} + 6$

35. The plane $ax + by + z = 6$ contains the entire line $r = (1, 1, 1) + t(1, -1, 2)$. Find the distance of the plane from the origin.

(A) $\frac{12\sqrt{62}}{31}$ (B) $\frac{6\sqrt{31}}{31}$

(C) $\frac{6\sqrt{62}}{31}$ (D) $\frac{3\sqrt{62}}{31}$

36. A plane $ax + by + z = 4$ passes through $(1, 1, 1)$ and is perpendicular to $x + 2y + 3z = 1$. Find the distance of $(0, 0, 1)$ from it.

(A) $\frac{2\sqrt{118}}{59}$ (B) $\frac{3\sqrt{106}}{106}$

(C) $\frac{3\sqrt{118}}{118}$ (D) $\frac{\sqrt{118}}{3}$

37. The tetrahedron with vertices O, $(t, 1, 1)$, $(1, t, 1)$ and $(1, 1, t)$ has volume $2/3$, where $t > 1$. Find $t^2 + 2$.

(A) 3 (B) 6

(C) 9 (D) 4

38. A tetrahedron has vertices $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$ and (t, t, t). Its volume is 1 and $t > 0$. Find t.

(A) $\frac{6}{11}$ (B) $\frac{11}{12}$

(C) $\frac{12}{11}$ (D) $\frac{18}{11}$

39. On the circle defined by $x^2 + y^2 + z^2 = 9$ and $x + y + z = 3$, find the largest possible x coordinate.

(A) $\frac{5}{2}$ (B) $1 + \sqrt{6}$

(C) 3 (D) 2

40. On the circle $(x - 1)^2 + y^2 + z^2 = 9$, $x + y + z = 1$, find the largest possible value of $2x - y + z$.

(A) $2 + \sqrt{42}$ (B) $2 + \sqrt{14}$

(C) $2 + 3\sqrt{6}$ (D) $\sqrt{42}$

41–50 / Preserved hard practice

41. The reflection of $(1,2,3)$ in $x+2y+2z=k$ lies on $x+y+z=0$. Find the distance between the point and its image.

(A) $\frac{13}{5}$ (B) $\frac{18}{5}$
(C) $\frac{28}{5}$ (D) $\frac{23}{5}$

42. A sphere passes through the origin, has centre $(1+t,0,t)$, and is tangent to $z=2$. Two spheres satisfy these conditions. Find the product of their radii.

(A) 13 (B) 14
(C) 15 (D) 12

43. Find the squared distance from $(1,0,1)$ to the line $x+y+z=1$, $x-y+2z=2$.

(A) $\frac{5}{14}$ (B) $\frac{33}{14}$
(C) $-\frac{9}{14}$ (D) $\frac{19}{14}$

44. The common perpendicular segment between lines $(1,0,1)+t(1,1,0)$ and $(0,1,0)+s(1,0,1)$ has midpoint M. Find $x_M+y_M+z_M$.

(A) $\frac{9}{2}$ (B) $\frac{5}{2}$
(C) $\frac{7}{2}$ (D) $\frac{11}{2}$

45. Two planes through the intersection of $x+2y-z=3$ and $2x-y+z=1$ are tangent to the unit sphere centred at the origin. Find the cosine of the acute angle between these tangent planes.

(A) $-\frac{31}{33}$ (B) $\frac{35}{33}$
(C) $\frac{68}{33}$ (D) $\frac{2}{33}$

46. Lines $(1,0,0)+t(a,1,1)$ tangent to the sphere $x^2+y^2+z^2=1/2$ have two possible real values of a. Find the distance between their tangency points.

(A) 1 (B) 2
(C) 0 (D) 3

47. A sphere has centre $(1,2,c)$. Its sections by $z=0$ and $z=2$ have radii $\sqrt{5}$ and 3, respectively. Find the area divided by π of its section by $x+y+z=3$.

(A) $\frac{23}{3}$ (B) $\frac{29}{3}$
(C) $\frac{20}{3}$ (D) $\frac{26}{3}$

48. Point (x,y,z) satisfies $x+y=1$ and $z=x^2-1$. Find the minimum of $x^2+y^2+z^2$.

(A) $3-\frac{3\cdot 2^{\frac{2}{3}}}{4}$ (B) $2-\frac{3\cdot 2^{\frac{2}{3}}}{4}$
(C) $4-\frac{3\cdot 2^{\frac{2}{3}}}{4}$ (D) $1-\frac{3\cdot 2^{\frac{2}{3}}}{4}$

49. For $t \in [0, 2\pi)$, consider planes $x\cos t + y\sin t + z = 0$ and $x\cos t - y\sin t + z = 0$. How many values of t make them nonparallel and their line of intersection inclined at 45° to the z-axis?

(A) 2 (B) 8

(C) 4 (D) 0

50. A ray starts at $(1, 1, 0)$ in direction $(1, -2, 2)$, reflects in the plane $x + y = 0$ with equal incident and reflected angles, and then reaches $z = 8$. Find the total distance travelled.

(A) 14 (B) 11

(C) 12 (D) 13

Answer key

3D Geometry

M-3DGE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**A**	21	**A**	31	**B**	41	**B**
02	**D**	12	**D**	22	**A**	32	**B**	42	**A**
03	**B**	13	**C**	23	**D**	33	**A**	43	**A**
04	**C**	14	**C**	24	**B**	34	**A**	44	**C**
05	**B**	15	**D**	25	**D**	35	**C**	45	**D**
06	**A**	16	**A**	26	**A**	36	**C**	46	**A**
07	**D**	17	**D**	27	**B**	37	**B**	47	**A**
08	**B**	18	**C**	28	**B**	38	**C**	48	**B**
09	**B**	19	**D**	29	**C**	39	**C**	49	**D**
10	**C**	20	**D**	30	**D**	40	**A**	50	**C**

Chapter 26

Statistics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A population of size 3 has values $a, a+2, a+4$. Find its variance.

(A) $\frac{14}{3}$ (B) $\frac{5}{3}$

(C) $\frac{11}{3}$ (D) $\frac{8}{3}$

02. Two equally sized populations have means $2, 6$ and variances $1, 1$. Find their combined variance.

(A) 5 (B) 6

(C) 7 (D) 4

03. A population has mean 2 and variance 3. For $Y = 2X - 5$, find $E(Y^2)$.

(A) 15 (B) 12

(C) 13 (D) 14

04. A population has mean 4 and variance 5. Find the mean of $(X-1)^2$.

(A) 16 (B) 15

(C) 14 (D) 13

05. The mean of $x, 2x, 3x$ is 6. Find their population variance.

(A) 6 (B) 8

(C) 7 (D) 5

06. For n real observations with mean m and variance v, the minimum over real a of $\sum_i (x_i - a)^2$ is

(A) $nv + 1$ (B) $nv - 1$

(C) $nv + 2$ (D) nv

07. The values $1, 3, 5$ have frequencies $1, k, 1$. If their variance is 2, find k.

(A) 1 (B) 2

(C) 4 (D) 3

08. If $\sum_{i=1}^{n}(x_i - 2)^2 = 20$ and $\sum_{i=1}^{n}(x_i - 3)^2 = 15$ for $n = 5$, find the mean.

(A) 2 (B) 3

(C) 4 (D) 5

09. Find the population variance of the real roots of $x^2 - 6x + 5 = 0$.

(A) 4 (B) 6

(C) 3 (D) 5

10. The first five terms of an arithmetic progression are $2, 5, 8, 11, 14$. Find their population variance.

(A) 18 (B) 19

(C) 20 (D) 17

11–40 / New middle-to-hard practice

11. Five observations are $1, 4, 7, x, y$, where $x < y$. Their mean is 5 and population variance is 6. Find y.

(A) 8 (B) 10

(C) 5 (D) 7

12. Six observations are $2, 3, 9, 10, x, y$. Their mean is 6 and population variance is $29/3$. Find xy.

(A) 16 (B) 48

(C) 40 (D) 32

13. Ten recorded observations have mean 20 and population variance 9. Correcting exactly one observation raises the mean to 21 and population variance to 16. Find the corrected value.

(A) 24 (B) 39

(C) 29 (D) 19

14. Eight recorded observations have mean 10 and population variance 4. Correcting one value changes the mean to $21/2$ and the population variance to $11/4$. Find the originally recorded value.

(A) 11 (B) 9

(C) 7 (D) 5

15. Group A has mean 4 and unknown population variance v. Group B has mean 10 and population variance 4. Their union has mean 6 and population variance 10. Find v.

(A) 2 (B) 1

(C) 5 (D) 4

16. Group A has mean 5 and population variance 4. Group B has unknown mean m and population variance 9. Their union has mean 8 and population variance 31/2. Find m.

(A) $\frac{35}{6}$ (B) 10

(C) 12 (D) 11

17. The values 1,3,5 have respective frequencies a,2a, b. Total frequency is 16 and mean is 7/2. Find the population variance.

(A) $\frac{29}{2}$ (B) $\frac{7}{4}$

(C) $\frac{9}{2}$ (D) $\frac{9}{4}$

18. The values 0,2,5 have unknown nonnegative integer frequencies a, b, c, with total frequency 20. Their mean is 31/10 and population variance 409/100. Find c.

(A) 12 (B) 4

(C) 6 (D) 10

19. Five real observations lie in $[0, 10]$, have mean 4, and include both 0 and 10. Find the greatest possible population variance minus the least possible population variance.

(A) $\frac{32}{3}$ (B) $\frac{80}{3}$

(C) 24 (D) $\frac{40}{3}$

20. Four real observations have mean 3, median 3 and range 4. Find the greatest possible population variance.

(A) 4 (B) 8

(C) 3 (D) 2

21. Eight observations have mean 5 and population variance 9. Two new observations are added without changing the mean; the new population variance is 8. Find the absolute difference between the two added observations.

(A) 8 (B) 4

(C) 2 (D) 6

22. Six observations have mean 4 and population variance 5. One is removed, leaving population variance 96/25 and a smaller mean. Find the removed value.

(A) 9 (B) 7

(C) 1 (D) 4

23. The observations are $0, 2, 4, 6, x$ with $x \geq 6$. Their mean absolute deviation about the median is 14/5. Find the population variance.

(A) $\frac{196}{25}$ (B) $\frac{156}{5}$

(C) $\frac{296}{25}$ (D) $\frac{144}{25}$

24. The six observations are $1, 3, 3, 5, x, y$ with $5 \le x \le y$. Mean absolute deviation about the median is $11/3$ and population variance is $62/3$. Find the range.

(A) 10 (B) 13

(C) 14 (D) 9

25. A population has mean 10, standard deviation 2 and median 9. It is transformed by $Y = aX + b$, with $a > 0$. Y has mean 24 and coefficient of variation 25%. Find its median.

(A) 21 (B) 18

(C) 27 (D) 24

26. A population has mean 8, population variance 9 and median 7. Under $Y = aX + b$ with $a < 0$, the new mean is 5 and coefficient of variation is 60%. Find its median.

(A) 20 (B) 6

(C) 4 (D) 8

27. A population has mean 12. The sum of deviations from 10 is 18 and the sum of squared deviations from 10 is 81. Find its population variance.

(A) 5 (B) 7

(C) 4 (D) 9

28. A population has more than ten observations and population variance 4. Its sum of deviations from 3 is 20 and sum of squared deviations from 3 is 100. Find its mean.

(A) 7 (B) 4

(C) 5 (D) 6

29. Two aligned populations x_i, y_i have means 2,3 and population variances 1,4, respectively. The average of $(x_i - y_i)^2$ is 8. Find the population variance of the paired sums $x_i + y_i$.

(A) 5 (B) 8

(C) 3 (D) 7

30. Aligned populations x and y have means 1,4 and population variances 2,5. The average of $(x_i + y_i)^2$ is 36. Find the population variance of $2x_i - y_i$.

(A) 13 (B) 9

(C) 5 (D) 17

31. Five observations form an increasing AP, have mean 6, and root-mean-square $2\sqrt{11}$. Find the product of the smallest and largest observations.

(A) 36 (B) 20

(C) 40 (D) 32

32. Four positive observations form an increasing GP. Their mean is 15/4 and their product is 64. Find the population variance.

(A) $\frac{15}{16}$ (B) $\frac{115}{4}$

(C) $\frac{85}{4}$ (D) $\frac{115}{16}$

33. Eight observations have mean 10 and population variance 9. After removing the smallest and largest, the remaining six have mean 29/3 and population variance 56/9. Find the removed largest observation.

(A) 16 (B) 15

(C) 14 (D) 7

34. Ten observations have mean 6 and population variance 8. Removing two leaves eight observations with mean 11/2 and population variance 17/2. Find the absolute difference between the removed observations.

(A) 8 (B) 16

(C) 4 (D) 2

35. The classes $[0, 10), [10, 20), [20, 30)$ have frequencies f,10, g with total 30 and interpolated grouped median 15. Using class midpoints for variance, find the population variance. Use median $L + 10(N/2 - C)/m$, where C is preceding frequency and m median-class frequency.

(A) $\frac{200}{3}$ (B) 50

(C) $\frac{100}{3}$ (D) 100

36. Classes $[0, 10), [10, 20), [20, 30)$ have frequencies f, g,10, total 40 and interpolated grouped median 15. Using midpoints for variance, find the population variance. The grouped median is $L + 10(N/2 - C)/m$.

(A) 25 (B) 100

(C) 50 (D) $\frac{200}{3}$

37. X takes only 0,1,3, has mean 1 and population variance 3/2. Find the variance of its conditional distribution given $X > 0$.

(A) 1 (B) 2

(C) $\frac{3}{2}$ (D) $\frac{1}{2}$

38. X takes only 0,2,4, has mean 2 and variance 2. Find the variance of its conditional distribution given $X \neq 2$.

(A) 2 (B) 8

(C) 4 (D) 1

39. Ten real observations lie in $[0, 10]$, have mean 5 and population variance 10. What is the largest possible number of observations equal to 5?

(A) 5 (B) 8

(C) 7 (D) 6

40. Twelve real observations lie in $[0, 6]$, have mean 3 and population variance 3. What is the largest possible number equal to 0?

(A) 3 (B) 4

(C) 2 (D) 6

41–50 / Preserved hard practice

41. A population of 4 observations has mean 5 and variance 2. Appending 9 produces a new population. Find its variance.

(A) $\frac{129}{25}$ (B) $\frac{104}{25}$

(C) $\frac{79}{25}$ (D) $\frac{154}{25}$

42. A population of 5 observations has mean 4 and variance 8. One observation is 8. After removing it, find the new population variance.

(A) 5 (B) 4

(C) 6 (D) 7

43. Two populations have sizes $2, 3$, means $1, 6$, and variances $4, 1$. Find the combined variance.

(A) $\frac{46}{5}$ (B) $\frac{36}{5}$

(C) $\frac{41}{5}$ (D) $\frac{51}{5}$

44. A population has mean 3, variance 4, and $n = 5$. The value 5 is replaced by 0. Find the new variance.

(A) 8 (B) 5

(C) 4 (D) 6

45. For three real observations, $\sum x_i = 0$ and $\sum_{i<j} x_i x_j = -6$. Find their population variance.

(A) 6 (B) 3

(C) 4 (D) 5

46. The roots of $x^3 - 6x^2 + 9x - 2 = 0$ are three real numbers. Find their population variance.

(A) 3 (B) 2

(C) 1 (D) 4

47. For $n \geq 2$, find the variance of the population $1, 2, \ldots, n$.

(A) $\frac{n^2}{12} - \frac{13}{12}$ (B) $\frac{n^2}{12} - \frac{1}{12}$

(C) $\frac{n^2}{12} + \frac{23}{12}$ (D) $\frac{n^2}{12} + \frac{11}{12}$

48. An integer is chosen uniformly from $\{0, 1, \ldots, 7\}$. Let X be the number of ones in its three-bit binary representation. Find $\mathrm{Var}(X)$.

(A) $-\frac{1}{4}$ (B) $\frac{7}{4}$

(C) $\frac{11}{4}$ (D) $\frac{3}{4}$

49. A fair coin is tossed four times. If X is the number of heads and $Y = 3X - 2$, find $E(Y^2)$.

(A) 25 (B) 26

(C) 24 (D) 27

50. Among all populations of three real numbers with sum 6, the least possible sum of their squares is

(A) 14 (B) 11

(C) 13 (D) 12

Answer key

Statistics

M-STAT / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**A**	21	**B**	31	**B**	41	**B**
02	**A**	12	**D**	22	**B**	32	**D**	42	**A**
03	**C**	13	**C**	23	**C**	33	**B**	43	**C**
04	**C**	14	**C**	24	**B**	34	**D**	44	**C**
05	**A**	15	**B**	25	**A**	35	**A**	45	**C**
06	**D**	16	**D**	26	**B**	36	**C**	46	**B**
07	**B**	17	**D**	27	**A**	37	**A**	47	**B**
08	**B**	18	**D**	28	**B**	38	**C**	48	**D**
09	**A**	19	**D**	29	**C**	39	**D**	49	**A**
10	**A**	20	**A**	30	**C**	40	**A**	50	**D**

Chapter 27

Probability

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Two fair dice are rolled. Given that the first die is larger than the second, find the probability that the sum is 7.

(A) $\frac{1}{5}$ (B) $\frac{4}{5}$

(C) $\frac{1}{10}$ (D) $\frac{3}{5}$

02. Three fair coins are tossed. Given that at least one head occurs, find the probability of exactly two heads.

(A) $\frac{3}{7}$ (B) $\frac{4}{7}$

(C) $\frac{3}{14}$ (D) $\frac{5}{7}$

03. In an experiment, Bag I has 2 red and 1 blue and Bag II has 1 red and 3 blue; the two bags are equally likely. Given a red draw, find the probability Bag I was selected.

(A) $\frac{4}{11}$ (B) $\frac{3}{11}$

(C) $\frac{8}{11}$ (D) $\frac{19}{22}$

04. Two machines produce 60% and 40% of items, with defect rates 2% and 5%. Given an item is defective, find the probability it came from the second machine.

(A) $\frac{5}{8}$ (B) $\frac{3}{8}$

(C) $\frac{13}{16}$ (D) $\frac{5}{16}$

05. If independent events A, B satisfy $P(A) = 1/2$, $P(B) = 1/3$, find the probability that exactly one occurs.

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

06. Three cards are drawn without replacement from a standard 52-card deck. Find the probability all three are aces.

(A) $\frac{5524}{5525}$ (B) $\frac{2763}{5525}$

(C) $\frac{1}{5525}$ (D) $\frac{1}{11050}$

07. Events A, B satisfy $P(A) = P(B) = 1/2$ and $P(A \cup B) = 3/4$. Find $P(A \mid B)$.

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{3}{4}$ (D) $\frac{1}{3}$

08. Events A, B are independent with $P(A \cup B) = 3/4$ and $P(A) = 1/2$. Find $P(B)$.

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

09. Two fair dice are rolled. Find the probability their product is divisible by 6.

(A) $\frac{7}{12}$ (B) $\frac{5}{24}$

(C) $\frac{17}{24}$ (D) $\frac{5}{12}$

10. Two distinct numbers are chosen uniformly as an unordered pair from $\{1, 2, 3, 4, 5\}$. Find the probability their sum is odd.

(A) $\frac{4}{5}$ (B) $\frac{3}{10}$

(C) $\frac{2}{5}$ (D) $\frac{3}{5}$

11–40 / New middle-to-hard practice

11. Urn I has 4 red and 2 blue balls; urn II has 2 red and 4 blue. The unknown probability of selecting urn I is q. A single red draw gives posterior probability 4/5 for urn I. In a fresh trial, an urn is chosen using the same q and two balls are drawn without replacement from it. Given both are red, find the posterior probability of urn I.

(A) $\frac{6}{13}$ (B) $\frac{12}{13}$

(C) $\frac{25}{26}$ (D) $\frac{1}{13}$

12. Urn I has 4 red and 1 blue; urn II has 2 red and 3 blue. Select urn I with probability 2/5, otherwise II. Draw two balls without replacement; one is red and one blue. What is the probability that a third draw from the same urn is red?

(A) $\frac{7}{13}$ (B) $\frac{10}{13}$

(C) $\frac{6}{13}$ (D) $\frac{7}{26}$

13. Urn A has 3 red and 2 blue balls; B has 2 red and 2 blue. One uniformly random ball is transferred from A to B. Two balls are then drawn without replacement from B and both are red. Find the probability that the transferred ball was red.

(A) $\frac{9}{22}$ (B) $\frac{2}{11}$

(C) $\frac{10}{11}$ (D) $\frac{9}{11}$

14. Urn A has 3 red and 2 blue balls; B has 2 red and 2 blue. Transfer one random ball from A to B, then draw two blue balls without replacement from B. What is the probability that the next ball drawn from B is red?

(A) $\frac{7}{18}$ (B) $\frac{7}{9}$

(C) $\frac{8}{9}$ (D) $\frac{2}{9}$

15. A machine is chosen once: A with probability 3/5, B with 2/5. Conditional on that choice, successive items have independent defect probabilities 1/10 and 3/10, respectively. Exactly one of its first two items is defective. Find the probability its third item is defective.

(A) $\frac{51}{230}$ (B) $\frac{281}{460}$

(C) $\frac{51}{460}$ (D) $\frac{179}{230}$

16. Two items independently choose machine A with probability 3/5 or B with 2/5. A and B defect probabilities are 1/10 and 3/10. Given exactly one of the two items is defective, find the probability that the two items came from different machines.

(A) $\frac{191}{246}$ (B) $\frac{34}{123}$

(C) $\frac{55}{123}$ (D) $\frac{68}{123}$

17. A binomial variable has mean 3 and variance 9/4. Find the probability it is even.

(A) $\frac{4097}{8192}$ (B) $\frac{12289}{16384}$

(C) $\frac{4095}{8192}$ (D) $\frac{4097}{16384}$

18. A binomial variable X has mean 4 and variance 3. Find $P(X = 1 \mid X \leq 1)$.

(A) $\frac{3}{19}$ (B) $\frac{35}{38}$

(C) $\frac{16}{19}$ (D) $\frac{8}{19}$

19. Each odd face of a die has the same probability, and each even face has the same probability. An odd result is more likely than an even result. Two independent rolls have an odd sum with probability 12/25. Find the probability their sum is 8.

(A) $\frac{1}{15}$ (B) $\frac{13}{15}$

(C) $\frac{2}{15}$ (D) $\frac{17}{30}$

20. Each odd face of a die is equally likely, and each even face is equally likely. Odd results are more likely overall. Two independent rolls have the same parity with probability 5/8. Given their sum is 8, find the probability both are odd.

(A) $\frac{6}{7}$ (B) $\frac{1}{7}$

(C) $\frac{3}{7}$ (D) $\frac{13}{14}$

21. Three fair dice show sum 10 and at least one die shows 2. Find the probability that all three faces are different.

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{3}$

22. Four fair dice have total 10. Find the probability all four faces are different.

(A) $\frac{3}{20}$ (B) $\frac{13}{20}$

(C) $\frac{7}{10}$ (D) $\frac{3}{10}$

23. Three cards are drawn without replacement from a standard deck. Given at least one king is drawn, find the probability that exactly two kings are drawn and both kings are red.

(A) $\frac{1213}{2402}$ (B) $\frac{1189}{1201}$

(C) $\frac{12}{1201}$ (D) $\frac{6}{1201}$

24. Three cards are drawn without replacement from a standard deck. Given exactly two are red, find the probability that exactly one king is drawn.

(A) $\frac{852}{4225}$ (B) $\frac{5077}{8450}$

(C) $\frac{426}{4225}$ (D) $\frac{3373}{4225}$

25. A coin has $P(H) = 2/3$ and independent tosses. Toss until either HH or TT first appears. Find the probability that HH appears first.

(A) $\frac{37}{42}$ (B) $\frac{16}{21}$

(C) $\frac{5}{21}$ (D) $\frac{8}{21}$

26. A coin is tossed independently until the first head. The probability that the stopping index is odd is 3/4. Given that index is odd, find the probability that it is greater than 3.

(A) $\frac{1}{162}$ (B) $\frac{41}{81}$

(C) $\frac{80}{81}$ (D) $\frac{1}{81}$

27. Events A and B satisfy $P(A) = 3/5$, $P(B) = 1/2$, and the probability that exactly one occurs is 1/2. Given exactly one occurs, find the probability that A does not occur.

(A) $\frac{7}{10}$ (B) $\frac{2}{5}$

(C) $\frac{1}{5}$ (D) $\frac{3}{5}$

28. Events A and B satisfy $P(A \mid B) = 3/4$, $P(B \mid A) = 1/2$, and $P(A \cup B) = 4/5$. Find $P(\text{exactly one of A,B} \mid A \cup B)$.

(A) $\frac{3}{7}$ (B) $\frac{4}{7}$

(C) $\frac{2}{7}$ (D) $\frac{11}{14}$

29. A random variable X takes only 0,1,2, has mean 6/5 and variance 14/25. For independent copies X, Y, find $P(X + Y = 2)$.

(A) $\frac{17}{25}$ (B) $\frac{8}{25}$

(C) $\frac{33}{50}$ (D) $\frac{4}{25}$

30. X takes only 0,1,2, with mean 1 and variance 1/2. For independent copies X, Y, find $P(XY = 0 \mid X + Y \geq 1)$.

(A) $\frac{7}{10}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{1}{5}$

31. An urn has r red and 3 blue balls, where r is a positive integer. The probability of two red draws without replacement is 1/5. Find the probability that three draws without replacement contain exactly two red balls.

(A) $\frac{11}{20}$ (B) $\frac{29}{40}$

(C) $\frac{9}{20}$ (D) $\frac{9}{40}$

32. An urn has r red and 4 blue balls. Its two-draw probability of equal colours without replacement is 3/7, and a single draw is red with probability at least 1/2. Find the probability at least two of three draws without replacement are red.

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

33. X, Y are independent and uniform on $[0, 1]$. Given $X+Y \leq 1$, find the probability that $|X-Y| \leq 1/2$.

(A) $\frac{3}{8}$ (B) $\frac{7}{8}$

(C) $\frac{3}{4}$ (D) $\frac{1}{4}$

34. X, Y are independent and uniform on $[0, 1]$. Given $X+Y \leq 1$, find the probability that $\max(X, Y) \leq 3/4$.

(A) $\frac{7}{8}$ (B) $\frac{7}{16}$

(C) $\frac{1}{8}$ (D) $\frac{15}{16}$

35. A uniformly random permutation of five labels has exactly two fixed points. Find the probability that labels 1 and 2 lie in the same cycle.

(A) $\frac{7}{10}$ (B) $\frac{3}{10}$

(C) $\frac{3}{20}$ (D) $\frac{13}{20}$

36. A uniformly random permutation of six labels is known to fix neither 1 nor 2. Find the probability that it swaps 1 and 2 with each other.

(A) $\frac{11}{21}$ (B) $\frac{20}{21}$

(C) $\frac{1}{21}$ (D) $\frac{1}{42}$

37. Choose a uniformly from $\{1, 2, 3, 4\}$ and b independently from $\{1, \ldots, 6\}$. Given that $x^2-2ax+b = 0$ has two distinct real roots, find the probability both roots exceed 1.

(A) $\frac{1}{15}$ (B) $\frac{14}{15}$

(C) $\frac{1}{30}$ (D) $\frac{8}{15}$

38. Choose a uniformly from $\{1,2,3,4\}$ and b independently from $\{1,\ldots,6\}$. Given that $x^2-ax+b=0$ has real roots, find the probability they coincide.

(A) $\frac{1}{7}$ (B) $\frac{9}{14}$
(C) $\frac{5}{7}$ (D) $\frac{2}{7}$

39. An item is defective with probability 1/10. A check is positive with probability 4/5 for defective items and 1/10 for good items. Two checks are independent conditional on the item status. Given exactly one check is positive, find the probability the item is defective.

(A) $\frac{81}{97}$ (B) $\frac{113}{194}$
(C) $\frac{8}{97}$ (D) $\frac{16}{97}$

40. An item is defective with probability 1/10. Each check is positive with probability 4/5 for defective items and 1/10 for good items. Checks are independent conditional on status. Given at least one of two checks is positive, find the probability the item is defective.

(A) $\frac{32}{89}$ (B) $\frac{16}{89}$
(C) $\frac{57}{89}$ (D) $\frac{121}{178}$

41–50 / Preserved hard practice

41. A bag contains 4 red and 3 blue balls. Three are drawn without replacement. Given that at least one red is drawn, find the probability of exactly two red.

(A) $\frac{8}{17}$ (B) $\frac{9}{17}$
(C) $\frac{13}{17}$ (D) $\frac{9}{34}$

42. Two fair dice are rolled repeatedly and independently. Find the probability that a sum of 7 occurs before a sum of 8.

(A) $\frac{5}{11}$ (B) $\frac{17}{22}$
(C) $\frac{6}{11}$ (D) $\frac{3}{11}$

43. Events A, B, C are mutually independent and each has probability 1/2. Given that at least one occurs, find the probability exactly two occur.

(A) $\frac{5}{7}$ (B) $\frac{3}{7}$
(C) $\frac{3}{14}$ (D) $\frac{4}{7}$

44. A permutation of $1,2,3,4$ is selected uniformly. Find the probability that no number remains in its original position.

(A) $\frac{5}{8}$ (B) $\frac{3}{8}$
(C) $\frac{11}{16}$ (D) $\frac{3}{16}$

45. A fair coin is tossed until either HH or HT first appears as a consecutive pair. Find the probability that HH appears first.

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{3}{4}$ (D) $\frac{1}{4}$

46. A real quadratic is formed as $x^2 - bx + c$, where b, c are chosen independently and uniformly from $\{1, 2, 3, 4\}$. Find the probability of two distinct real roots.

(A) $\frac{11}{16}$ (B) $\frac{5}{16}$

(C) $\frac{5}{32}$ (D) $\frac{21}{32}$

47. Choose a, b independently and uniformly from $\{0, 1, 2\}$. Find the probability that $\begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix}$ is singular.

(A) $\frac{8}{9}$ (B) $\frac{1}{18}$

(C) $\frac{5}{9}$ (D) $\frac{1}{9}$

48. Let X be the number of heads in 3 independent fair tosses. Find $E[(X - 1)^2]$.

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) 0 (D) 1

49. A number a is chosen uniformly from $\{0, 1, 2, 3, 4\}$. Find the probability that $f(x) = x^3 - 3x + a$ has three distinct real zeros.

(A) $\frac{3}{5}$ (B) $\frac{2}{5}$

(C) $\frac{7}{10}$ (D) $\frac{1}{5}$

50. A coin has head probability $p \in [0, 1]$. For three independent tosses, the probability of exactly one head is greatest at $p =$

(A) $\frac{1}{6}$ (B) $\frac{1}{4}$

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Answer key

Probability

M-PROB / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**B**	21	**B**	31	**C**	41	**B**
02	**A**	12	**A**	22	**D**	32	**C**	42	**C**
03	**C**	13	**D**	23	**C**	33	**C**	43	**B**
04	**A**	14	**B**	24	**A**	34	**A**	44	**B**
05	**D**	15	**A**	25	**B**	35	**B**	45	**A**
06	**C**	16	**D**	26	**D**	36	**C**	46	**B**
07	**A**	17	**A**	27	**B**	37	**A**	47	**D**
08	**C**	18	**C**	28	**B**	38	**D**	48	**D**
09	**D**	19	**C**	29	**B**	39	**D**	49	**B**
10	**D**	20	**A**	30	**C**	40	**A**	50	**D**

Chapter 28

Trigonometry

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The number of solutions of $2\sin^2 x - 3\sin x + 1 = 0$ in $[0, 2\pi)$ is

(A) 2 (B) 3

(C) 6 (D) 4

02. The number of solutions of $\cos 2x = \cos x$ in $[0, 2\pi)$ is

(A) 3 (B) 2

(C) 4 (D) 5

03. If $\tan A \tan B = 1/8$ and $A + B = \pi/4$, find $\tan A + \tan B$.

(A) $\frac{5}{4}$ (B) $\frac{1}{4}$

(C) $\frac{7}{8}$ (D) $\frac{4}{3}$

04. If $\sin x + \sin y$ is maximised subject to $x + y = \pi/3$, its maximum is

(A) 3 (B) 2

(C) 0 (D) 1

05. If $\sin x + \cos x = \sqrt{3/2}$, find $\sin^4 x + \cos^4 x$.

(A) $\frac{15}{16}$ (B) $\frac{1}{2}$

(C) $\frac{3}{4}$ (D) $\frac{7}{8}$

06. Where defined, find $\frac{\sin 3x}{\sin x}$ when $\cos 2x = 1/3$.

(A) $\frac{1}{3}$ (B) $\frac{7}{3}$

(C) $\frac{5}{3}$ (D) $\frac{2}{3}$

07. For $x \in \mathbb{R}$, the largest value of $2\sin x \cos x + \cos 2x$ is

(A) $\sqrt{2}$ (B) 3

(C) 2 (D) 1

08. The roots of $t^2 - \sqrt{3/2}\,t + 1/4 = 0$ are $\sin x$ and $\cos x$ for some real x. Find $\sin 2x$.

(A) $\frac{1}{4}$ (B) $-\frac{1}{2}$

(C) 1 (D) $\frac{1}{2}$

09. Let $a_n = \sin(n\pi/2)$. Find $\sum_{n=1}^{100} a_n$.

(A) 0 (B) -1

(C) 1 (D) 2

10. For real x, the coefficient of t^2 in $(1 + t\sin x)^4$ is 3. Find $\sin^2 x$.

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$

(C) $\frac{1}{4}$ (D) 1

11–40 / New middle-to-hard practice

11. How many $x \in [0, 2\pi)$ satisfy $\sin^4 x + \cos^4 x + \sin^6 x + \cos^6 x = 11/8$?

(A) 12 (B) 8

(C) 4 (D) 6

12. Find the sum of all solutions in $[0, \pi]$ of $2(\sin^6 x + \cos^6 x) - (\sin^4 x + \cos^4 x) = 3/4$.

(A) π (B) 3π

(C) 2π (D) $\frac{3\pi}{2}$

13. Find the sum of all solutions in $[0, 2\pi]$ of $\cos x \cos(\pi/3 + x)\cos(\pi/3 - x) = 1/8$.

(A) 4π (B) 3π

(C) 6π (D) 12π

14. How many distinct solutions of $\sin 7x = \sin 3x$ lie in $[0, 2\pi]$?

(A) 14 (B) 15

(C) 13 (D) 11

15. In triangle ABC, $\tan(A/2) = 1/2$ and $\tan(B/2) = 1/3$. Find the ratio of the side opposite C to the side opposite A.

(A) $\frac{5}{3}$ (B) $\frac{4}{5}$

(C) $\frac{3}{2}$ (D) $\frac{5}{4}$

16. In triangle ABC, $\tan(A/2) = 2/5$ and $\tan(B/2) = 1/2$. Find $\cos C$.

(A) $\frac{17}{145}$ (B) $\frac{17}{81}$

(C) $\frac{64}{145}$ (D) $-\frac{17}{145}$

17. If $\sin x + \cos x = \sqrt{3/2}$ and $0 < x < \pi/4$, find $\sin 3x$.

(A) $-\frac{\sqrt{2}}{2}$ (B) $\frac{1}{2}$

(C) $\frac{\sqrt{2}}{2}$ (D) $\frac{\sqrt{3}}{2}$

18. For $0 < x < \pi/2$, $2\sin x - \cos x = 1$. Find $\cos 2x$.

(A) $-\frac{7}{25}$ (B) $-\frac{24}{25}$

(C) $\frac{24}{25}$ (D) $\frac{7}{25}$

19. Find the greatest possible value of $\sin x + \cos 2x$.

(A) $\frac{9}{8}$ (B) $\frac{7}{8}$

(C) 1 (D) $\frac{5}{4}$

20. For $0 \le x \le \pi/2$ with $\sin x \ge 9/10$, find the greatest value of $3\sin x + 4\cos x$.

(A) $\frac{2\sqrt{19}}{5} + \frac{9}{5}$ (B) $\frac{\sqrt{19}}{5} + \frac{27}{10}$

(C) 5 (D) $\frac{2\sqrt{19}}{5} + \frac{27}{10}$

21. Find the least positive period of $|\sin 2x| + \cos 3x$.

(A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$

(C) 2π (D) π

22. Find the least positive period of $|\sin(3x/2)| + \cos(5x/2)$.

(A) $\frac{2\pi}{3}$ (B) 4π

(C) $\frac{4\pi}{5}$ (D) 2π

23. If $\sin^4 x + \cos^4 x = 5/8$, find $\sin^8 x + \cos^8 x$.

(A) $\frac{73}{128}$ (B) $\frac{41}{64}$

(C) $\frac{25}{64}$ (D) $\frac{41}{128}$

24. If $\sin^6 x + \cos^6 x = 13/16$ and $\sin 2x < 0$, find $(\sin x + \cos x)^4$.

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$

(C) $\frac{1}{4}$ (D) $\frac{9}{4}$

25. The angles A, B, C of a triangle are in AP, $A < C$, and its opposite sides satisfy $a + c = \sqrt{3}\,b$. Find c/a.

(A) $\frac{3}{2}$ (B) 2

(C) $\sqrt{3}$ (D) $\sqrt{2}$

26. A triangle has angles in ratio $1:2:2$. Its greatest side divided by its least side is $(k+1)/(k-1)$, $k>1$. Find k.

(A) $1+\sqrt{5}$ (B) $2+\sqrt{5}$

(C) $2-\sqrt{5}$ (D) $\sqrt{5}$

27. How many $x \in [0, 2\pi)$ satisfy $9^{\sin^2 x} + 9^{\cos^2 x} = 6$?

(A) 2 (B) 4

(C) 6 (D) 8

28. How many $x \in [0, 2\pi)$ satisfy $81^{\sin^2 x} + 81^{\cos^2 x} = 30$?

(A) 8 (B) 4

(C) 6 (D) 12

29. Find the sum of all solutions in $[0, 2\pi)$ of $\sin 2x / \sin x = 2\cos^2 x$, considering only points where the expression is defined.

(A) π (B) 0

(C) 2π (D) 3π

30. How many solutions of $|\tan x| = \tan x + \sec x$ lie in $[0, 2\pi)$ where both sides are defined?

(A) 2 (B) 1

(C) 0 (D) 4

31. In the identity $\cos^2 x \sin 3x = a_1 \sin x + a_3 \sin 3x + a_5 \sin 5x$, find $a_1 + 2a_3 + 3a_5$.

(A) 2 (B) $\frac{5}{2}$

(C) 1 (D) $\frac{3}{2}$

32. In $\sin^2 x \cos 2x = a_0 + a_2 \cos 2x + a_4 \cos 4x$, find $a_0 + 2a_2 + 4a_4$.

(A) 0 (B) $-\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $-\frac{1}{4}$

33. The positive numbers $\tan A, \tan B$ are the roots of $t^2 - 3t + 1 = 0$, with $A, B \in (0, \pi/2)$. Find $\cos 2(A-B)$.

(A) $-\frac{1}{9}$ (B) $-\frac{5}{9}$

(C) $\frac{4}{9}$ (D) $\frac{1}{9}$

34. For acute A, B, $\tan A + \tan B = 4$ and $\tan A \tan B = 2$. Find $\cos(A-B)$.

(A) $\frac{\sqrt{17}}{17}$ (B) $\frac{3\sqrt{17}}{17}$

(C) $\frac{4\sqrt{17}}{17}$ (D) $-\frac{3\sqrt{17}}{17}$

35. For real t, evaluate $\sum_{k=0}^{6} \sin^2(t + k\pi/7)\cos^2(t + k\pi/7)$.

(A) $\frac{7}{16}$ (B) $\frac{7}{4}$

(C) $\frac{3}{4}$ (D) $\frac{7}{8}$

36. Evaluate $\prod_{k=1}^{4} 2\sin(k\pi/5)$.

(A) $\sqrt{5}$ (B) 5

(C) $\frac{5}{4}$ (D) 25

37. Choose X uniformly from $[0, 2\pi)$. Given $\sin X > 0$, find the probability that $\sin X + \cos X \geq \sqrt{3/2}$.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{6}$ (D) $\frac{1}{2}$

38. Choose X uniformly from $[0, 2\pi)$. Given $\cos 2X > 0$, find the probability that $\sin X > 1/2$.

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$

(C) $\frac{1}{12}$ (D) $\frac{1}{6}$

39. Find the least real a for which $a + \sin x + \cos 2x \geq 0$ for every real x.

(A) 3 (B) 2

(C) $\frac{9}{8}$ (D) 1

40. Find the least real a for which $a + (\sin x + \cos x)^2 - 3(\sin x + \cos x) \geq 0$ for every real x.

(A) $2 + 3\sqrt{2}$ (B) $-2 + 3\sqrt{2}$

(C) $\frac{9}{4}$ (D) $2 - \sqrt{2}$

41–50 / Preserved hard practice

41. How many solutions does $\sin 3x = \cos 2x$ have in $[0, 2\pi)$?

(A) 7 (B) 6

(C) 4 (D) 5

42. How many solutions does $\sin x + \sin 3x = \cos x$ have in $[0, 2\pi)$?

(A) 7 (B) 5

(C) 6 (D) 8

43. How many solutions does $|\sin x| + |\cos x| = \sqrt{3/2}$ have in $[0, 2\pi)$?

(A) 9 (B) 7

(C) 10 (D) 8

44. Find $\sum_{k=1}^{6} \cos(2k\pi/7)$.

(A) 1 (B) -2

(C) 0 (D) -1

45. Find $\sin(\pi/7)\sin(2\pi/7)\sin(3\pi/7)$.

(A) $\frac{\sqrt{7}}{4}$ (B) $\frac{\sqrt{7}}{8}$

(C) $\frac{7}{8}$ (D) $\frac{7}{64}$

46. An integer k is chosen uniformly from $\{0, 1, \ldots, 11\}$. Find the probability $\cos(k\pi/6) > 0$.

(A) $\frac{7}{12}$ (B) $\frac{1}{2}$

(C) $\frac{5}{12}$ (D) $\frac{1}{3}$

47. Find $\sum_{n=1}^{\infty} 2^{-n}\cos(n\pi)$.

(A) $\frac{1}{3}$ (B) -1

(C) $-\frac{1}{3}$ (D) $-\frac{1}{2}$

48. Let z run over all six roots of $z^6 = 1$. Find the sum of the positive imaginary parts of these roots.

(A) $\sqrt{3}$ (B) 1

(C) $\frac{\sqrt{3}}{2}$ (D) $2\sqrt{3}$

49. If the roots of $t^2 - t + k = 0$ can be $\sin^2 x, \cos^2 x$, the greatest possible k is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) 1 (D) 0

50. The maximum of $f(x) = \sin x \cos 2x$ on $[0, \pi/2]$ is

(A) $\frac{\sqrt{6}}{9}$ (B) $\frac{\sqrt{6}}{6}$

(C) $\frac{1}{3}$ (D) $\frac{\sqrt{3}}{9}$

Answer key

Trigonometry

M-TRIG / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**C**	31	**A**	41	**D**
02	**A**	12	**C**	22	**B**	32	**D**	42	**C**
03	**C**	13	**C**	23	**D**	33	**A**	43	**D**
04	**D**	14	**C**	24	**C**	34	**B**	44	**D**
05	**D**	15	**D**	25	**B**	35	**D**	45	**B**
06	**C**	16	**A**	26	**B**	36	**B**	46	**C**
07	**A**	17	**C**	27	**B**	37	**A**	47	**C**
08	**D**	18	**A**	28	**A**	38	**D**	48	**A**
09	**A**	19	**A**	29	**C**	39	**B**	49	**B**
10	**A**	20	**D**	30	**B**	40	**B**	50	**A**

Chapter 29

Inverse Trigonometry

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Find $\arctan(1/2) + \arctan(1/3)$.

(A) $\frac{3\pi}{4}$ (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

02. Find $\arctan 2 + \arctan 3$.

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$

(C) $\frac{3\pi}{4}$ (D) $-\frac{\pi}{4}$

03. Find $\arctan 2 - \arctan 1$.

(A) $\operatorname{atan}\left(\frac{1}{2}\right)$ (B) $\operatorname{atan}\left(\frac{1}{3}\right)$

(C) $\frac{\pi}{4}$ (D) $\operatorname{atan}(3)$

04. Find $\cos(\arcsin(3/5) + \arccos(5/13))$.

(A) $\frac{16}{65}$ (B) $-\frac{56}{65}$

(C) $-\frac{16}{65}$ (D) $\frac{56}{65}$

05. For $x < 0$, $\arctan x + \arctan(1/x)$ equals

(A) $\frac{\pi}{2}$ (B) $-\frac{\pi}{2}$

(C) $-\pi$ (D) 0

06. The real equation $\arcsin x + \arccos(2x) = \pi/2$ has solution $x =$

(A) 1 (B) 0

(C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

07. If $\arccos x = 2\arcsin x$, find x.

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{2}}{2}$

(C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

08. Find $\sin(\arcsin(1/3) + \arcsin(2/3))$.

(A) $\frac{1}{3}$ (B) $-\frac{4\sqrt{2}}{9} + \frac{\sqrt{5}}{9}$

(C) $\frac{\sqrt{5}}{9} + \frac{4\sqrt{2}}{9}$ (D) $\frac{\sqrt{5}}{9} + \frac{2\sqrt{2}}{9}$

09. Find $\sum_{k=1}^{3} (\arctan(k+1) - \arctan k)$.

(A) $\operatorname{atan}(3)$ (B) $-\frac{\pi}{4} + \operatorname{atan}(4)$

(C) $-\operatorname{atan}(2) + \operatorname{atan}(4)$ (D) $\frac{\pi}{4} + \operatorname{atan}(4)$

10. An integer x is selected uniformly from $\{-2, -1, 0, 1, 2\}$. Find the probability that $\arcsin(x/2) \geq 0$.

(A) $\frac{2}{5}$ (B) $\frac{1}{5}$

(C) $\frac{4}{5}$ (D) $\frac{3}{5}$

11–40 / New middle-to-hard practice

11. How many real $x \in [0, 4\pi]$ satisfy $\arcsin(\sin x) = \arccos(\cos x) - \pi/2$?

(A) 2 (B) 3

(C) 4 (D) 5

12. Find the sum of all solutions in $[0, 2\pi)$ of $\arctan(\tan 2x) = \arcsin(\sin x)$, using only points where the displayed tangent is defined.

(A) 3π (B) 2π

(C) $\frac{5\pi}{2}$ (D) 4π

13. Find the real solution of $\arctan x + \arctan(2x) = 3\pi/4$.

(A) $\frac{3}{4} + \frac{\sqrt{17}}{4}$ (B) $-\frac{3}{4} + \frac{\sqrt{17}}{4}$

(C) $\frac{3}{2} + \frac{\sqrt{17}}{2}$ (D) $\frac{3}{4} - \frac{\sqrt{17}}{4}$

14. Find the real solution of $\arctan x + \arctan(3x) = -3\pi/4$.

(A) $-\frac{\sqrt{7}}{2} - 1$ (B) $-\frac{\sqrt{7}}{3} - \frac{2}{3}$

(C) $-\frac{2}{3} + \frac{\sqrt{7}}{3}$ (D) $\frac{2}{3} + \frac{\sqrt{7}}{3}$

15. Find the real solution of $\arcsin x + \arcsin(2x) = \pi/3$.

(A) $\frac{\sqrt{21}}{14}$ (B) $\frac{\sqrt{3}}{5}$

(C) $\frac{\sqrt{21}}{7}$ (D) $-\frac{\sqrt{21}}{14}$

16. Find the real solution of $\arccos x + \arccos(3x) = 2\pi/3$.

(A) $\frac{\sqrt{39}}{13}$ (B) $-\frac{\sqrt{39}}{26}$

(C) $\frac{\sqrt{3}}{7}$ (D) $\frac{\sqrt{39}}{26}$

17. Let $x = k/10$, with integer $-10 \le k \le 10$. For how many k does $\arcsin(3x - 4x^3) = 3\arcsin x$ hold?

(A) 11 (B) 10

(C) 12 (D) 21

18. Let $x = k/10$, with integer $-10 \le k \le 10$. For how many k does $\arccos(4x^3 - 3x) = 3\arccos x$ hold?

(A) 5 (B) 6

(C) 11 (D) 10

19. Find the minimum of $(\arcsin x)^2 + 2(\arccos x)^2$ for $-1 \le x \le 1$.

(A) $\frac{\pi^2}{6}$ (B) $\frac{\pi^2}{4}$

(C) $\frac{\pi^2}{12}$ (D) $\frac{\pi^2}{8}$

20. Find the minimum of $3(\arcsin x)^2 + (\arccos x)^2$ for $1/2 \le x \le 1$.

(A) $\frac{3\pi^2}{16}$ (B) $\frac{\pi^2}{6}$

(C) $\frac{\pi^2}{4}$ (D) $\frac{7\pi^2}{36}$

21. A positive integer n satisfies $\tan\left(\sum_{k=1}^{n} \arctan(1/(k^2+k+1))\right) = 5/6$. Find n.

(A) 8 (B) 12

(C) 9 (D) 10

22. Evaluate $\tan\left(\sum_{k=1}^{\infty} \arctan(2/(k^2+2k+2))\right)$.

(A) -3 (B) $\frac{1}{3}$

(C) 1 (D) 3

23. For $x > 0$, solve $\arctan x + \arctan(2x) + \arctan(3x) = \pi$.

(A) $\sqrt{2}$ (B) 1

(C) $\frac{1}{2}$ (D) 0

24. For $x > 0$, $\arctan x + \arctan(2x) + \arctan(4x) = \pi$. Find x^2.

(A) $\frac{7}{4}$ (B) $\frac{7}{8}$

(C) $\frac{8}{7}$ (D) $\frac{1}{8}$

25. How many integers $k \in [-5, 5]$ belong to the real domain of $\arcsin((k-1)/(k+1)) + \ln(4-k)$?

(A) 5 (B) 3

(C) 6 (D) 4

26. How many quarter-integers $x = k/4$ belong to the real domain of $\arccos((x^2-5)/(x^2+1)) + \arcsin(\log_2 x)$?

(A) 3 (B) 2

(C) 4 (D) 5

27. How many integers $x \in [-5, 5]$ satisfy $\arctan((x-1)/(x+1)) + \arctan((x+1)/(x-1)) = \pi/2$?

(A) 10 (B) 6

(C) 9 (D) 8

28. Let $x = k/3$ with integer $-6 \le k \le 9$. How many k make $\arctan((x-2)/(x+1)) + \arctan((x+1)/(x-2)) = -\pi/2$?

(A) 8 (B) 10

(C) 9 (D) 7

29. Find $\cos(2(\arctan(3/4) + \arctan(5/12)))$.

(A) $\frac{3696}{4225}$ (B) $\frac{2047}{4225}$

(C) $-\frac{2047}{4225}$ (D) $-\frac{3696}{4225}$

30. Find $\sin(2(\arcsin(5/13) + \arccos(3/5)))$.

(A) $\frac{3713}{4225}$ (B) $\frac{2016}{4225}$

(C) $\frac{1008}{4225}$ (D) $-\frac{2016}{4225}$

31. Let $x = k/20$, $k \in \{0, 1, \ldots, 20\}$. For how many k is $\arcsin(2x-1) > \arccos x$?

(A) 6 (B) 5

(C) 4 (D) 3

32. Let $x = k/4$ with integer $-4 \le k \le 4$. For how many k do $\arccos x \ge 2\arcsin x$ and $\log_2(x+1) \ge -1$ both hold?

(A) 4 (B) 6

(C) 7 (D) 5

33. For $\theta = k\pi/6$, $k = 0, 1, \ldots, 6$, sum all seven values of $\arccos((\sqrt{3}/2)\sin\theta + (1/2)\cos\theta)$.

(A) $\frac{13\pi}{6}$ (B) $\frac{5\pi}{2}$

(C) $\frac{11\pi}{6}$ (D) $\frac{7\pi}{6}$

34. How many $x \in [-1, 1]$ satisfy $\arccos((3/5)\sqrt{1-x^2} + (4/5)x) = \arccos x + \arctan(3/4)$?

(A) 0 (B) 3

(C) 1 (D) 2

35. Find the positive real x satisfying $\arctan(\log_2 x) + \arctan(\log_4 x) = \pi/4$.

(A) $2^{-\frac{3}{2}+\frac{\sqrt{17}}{2}}$ (B) $2^{-\frac{\sqrt{17}}{2}-\frac{3}{2}}$

(C) $2^{\frac{3}{2}+\frac{\sqrt{17}}{2}}$ (D) $2^{-\frac{3}{4}+\frac{\sqrt{17}}{4}}$

36. Find the positive real x satisfying $\arcsin(\log_2 x) + \arcsin(\log_2(x/2)) = \pi/6$.

(A) $2^{\frac{1}{2}+\frac{\sqrt{5+3\sqrt{3}}}{2}}$ (B) $2^{\frac{1}{2}-\frac{\sqrt{-5+3\sqrt{3}}}{2}}$

(C) $2^{\frac{\sqrt{-5+3\sqrt{3}}}{4}+\frac{1}{4}}$ (D) $2^{\frac{\sqrt{-5+3\sqrt{3}}}{2}+\frac{1}{2}}$

37. Let $z = (1+i)^8/(1-i)^5$, with $i^2 = -1$. Find $\arccos(\operatorname{Re} z/|z|) + \arcsin(\operatorname{Im} z/|z|)$.

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$

(C) 0 (D) $-\frac{\pi}{2}$

38. Let $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ and $B = A^5$. Find $\arctan(B_{12}/B_{11})$.

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$

(C) $\frac{3\pi}{4}$ (D) $-\frac{3\pi}{4}$

39. Choose a, b independently and uniformly from $\{1,2,3,4\}$. Given $a \neq b$, find the probability that $\arctan a + \arctan b > 3\pi/4$.

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{3}{8}$ (D) $\frac{1}{4}$

40. Choose a, b independently and uniformly from $\{1,2,3,4,5\}$. Given $a \neq b$, find the probability that $\arcsin(a/5) + \arcsin(b/5) > \pi/2$.

(A) $\frac{3}{10}$ (B) $\frac{8}{25}$

(C) $\frac{1}{2}$ (D) $\frac{2}{5}$

41–50 / Preserved hard practice

41. The positive solution of $\arctan x + \arctan 2x = \pi/4$ is

(A) $\frac{1}{3}$ (B) $\frac{3}{4} + \frac{\sqrt{17}}{4}$

(C) $-\frac{3}{4} + \frac{\sqrt{17}}{4}$ (D) $-\frac{3}{2} + \frac{\sqrt{17}}{2}$

42. The positive solution of $\arctan x + \arctan 2x + \arctan 3x = \pi$ is

(A) $\sqrt{3}$ (B) 2

(C) 1 (D) $\frac{1}{2}$

43. If $\arcsin(2x) + \arcsin x = \pi/2$, find x.

(A) $\frac{1}{3}$ (B) $\frac{\sqrt{5}}{5}$

(C) $-\frac{\sqrt{5}}{5}$ (D) $\frac{1}{2}$

44. For $x > 0$, $\arccos x + \arccos 2x = 2\pi/3$. Find x.

(A) $\frac{\sqrt{21}}{14}$ (B) $\frac{\sqrt{7}}{4}$

(C) $\frac{\sqrt{3}}{4}$ (D) $\frac{\sqrt{21}}{7}$

45. For $0 < x < 1$, the expression $\arctan \frac{2x}{1-x^2}$ equals

(A) $\arctan x$ (B) $\pi - 2\arctan x$

(C) $2\arctan x$ (D) $2\arctan x - \pi$

46. If $f(x) = \arcsin\left(2x/(1+x^2)\right)$ for $0 < x < 1$, find $f\prime(1/2)$.

(A) $\frac{4}{5}$ (B) $-\frac{8}{5}$

(C) $\frac{8}{5}$ (D) $\frac{16}{5}$

47. Evaluate $\int_0^1 \arctan x \, dx$.

(A) $-\frac{\log(2)}{2} + \frac{\pi}{4}$ (B) $\frac{\log(2)}{2} + \frac{\pi}{4}$

(C) $\frac{\log(2)}{2}$ (D) $-\log(2) + \frac{\pi}{2}$

48. How many integers x satisfy $\arcsin(x/3) \leq \pi/6$ within its real domain?

(A) 5 (B) 4

(C) 6 (D) 3

49. If the real roots of $t^2 - 3t + 1 = 0$ are a, b, find $\arctan a + \arctan b$.

(A) 0 (B) $\frac{\pi}{4}$

(C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

50. Three distinct numbers a, b, c are selected uniformly as an unordered triple from $\{1/2, 1, 2, 3, 4\}$. Find the probability $\arctan a + \arctan b + \arctan c > \pi$.

(A) $\frac{1}{5}$ (B) $\frac{3}{10}$

(C) $\frac{1}{2}$ (D) $\frac{2}{5}$

Answer key

Inverse Trigonometry

M-INTR / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**D**	31	**C**	41	**C**
02	**C**	12	**A**	22	**D**	32	**D**	42	**C**
03	**B**	13	**A**	23	**B**	33	**A**	43	**B**
04	**C**	14	**B**	24	**B**	34	**C**	44	**A**
05	**B**	15	**A**	25	**D**	35	**A**	45	**C**
06	**B**	16	**D**	26	**A**	36	**D**	46	**C**
07	**C**	17	**A**	27	**D**	37	**A**	47	**A**
08	**C**	18	**B**	28	**A**	38	**B**	48	**A**
09	**B**	19	**A**	29	**C**	39	**B**	49	**D**
10	**D**	20	**D**	30	**B**	40	**D**	50	**B**

Chapter 30

Linear Programming

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For $x, y \geq 0$ subject to $2x + y \leq 10, \quad x + 2y \leq 8$, find the maximum value of $4x + 3y$.

(A) 23 (B) 21

(C) 24 (D) 22

02. For $x, y \geq 0$ subject to $x + y \leq 6, \quad 2x + y \leq 9$, find the maximum value of $5x + 4y$.

(A) 26 (B) 28

(C) 27 (D) 29

03. For $x, y \geq 0$ subject to $2x + y \leq 12, \quad x + 2y \leq 12, \quad x \leq 5$, find the maximum value of $7x + 3y$.

(A) 40 (B) 43

(C) 41 (D) 42

04. For $x, y \geq 0$ subject to $x + y \leq 7, \quad 2x + 3y \leq 18$, find the maximum value of $3x + 4y$.

(A) 27 (B) 25

(C) 26 (D) 24

05. For $x, y \geq 0$ subject to $-x - 2y \leq -8, \quad x \leq 6, \quad y \leq 5$, find the minimum value of $3x + 5y$.

(A) 19 (B) 21

(C) 22 (D) 20

06. For $x, y \geq 0$ subject to $x + y \leq 5, \quad 3x + y \leq 9$, the maximum of $4x + 2y$ occurs at a point with $x =$

(A) 1 (B) 3

(C) 4 (D) 2

07. For $x, y \geq 0$ subject to $x + y \leq 8, \quad 3x + 2y \leq 18, \quad y \leq 6$, find the maximum value of $6x + 5y$.

(A) 44 (B) 41

(C) 43 (D) 42

08. For $x, y \geq 0$, $x + y \leq 4$, the maximum of $ax + y$ equals 12 with $a > 0$. Find a.

(A) 3 (B) 5

(C) 4 (D) 2

09. For $x, y \geq 0$, maximise $kx + y$ subject to $x + y \leq 4$. The vertex $(4, 0)$ is the unique optimum exactly when $k > c$. Find c.

(A) 2 (B) 3

(C) 1 (D) 0

10. For $x, y \geq 0$ subject to $x + y \leq 5, \quad 2x + y \leq 8$, find the maximum value of $3x + 2y$.

(A) 15 (B) 14

(C) 12 (D) 13

11–40 / New middle-to-hard practice

11. Let $x, y \geq 0$, $x + y \leq 6$, and $2x + y \leq 8$. For $\lambda = k/4$ with integer $0 \leq k \leq 12$, how many λ make $(2, 4)$ the unique maximizer of $\lambda x + y$?

(A) 2 (B) 3

(C) 5 (D) 4

12. For $x, y \geq 0$, $x + y \leq 6$, $2x + y \leq 8$, how many integers $\lambda \in \{0, 1, \ldots, 8\}$ make $(4, 0)$ the unique maximizer of $\lambda x + 2y$?

(A) 4 (B) 5

(C) 3 (D) 6

13. Over $x, y \geq 0$, $x + y \leq 6$, $2x + y \leq 8$, the maximum of $\lambda x + 3y$ is 24 and is attained at infinitely many points. Find the sum of the x coordinates of the endpoints of the optimal edge.

(A) 8 (B) 6

(C) 2 (D) 4

14. Over $x, y \geq 0$, $x + 2y \leq 10$, $2x + y \leq 8$, the maximum of $\lambda x + 2y$ is attained along an edge and equals 16. Find λ.

(A) 3 (B) 4

(C) 1 (D) 2

15. For $x, y \geq 0$, $x + 2y \leq p$, and $2x + y \leq 12$, the maximum of $3x + 4y$ is 28. Find p.

(A) 12 (B) 14

(C) 16 (D) 10

16. For $x, y \geq 0$, $x + 2y \leq p$, $2x + y \leq 12$, the maximum of $5x + 3y$ is 34. Find $x + y$ at the optimum.

(A) 12 (B) 9

(C) 10 (D) 8

17. For $x, y \geq 0$, $x + 2y \leq 8$, $2x + y \leq 10$, how many integers $k \in [0, 20]$ make the extra constraint $3x + 2y \leq k$ redundant?

(A) 6 (B) 7

(C) 4 (D) 5

18. For $x, y \geq 0$, $x + 2y \leq 8$, $2x + y \leq 10$, let $a = k/3$ with integer $0 \leq k \leq 12$. For how many a does every feasible point satisfy $ax + 2y \leq 12$?

(A) 8 (B) 6

(C) 9 (D) 7

19. For $x, y \geq 0$, $x + y \geq 4$, $2x + y \geq 6$, how many integers $\lambda \in [0, 8]$ make $(2, 2)$ the unique minimizer of $\lambda x + 3y$?

(A) 3 (B) 4

(C) 2 (D) 1

20. Over $x, y \geq 0$, $x + y \geq 4$, $2x + y \geq 6$, first minimize $2x + y$. Among all its minimizers, find the least value of $4x^2 + y^2$.

(A) 18 (B) 8

(C) 20 (D) 36

21. The set $x, y \geq 0$, $2x + y \leq 6$, $x + 2y \leq 6$, $x + y \geq a$ contains exactly one point. Find a.

(A) 3 (B) 6

(C) 5 (D) 4

22. Find the area of the feasible set $x, y \geq 0$, $2x + y \leq 6$, $x + 2y \leq 6$, $x + y \geq 3$.

(A) $\frac{9}{2}$ (B) 2

(C) $\frac{3}{2}$ (D) 3

23. For $x, y \geq 0$, $2x + y \leq 7$ and $x + 2y \leq 7$, let C be the continuous maximum of $3x + 4y$ and I its maximum when x, y are integers. Find $C - I$.

(A) $\frac{5}{3}$ (B) $\frac{7}{3}$

(C) 2 (D) $\frac{4}{3}$

24. For $x, y \geq 0$, $2x + y \leq 8$ and $x + 2y \leq 8$, let C be the continuous maximum of $3x + 4y$ and I its maximum when x, y are integers. Find $C - I$.

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$

(C) $\frac{5}{3}$ (D) 1

25. Find the area of $x, y \geq 0$, $y \leq 4$, $x - y \leq 2$, and $x + y \geq 3$.

(A) $\frac{39}{4}$ (B) $\frac{55}{4}$

(C) $\frac{47}{4}$ (D) $\frac{63}{4}$

26. Over $x, y \geq 0$, $y \leq 4$, $x - y \leq 2$, $x + y \geq 3$, find the greatest value of $2x - y$.

(A) 8 (B) 10

(C) 7 (D) 9

27. A batch uses nonnegative amounts x and y, with $x + y \leq 100$, $x \geq 0.4(x + y)$, $y \geq 0.2(x + y)$, and $2x + y \leq 140$. Find the maximum of $3x + 2y$.

(A) 260 (B) 240

(C) 220 (D) 280

28. A batch uses nonnegative amounts x and y, with $x + y \leq 100$, $y \geq 0.2(x + y)$ and $x + 2y \leq 130$. At the maximum of $5x + 2y$, how much of the capacity 130 is unused?

(A) 10 (B) 0

(C) 20 (D) 30

29. For $x, y \geq 0$, $x - y \leq 2$, $y - x \leq 1$, how many integers $k \in [-5, 5]$ make $kx - 3y$ have a unique maximum?

(A) 6 (B) 9

(C) 7 (D) 8

30. Over $x, y \geq 0$, $x - y \leq 2$, $y - x \leq 1$, maximize $4x - 4y$. Among all its maximizers, find the least squared distance to $(0, 3)$.

(A) 16 (B) 10

(C) 13 (D) $\frac{25}{2}$

31. For $x, y \geq 0$, $x + 2y \leq 10$, $3x + y \leq 12$, find the maximum of $5x + 5y$.

(A) 32 (B) 30

(C) 34 (D) 35

32. For $x, y \geq 0$, $2x + y \leq a$ and $x + 3y \leq b$, the unique optimum of $5x + 5y$ is $(3, 2)$. Find the maximum of $2x + 7y$ over the same region.

(A) 21 (B) 23

(C) 24 (D) 20

33. Find the area of $x, y \geq 0$, $x + y \leq 4$, and $|x - y| \leq 2$.

(A) 12 (B) 6

(C) 8 (D) 4

34. For $x, y \geq 0$, $2 \leq x + y \leq 6$, $|x - y| \leq 2$, find the maximum of $3x + y$.

(A) 16 (B) 14

(C) 18 (D) 12

35. For $x, y \geq 0$, $x + y \leq 6$, $x \leq 4$, $y \leq 4$, first maximize $2x + 2y$. Among its maximizers, find the greatest value of xy.

(A) 16 (B) 8

(C) 9 (D) 12

36. For $x, y \geq 0$, $x + y \leq 6$, $x \leq 4$, $y \leq 4$, first maximize $x + y$. Among its maximizers, find the least value of $(x - 1)^2 + (y - 2)^2$.

(A) 5 (B) $\frac{9}{2}$

(C) 4 (D) $\frac{7}{2}$

37. A feasible triangle is bounded only by $x, y \geq 0$ and $ax + by \leq 12$, with $a, b > 0$. Its maximum x coordinate is 4 and maximum y coordinate is 6. After imposing $x \geq 1$, find the maximum of $2x + 3y$.

(A) $\frac{31}{2}$ (B) 16

(C) 18 (D) 15

38. A first-quadrant feasible triangle has its non-axis edge $ax + by \leq 18$, with $a, b > 0$. The largest x coordinate is 6 and largest y coordinate is 9. Add the condition $y \geq 2x$. Find the maximum of $4x + y$.

(A) $\frac{90}{7}$ (B) $\frac{108}{7}$

(C) $\frac{120}{7}$ (D) 18

39. Choose a, b independently and uniformly from $\{1, 2, 3, 4, 5\}$. Given that $a + b$ is odd, find the probability that $(2, 4)$ uniquely maximizes $ax + by$ over $x, y \geq 0$, $x + y \leq 6$, $2x + y \leq 8$.

(A) $\frac{1}{5}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{4}{25}$

40. Choose a, b independently and uniformly from $\{1, 2, 3, 4, 5\}$. Find the probability that $ax + by$ has infinitely many maximizers over $x, y \geq 0$, $x + y \leq 6$, $2x + y \leq 8$.

(A) $\frac{7}{25}$ (B) $\frac{2}{25}$

(C) $\frac{8}{25}$ (D) $\frac{1}{5}$

41–50 / Preserved hard practice

41. For $x, y \geq 0$ subject to $2x + 3y \leq 17$, $4x + y \leq 19$, $x + y \leq 8$, find the maximum value of $7x + 11y$.

(A) $\frac{193}{3}$ (B) $\frac{187}{3}$

(C) $\frac{184}{3}$ (D) $\frac{190}{3}$

42. For $x, y \geq 0$ subject to $3x + 2y \leq 18$, $x + 3y \leq 18$, $x \leq 5$, find the maximum value of $11x + 8y$.

(A) $\frac{486}{7}$ (B) $\frac{500}{7}$

(C) $\frac{479}{7}$ (D) $\frac{493}{7}$

43. For $x, y \geq 0$ subject to $-2x - y \leq -8$, $-x - 3y \leq -9$, $x \leq 8$, $y \leq 8$, find the minimum value of $5x + 7y$.

(A) 29 (B) 28

(C) 31 (D) 30

44. For $x, y \geq 0$, maximise $kx + 2y$ subject to $x + y \leq 6$ and $2x + y \leq 8$. The intersection $(2, 4)$ is the unique optimum for $a < k < b$. Find $a + b$.

(A) 5 (B) 6

(C) 8 (D) 7

45. For nonnegative integers x, y with $2x + 3y \leq 11$, find the maximum of $3x + 4y$.

(A) 17 (B) 15

(C) 18 (D) 16

46. Two products use resources according to $2x + y \leq 10$ and $x + 3y \leq 15$, with $x, y \geq 0$. Profit is $4x + 5y$. Find the optimal profit.

(A) 31 (B) 33

(C) 34 (D) 32

47. A feasible region has vertices $(0, 0), (4, 0), (3, 2), (0, 3)$. Find the area of the region.

(A) $\frac{19}{2}$ (B) $\frac{17}{2}$

(C) $\frac{21}{2}$ (D) $\frac{15}{2}$

48. For $x, y \geq 0$, $x + y \leq 4$, the maximum of xy is

(A) 5 (B) 3

(C) 4 (D) 6

49. Choose uniformly an integer point in $x, y \geq 0$, $x + y \leq 4$. Find the probability that it maximises $x + y$.

(A) $\frac{1}{3}$ (B) $\frac{7}{3}$

(C) $\frac{4}{3}$ (D) $-\frac{2}{3}$

50. For $x, y \geq 0$ with $x + y \leq 2$, the determinant of $\begin{pmatrix} x & 1 \\ 1 & y \end{pmatrix}$ is maximised at value

(A) 2 (B) 1

(C) 0 (D) -1

Answer key

Linear Programming

M-LPP / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**B**	21	**D**	31	**A**	41	**B**
02	**C**	12	**A**	22	**C**	32	**A**	42	**A**
03	**C**	13	**B**	23	**D**	33	**B**	43	**A**
04	**B**	14	**B**	24	**B**	34	**B**	44	**B**
05	**D**	15	**A**	25	**C**	35	**C**	45	**D**
06	**D**	16	**C**	26	**A**	36	**B**	46	**D**
07	**D**	17	**D**	27	**B**	37	**A**	47	**B**
08	**A**	18	**D**	28	**A**	38	**B**	48	**C**
09	**C**	19	**C**	29	**C**	39	**C**	49	**A**
10	**D**	20	**A**	30	**D**	40	**A**	50	**C**

Chapter 31

Mathematical Modelling

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Two pipes fill a tank in 6 h and 3 h separately. A leak empties a full tank in 6 h. With all three open at constant rates, find the fill time from empty, in hours.

(A) 4 (B) 3

(C) 2 (D) 5

02. A train 180 m long passes a pole in 12 s. At the same speed, how many seconds does it take to pass a 120 m platform?

(A) 21 (B) 20

(C) 19 (D) 22

03. A 10-litre mixture is 20% acid. How many litres of 50% acid solution must be added to obtain 30% acid, assuming additive volumes?

(A) 5 (B) 7

(C) 4 (D) 6

04. A tank contains 100 L of pure water. Each cycle removes 20 L of well-mixed liquid and replaces it with 20 L of dye solution that is 50% dye. How many litres of dye are present after two cycles?

(A) 18 (B) 19

(C) 17 (D) 20

05. A number of bacteria satisfies $N_{k+1} = 2N_k + 3$ with $N_0 = 1$. Find N_3.

(A) 28 (B) 29

(C) 31 (D) 30

06. A ride costs Rs $a + bk$ for k km. Fares are Rs 70 at 3 km and Rs 110 at 7 km. Find a, in rupees.

(A) 40 (B) 39

(C) 41 (D) 42

07. A revenue model is $R(p) = p(120 - 3p)$ for $0 \le p \le 40$. Find the price that maximises revenue.

(A) 20 (B) 22

(C) 21 (D) 19

08. A savings account receives Rs 100 at the end of each year and earns 10% annually. Starting at zero, find the balance just after the third deposit, in rupees.

(A) 333 (B) 331

(C) 332 (D) 330

09. A fair six-sided die is rolled for a game. The player receives Rs 12 on a six and Rs 0 otherwise. If entry costs Rs 3, find expected net gain, in rupees.

(A) -2 (B) -1

(C) 0 (D) 1

10. A rectangle has area $24\,\mathrm{m}^2$ and perimeter 20 m. Find the positive difference between its length and width, in metres.

(A) 1 (B) 4

(C) 2 (D) 3

11–40 / New middle-to-hard practice

11. A counter reads $N_n = Ar^n + B$, with $A > 0, r > 1$ and a constant background B. Its readings at n=0,1,2 are $120, 220, 420$. True background-free population at n=4 is

(A) 1620 (B) 1600

(C) 1680 (D) 1640

12. A decay meter reads $M(t) = Aq^{t/2} + B$, with $0 < q < 1$. Readings at times $0, 2, 4$ hours are $90, 50, 30$. Reading at 8 hours is

(A) 25 (B) 20

(C) 10 (D) 15

13. A 100 L tank initially contains 20% dye solution. Remove 20 L of mixed liquid and replace it with water. Then remove x L of mixed liquid and replace it with 50% dye solution. Final dye concentration is 30%. Find x, in L.

(A) $\frac{600}{17}$ (B) $\frac{700}{17}$

(C) 40 (D) 50

14. A full tank of unknown capacity V initially holds water. Twice, remove 10 L of mixed liquid and replace it with 40% dye solution. It then holds 7.5 L of dye. Remove 20 L of mixed liquid and replace with water. Final dye volume, in L, is

(A) $\frac{15}{4}$ (B) $\frac{15}{2}$

(C) $\frac{25}{4}$ (D) $\frac{45}{8}$

15. In successive ten-minute intervals a tank volume obeys $V_{n+1} = aV_n + b$, with constants $0 < a < 1$. Starting empty, volumes are 40 L after one interval and 64 L after two. Least number of intervals needed to exceed 90 L is

(A) 5 (B) 3

(C) 6 (D) 4

16. With a leak open, pipe A alone fills a tank in 12 h, B alone in 8 h, and both together in 4 h. All rates are constant. First fill half the tank with A and B but the leak closed; then close A and open the leak until full. Total time, in h, is

(A) $\frac{36}{7}$ (B) $\frac{40}{7}$

(C) $\frac{44}{7}$ (D) 6

17. Linear demand is 80 units at price 20 and 60 units at price 30. Unit cost is 10 and fixed cost 200. Prices may vary continuously while demand is nonnegative. Greatest profit is

(A) 1250 (B) 1400

(C) 1000 (D) 1050

18. Demand is $q = 100 - p$. Sales may not exceed 30 units; unit cost is 20 and fixed cost 100. Price must be a nonnegative multiple of 3. Demand must be nonnegative and all demand is served. Profit-maximising price is

(A) 60 (B) 72

(C) 75 (D) 69

19. A fare is $a + b\min(d, 5) + c\max(d - 5, 0)$ for distance d in km. Fares at 3,7,9 km are 76,116,132. Distance giving fare 88 is

(A) $\frac{7}{2}$ (B) 6

(C) 4 (D) $\frac{9}{2}$

20. An electricity bill has fixed charge f, rate a per unit for the first 100 units, and rate b thereafter. Bills for 80,120,160 units are 260,360,520. Greatest integer consumption with bill at most 250 is

(A) 80 (B) 70

(C) 75 (D) 60

21. A train passes a person walking towards it at 2 m/s in 10 s, and a stationary 180 m platform in 20 s. It then overtakes a 180 m train moving at 10 m/s in the same direction. Complete overtaking time, in s, is

(A) 40 (B) 44

(C) 30 (D) 20

22. Two trains take 14 s to cross in opposite directions and 70 s to pass in the same direction. The faster train is 150 m long and passes a pole in 10 s. Time for the slower train to pass a stationary 100 m platform, in s, is

(A) 30 (B) 35

(C) 25 (D) 20

23. A test detects 90% of defective items and falsely flags 10% of sound items. Overall 14% are flagged. Replacing a defective flagged item costs 1000, while a sound flagged item costs nothing. Expected replacement cost conditional on a flag is

(A) 50 (B) $\frac{1000}{7}$

(C) $\frac{2250}{7}$ (D) 450

24. Two machines fail independently, with probabilities p and q where $p \geq q$. Probability neither fails is 0.72 and probability both fail is 0.02. Given at least one fails, probability the first fails is

(A) $\frac{5}{7}$ (B) $\frac{2}{7}$

(C) $\frac{4}{7}$ (D) $\frac{1}{2}$

25. A savings balance starts at zero. Each year it earns 10% and then receives the same deposit D. Just after year 2 the balance is 2100. Immediately after the year 3 deposit,1000 is withdrawn. Balance just after the year 4 deposit is

(A) 4641 (B) 3641

(C) 3410 (D) 3541

26. A 10000 loan earns 10% annual interest before the same annual repayment P. Balance after two repayments is 7900. Immediately after repayment 3, an extra 1000 is paid. Balance after repayment 4 is

(A) 4459 (B) 4259

(C) 5359 (D) 4369

27. On a 400 m track, two runners starting together meet after 40 s when running oppositely, and after 200 s when running in the same direction. The faster starts 100 m behind the slower in the same direction. Distance covered by the faster runner before the first catch, in m, is

(A) 400 (B) 200

(C) 300 (D) 250

28. Runners A, B start together on a 400 m track in the same direction. A runs at 6 m/s but rests 10 s immediately after each completed lap; B runs continuously at 4 m/s. First subsequent meeting occurs at time, in s,

(A) 200 (B) 300

(C) 290 (D) 230

29. Two machines process every item in order A then B, with times 2 and 3 min per item. They start empty at t=0; transfer is immediate. B must pause 5 min after each four completed items. A may continue into unlimited intermediate storage. Greatest number completed by 50 min is

(A) 14 (B) 11

(C) 12 (D) 13

30. Machines A, B can make at most 40 and 30 raw units per hour. Good fractions are 90% and 80%. Inspection capacity is 50 raw units per hour, and at least 16 good units per hour must come from B. Fractional rates are allowed. Greatest total good-unit rate is

(A) 42 (B) 44

(C) 46 (D) 43

31. A rectangular floor has perimeter 40 m and area 96 square metres. A uniform inner border leaves half the original area uncovered. Border width, in m, is

(A) $5 - \sqrt{13}$ (B) $5 - \sqrt{19}$

(C) 2 (D) $\sqrt{13} + 5$

32. An open box is made by cutting squares of side x from all corners of a 20 cm square sheet and folding up the sides. Manufacturing requires $4 \leq x \leq 8$ cm. Greatest box volume, in cubic cm, is

(A) 576 (B) 512

(C) 588 (D) 384

33. A junction receives 60 external vehicles per minute. Fraction p of all arrivals enters a loop and returns; all others exit. In steady flow the loop counter reads 100 vehicles per minute. Divert one-quarter of the would-be loop vehicles permanently out, and increase external inflow to 80. New steady loop flow is

(A) $\frac{1000}{17}$ (B) 100

(C) 75 (D) $\frac{1200}{17}$

34. A road receives uniform traffic. During 30 s of red, its queue grows by 15 vehicles. During 20 s of green, a persistent queue would fall from 40 to 10 vehicles. Consider a cycle starting empty:30 s red then 20 s green; service is immediate once the queue clears. Mean queue waiting time per arriving vehicle, in s, is

(A) 10 (B) 12

(C) 18 (D) 15

35. Mixing 2 L of solution A with 3 L of B gives 34% solute; mixing 3 L of A with 2 L of B gives 26%. Volumes add. To make 10 L of 20% solution, volume of A needed, in L, is

(A) 8 (B) 5

(C) $\frac{13}{2}$ (D) $\frac{15}{2}$

36. A full tank initially contains 30% salt solution by mass. Remove 10 kg of well-mixed solution and replace with 10 kg water, reducing concentration to 24%. Then evaporate 10 kg of water without losing salt. Water mass needed to dilute the remainder to 20%, in kg, is

(A) 15 (B) 25

(C) 10 (D) 20

37. A population follows $N(t) = 1000/(1 + Ae^{-rt})$, with $A, r > 0$. Values are $N(0) = 200$ and $N(2) = 500$. Find $N(6)$.

(A) $\frac{16000}{17}$ (B) 800

(C) $\frac{16000}{19}$ (D) 900

38. At integer times a logistic model has $N_n = K/(1 + Aq^n)$, where $K, A > 0$ and $0 < q < 1$. Observations are $N_0 = 20, N_1 = 30, N_2 = 40$. Find N_3.

(A) 60 (B) 48

(C) 45 (D) 50

39. A group of 230 people must travel in at most 5 buses. Large buses seat 50 and cost 100 each; small buses seat 30 and cost 70. Whole buses only, and empty seats are permitted. Minimum total cost is

(A) 480 (B) 460

(C) 470 (D) 500

40. One unit of food A supplies 3 protein units and 1 iron unit at cost 4. One unit of B supplies 1 protein and 2 iron at cost 3. Buy nonnegative whole units to supply at least 10 protein and 8 iron. Minimum cost is

(A) 19 (B) 21

(C) 18 (D) 20

41–50 / Preserved hard practice

41. A boat covers 24 km downstream and 40 km upstream in 7 h; it covers 36 km downstream and 24 km upstream in 6 h. Water current is constant. Find the boat speed in still water, in km/h.

(A) 12 (B) 9

(C) 10 (D) 11

42. Two workers together complete a job in 12 days. If the first works alone for 4 days and the second alone for 6 days, exactly 2/5 of the job is completed. Find the first worker's solo completion time, in days.

(A) 19 (B) 21

(C) 20 (D) 22

43. A car covers a fixed distance at a constant speed. Increasing the speed by 15 km/h reduces the time by 1 h; reducing it by 15 km/h increases the time by 2 h. Find the distance in km.

(A) 179 (B) 182

(C) 180 (D) 181

44. A 100 L tank initially contains pure water. It is filled with 10 L of pure dye while 10 L of well-mixed liquid are removed continuously, at equal constant rates over one hour. Assuming perfect mixing and unchanged volume, how many litres of dye are present after one hour?

(A) $99 - \frac{100}{e^{\frac{1}{10}}}$ (B) $101 - \frac{100}{e^{\frac{1}{10}}}$

(C) $100 - \frac{100}{e^{\frac{1}{10}}}$ (D) $102 - \frac{100}{e^{\frac{1}{10}}}$

45. A retailer sells integer n units at price Rs $(100 - n)$ each, where $0 \le n \le 100$. Total cost is Rs $(20n + 300)$ for $n \le 30$, and Rs $(15n + 500)$ for $n \ge 31$. Find the maximum profit in rupees.

(A) 1305 (B) 1308

(C) 1307 (D) 1306

46. A closed cylinder of volume $16\pi\,\text{cm}^3$ is built with uniform material cost per unit area. At the least surface area, find its radius in cm.

(A) 2 (B) 4

(C) 3 (D) 1

47. A quantity obeys $Q\prime = 3Q$ and $Q(0) = 2$. Find the time when $Q = 18$.

(A) $\frac{2\log(3)}{3}$ (B) $-1 + \frac{2\log(3)}{3}$

(C) $\frac{2\log(3)}{3} + 2$ (D) $\frac{2\log(3)}{3} + 1$

48. A blend uses divisible amounts $x, y \ge 0$ of two ingredients. Each unit supplies respectively $(2, 1)$ and $(1, 3)$ units of two nutrients. At least 8 and 9 units of these nutrients are required. Ingredient costs are Rs 5 and Rs 4 per unit. Find the minimum total cost.

(A) 24 (B) 22

(C) 25 (D) 23

49. A particle has velocity $v(t) = t^3 - 6t^2 + 8t$ m/s for $0 \le t \le 5$. Find its total distance travelled, in metres.

(A) $\frac{61}{4}$ (B) $\frac{53}{4}$

(C) $\frac{57}{4}$ (D) $\frac{65}{4}$

50. A device has three independent components, each working with probability p. It works if at least two components work. If its working probability is $1/2$ and $0 < p < 1$, find p.

(A) $\frac{5}{2}$ (B) $-\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{1}{2}$

Answer key

Mathematical Modelling

M-MMOD / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**A**	31	**A**	41	**C**
02	**B**	12	**D**	22	**A**	32	**A**	42	**C**
03	**A**	13	**B**	23	**C**	33	**D**	43	**C**
04	**A**	14	**D**	24	**A**	34	**B**	44	**C**
05	**B**	15	**A**	25	**D**	35	**D**	45	**D**
06	**A**	16	**B**	26	**B**	36	**D**	46	**A**
07	**A**	17	**D**	27	**C**	37	**A**	47	**A**
08	**B**	18	**B**	28	**C**	38	**B**	48	**D**
09	**B**	19	**C**	29	**C**	39	**C**	49	**C**
10	**C**	20	**B**	30	**D**	40	**D**	50	**D**

Chapter 32

Heights & Distances

50 questions / Single correct / Four options

Optional enrichment practice. Heights and Distances is retained as an observed engine unit for applied trigonometry enrichment. It is not asserted as a separately named official syllabus unit.

01–10 / Selected middle practice

01. A tower is seen at elevation 30°. After approaching its foot by 20 m on level ground, elevation is 60°. Height in metres is

(A) $2+10\sqrt{3}$ (B) $-1+10\sqrt{3}$

(C) $1+10\sqrt{3}$ (D) $10\sqrt{3}$

02. A tower is seen at elevation 30°, then 60° after moving 20 m toward it. Initial distance to foot in metres is

(A) 32 (B) 29

(C) 31 (D) 30

03. A tower is seen at elevation 45°. Moving 10 m toward its foot raises elevation to 60°. Height in metres is

(A) $5\sqrt{3}+14$ (B) $5\sqrt{3}+17$

(C) $5\sqrt{3}+16$ (D) $5\sqrt{3}+15$

04. From a ground point, elevation of a tower top is 60°. Retreating 10 m gives 45°. Initial horizontal distance in metres is

(A) $7+5\sqrt{3}$ (B) $5+5\sqrt{3}$

(C) $4+5\sqrt{3}$ (D) $6+5\sqrt{3}$

05. A balloon at height 30 m is seen at 30°. It moves horizontally toward the observer at unchanged height until elevation 60°. Horizontal distance moved in metres is

(A) $20\sqrt{3}$ (B) $-1+20\sqrt{3}$

(C) $1+20\sqrt{3}$ (D) $2+20\sqrt{3}$

06. A ladder 10 m initially reaches height 8 m. Its foot is moved until reached height is 6 m. Foot moves outward by how many metres?

(A) 3 (B) 2

(C) 1 (D) 4

07. A tower top is seen from two level ground points on opposite sides of its foot, at elevations 30° and 60°. Points are 40 m apart. Height in metres is

(A) $2+10\sqrt{3}$ (B) $1+10\sqrt{3}$

(C) $10\sqrt{3}$ (D) $-1+10\sqrt{3}$

08. A tower top is seen at 45° and 30° from points on opposite sides of its foot, 20 m apart. Height in metres is

(A) $-8+10\sqrt{3}$ (B) $-9+10\sqrt{3}$

(C) $-11+10\sqrt{3}$ (D) $-10+10\sqrt{3}$

09. A tower 10 m high is viewed from a ground point at distance d. If elevation satisfies $\tan\theta = 2\sin\theta$ with $0 < \theta < \pi/2$, distance in metres is

(A) $2+\frac{10\sqrt{3}}{3}$ (B) $-1+\frac{10\sqrt{3}}{3}$

(C) $\frac{10\sqrt{3}}{3}$ (D) $1+\frac{10\sqrt{3}}{3}$

10. An observer distance from a 10 m tower is chosen uniformly as an integer from 10 to 20 m on level ground. Find the probability that the top elevation is at least 30°.

(A) $\frac{4}{5}$ (B) $\frac{7}{11}$

(C) $\frac{8}{11}$ (D) $\frac{9}{11}$

11–40 / New middle-to-hard practice

11. An observer remains at unknown fixed eye height above level ground. Initially the tower top has elevation 45° and its foot depression 30°. Moving 20 m towards the foot raises the top elevation to 60°. Tower height, in m, is

(A) $20+20\sqrt{3}$ (B) $10\sqrt{3}+30$

(C) $30\sqrt{3}$ (D) $20\sqrt{3}+40$

12. From a horizontal platform, an observer sees a tower foot at depression 45° and its top at elevation 30°. Moving 10 m towards the foot at unchanged eye height makes top elevation 45°. Original eye height above ground, in m, is

(A) $10+10\sqrt{3}$ (B) $5+5\sqrt{3}$

(C) 20 (D) $5\sqrt{3}+15$

13. At A, a tower top has elevation 45°. The observer walks 20 m along a straight ground path making 30° with the direction from A to the foot. At B elevation is 60°. B is reached before the path's closest point to the foot. Tower height, in m, is

(A) 30 (B) $20\sqrt{3}$

(C) 40 (D) $10\sqrt{3}$

14. A tower top is at elevation 60° from A. Walk 12 m on level ground perpendicular to the direction from A to the foot, reaching B where elevation is 30°. Tower height, in m, is

(A) 6 (B) $3\sqrt{6}$

(C) $12\sqrt{2}$ (D) $6\sqrt{3}$

15. A car moves uniformly along a ground line through a tower foot. The top has elevation 60°, then 45° four seconds later, with the foot crossed between these observations. Additional time until the receding elevation is 30°, in seconds, is

(A) $4\sqrt{3}$ (B) 4

(C) $-4+4\sqrt{3}$ (D) $-12+8\sqrt{3}$

16. A balloon at constant height is initially horizontally between observers A and B, 40 m apart. Its elevations are 45° at A and 60° at B. It then travels uniformly along AB towards B and beyond. After 10 s its elevation at A is 30°. Speed, in m/s, is

(A) $-12+8\sqrt{3}$ (B) $12-4\sqrt{3}$

(C) $4\sqrt{3}$ (D) $2\sqrt{3}$

17. A window is 18 m horizontally from another building. The lines to that building's foot and top are perpendicular. Top elevation is twice foot depression. Height of the other building, in m, is

(A) $24\sqrt{3}$ (B) $12\sqrt{3}$

(C) $18\sqrt{3}$ (D) 36

18. A window is 12 m horizontally from a 26 m building. Lines of sight from the window to its foot and top are perpendicular. The window is below half the building height. Window height above ground, in m, is

(A) 8 (B) 12

(C) 18 (D) 13

19. At horizontal distance 20 m, the upper three-quarters of a vertical pole subtends $\arctan(3/5)$. There are two possible positive pole heights. Larger height, in m, is

(A) 80 (B) 40

(C) 100 (D) 20

20. From 12 m away, the upper half of a vertical pole subtends $\arctan(1/3)$. Product of the two possible positive heights, in square metres, is

(A) 576 (B) 432

(C) 288 (D) 144

21. From the top of a 30 m tower, elevation of a second tower's top is θ and depression of its foot is 2θ. The second tower is 40 m high, and $0 < \theta < 45^\circ$. Horizontal separation, in m, is

(A) 10 (B) 30

(C) $10\sqrt{3}$ (D) $20\sqrt{3}$

22. Towers on level ground are 12 m apart. The taller is 22 m high. From the shorter top, elevation to the taller top is twice depression to its foot; the depression is below 45°. Shorter height, in m, is

(A) 8 (B) 10

(C) 12 (D) 6

23. A tower foot is the midpoint of BC in an isosceles triangle ABC, with AB=AC=20 m. Tower-top elevations at B, C are 60°, and at A 45°. Tower height, in m, is

(A) $10\sqrt{3}$ (B) 10

(C) 20 (D) $20\sqrt{3}$

24. Ground triangle ABC is right-angled at B, with AB=12 m and BC=5 m. A tower foot O lies strictly inside AB. Top elevations at A and B are 45° and 60°. Square of the tangent of its elevation at C is

(A) $\frac{26352}{13009} - \frac{5400\sqrt{3}}{13009}$ (B) $\frac{3}{4}$

(C) $\frac{432}{25} - \frac{216\sqrt{3}}{25}$ (D) $\frac{57456}{68881} - \frac{5400\sqrt{3}}{68881}$

25. Two ground stations A, B are 20 m apart and both see a tower top at elevation 45°. The angle between AB and the direction from A to its foot O is 60°. Perpendicular distance of O from AB, in m, is

(A) 20 (B) $10\sqrt{3}$

(C) 10 (D) $20\sqrt{3}$

26. Tower-top elevations at ground stations A, B are 30° and 45°. At the foot O, ground directions OA and OB are perpendicular; AB=40 m. Perpendicular distance from O to AB, in m, is

(A) $20\sqrt{3}$ (B) 20

(C) $10\sqrt{3}$ (D) 10

27. An observer and tower are 30 m apart horizontally across flat water. The tower-top reflection point is 3 m from the observer. After water falls 1 m, it is 4 m from the observer; eye and tower top remain fixed. Original tower-top height above water, in m, is

(A) $\frac{99}{5}$ (B) $\frac{66}{5}$

(C) 22 (D) 18

28. An observer's eye is 3 m above flat water. A tower-top reflection is seen 4 m horizontally from the observer towards the tower. Direct top elevation has tangent 1/2. Tower-top height above water, in m, is

(A) 9 (B) 12

(C) 15 (D) 18

29. A 6 m vertical pole casts a $6\sqrt{3}$ m shadow on horizontal ground. At the same instant another vertical pole casts a $6\sqrt{2}$ m shadow uphill along a plane inclined at 45° in the shadow direction, away from the Sun. Pole height, in m, is

(A) 12 (B) $6\sqrt{3}$

(C) $6 - 2\sqrt{3}$ (D) $2\sqrt{3} + 6$

30. A pole's shadow tip relative to its foot is initially $(6, 8)$ m, at solar elevation $30°$. Later solar elevation is $60°$ and the shadow direction has rotated $90°$. Distance between the two shadow-tip positions, in m, is

(A) $10\sqrt{2}$ (B) $\frac{10\sqrt{10}}{3}$

(C) $\frac{20}{3}$ (D) $\frac{40}{3}$

31. A straight tower leans $30°$ from vertical in the vertical plane of two observers on opposite sides of its foot. Their ground separation is 30 m and top elevations $45°, 30°$. The top projection lies between them. Actual tower length, in m, is

(A) $-15 + 15\sqrt{3}$ (B) 20

(C) $30 - 10\sqrt{3}$ (D) $10\sqrt{3}$

32. A tower foot is 40 m east of A and 20 m west of B on one ground line. Its top leans towards A and projects between A and the foot. Top elevations from A, B are $45°, 60°$. Horizontal displacement of top from foot, in m, is

(A) $50 - 20\sqrt{3}$ (B) $-50 + 30\sqrt{3}$

(C) $-30 + 20\sqrt{3}$ (D) 10

33. A plane flies horizontally towards an observer at fixed height and constant speed. When elevation is $45°$, it increases at $1/20$ rad/s. Time needed for elevation to rise from $30°$ to $60°$, in seconds, is

(A) $10\sqrt{3}$ (B) 20

(C) $\frac{20\sqrt{3}}{3}$ (D) $\frac{40\sqrt{3}}{3}$

34. A bird approaches an observer horizontally at constant height and speed. Initially elevation is $45°$; one second later its tangent is 2. The bird has not passed overhead. Angular rate then, in rad/s, is

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) $\frac{2}{5}$ (D) $\frac{1}{5}$

35. A straight hillside rises at $30°$. A vertical tower stands on it ahead of an observer. Top elevation is initially $45°$; after walking 10 m uphill towards the tower it becomes $60°$. Tower height above its hillside foot, in m, is

(A) $\frac{10\sqrt{3}}{3}$ (B) $5\sqrt{3}$

(C) $5 + 5\sqrt{3}$ (D) 10

36. A staircase rises 3 m per 4 m horizontal advance directly towards a tower. From its foot a tower top has elevation $45°$; after travelling 10 m along the staircase elevation is $60°$. Initial horizontal distance to the tower, in m, is

(A) $6 + 4\sqrt{3}$ (B) $2\sqrt{3} + 8$

(C) $\sqrt{3} + 9$ (D) 12

37. An observer's distance in metres from a 10 m tower is uniformly chosen from integers 1 through 20. Probability that top elevation lies from 30° to 60°, inclusive, is

(A) $\frac{3}{5}$ (B) $\frac{11}{20}$

(C) $\frac{1}{2}$ (D) $\frac{7}{10}$

38. A point is uniform by area in the ground disc of radius 20 m centred at a 10 m tower foot. Probability that top elevation lies from 30° to 60° is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

39. A pole segment extends from height 2 m to 8 m above level ground. From a ground point at variable positive distance x, its subtended angle is θ. Greatest possible $\sin\theta$ is

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$

(C) $\frac{4}{5}$ (D) $\frac{3}{4}$

40. An observer's eye is 3 m above ground. A vertical flag occupies heights 6 to 15 m. Its horizontal viewing distance x is restricted to $2 \le x \le 5$ m. Greatest tangent of the angle subtended by the flag is

(A) $\frac{9}{20}$ (B) $\frac{9}{5}$

(C) $\frac{3}{4}$ (D) $\frac{45}{61}$

41–50 / Preserved hard practice

41. An observer moves toward a tower along level ground in equal successive steps. The first two elevations are 30° and 45°, and the third position is still before the tower foot. Find the third elevation in degrees.

(A) 77 (B) 75

(C) 76 (D) 74

42. Two towers stand on level ground. At a point between them their top elevations are 30° and 60°, respectively. Moving 10 m toward the first tower makes both elevations 45°. Find the distance between the tower feet, in metres.

(A) $10\sqrt{3}+22$ (B) $10\sqrt{3}+20$

(C) $10\sqrt{3}+21$ (D) $10\sqrt{3}+19$

43. The sun elevation increases from α to $\alpha+30°$, with both angles between 0° and 90°. A vertical pole shadow becomes one quarter of its earlier length. Find the product of all possible values of $\tan\alpha$.

(A) $\frac{5}{4}$ (B) $\frac{9}{4}$

(C) $\frac{1}{4}$ (D) $-\frac{3}{4}$

44. A vertical tower stands on a hill. From a ground point at the hill base level, the tower base and top have elevations 30° and 45°. After moving 20 m horizontally toward the tower, the top elevation is 60°. Find the tower height in metres.

(A) 22 (B) 21

(C) 19 (D) 20

45. A ladder rests against a vertical wall on level ground. Moving its foot 3 m outward lowers its top 1 m. Moving the foot a further 3 m lowers the top a further 2 m. Find the ladder length in metres.

(A) $\frac{5\sqrt{26}}{2}$ (B) $2+\frac{5\sqrt{26}}{2}$

(C) $1+\frac{5\sqrt{26}}{2}$ (D) $-1+\frac{5\sqrt{26}}{2}$

46. Ground points A, B are 20 m apart, and $OA \perp OB$, where O is a tower foot. Its top elevations from A, B are $30°, 60°$. Find the tangent of its elevation from the midpoint of AB.

(A) $\frac{\sqrt{30}}{5}$ (B) $1+\frac{\sqrt{30}}{5}$

(C) $-1+\frac{\sqrt{30}}{5}$ (D) $\frac{\sqrt{30}}{5}+2$

47. ABC is an equilateral triangle of side 20 m on level ground. A tower foot O lies on AB. Its top elevations from A, B are $30°, 60°$. Find the tangent of its top elevation from C.

(A) $\frac{\sqrt{39}}{13}+1$ (B) $\frac{\sqrt{39}}{13}$

(C) $\frac{\sqrt{39}}{13}+2$ (D) $-1+\frac{\sqrt{39}}{13}$

48. A vertical tower of height 10 m stands inside a horizontal circular platform. The greatest and least top elevations seen from the rim are $60°$ and $30°$. Find the length, in metres, of the chord joining the two rim points from which elevation is $45°$.

(A) 21 (B) 22

(C) 19 (D) 20

49. A 3 m high fence stands 4 m from a vertical building wall on level ground. A ladder has its foot beyond the fence, passes over its top, and touches the building wall. Find the minimum possible ladder length, in metres.

(A) $\left(3^{\frac{2}{3}}+2\sqrt[3]{2}\right)^{\frac{3}{2}}$ (B) $2+\left(3^{\frac{2}{3}}+2\sqrt[3]{2}\right)^{\frac{3}{2}}$

(C) $-1+\left(3^{\frac{2}{3}}+2\sqrt[3]{2}\right)^{\frac{3}{2}}$ (D) $1+\left(3^{\frac{2}{3}}+2\sqrt[3]{2}\right)^{\frac{3}{2}}$

50. A regular octagonal platform has circumradius 10 m. A vertical tower of height $10\sqrt{2}$ m stands at one vertex. Two observer positions are chosen uniformly as distinct vertices among the other seven. Find the probability that the sum of their top elevation angles is at least $90°$.

(A) $\frac{3}{7}$ (B) $\frac{4}{7}$

(C) $\frac{11}{21}$ (D) $\frac{2}{3}$

Answer key

Heights & Distances

M-HTDS / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**D**	21	**C**	31	**C**	41	**B**
02	**D**	12	**D**	22	**D**	32	**B**	42	**B**
03	**D**	13	**B**	23	**A**	33	**C**	43	**C**
04	**B**	14	**B**	24	**A**	34	**C**	44	**D**
05	**A**	15	**D**	25	**B**	35	**A**	45	**A**
06	**B**	16	**A**	26	**C**	36	**C**	46	**A**
07	**C**	17	**A**	27	**A**	37	**A**	47	**B**
08	**D**	18	**A**	28	**C**	38	**B**	48	**D**
09	**C**	19	**A**	29	**D**	39	**B**	49	**A**
10	**C**	20	**C**	30	**B**	40	**D**	50	**B**